

Circular_Window_Rebound_MAXQDA_final

Regular maintenance and repair of window systems in the existing building stock provide both resource and energy savings. While planning the maintenance strategy of window systems that offer benefit in terms of two different aspects, first of all, the possible damage types should be known. Then, what is the damage to the scope of this study? The question was investigated, and the causes and types of defects in the wooden windows were listed. The proposed maintenance planning method is based on 5 tools. The tools that can be used for maintenance planning are listed in the literature. First, the defect levels of the wood windows were specified and then the AHP method was used to select among the alternative maintenance according to the severity of the defect. The study was planned according to the 6 Sigma tool. Firstly, the causes of the defects were investigated and then these defects were evaluated by assigning numbers to each defect by experts. Finally, the AHP method was used to select the right alternative among proposed maintenance alternatives according to the level of defects. By using this simple method while choosing the window maintenance strategy according to the demands, wrong decision-making will be prevented.

Code: • Circular strategy > Maintenance Weight score: 0
 Search A - Window maintenance and component repair > A-GS-03, p. 2
 Created: 22.6.2026 r. 12:54 by user, Modified: 22.6.2026 r. 12:54 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)

Regular maintenance and repair of window systems in the existing building stock provide both resource and energy savings. While planning the maintenance strategy of window systems that offer benefit in terms of two different aspects, first of all, the possible damage types should be known. Then, what is the damage to the scope of this study? The question was investigated, and the causes and types of defects in the wooden windows were listed. The proposed maintenance planning method is based on 5 tools. The tools that can be used for maintenance planning are listed in the literature. First, the defect levels of the wood windows were specified and then the AHP method was used to select among the alternative maintenance according to the severity of the defect. The study was planned according to the 6 Sigma tool. Firstly, the causes of the defects were investigated and then these defects were evaluated by assigning numbers to each defect by experts. Finally, the AHP method was used to select the right alternative among proposed maintenance alternatives according to the level of defects. By using this simple method while choosing the window maintenance strategy according to the demands, wrong decision-making will be prevented.

Code: • Mitigation > Maintenance planning Weight score: 0
 Search A - Window maintenance and component repair > A-GS-03, p. 2
 Created: 22.6.2026 r. 12:54 by user, Modified: 22.6.2026 r. 12:54 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Repair (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)

Regular maintenance and repair of window systems in the existing building stock provide both resource and energy savings. While planning the maintenance strategy of window systems that offer benefit in terms of two different aspects, first of all, the possible damage types should be known. Then, what is the damage to the scope of this study? The question was investigated, and the causes

and types of defects in the wooden windows were listed. The proposed maintenance planning method is based on 5 tools. The tools that can be used for maintenance planning are listed in the literature. First, the defect levels of the wood windows were specified and then the AHP method was used to select among the alternative maintenance according to the severity of the defect. The study was planned according to the 6 Sigma tool. Firstly, the causes of the defects were investigated and then these defects were evaluated by assigning numbers to each defect by experts. Finally, the AHP method was used to select the right alternative among proposed maintenance alternatives according to the level of defects. By using this simple method while choosing the window maintenance strategy according to the demands, wrong decision-making will be prevented.

Code: • Negative mechanism > Premature replacement Weight score: 0

Search A - Window maintenance and component repair > A-GS-03, p. 2

Created: 22.6.2026 r. 12:55 by user, Modified: 22.6.2026 r. 12:55 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Repair (Weight: 0)

Obsolescence > Physical (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Mitigation > Maintenance planning (Weight: 0)

Regular maintenance and repair of window systems in the existing building stock provide both resource and energy savings. While planning the maintenance strategy of window systems that offer benefit in terms of two different aspects, first of all, the possible damage types should be known. Then, what is the damage to the scope of this study? The question was investigated, and the causes and types of defects in the wooden windows were listed. The proposed maintenance planning method is based on 5 tools. The tools that can be used for maintenance planning are listed in the literature. First, the defect levels of the wood windows were specified and then the AHP method was used to select among the alternative maintenance according to the severity of the defect. The study was planned according to the 6 Sigma tool. Firstly, the causes of the defects were investigated and then these defects were evaluated by assigning numbers to each defect by experts. Finally, the AHP method was used to select the right alternative among proposed maintenance alternatives according to the level of defects. By using this simple method while choosing the window maintenance strategy according to the demands, wrong decision-making will be prevented.

Code: • Obsolescence > Physical Weight score: 0

Search A - Window maintenance and component repair > A-GS-03, p. 2

Created: 22.6.2026 r. 12:54 by user, Modified: 22.6.2026 r. 12:54 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Repair (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Mitigation > Maintenance planning (Weight: 0)

Regular maintenance and repair of window systems in the existing building stock provide both resource and energy savings. While planning the maintenance strategy of window systems that offer benefit in terms of two different aspects, first of all, the possible damage types should be known. Then, what is the damage to the scope of this study? The question was investigated, and the causes and types of defects in the wooden windows were listed. The proposed maintenance planning method is based on 5 tools. The tools that can be used for maintenance planning are listed in the literature. First, the defect levels of the wood windows were specified and then the AHP method was used to select among the alternative maintenance according to the severity of the defect. The study was planned according to the 6 Sigma tool. Firstly, the causes of the defects were investigated and then these defects were evaluated by assigning numbers to each defect by experts. Finally, the AHP method was used to select the right alternative among proposed maintenance alternatives according

to the level of defects. By using this simple method while choosing the window maintenance strategy according to the demands, wrong decision-making will be prevented.

Code: • Circular strategy > Repair Weight score: 0
 Search A - Window maintenance and component repair > A-GS-03, p. 2
 Created: 22.6.2026 r. 12:54 by user, Modified: 22.6.2026 r. 12:54 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)

Regular maintenance and repair of window systems in the existing building stock provide both resource and energy savings. While planning the maintenance strategy of window systems that offer benefit in terms of two different aspects, first of all, the possible damage types should be known. Then, what is the damage to the scope of this study? The question was investigated, and the causes and types of defects in the wooden windows were listed. The proposed maintenance planning method is based on 5 tools. The tools that can be used for maintenance planning are listed in the literature. First, the defect levels of the wood windows were specified and then the AHP method was used to select among the alternative maintenance according to the severity of the defect. The study was planned according to the 6 Sigma tool. Firstly, the causes of the defects were investigated and then these defects were evaluated by assigning numbers to each defect by experts. Finally, the AHP method was used to select the right alternative among proposed maintenance alternatives according to the level of defects. By using this simple method while choosing the window maintenance strategy according to the demands, wrong decision-making will be prevented.

Code: • Enabling condition > Repair information Weight score: 0
 Search A - Window maintenance and component repair > A-GS-03, p. 2
 Created: 22.6.2026 r. 12:54 by user, Modified: 22.6.2026 r. 12:54 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Repair (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)

Defects in window systems can be diagnosed not only by complex tools, tests, etc. but also by observation-based inspections with simple-to-use hand tools. The accuracy of this method can be proved by different research tools. According to the results, the defects can be rehabilitated with maintenance actions that will increase the comfort of the user and meet the expectations, and the possible damages can be prevented or postponed.

Code: • Circular strategy > Maintenance Weight score: 0
 Search A - Window maintenance and component repair > A-GS-03, p. 3
 Created: 23.6.2026 r. 8:43 by user, Modified: 23.6.2026 r. 8:43 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)

Defects in window systems can be diagnosed not only by complex tools, tests, etc. but also by observation-based inspections with simple-to-use hand tools. The accuracy of this method can be proved by different research tools. According to the results, the defects can be rehabilitated with

maintenance actions that will increase the comfort of the user and meet the expectations, and the possible damages can be prevented or postponed.

Code: • Enabling condition > Repair information Weight score: 0
Search A - Window maintenance and component repair > A-GS-03, p. 3
Created: 23.6.2026 r. 8:44 by user, Modified: 23.6.2026 r. 8:44 by user
Other codes assigned to segment:
Circular strategy > Maintenance (Weight: 0)
Circular strategy > Repair (Weight: 0)
Obsolescence > Physical (Weight: 0)
Negative mechanism > Premature replacement (Weight: 0)
Mitigation > Maintenance planning (Weight: 0)

Defects in window systems can be diagnosed not only by complex tools, tests, etc. but also by observation-based inspections with simple-to-use hand tools. The accuracy of this method can be proved by different research tools. According to the results, the defects can be rehabilitated with maintenance actions that will increase the comfort of the user and meet the expectations, and the possible damages can be prevented or postponed.

Code: • Negative mechanism > Premature replacement Weight score: 0
Search A - Window maintenance and component repair > A-GS-03, p. 3
Created: 23.6.2026 r. 8:44 by user, Modified: 23.6.2026 r. 8:44 by user
Other codes assigned to segment:
Circular strategy > Maintenance (Weight: 0)
Circular strategy > Repair (Weight: 0)
Obsolescence > Physical (Weight: 0)
Enabling condition > Repair information (Weight: 0)
Mitigation > Maintenance planning (Weight: 0)

Defects in window systems can be diagnosed not only by complex tools, tests, etc. but also by observation-based inspections with simple-to-use hand tools. The accuracy of this method can be proved by different research tools. According to the results, the defects can be rehabilitated with maintenance actions that will increase the comfort of the user and meet the expectations, and the possible damages can be prevented or postponed.

Code: • Obsolescence > Physical Weight score: 0
Search A - Window maintenance and component repair > A-GS-03, p. 3
Created: 23.6.2026 r. 8:43 by user, Modified: 23.6.2026 r. 8:43 by user
Other codes assigned to segment:
Circular strategy > Maintenance (Weight: 0)
Circular strategy > Repair (Weight: 0)
Enabling condition > Repair information (Weight: 0)
Negative mechanism > Premature replacement (Weight: 0)
Mitigation > Maintenance planning (Weight: 0)

Defects in window systems can be diagnosed not only by complex tools, tests, etc. but also by observation-based inspections with simple-to-use hand tools. The accuracy of this method can be proved by different research tools. According to the results, the defects can be rehabilitated with maintenance actions that will increase the comfort of the user and meet the expectations, and the possible damages can be prevented or postponed.

Code: • Mitigation > Maintenance planning Weight score: 0
Search A - Window maintenance and component repair > A-GS-03, p. 3
Created: 23.6.2026 r. 8:44 by user, Modified: 23.6.2026 r. 8:44 by user
Other codes assigned to segment:
Circular strategy > Maintenance (Weight: 0)
Circular strategy > Repair (Weight: 0)
Obsolescence > Physical (Weight: 0)

Enabling condition > Repair information (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)

Defects in window systems can be diagnosed not only by complex tools, tests, etc. but also by observation-based inspections with simple-to-use hand tools. The accuracy of this method can be proved by different research tools. According to the results, the defects can be rehabilitated with maintenance actions that will increase the comfort of the user and meet the expectations, and the possible damages can be prevented or postponed.

Code: • Circular strategy > Repair Weight score: 0
 Search A - Window maintenance and component repair > A-GS-03, p. 3
 Created: 23.6.2026 r. 8:43 by user, Modified: 23.6.2026 r. 8:43 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)

Defect Locations Maintenance Recommendations

HY1.Frame Components Maintenance

A1- Repair or replacement of EPDM, silicone, strip, etc. sealing materials A2- Repair / replacement of fasteners such as nail screws A3- Replacing the glazing bead
 A4- Repair / replacement of hinges, opening / closing mechanisms

HY2. Frame Maintenance

B1- General cleaning of the frame
 B2- Repairing damaged areas according to the material of the frame B3- Repair of drains and drainage B4- Removal of damaged parts of glass

HY3.Glass Maintenance

C1-Glass replacement C2-Glass repair
 C3-Installation of ventilation system

HY4. Frame-Lintel Maintenance

L1- Silicone squeezing between frame-lintel L2- Insulation if thinner than lintel wall L3- Strengthening frame fixings

L4- Placing raincoats / profiles to remove rainwater etc. HY5.Frame-Sills Maintenance

D1-Silicone squeezing between case-windowsill

D2- Installation of profile / raincoat to reduce water leakage between frame (case) and windowsill

Code: • Circular strategy > Maintenance Weight score: 0
 Search A - Window maintenance and component repair > A-GS-03, p. 11
 Created: 23.6.2026 r. 8:50 by user, Modified: 23.6.2026 r. 8:50 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Component replacement (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)

Defect Locations Maintenance Recommendations

HY1.Frame Components Maintenance

A1- Repair or replacement of EPDM, silicone, strip, etc. sealing materials A2- Repair / replacement of fasteners such as nail screws A3- Replacing the glazing bead
 A4- Repair / replacement of hinges, opening / closing mechanisms

HY2. Frame Maintenance

B1- General cleaning of the frame

B2- Repairing damaged areas according to the material of the frame B3- Repair of drains and drainage B4- Removal of damaged parts of glass
 HY3. Glass Maintenance
 C1- Glass replacement C2- Glass repair
 C3- Installation of ventilation system
 HY4. Frame- Lintel Maintenance
 L1- Silicone squeezing between frame-lintel L2- Insulation if thinner than lintel wall L3- Strengthening frame fixings
 L4- Placing raincoats / profiles to remove rainwater etc. HY5. Frame-Sills Maintenance
 D1- Silicone squeezing between case-windowsill
 D2- Installation of profile / raincoat to reduce water leakage between frame (case) and windowsill

Code: • Mitigation > Maintenance planning Weight score: 0
 Search A - Window maintenance and component repair > A-GS-03, p. 11
 Created: 23.6.2026 r. 8:51 by user, Modified: 23.6.2026 r. 8:51 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Component replacement (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Enabling condition > Repair information (Weight: 0)

Defect Locations Maintenance Recommendations

HY1. Frame Components Maintenance
 A1- Repair or replacement of EPDM, silicone, strip, etc. sealing materials A2- Repair / replacement of fasteners such as nail screws A3- Replacing the glazing bead
 A4- Repair / replacement of hinges, opening / closing mechanisms
 HY2. Frame Maintenance
 B1- General cleaning of the frame
 B2- Repairing damaged areas according to the material of the frame B3- Repair of drains and drainage B4- Removal of damaged parts of glass
 HY3. Glass Maintenance
 C1- Glass replacement C2- Glass repair
 C3- Installation of ventilation system
 HY4. Frame- Lintel Maintenance
 L1- Silicone squeezing between frame-lintel L2- Insulation if thinner than lintel wall L3- Strengthening frame fixings
 L4- Placing raincoats / profiles to remove rainwater etc. HY5. Frame-Sills Maintenance
 D1- Silicone squeezing between case-windowsill
 D2- Installation of profile / raincoat to reduce water leakage between frame (case) and windowsill

Code: • Enabling condition > Repair information Weight score: 0
 Search A - Window maintenance and component repair > A-GS-03, p. 11
 Created: 23.6.2026 r. 8:51 by user, Modified: 23.6.2026 r. 8:51 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Component replacement (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)

Defect Locations Maintenance Recommendations

HY1. Frame Components Maintenance
 A1- Repair or replacement of EPDM, silicone, strip, etc. sealing materials A2- Repair / replacement of fasteners such as nail screws A3- Replacing the glazing bead

A4- Repair / replacement of hinges, opening / closing mechanisms

HY2. Frame Maintenance

B1- General cleaning of the frame

B2- Repairing damaged areas according to the material of the frame B3- Repair of drains and drainage B4- Removal of damaged parts of glass

HY3.Glass Maintenance

C1-Glass replacement C2-Glass repair

C3-Installation of ventilation system

HY4. Frame-Lintel Maintenance

L1- Silicone squeezing between frame-lintel L2- Insulation if thinner than lintel wall L3- Strengthening frame fixings

L4- Placing raincoats / profiles to remove rainwater etc. HY5.Frame-Sills Maintenance

D1-Silicone squeezing between case-windowsill

D2- Installation of profile / raincoat to reduce water leakage between frame (case) and windowsill

Code: • [Obsolescence](#) > [Physical](#) Weight score: 0

[Search A - Window maintenance and component repair](#) > [A-GS-03](#), p. 11

[Created: 23.6.2026 r. 8:50 by user, Modified: 23.6.2026 r. 8:50 by user](#)

Other codes assigned to segment:

[Circular strategy](#) > [Maintenance](#) (Weight: 0)

[Circular strategy](#) > [Repair](#) (Weight: 0)

[Circular strategy](#) > [Component replacement](#) (Weight: 0)

[Enabling condition](#) > [Repair information](#) (Weight: 0)

[Mitigation](#) > [Maintenance planning](#) (Weight: 0)

Defect Locations Maintenance Recommendations

HY1.Frame Components Maintenance

A1- Repair or replacement of EPDM, silicone, strip, etc. sealing materials A2- Repair / replacement of fasteners such as nail screws A3- Replacing the glazing bead

A4- Repair / replacement of hinges, opening / closing mechanisms

HY2. Frame Maintenance

B1- General cleaning of the frame

B2- Repairing damaged areas according to the material of the frame B3- Repair of drains and drainage B4- Removal of damaged parts of glass

HY3.Glass Maintenance

C1-Glass replacement C2-Glass repair

C3-Installation of ventilation system

HY4. Frame-Lintel Maintenance

L1- Silicone squeezing between frame-lintel L2- Insulation if thinner than lintel wall L3- Strengthening frame fixings

L4- Placing raincoats / profiles to remove rainwater etc. HY5.Frame-Sills Maintenance

D1-Silicone squeezing between case-windowsill

D2- Installation of profile / raincoat to reduce water leakage between frame (case) and windowsill

Code: • [Circular strategy](#) > [Repair](#) Weight score: 0

[Search A - Window maintenance and component repair](#) > [A-GS-03](#), p. 11

[Created: 23.6.2026 r. 8:50 by user, Modified: 23.6.2026 r. 8:50 by user](#)

Other codes assigned to segment:

[Circular strategy](#) > [Maintenance](#) (Weight: 0)

[Circular strategy](#) > [Component replacement](#) (Weight: 0)

[Obsolescence](#) > [Physical](#) (Weight: 0)

[Enabling condition](#) > [Repair information](#) (Weight: 0)

[Mitigation](#) > [Maintenance planning](#) (Weight: 0)

Defect Locations Maintenance Recommendations

HY1.Frame Components Maintenance

A1- Repair or replacement of EPDM, silicone, strip, etc. sealing materials A2- Repair / replacement of fasteners such as nail screws A3- Replacing the glazing bead

A4- Repair / replacement of hinges, opening / closing mechanisms

HY2. Frame Maintenance

B1- General cleaning of the frame

B2- Repairing damaged areas according to the material of the frame B3- Repair of drains and drainage B4- Removal of damaged parts of glass

HY3.Glass Maintenance

C1-Glass replacement C2-Glass repair

C3-Installation of ventilation system

HY4. Frame-Lintel Maintenance

L1- Silicone squeezing between frame-lintel L2- Insulation if thinner than lintel wall L3- Strengthening frame fixings

L4- Placing raincoats / profiles to remove rainwater etc. HY5.Frame-Sills Maintenance

D1-Silicone squeezing between case-windowsill

D2- Installation of profile / raincoat to reduce water leakage between frame (case) and windowsill

Code: • Circular strategy > Component replacement Weight score: 0

Search A - Window maintenance and component repair > A-GS-03, p. 11

Created: 23.6.2026 r. 8:50 by user, Modified: 23.6.2026 r. 8:50 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Repair (Weight: 0)

Obsolescence > Physical (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Mitigation > Maintenance planning (Weight: 0)

According to the score obtained as a result of the visual evaluations, the selection is made by applying the AHP method while planning maintenance for the relevant window. The AHP method is used in the selection of maintenance alternatives. The AHP method is one of the most widely used multi-criteria decision-making methods where the decision (selection and prioritization of alternatives) is based on a variety of tangible and intangible criteria (sub-criteria). The complex problem-solving process is divided into a hierarchical structure consisting of the purpose of the problem, criteria, sub-criteria, and alternatives. When designing the AHP hierarchical tree, the objective is to develop a general framework that meets the needs of analysts to solve the problem of selecting the best maintenance policy. AHP begins by dividing the complex multi-criteria problem into a hierarchy where each level contains several manageable elements that can be divided into another group of elements [22]. The AHP hierarchy developed in this study is based on the selection of the performance characteristics in the window system and the appropriate maintenance alternatives. Tables 9, 10 and 11 are used for selection of the best alternative.

Code: • Circular strategy > Maintenance Weight score: 0

Search A - Window maintenance and component repair > A-GS-03, p. 12

Created: 23.6.2026 r. 8:47 by user, Modified: 23.6.2026 r. 8:47 by user

Other codes assigned to segment:

Obsolescence > Physical (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Mitigation > Maintenance planning (Weight: 0)

According to the score obtained as a result of the visual evaluations, the selection is made by applying the AHP method while planning maintenance for the relevant window. The AHP method is used in the selection of maintenance alternatives. The AHP method is one of the most widely used multi-criteria decision-making methods where the decision (selection and prioritization of

alternatives) is based on a variety of tangible and intangible criteria (sub-criteria). The complex problem-solving process is divided into a hierarchical structure consisting of the purpose of the problem, criteria, sub-criteria, and alternatives. When designing the AHP hierarchical tree, the objective is to develop a general framework that meets the needs of analysts to solve the problem of selecting the best maintenance policy. AHP begins by dividing the complex multi-criteria problem into a hierarchy where each level contains several manageable elements that can be divided into another group of elements [22]. The AHP hierarchy developed in this study is based on the selection of the performance characteristics in the window system and the appropriate maintenance alternatives. Tables 9, 10 and 11 are used for selection of the best alternative.

Code: • Enabling condition > Repair information Weight score: 0
 Search A - Window maintenance and component repair > A-GS-03, p. 12
 Created: 23.6.2026 r. 8:47 by user, Modified: 23.6.2026 r. 8:47 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)

According to the score obtained as a result of the visual evaluations, the selection is made by applying the AHP method while planning maintenance for the relevant window. The AHP method is used in the selection of maintenance alternatives. The AHP method is one of the most widely used multi-criteria decision-making methods where the decision (selection and prioritization of alternatives) is based on a variety of tangible and intangible criteria (sub-criteria). The complex problem-solving process is divided into a hierarchical structure consisting of the purpose of the problem, criteria, sub-criteria, and alternatives. When designing the AHP hierarchical tree, the objective is to develop a general framework that meets the needs of analysts to solve the problem of selecting the best maintenance policy. AHP begins by dividing the complex multi-criteria problem into a hierarchy where each level contains several manageable elements that can be divided into another group of elements [22]. The AHP hierarchy developed in this study is based on the selection of the performance characteristics in the window system and the appropriate maintenance alternatives. Tables 9, 10 and 11 are used for selection of the best alternative.

Code: • Mitigation > Maintenance planning Weight score: 0
 Search A - Window maintenance and component repair > A-GS-03, p. 12
 Created: 23.6.2026 r. 8:47 by user, Modified: 23.6.2026 r. 8:47 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Enabling condition > Repair information (Weight: 0)

According to the score obtained as a result of the visual evaluations, the selection is made by applying the AHP method while planning maintenance for the relevant window. The AHP method is used in the selection of maintenance alternatives. The AHP method is one of the most widely used multi-criteria decision-making methods where the decision (selection and prioritization of alternatives) is based on a variety of tangible and intangible criteria (sub-criteria). The complex problem-solving process is divided into a hierarchical structure consisting of the purpose of the problem, criteria, sub-criteria, and alternatives. When designing the AHP hierarchical tree, the objective is to develop a general framework that meets the needs of analysts to solve the problem of selecting the best maintenance policy. AHP begins by dividing the complex multi-criteria problem into a hierarchy where each level contains several manageable elements that can be divided into another group of elements [22]. The AHP hierarchy developed in this study is based on the selection of the performance characteristics in the window system and the appropriate maintenance alternatives. Tables 9, 10 and 11 are used for selection of the best alternative.

Code: • Obsolescence > Physical Weight score: 0

Search A - Window maintenance and component repair > A-GS-03, p. 12

Created: 23.6.2026 r. 8:47 by user, Modified: 23.6.2026 r. 8:47 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Mitigation > Maintenance planning (Weight: 0)

The most governing element of the window is the insulated glazing unit (IGU). There are many developments concerning coatings to add functions to the glass, such as solar control or self-cleaning windows but they contain critical materials such as cobalt, copper and titanium. Added elements such as coatings, foils for laminated glass and the sealant prohibit the IGU from being recyclable, as the prevailing glass industry requires high quality and clean ingredients only. This makes the IGU a finite, single life product resulting in almost 125.000 tonnes of post-consumer glass waste each year in the Netherlands.

The IGU works optimally as an insulating element as long as there is dry (argon) gas inside the glazing panes. However, the life span of current IGUs is just around 15-20 years and is dependent on the butyl seal which is just 1% of the costs and 0.1% of the weight. During its life span the seal starts failing by allowing water molecules inside the cavity. The panel as a result starts building up water vapor and the coating inside will start to corrode, the glass shows fogging and the thermal performance drops down due to outgassing of the panel. In the current design of the IGU no refurbishment is possible, meaning that the glass panes and the spacer will not be re-used, but instead end up as landfill while these materials exceed the life span of the sealant by a large margin (Veer, 2016).

Code: • Enabling condition > Component accessibility Weight score: 0

Search A - Window maintenance and component repair > A-GS-06, p. 5

Created: 23.6.2026 r. 8:56 by user, Modified: 23.6.2026 r. 8:56 by user

Other codes assigned to segment:

Circular strategy > Component replacement (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)

Obsolescence > Physical (Weight: 0)

Obsolescence > Technical (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Negative mechanism > Lock-in (Weight: 0)

Mitigation > Design intervention (Weight: 0)

The most governing element of the window is the insulated glazing unit (IGU). There are many developments concerning coatings to add functions to the glass, such as solar control or self-cleaning windows but they contain critical materials such as cobalt, copper and titanium. Added elements such as coatings, foils for laminated glass and the sealant prohibit the IGU from being recyclable, as the prevailing glass industry requires high quality and clean ingredients only. This makes the IGU a finite, single life product resulting in almost 125.000 tonnes of post-consumer glass waste each year in the Netherlands.

The IGU works optimally as an insulating element as long as there is dry (argon) gas inside the glazing panes. However, the life span of current IGUs is just around 15-20 years and is dependent on the butyl seal which is just 1% of the costs and 0.1% of the weight. During its life span the seal starts failing by allowing water molecules inside the cavity. The panel as a result starts building up water vapor and the coating inside will start to corrode, the glass shows fogging and the thermal performance drops down due to outgassing of the panel. In the current design of the IGU no refurbishment is possible, meaning that the glass panes and the spacer will not be re-used, but instead end up as landfill while these materials exceed the life span of the sealant by a large margin (Veer, 2016).

Code: • Negative mechanism > Lock-in Weight score: 0

Search A - Window maintenance and component repair > A-GS-06, p. 5

Created: 23.6.2026 r. 8:57 by user, Modified: 23.6.2026 r. 8:57 by user

Other codes assigned to segment:

- Circular strategy > Component replacement (Weight: 0)
- Circular strategy > Refurbishment (Weight: 0)
- Obsolescence > Physical (Weight: 0)
- Obsolescence > Technical (Weight: 0)
- Enabling condition > Component accessibility (Weight: 0)
- Negative mechanism > Premature replacement (Weight: 0)
- Mitigation > Design intervention (Weight: 0)

The most governing element of the window is the insulated glazing unit (IGU). There are many developments concerning coatings to add functions to the glass, such as solar control or self-cleaning windows but they contain critical materials such as cobalt, copper and titanium. Added elements such as coatings, foils for laminated glass and the sealant prohibit the IGU from being recyclable, as the prevailing glass industry requires high quality and clean ingredients only. This makes the IGU a finite, single life product resulting in almost 125.000 tonnes of post-consumer glass waste each year in the Netherlands.

The IGU works optimally as an insulating element as long as there is dry (argon) gas inside the glazing panes. However, the life span of current IGUs is just around 15-20 years and is dependent on the butyl seal which is just 1% of the costs and 0.1% of the weight. During its life span the seal starts failing by allowing water molecules inside the cavity. The panel as a result starts building up water vapor and the coating inside will start to corrode, the glass shows fogging and the thermal performance drops down due to outgassing of the panel. In the current design of the IGU no refurbishment is possible, meaning that the glass panes and the spacer will not be re-used, but instead end up as landfill while these materials exceed the life span of the sealant by a large margin (Veer, 2016).

Code: • Circular strategy > Refurbishment Weight score: 0

Search A - Window maintenance and component repair > A-GS-06, p. 5

Created: 23.6.2026 r. 8:56 by user, Modified: 23.6.2026 r. 8:56 by user

Other codes assigned to segment:

- Circular strategy > Component replacement (Weight: 0)
- Obsolescence > Physical (Weight: 0)
- Obsolescence > Technical (Weight: 0)
- Enabling condition > Component accessibility (Weight: 0)
- Negative mechanism > Premature replacement (Weight: 0)
- Negative mechanism > Lock-in (Weight: 0)
- Mitigation > Design intervention (Weight: 0)

The most governing element of the window is the insulated glazing unit (IGU). There are many developments concerning coatings to add functions to the glass, such as solar control or self-cleaning windows but they contain critical materials such as cobalt, copper and titanium. Added elements such as coatings, foils for laminated glass and the sealant prohibit the IGU from being recyclable, as the prevailing glass industry requires high quality and clean ingredients only. This makes the IGU a finite, single life product resulting in almost 125.000 tonnes of post-consumer glass waste each year in the Netherlands.

The IGU works optimally as an insulating element as long as there is dry (argon) gas inside the glazing panes. However, the life span of current IGUs is just around 15-20 years and is dependent on the butyl seal which is just 1% of the costs and 0.1% of the weight. During its life span the seal starts failing by allowing water molecules inside the cavity. The panel as a result starts building up water vapor and the coating inside will start to corrode, the glass shows fogging and the thermal performance drops down due to outgassing of the panel. In the current design of the IGU no refurbishment is possible, meaning that the glass panes and the spacer will not be re-used, but instead end up as landfill while these materials exceed the life span of the sealant by a large margin (Veer, 2016).

Code: • [Obsolescence > Technical](#) Weight score: 0
 Search A - Window maintenance and component repair > A-GS-06, p. 5
 Created: 23.6.2026 r. 8:56 by user, Modified: 23.6.2026 r. 8:56 by user
 Other codes assigned to segment:
 [Circular strategy > Component replacement](#) (Weight: 0)
 [Circular strategy > Refurbishment](#) (Weight: 0)
 [Obsolescence > Physical](#) (Weight: 0)
 [Enabling condition > Component accessibility](#) (Weight: 0)
 [Negative mechanism > Premature replacement](#) (Weight: 0)
 [Negative mechanism > Lock-in](#) (Weight: 0)
 [Mitigation > Design intervention](#) (Weight: 0)

The most governing element of the window is the insulated glazing unit (IGU). There are many developments concerning coatings to add functions to the glass ,such as solar control or self-cleaning windows but they contain critical materials such as cobalt, copper and titaninum. Added elements such as coatings, foils for laminated glass and the sealant prohibit the IGU from being recyclable, as the prevailing glass industry requires high quality and clean ingredients only. This makes the IGU a finite, single life product resulting in almost 125.000 tonnes of post-consumer glass waste each year in the Netherlands.

The IGU works optimally as an insulating element as long as there is dry (argon) gas inside the glazing panes. However, the life span of current IGUs is just around 15-20 years and is dependent on the butyl seal which is just 1% of the costs and 0.1% of the weight. During its life span the seal starts failing by allowing water molecules inside the cavity. The panel as a result starts building up water vapor and the coating inside will start to corrode, the glass shows fogging and the thermal performance drops down due to outgassing of the panel. In the current design of the IGU no refurbishment is possible, meaning that the glass panes and the spacer will not be re-used, but instead end up as landfill while these materials exceed the life span of the sealant by a large margin (Veer,2016).

Code: • [Mitigation > Design intervention](#) Weight score: 0
 Search A - Window maintenance and component repair > A-GS-06, p. 5
 Created: 23.6.2026 r. 8:57 by user, Modified: 23.6.2026 r. 8:57 by user
 Other codes assigned to segment:
 [Circular strategy > Component replacement](#) (Weight: 0)
 [Circular strategy > Refurbishment](#) (Weight: 0)
 [Obsolescence > Physical](#) (Weight: 0)
 [Obsolescence > Technical](#) (Weight: 0)
 [Enabling condition > Component accessibility](#) (Weight: 0)
 [Negative mechanism > Premature replacement](#) (Weight: 0)
 [Negative mechanism > Lock-in](#) (Weight: 0)

The most governing element of the window is the insulated glazing unit (IGU). There are many developments concerning coatings to add functions to the glass ,such as solar control or self-cleaning windows but they contain critical materials such as cobalt, copper and titaninum. Added elements such as coatings, foils for laminated glass and the sealant prohibit the IGU from being recyclable, as the prevailing glass industry requires high quality and clean ingredients only. This makes the IGU a finite, single life product resulting in almost 125.000 tonnes of post-consumer glass waste each year in the Netherlands.

The IGU works optimally as an insulating element as long as there is dry (argon) gas inside the glazing panes. However, the life span of current IGUs is just around 15-20 years and is dependent on the butyl seal which is just 1% of the costs and 0.1% of the weight. During its life span the seal starts failing by allowing water molecules inside the cavity. The panel as a result starts building up water vapor and the coating inside will start to corrode, the glass shows fogging and the thermal performance drops down due to outgassing of the panel. In the current design of the IGU no refurbishment is possible, meaning that the glass panes and the spacer will not be re-used, but

instead end up as landfill while these materials exceed the life span of the sealant by a large margin (Veer,2016).

Code: • [Obsolescence > Physical](#) Weight score: 0

[Search A - Window maintenance and component repair > A-GS-06, p. 5](#)

Created: 23.6.2026 r. 8:56 by user, Modified: 23.6.2026 r. 8:56 by user

Other codes assigned to segment:

[Circular strategy > Component replacement](#) (Weight: 0)

[Circular strategy > Refurbishment](#) (Weight: 0)

[Obsolescence > Technical](#) (Weight: 0)

[Enabling condition > Component accessibility](#) (Weight: 0)

[Negative mechanism > Premature replacement](#) (Weight: 0)

[Negative mechanism > Lock-in](#) (Weight: 0)

[Mitigation > Design intervention](#) (Weight: 0)

The most governing element of the window is the insulated glazing unit (IGU). There are many developments concerning coatings to add functions to the glass ,such as solar control or self-cleaning windows but they contain critical materials such as cobalt, copper and titaninum. Added elements such as coatings, foils for laminated glass and the sealant prohibit the IGU from being recyclable, as the prevailing glass industry requires high quality and clean ingredients only. This makes the IGU a finite, single life product resulting in almost 125.000 tonnes of post-consumer glass waste each year in the Netherlands.

The IGU works optimally as an insulating element as long as there is dry (argon) gas inside the glazing panes. However, the life span of current IGUs is just around 15-20 years and is dependent on the butyl seal which is just 1% of the costs and 0.1% of the weight. During its life span the seal starts failing by allowing water molecules inside the cavity. The panel as a result starts building up water vapor and the coating inside will start to corrode, the glass shows fogging and the thermal performance drops down due to outgassing of the panel. In the current design of the IGU no refurbishment is possible, meaning that the glass panes and the spacer will not be re-used, but instead end up as landfill while these materials exceed the life span of the sealant by a large margin (Veer,2016).

Code: • [Circular strategy > Component replacement](#) Weight score: 0

[Search A - Window maintenance and component repair > A-GS-06, p. 5](#)

Created: 23.6.2026 r. 8:56 by user, Modified: 23.6.2026 r. 8:56 by user

Other codes assigned to segment:

[Circular strategy > Refurbishment](#) (Weight: 0)

[Obsolescence > Physical](#) (Weight: 0)

[Obsolescence > Technical](#) (Weight: 0)

[Enabling condition > Component accessibility](#) (Weight: 0)

[Negative mechanism > Premature replacement](#) (Weight: 0)

[Negative mechanism > Lock-in](#) (Weight: 0)

[Mitigation > Design intervention](#) (Weight: 0)

The most governing element of the window is the insulated glazing unit (IGU). There are many developments concerning coatings to add functions to the glass ,such as solar control or self-cleaning windows but they contain critical materials such as cobalt, copper and titaninum. Added elements such as coatings, foils for laminated glass and the sealant prohibit the IGU from being recyclable, as the prevailing glass industry requires high quality and clean ingredients only. This makes the IGU a finite, single life product resulting in almost 125.000 tonnes of post-consumer glass waste each year in the Netherlands.

The IGU works optimally as an insulating element as long as there is dry (argon) gas inside the glazing panes. However, the life span of current IGUs is just around 15-20 years and is dependent on the butyl seal which is just 1% of the costs and 0.1% of the weight. During its life span the seal starts failing by allowing water molecules inside the cavity. The panel as a result starts building up water vapor and the coating inside will start to corrode, the glass shows fogging and the thermal

performance drops down due to outgassing of the panel. In the current design of the IGU no refurbishment is possible, meaning that the glass panes and the spacer will not be re-used, but instead end up as landfill while these materials exceed the life span of the sealant by a large margin (Veer,2016).

Code: • Negative mechanism > Premature replacement Weight score: 0

Search A - Window maintenance and component repair > A-GS-06, p. 5

Created: 23.6.2026 r. 8:56 by user, Modified: 23.6.2026 r. 8:56 by user

Other codes assigned to segment:

Circular strategy > Component replacement (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)

Obsolescence > Physical (Weight: 0)

Obsolescence > Technical (Weight: 0)

Enabling condition > Component accessibility (Weight: 0)

Negative mechanism > Lock-in (Weight: 0)

Mitigation > Design intervention (Weight: 0)

The glass industry was one of the first with a high degree of recyclability from its products. Crushed glass can be recurrently recycled into similar products, with almost the same quality and purity (Silva, Brito, Lye, & Dhir, 2017). However, float glass is still a linear economy ending at land fill after usage. Although the embodied energy of glass is relatively low in comparison with typical building materials, the recycling process of glass still takes a lot of energy due to the three following reasons: impurities, different recipes of glass and adhesives. Therefore, despite the potentials, most of the discarded float glass (in the Netherlands already 125000 tonnes a year) ,fails to pass the high quality standards of the prevailing glass industry making it a finite, single used product.– (Bristogianni, Oikonomopoulou, Justino de Lima, Veer, & Nijse, 2018).

Code: • Circular strategy > Recycling Weight score: 0

Search A - Window maintenance and component repair > A-GS-06, p. 9

Created: 23.6.2026 r. 9:04 by user, Modified: 23.6.2026 r. 9:04 by user

Other codes assigned to segment:

Obsolescence > Technical (Weight: 0)

Negative mechanism > Increased material complexity (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Design intervention (Weight: 0)

The glass industry was one of the first with a high degree of recyclability from its products. Crushed glass can be recurrently recycled into similar products, with almost the same quality and purity (Silva, Brito, Lye, & Dhir, 2017). However, float glass is still a linear economy ending at land fill after usage. Although the embodied energy of glass is relatively low in comparison with typical building materials, the recycling process of glass still takes a lot of energy due to the three following reasons: impurities, different recipes of glass and adhesives. Therefore, despite the potentials, most of the discarded float glass (in the Netherlands already 125000 tonnes a year) ,fails to pass the high quality standards of the prevailing glass industry making it a finite, single used product.– (Bristogianni, Oikonomopoulou, Justino de Lima, Veer, & Nijse, 2018).

Code: • Mitigation > Design intervention Weight score: 0

Search A - Window maintenance and component repair > A-GS-06, p. 9

Created: 23.6.2026 r. 9:05 by user, Modified: 23.6.2026 r. 9:05 by user

Other codes assigned to segment:

Circular strategy > Recycling (Weight: 0)

Obsolescence > Technical (Weight: 0)

Negative mechanism > Increased material complexity (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

The glass industry was one of the first with a high degree of recyclability from its products. Crushed glass can be recurrently recycled into similar products, with almost the same quality and purity (Silva,

Brito, Lye, & Dhir, 2017). However, float glass is still a linear economy ending at land fill after usage. Although the embodied energy of glass is relatively low in comparison with typical building materials, the recycling process of glass still takes a lot of energy due to the three following reasons: impurities, different recipes of glass and adhesives. Therefore, despite the potentials, most of the discarded float glass (in the Netherlands already 125000 tonnes a year) ,fails to pass the high quality standards of the prevailing glass industry making it a finite, single used product.– (Bristogianni, Oikonimopoulou, Justino de Lima, Veer, & Nijse, 2018).

Code: • Negative mechanism > Burden shifting Weight score: 0

Search A - Window maintenance and component repair > A-GS-06, p. 9

Created: 23.6.2026 r. 9:05 by user, Modified: 23.6.2026 r. 9:05 by user

Other codes assigned to segment:

Circular strategy > Recycling (Weight: 0)

Obsolescence > Technical (Weight: 0)

Negative mechanism > Increased material complexity (Weight: 0)

Mitigation > Design intervention (Weight: 0)

The glass industry was one of the first with a high degree of recyclability from its products. Crushed glass can be recurrently recycled into similar products, with almost the same quality and purity (Silva, Brito, Lye, & Dhir, 2017). However, float glass is still a linear economy ending at land fill after usage. Although the embodied energy of glass is relatively low in comparison with typical building materials, the recycling process of glass still takes a lot of energy due to the three following reasons: impurities, different recipes of glass and adhesives. Therefore, despite the potentials, most of the discarded float glass (in the Netherlands already 125000 tonnes a year) ,fails to pass the high quality standards of the prevailing glass industry making it a finite, single used product.– (Bristogianni, Oikonimopoulou, Justino de Lima, Veer, & Nijse, 2018).

Code: • Obsolescence > Technical Weight score: 0

Search A - Window maintenance and component repair > A-GS-06, p. 9

Created: 23.6.2026 r. 9:04 by user, Modified: 23.6.2026 r. 9:04 by user

Other codes assigned to segment:

Circular strategy > Recycling (Weight: 0)

Negative mechanism > Increased material complexity (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Design intervention (Weight: 0)

The glass industry was one of the first with a high degree of recyclability from its products. Crushed glass can be recurrently recycled into similar products, with almost the same quality and purity (Silva, Brito, Lye, & Dhir, 2017). However, float glass is still a linear economy ending at land fill after usage. Although the embodied energy of glass is relatively low in comparison with typical building materials, the recycling process of glass still takes a lot of energy due to the three following reasons: impurities, different recipes of glass and adhesives. Therefore, despite the potentials, most of the discarded float glass (in the Netherlands already 125000 tonnes a year) ,fails to pass the high quality standards of the prevailing glass industry making it a finite, single used product.– (Bristogianni, Oikonimopoulou, Justino de Lima, Veer, & Nijse, 2018).

Code: • Negative mechanism > Increased material complexity Weight score: 0

Search A - Window maintenance and component repair > A-GS-06, p. 9

Created: 23.6.2026 r. 9:04 by user, Modified: 23.6.2026 r. 9:04 by user

Other codes assigned to segment:

Circular strategy > Recycling (Weight: 0)

Obsolescence > Technical (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Design intervention (Weight: 0)

Problem statement

The life span of current Insulated Glass Unit (IGU) is around 15-20 years, after this the butyl seal which is just 1% of the costs and 0.1% of the weight, starts allowing water molecules getting inside. The panel as a result starts building up water vapour and the coating inside will start to corrode, the glass shows fogging, and the thermal performance drops down due to outgassing of the panel. In the current design of the IGU no refurbishment is possible meaning that the glass panes and the spacers will not be re-used but instead end up at landfill while these materials exceed this life span of the sealant by far(Veer,2016).

IVII|

The idea in reaction to this problem is to design a Insulated glass unit that is resealable. By just changing the edge sealant every ten years, the life span of the total panel could be extended to a life time of 100 years which results in saving materials, embodied energy and money.

Code: • [Obsolescence > Physical](#) Weight score: 0

Search A - Window maintenance and component repair > A-GS-06, p. 9 - 10

Created: 23.6.2026 r. 9:00 by user, Modified: 23.6.2026 r. 9:00 by user

Other codes assigned to segment:

[Circular strategy > Component replacement](#) (Weight: 0)

[Circular strategy > Refurbishment](#) (Weight: 0)

[Obsolescence > Technical](#) (Weight: 0)

[Enabling condition > Component accessibility](#) (Weight: 0)

[Enabling condition > Spare parts](#) (Weight: 0)

[Negative mechanism > Premature replacement](#) (Weight: 0)

[Negative mechanism > Lock-in](#) (Weight: 0)

[Mitigation > Design intervention](#) (Weight: 0)

Problem statement

The life span of current Insulated Glass Unit (IGU) is around 15-20 years, after this the butyl seal which is just 1% of the costs and 0.1% of the weight, starts allowing water molecules getting inside. The panel as a result starts building up water vapour and the coating inside will start to corrode, the glass shows fogging, and the thermal performance drops down due to outgassing of the panel. In the current design of the IGU no refurbishment is possible meaning that the glass panes and the spacers will not be re-used but instead end up at landfill while these materials exceed this life span of the sealant by far(Veer,2016).

IVII|

The idea in reaction to this problem is to design a Insulated glass unit that is resealable. By just changing the edge sealant every ten years, the life span of the total panel could be extended to a life time of 100 years which results in saving materials, embodied energy and money.

Code: • [Negative mechanism > Premature replacement](#) Weight score: 0

Search A - Window maintenance and component repair > A-GS-06, p. 9 - 10

Created: 23.6.2026 r. 9:01 by user, Modified: 23.6.2026 r. 9:01 by user

Other codes assigned to segment:

[Circular strategy > Component replacement](#) (Weight: 0)

[Circular strategy > Refurbishment](#) (Weight: 0)

[Obsolescence > Physical](#) (Weight: 0)

[Obsolescence > Technical](#) (Weight: 0)

[Enabling condition > Component accessibility](#) (Weight: 0)

[Enabling condition > Spare parts](#) (Weight: 0)

[Negative mechanism > Lock-in](#) (Weight: 0)

[Mitigation > Design intervention](#) (Weight: 0)

Problem statement

The life span of current Insulated Glass Unit (IGU) is around 15-20 years, after this the butyl seal which is just 1% of the costs and 0.1% of the weight, starts allowing water molecules getting inside. The panel as a result starts building up water vapour and the coating inside will start to corrode, the

glass shows fogging, and the thermal performance drops down due to outgassing of the panel. In the current design of the IGU no refurbishment is possible meaning that the glass panes and the spacers will not be re-used but instead end up at landfill while these materials exceed this life span of the sealant by far(Veer,2016).

IVII|

The idea in reaction to this problem is to design a Insulated glass unit that is resealable. By just changing the edge sealant every ten years, the life span of the total panel could be extended to a life time of 100 years which results in saving materials, embodied energy and money.

Code: • Negative mechanism > Lock-in Weight score: 0

Search A - Window maintenance and component repair > A-GS-06, p. 9 - 10

Created: 23.6.2026 r. 9:01 by user, Modified: 23.6.2026 r. 9:01 by user

Other codes assigned to segment:

Circular strategy > Component replacement (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)

Obsolescence > Physical (Weight: 0)

Obsolescence > Technical (Weight: 0)

Enabling condition > Component accessibility (Weight: 0)

Enabling condition > Spare parts (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Mitigation > Design intervention (Weight: 0)

The idea in reaction to this problem is to design a Insulated glass unit that is resealable. By just changing the edge sealant every ten years, the life span of the total panel could be extended to a life time of 100 years which results in saving materials, embodied energy and money.

Goal thesis

The goal of the thesis is to design a different edge seal system for an IGU that is easy to replace and assures the functions of the product for at least ten years. By the ease of replacement the design should enhance refurbishment in order to give the IGU a life span >100 years.

Code: • Enabling condition > Spare parts Weight score: 0

Search A - Window maintenance and component repair > A-GS-06, p. 10

Created: 23.6.2026 r. 9:03 by user, Modified: 23.6.2026 r. 9:03 by user

Other codes assigned to segment:

Circular strategy > Component replacement (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)

Obsolescence > Physical (Weight: 0)

Obsolescence > Technical (Weight: 0)

Enabling condition > Component accessibility (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Negative mechanism > Lock-in (Weight: 0)

Mitigation > Design intervention (Weight: 0)

The idea in reaction to this problem is to design a Insulated glass unit that is resealable. By just changing the edge sealant every ten years, the life span of the total panel could be extended to a life time of 100 years which results in saving materials, embodied energy and money.

Goal thesis

The goal of the thesis is to design a different edge seal system for an IGU that is easy to replace and assures the functions of the product for at least ten years. By the ease of replacement the design should enhance refurbishment in order to give the IGU a life span >100 years.

Code: • Mitigation > Design intervention Weight score: 0

Search A - Window maintenance and component repair > A-GS-06, p. 10

Created: 23.6.2026 r. 9:03 by user, Modified: 23.6.2026 r. 9:03 by user

Other codes assigned to segment:

Circular strategy > Component replacement (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)

Obsolescence > Physical (Weight: 0)

Obsolescence > Technical (Weight: 0)
 Enabling condition > Component accessibility (Weight: 0)
 Enabling condition > Spare parts (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)

The idea in reaction to this problem is to design a Insulated glass unit that is resealable. By just changing the edge sealant every ten years, the life span of the total panel could be extended to a life time of 100 years which results in saving materials, embodied energy and money.

Goal thesis

The goal of the thesis is to design a different edge seal system for an IGU that is easy to replace and assures the functions of the product for at least ten years. By the ease of replacement the design should enhance refurbishment in order to give the IGU a life span >100 years.

Code: • Obsolescence > Technical Weight score: 0
 Search A - Window maintenance and component repair > A-GS-06, p. 10
 Created: 23.6.2026 r. 9:02 by user, Modified: 23.6.2026 r. 9:02 by user
 Other codes assigned to segment:
 Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Enabling condition > Component accessibility (Weight: 0)
 Enabling condition > Spare parts (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

The idea in reaction to this problem is to design a Insulated glass unit that is resealable. By just changing the edge sealant every ten years, the life span of the total panel could be extended to a life time of 100 years which results in saving materials, embodied energy and money.

Goal thesis

The goal of the thesis is to design a different edge seal system for an IGU that is easy to replace and assures the functions of the product for at least ten years. By the ease of replacement the design should enhance refurbishment in order to give the IGU a life span >100 years.

Code: • Circular strategy > Refurbishment Weight score: 0
 Search A - Window maintenance and component repair > A-GS-06, p. 10
 Created: 23.6.2026 r. 9:02 by user, Modified: 23.6.2026 r. 9:02 by user
 Other codes assigned to segment:
 Circular strategy > Component replacement (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Enabling condition > Component accessibility (Weight: 0)
 Enabling condition > Spare parts (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

The idea in reaction to this problem is to design a Insulated glass unit that is resealable. By just changing the edge sealant every ten years, the life span of the total panel could be extended to a life time of 100 years which results in saving materials, embodied energy and money.

Goal thesis

The goal of the thesis is to design a different edge seal system for an IGU that is easy to replace and assures the functions of the product for at least ten years. By the ease of replacement the design should enhance refurbishment in order to give the IGU a life span >100 years.

Code: • Circular strategy > Component replacement Weight score: 0
 Search A - Window maintenance and component repair > A-GS-06, p. 10

Created: 23.6.2026 r. 9:02 by user, Modified: 23.6.2026 r. 9:02 by user

Other codes assigned to segment:

- Circular strategy > Refurbishment (Weight: 0)
- Obsolescence > Physical (Weight: 0)
- Obsolescence > Technical (Weight: 0)
- Enabling condition > Component accessibility (Weight: 0)
- Enabling condition > Spare parts (Weight: 0)
- Negative mechanism > Premature replacement (Weight: 0)
- Negative mechanism > Lock-in (Weight: 0)
- Mitigation > Design intervention (Weight: 0)

The idea in reaction to this problem is to design a Insulated glass unit that is resealable. By just changing the edge sealant every ten years, the life span of the total panel could be extended to a life time of 100 years which results in saving materials, embodied energy and money.

Goal thesis

The goal of the thesis is to design a different edge seal system for an IGU that is easy to replace and assures the functions of the product for at least ten years. By the ease of replacement the design should enhance refurbishment in order to give the IGU a life span >100 years.

Code: • Enabling condition > Component accessibility Weight score: 0

Search A - Window maintenance and component repair > A-GS-06, p. 10

Created: 23.6.2026 r. 9:02 by user, Modified: 23.6.2026 r. 9:02 by user

Other codes assigned to segment:

- Circular strategy > Component replacement (Weight: 0)
- Circular strategy > Refurbishment (Weight: 0)
- Obsolescence > Physical (Weight: 0)
- Obsolescence > Technical (Weight: 0)
- Enabling condition > Spare parts (Weight: 0)
- Negative mechanism > Premature replacement (Weight: 0)
- Negative mechanism > Lock-in (Weight: 0)
- Mitigation > Design intervention (Weight: 0)

The three steps picture(Figure 2.6.0.1) shows how easy it is to change the IGU: The first step is to take out the four EPDM gaskets this can be done with a simple push with e.g. a screwdriver to tilt the EPDM gaskets and pull them out.

Once the EPDM gaskets are removed, the glazing bead can be taken out as well which leaves the glazing panel the only thing to remove. The glazing panels can be picked up with a suction lifter that clamps on the glass (figure2.6.0.2).

Code: • Mitigation > Design intervention Weight score: 0

Search A - Window maintenance and component repair > A-GS-06, p. 29

Created: 23.6.2026 r. 9:06 by user, Modified: 23.6.2026 r. 9:06 by user

Other codes assigned to segment:

- Circular strategy > Component replacement (Weight: 0)
- Enabling condition > Component accessibility (Weight: 0)
- Enabling condition > Disassembly (Weight: 0)
- Enabling condition > Repair information (Weight: 0)

The three steps picture(Figure 2.6.0.1) shows how easy it is to change the IGU: The first step is to take out the four EPDM gaskets this can be done with a simple push with e.g. a screwdriver to tilt the EPDM gaskets and pull them out.

Once the EPDM gaskets are removed, the glazing bead can be taken out as well which leaves the glazing panel the only thing to remove. The glazing panels can be picked up with a suction lifter that clamps on the glass (figure2.6.0.2).

Code: • Enabling condition > Disassembly Weight score: 0

Search A - Window maintenance and component repair > A-GS-06, p. 29

Created: 23.6.2026 r. 9:06 by user, Modified: 23.6.2026 r. 9:06 by user

Other codes assigned to segment:

Circular strategy > Component replacement (Weight: 0)
 Enabling condition > Component accessibility (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

The three steps picture(Figure 2.6.0.1) shows how easy it is to change the IGU: The first step is to take out the four EPDM gaskets this can be done with a simple push with e.g. a screwdriver to tilt the EPDM gaskets and pull them out.

Once the EPDM gaskets are removed, the glazing bead can be taken out as well which leaves the glazing panel the only thing to remove. The glazing panels can be picked up with a suction lifter that clamps on the glass (figure2.6.0.2).

Code: • Circular strategy > Component replacement Weight score: 0

Search A - Window maintenance and component repair > A-GS-06, p. 29

Created: 23.6.2026 r. 9:06 by user, Modified: 23.6.2026 r. 9:06 by user

Other codes assigned to segment:

Enabling condition > Component accessibility (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

The three steps picture(Figure 2.6.0.1) shows how easy it is to change the IGU: The first step is to take out the four EPDM gaskets this can be done with a simple push with e.g. a screwdriver to tilt the EPDM gaskets and pull them out.

Once the EPDM gaskets are removed, the glazing bead can be taken out as well which leaves the glazing panel the only thing to remove. The glazing panels can be picked up with a suction lifter that clamps on the glass (figure2.6.0.2).

Code: • Enabling condition > Repair information Weight score: 0

Search A - Window maintenance and component repair > A-GS-06, p. 29

Created: 23.6.2026 r. 9:06 by user, Modified: 23.6.2026 r. 9:06 by user

Other codes assigned to segment:

Circular strategy > Component replacement (Weight: 0)
 Enabling condition > Component accessibility (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

The three steps picture(Figure 2.6.0.1) shows how easy it is to change the IGU: The first step is to take out the four EPDM gaskets this can be done with a simple push with e.g. a screwdriver to tilt the EPDM gaskets and pull them out.

Once the EPDM gaskets are removed, the glazing bead can be taken out as well which leaves the glazing panel the only thing to remove. The glazing panels can be picked up with a suction lifter that clamps on the glass (figure2.6.0.2).

Code: • Enabling condition > Component accessibility Weight score: 0

Search A - Window maintenance and component repair > A-GS-06, p. 29

Created: 23.6.2026 r. 9:06 by user, Modified: 23.6.2026 r. 9:06 by user

Other codes assigned to segment:

Circular strategy > Component replacement (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

One of the primary reasons that historic steel windows are often replaced is the presumption that they cannot be made more energy efficient. Another is the fact that there are myriad replacement options on the market, including aluminum and new steel windows. However, the sightlines,

proportions of elements, profiles, and shadow lines of many of the modern replacements do not match the details of the original windows. Hence, the substitution of replacement windows can result in the loss of the original character of the building.

Code: • Mitigation > Lifecycle verification Weight score: 0
Search A - Window maintenance and component repair > A-GS-08, p. 1
Created: 23.6.2026 r. 9:12 by user, Modified: 23.6.2026 r. 9:12 by user
Other codes assigned to segment:
Circular strategy > Repair (Weight: 0)
Circular strategy > Refurbishment (Weight: 0)
Obsolescence > Functional (Weight: 0)
Obsolescence > Aesthetic (Weight: 0)
Negative mechanism > Premature replacement (Weight: 0)

One of the primary reasons that historic steel windows are often replaced is the presumption that they cannot be made more energy efficient. Another is the fact that there are myriad replacement options on the market, including aluminum and new steel windows. However, the sightlines, proportions of elements, profiles, and shadow lines of many of the modern replacements do not match the details of the original windows. Hence, the substitution of replacement windows can result in the loss of the original character of the building.

Code: • Circular strategy > Repair Weight score: 0
Search A - Window maintenance and component repair > A-GS-08, p. 1
Created: 23.6.2026 r. 9:11 by user, Modified: 23.6.2026 r. 9:11 by user
Other codes assigned to segment:
Circular strategy > Refurbishment (Weight: 0)
Obsolescence > Functional (Weight: 0)
Obsolescence > Aesthetic (Weight: 0)
Negative mechanism > Premature replacement (Weight: 0)
Mitigation > Lifecycle verification (Weight: 0)

One of the primary reasons that historic steel windows are often replaced is the presumption that they cannot be made more energy efficient. Another is the fact that there are myriad replacement options on the market, including aluminum and new steel windows. However, the sightlines, proportions of elements, profiles, and shadow lines of many of the modern replacements do not match the details of the original windows. Hence, the substitution of replacement windows can result in the loss of the original character of the building.

Code: • Obsolescence > Aesthetic Weight score: 0
Search A - Window maintenance and component repair > A-GS-08, p. 1
Created: 23.6.2026 r. 9:11 by user, Modified: 23.6.2026 r. 9:11 by user
Other codes assigned to segment:
Circular strategy > Repair (Weight: 0)
Circular strategy > Refurbishment (Weight: 0)
Obsolescence > Functional (Weight: 0)
Negative mechanism > Premature replacement (Weight: 0)
Mitigation > Lifecycle verification (Weight: 0)

One of the primary reasons that historic steel windows are often replaced is the presumption that they cannot be made more energy efficient. Another is the fact that there are myriad replacement options on the market, including aluminum and new steel windows. However, the sightlines, proportions of elements, profiles, and shadow lines of many of the modern replacements do not match the details of the original windows. Hence, the substitution of replacement windows can result in the loss of the original character of the building.

Code: • Circular strategy > Refurbishment Weight score: 0
Search A - Window maintenance and component repair > A-GS-08, p. 1
Created: 23.6.2026 r. 9:11 by user, Modified: 23.6.2026 r. 9:11 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Obsolescence > Aesthetic (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

One of the primary reasons that historic steel windows are often replaced is the presumption that they cannot be made more energy efficient. Another is the fact that there are myriad replacement options on the market, including aluminum and new steel windows. However, the sightlines, proportions of elements, profiles, and shadow lines of many of the modern replacements do not match the details of the original windows. Hence, the substitution of replacement windows can result in the loss of the original character of the building.

Code: • Obsolescence > Functional Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 1

Created: 23.6.2026 r. 9:11 by user, Modified: 23.6.2026 r. 9:11 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Obsolescence > Aesthetic (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

One of the primary reasons that historic steel windows are often replaced is the presumption that they cannot be made more energy efficient. Another is the fact that there are myriad replacement options on the market, including aluminum and new steel windows. However, the sightlines, proportions of elements, profiles, and shadow lines of many of the modern replacements do not match the details of the original windows. Hence, the substitution of replacement windows can result in the loss of the original character of the building.

Code: • Negative mechanism > Premature replacement Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 1

Created: 23.6.2026 r. 9:11 by user, Modified: 23.6.2026 r. 9:11 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Obsolescence > Aesthetic (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Beyond the aesthetic benefits of restoring steel windows, the repair and/or retrofit of existing steel windows is often more economical than complete replacement. Some purchasing ramifications, such as trade tariffs or carbon credits, should also be considered in this decision. Nevertheless, there are situations in which window replacement makes sense, such as in buildings where the original windows are not character-defining features and good replacements with higher energy efficiencies are available. A new purchase can also be considered when the windows have deteriorated to the point where restoration (and potential abatement of any hazardous materials found) would not be economically feasible.

Code: • Circular strategy > Repair Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 2

Created: 23.6.2026 r. 9:12 by user, Modified: 23.6.2026 r. 9:12 by user

Other codes assigned to segment:

Circular strategy > Upgrade (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Obsolescence > Economic (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)
Mitigation > Lifecycle verification (Weight: 0)

Beyond the aesthetic benefits of restoring steel windows, the repair and/or retrofit of existing steel windows is often more economical than complete replacement. Some purchasing ramifications, such as trade tariffs or carbon credits, should also be considered in this decision. Nevertheless, there are situations in which window replacement makes sense, such as in buildings where the original windows are not character-defining features and good replacements with higher energy efficiencies are available. A new purchase can also be considered when the windows have deteriorated to the point where restoration (and potential abatement of any hazardous materials found) would not be economically feasible.

Code: • Mitigation > Lifecycle verification Weight score: 0
Search A - Window maintenance and component repair > A-GS-08, p. 2
Created: 23.6.2026 r. 9:13 by user, Modified: 23.6.2026 r. 9:13 by user
Other codes assigned to segment:
Circular strategy > Repair (Weight: 0)
Circular strategy > Upgrade (Weight: 0)
Circular strategy > Refurbishment (Weight: 0)
Obsolescence > Physical (Weight: 0)
Obsolescence > Economic (Weight: 0)
Negative mechanism > Premature replacement (Weight: 0)

Beyond the aesthetic benefits of restoring steel windows, the repair and/or retrofit of existing steel windows is often more economical than complete replacement. Some purchasing ramifications, such as trade tariffs or carbon credits, should also be considered in this decision. Nevertheless, there are situations in which window replacement makes sense, such as in buildings where the original windows are not character-defining features and good replacements with higher energy efficiencies are available. A new purchase can also be considered when the windows have deteriorated to the point where restoration (and potential abatement of any hazardous materials found) would not be economically feasible.

Code: • Circular strategy > Upgrade Weight score: 0
Search A - Window maintenance and component repair > A-GS-08, p. 2
Created: 23.6.2026 r. 9:13 by user, Modified: 23.6.2026 r. 9:13 by user
Other codes assigned to segment:
Circular strategy > Repair (Weight: 0)
Circular strategy > Refurbishment (Weight: 0)
Obsolescence > Physical (Weight: 0)
Obsolescence > Economic (Weight: 0)
Negative mechanism > Premature replacement (Weight: 0)
Mitigation > Lifecycle verification (Weight: 0)

Beyond the aesthetic benefits of restoring steel windows, the repair and/or retrofit of existing steel windows is often more economical than complete replacement. Some purchasing ramifications, such as trade tariffs or carbon credits, should also be considered in this decision. Nevertheless, there are situations in which window replacement makes sense, such as in buildings where the original windows are not character-defining features and good replacements with higher energy efficiencies are available. A new purchase can also be considered when the windows have deteriorated to the point where restoration (and potential abatement of any hazardous materials found) would not be economically feasible.

Code: • Obsolescence > Physical Weight score: 0
Search A - Window maintenance and component repair > A-GS-08, p. 2
Created: 23.6.2026 r. 9:13 by user, Modified: 23.6.2026 r. 9:13 by user
Other codes assigned to segment:
Circular strategy > Repair (Weight: 0)

Circular strategy > Upgrade (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Obsolescence > Economic (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Beyond the aesthetic benefits of restoring steel windows, the repair and/or retrofit of existing steel windows is often more economical than complete replacement. Some purchasing ramifications, such as trade tariffs or carbon credits, should also be considered in this decision. Nevertheless, there are situations in which window replacement makes sense, such as in buildings where the original windows are not character-defining features and good replacements with higher energy efficiencies are available. A new purchase can also be considered when the windows have deteriorated to the point where restoration (and potential abatement of any hazardous materials found) would not be economically feasible.

Code: • Negative mechanism > Premature replacement Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 2

Created: 23.6.2026 r. 9:13 by user, Modified: 23.6.2026 r. 9:13 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Obsolescence > Economic (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Beyond the aesthetic benefits of restoring steel windows, the repair and/or retrofit of existing steel windows is often more economical than complete replacement. Some purchasing ramifications, such as trade tariffs or carbon credits, should also be considered in this decision. Nevertheless, there are situations in which window replacement makes sense, such as in buildings where the original windows are not character-defining features and good replacements with higher energy efficiencies are available. A new purchase can also be considered when the windows have deteriorated to the point where restoration (and potential abatement of any hazardous materials found) would not be economically feasible.

Code: • Obsolescence > Economic Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 2

Created: 23.6.2026 r. 9:13 by user, Modified: 23.6.2026 r. 9:13 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Beyond the aesthetic benefits of restoring steel windows, the repair and/or retrofit of existing steel windows is often more economical than complete replacement. Some purchasing ramifications, such as trade tariffs or carbon credits, should also be considered in this decision. Nevertheless, there are situations in which window replacement makes sense, such as in buildings where the original windows are not character-defining features and good replacements with higher energy efficiencies are available. A new purchase can also be considered when the windows have deteriorated to the point where restoration (and potential abatement of any hazardous materials found) would not be economically feasible.

Code: • Circular strategy > Refurbishment Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 2

Created: 23.6.2026 r. 9:12 by user, Modified: 23.6.2026 r. 9:12 by user

Other codes assigned to segment:

- Circular strategy > Repair (Weight: 0)
- Circular strategy > Upgrade (Weight: 0)
- Obsolescence > Physical (Weight: 0)
- Obsolescence > Economic (Weight: 0)
- Negative mechanism > Premature replacement (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

When analyzing a project, the first step is to determine whether the windows should be restored or replaced. It is essential to work with a preservation professional familiar with the style and period of the building to assess whether the steel windows are a character-defining feature whose loss would diminish the aesthetic or confuse the building's original design intent. Major differences between historic steel windows and new 3

PRACTICE POINTS 29

steel or aluminum replacements relate to the scale and dimensions of the new window members—stiles, muntins, mullions, shadow lines, and so on. Even if the building's use changes (e.g., from manufacturing to residential), if the windows are central to its historic character, they should be retained and restored.

Code: • Circular strategy > Refurbishment Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 2 - 3

Created: 23.6.2026 r. 9:15 by user, Modified: 23.6.2026 r. 9:15 by user

Other codes assigned to segment:

- Obsolescence > Aesthetic (Weight: 0)
- Enabling condition > Repair information (Weight: 0)
- Enabling condition > Skilled labour (Weight: 0)
- Enabling condition > User acceptance (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

When analyzing a project, the first step is to determine whether the windows should be restored or replaced. It is essential to work with a preservation professional familiar with the style and period of the building to assess whether the steel windows are a character-defining feature whose loss would diminish the aesthetic or confuse the building's original design intent. Major differences between historic steel windows and new 3

PRACTICE POINTS 29

steel or aluminum replacements relate to the scale and dimensions of the new window members—stiles, muntins, mullions, shadow lines, and so on. Even if the building's use changes (e.g., from manufacturing to residential), if the windows are central to its historic character, they should be retained and restored.

Code: • Enabling condition > Skilled labour Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 2 - 3

Created: 23.6.2026 r. 9:15 by user, Modified: 23.6.2026 r. 9:15 by user

Other codes assigned to segment:

- Circular strategy > Refurbishment (Weight: 0)
- Obsolescence > Aesthetic (Weight: 0)
- Enabling condition > Repair information (Weight: 0)
- Enabling condition > User acceptance (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

When analyzing a project, the first step is to determine whether the windows should be restored or replaced. It is essential to work with a preservation professional familiar with the style and period of the building to assess whether the steel windows are a character-defining feature whose loss would diminish the aesthetic or confuse the building's original design intent. Major differences between historic steel windows and new 3

PRACTICE POINTS 29

steel or aluminum replacements relate to the scale and dimensions of the new window members—stiles, muntins, mullions, shadow lines, and so on. Even if the building's use changes (e.g., from manufacturing to residential), if the windows are central to its historic character, they should be retained and restored.

Code: • Enabling condition > User acceptance Weight score: 0
 Search A - Window maintenance and component repair > A-GS-08, p. 2 - 3
 Created: 23.6.2026 r. 9:16 by user, Modified: 23.6.2026 r. 9:16 by user
 Other codes assigned to segment:
 Circular strategy > Refurbishment (Weight: 0)
 Obsolescence > Aesthetic (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Skilled labour (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

When analyzing a project, the first step is to determine whether the windows should be restored or replaced. It is essential to work with a preservation professional familiar with the style and period of the building to assess whether the steel windows are a character-defining feature whose loss would diminish the aesthetic or confuse the building's original design intent. Major differences between historic steel windows and new 3

PRACTICE POINTS 29

steel or aluminum replacements relate to the scale and dimensions of the new window members—stiles, muntins, mullions, shadow lines, and so on. Even if the building's use changes (e.g., from manufacturing to residential), if the windows are central to its historic character, they should be retained and restored.

Code: • Obsolescence > Aesthetic Weight score: 0
 Search A - Window maintenance and component repair > A-GS-08, p. 2 - 3
 Created: 23.6.2026 r. 9:16 by user, Modified: 23.6.2026 r. 9:16 by user
 Other codes assigned to segment:
 Circular strategy > Refurbishment (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Skilled labour (Weight: 0)
 Enabling condition > User acceptance (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

When analyzing a project, the first step is to determine whether the windows should be restored or replaced. It is essential to work with a preservation professional familiar with the style and period of the building to assess whether the steel windows are a character-defining feature whose loss would diminish the aesthetic or confuse the building's original design intent. Major differences between historic steel windows and new 3

PRACTICE POINTS 29

steel or aluminum replacements relate to the scale and dimensions of the new window members—stiles, muntins, mullions, shadow lines, and so on. Even if the building's use changes (e.g., from manufacturing to residential), if the windows are central to its historic character, they should be retained and restored.

Code: • Mitigation > Lifecycle verification Weight score: 0
 Search A - Window maintenance and component repair > A-GS-08, p. 2 - 3
 Created: 23.6.2026 r. 9:16 by user, Modified: 23.6.2026 r. 9:16 by user
 Other codes assigned to segment:
 Circular strategy > Refurbishment (Weight: 0)
 Obsolescence > Aesthetic (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Skilled labour (Weight: 0)
 Enabling condition > User acceptance (Weight: 0)

When analyzing a project, the first step is to determine whether the windows should be restored or replaced. It is essential to work with a preservation professional familiar with the style and period of the building to assess whether the steel windows are a character-defining feature whose loss would diminish the aesthetic or confuse the building's original design intent. Major differences between historic steel windows and new 3

PRACTICE POINTS 29

steel or aluminum replacements relate to the scale and dimensions of the new window members—stiles, muntins, mullions, shadow lines, and so on. Even if the building's use changes (e.g., from manufacturing to residential), if the windows are central to its historic character, they should be retained and restored.

Code: • Enabling condition > Repair information Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 2 - 3

Created: 23.6.2026 r. 9:15 by user, Modified: 23.6.2026 r. 9:15 by user

Other codes assigned to segment:

Circular strategy > Refurbishment (Weight: 0)

Obsolescence > Aesthetic (Weight: 0)

Enabling condition > Skilled labour (Weight: 0)

Enabling condition > User acceptance (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Surface rust should not deter restoration, as it usually looks worse than it is because oxidized steel is much thicker than the original metal. Unless severe corrosion has caused major loss of material and/or complete loss of frame and sash members, restoration remains a viable option. Even when surface corrosion progresses to "rust jacking," where expansion causes displacement, restoration can often still be successful and economical.

Code: • Mitigation > Maintenance planning Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 3

Created: 23.6.2026 r. 9:18 by user, Modified: 23.6.2026 r. 9:18 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)

Obsolescence > Physical (Weight: 0)

Enabling condition > Skilled labour (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Surface rust should not deter restoration, as it usually looks worse than it is because oxidized steel is much thicker than the original metal. Unless severe corrosion has caused major loss of material and/or complete loss of frame and sash members, restoration remains a viable option. Even when surface corrosion progresses to "rust jacking," where expansion causes displacement, restoration can often still be successful and economical.

Code: • Negative mechanism > Premature replacement Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 3

Created: 23.6.2026 r. 9:17 by user, Modified: 23.6.2026 r. 9:17 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)

Obsolescence > Physical (Weight: 0)

Enabling condition > Skilled labour (Weight: 0)

Mitigation > Maintenance planning (Weight: 0)

Surface rust should not deter restoration, as it usually looks worse than it is because oxidized steel is much thicker than the original metal. Unless severe corrosion has caused major loss of material and/or complete loss of frame and sash members, restoration remains a viable option. Even when

surface corrosion progresses to “rust jacking,” where expansion causes displacement, restoration can often still be successful and economical.

Code: • Circular strategy > Repair Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 3

Created: 23.6.2026 r. 9:17 by user, Modified: 23.6.2026 r. 9:17 by user

Other codes assigned to segment:

Circular strategy > Refurbishment (Weight: 0)

Obsolescence > Physical (Weight: 0)

Enabling condition > Skilled labour (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Mitigation > Maintenance planning (Weight: 0)

Surface rust should not deter restoration, as it usually looks worse than it is because oxidized steel is much thicker than the original metal. Unless severe corrosion has caused major loss of material and/or complete loss of frame and sash members, restoration remains a viable option. Even when surface corrosion progresses to “rust jacking,” where expansion causes displacement, restoration can often still be successful and economical.

Code: • Circular strategy > Refurbishment Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 3

Created: 23.6.2026 r. 9:17 by user, Modified: 23.6.2026 r. 9:17 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)

Obsolescence > Physical (Weight: 0)

Enabling condition > Skilled labour (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Mitigation > Maintenance planning (Weight: 0)

Surface rust should not deter restoration, as it usually looks worse than it is because oxidized steel is much thicker than the original metal. Unless severe corrosion has caused major loss of material and/or complete loss of frame and sash members, restoration remains a viable option. Even when surface corrosion progresses to “rust jacking,” where expansion causes displacement, restoration can often still be successful and economical.

Code: • Enabling condition > Skilled labour Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 3

Created: 23.6.2026 r. 9:17 by user, Modified: 23.6.2026 r. 9:17 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)

Obsolescence > Physical (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Mitigation > Maintenance planning (Weight: 0)

Surface rust should not deter restoration, as it usually looks worse than it is because oxidized steel is much thicker than the original metal. Unless severe corrosion has caused major loss of material and/or complete loss of frame and sash members, restoration remains a viable option. Even when surface corrosion progresses to “rust jacking,” where expansion causes displacement, restoration can often still be successful and economical.

Code: • Obsolescence > Physical Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 3

Created: 23.6.2026 r. 9:17 by user, Modified: 23.6.2026 r. 9:17 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)

Enabling condition > Skilled labour (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)
Mitigation > Maintenance planning (Weight: 0)

IGU and VIG retrofit. In some cases, it is possible to replace existing monolithic glass with a more efficient insulated glass unit (IGU) or vacuuminsulated glass (VIG), depending on the dimensions of the existing window system (Fig. 7). First, confirm that the glazing pocket has sufficient depth to accommodate the proposed unit. If the space is adequate, the next consideration is weight: IGUs can weigh two to three times more than monolithic glass, which may place significant stress on the supporting hinges of the sash. In casement windows, wider units relative to their height exert increased stress on operating hinges, potentially affecting long-term functionality and durability.

Code: • Enabling condition > Component accessibility Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 8

Created: 23.6.2026 r. 9:19 by user, Modified: 23.6.2026 r. 9:19 by user

Other codes assigned to segment:

Circular strategy > Upgrade (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)

Obsolescence > Functional (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Negative mechanism > Increased material complexity (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

IGU and VIG retrofit. In some cases, it is possible to replace existing monolithic glass with a more efficient insulated glass unit (IGU) or vacuuminsulated glass (VIG), depending on the dimensions of the existing window system (Fig. 7). First, confirm that the glazing pocket has sufficient depth to accommodate the proposed unit. If the space is adequate, the next consideration is weight: IGUs can weigh two to three times more than monolithic glass, which may place significant stress on the supporting hinges of the sash. In casement windows, wider units relative to their height exert increased stress on operating hinges, potentially affecting long-term functionality and durability.

Code: • Negative mechanism > Increased material complexity Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 8

Created: 23.6.2026 r. 9:20 by user, Modified: 23.6.2026 r. 9:20 by user

Other codes assigned to segment:

Circular strategy > Upgrade (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)

Obsolescence > Functional (Weight: 0)

Enabling condition > Component accessibility (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

IGU and VIG retrofit. In some cases, it is possible to replace existing monolithic glass with a more efficient insulated glass unit (IGU) or vacuuminsulated glass (VIG), depending on the dimensions of the existing window system (Fig. 7). First, confirm that the glazing pocket has sufficient depth to accommodate the proposed unit. If the space is adequate, the next consideration is weight: IGUs can weigh two to three times more than monolithic glass, which may place significant stress on the supporting hinges of the sash. In casement windows, wider units relative to their height exert increased stress on operating hinges, potentially affecting long-term functionality and durability.

Code: • Circular strategy > Refurbishment Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 8

Created: 23.6.2026 r. 9:19 by user, Modified: 23.6.2026 r. 9:19 by user

Other codes assigned to segment:

- Circular strategy > Upgrade (Weight: 0)
- Obsolescence > Functional (Weight: 0)
- Enabling condition > Component accessibility (Weight: 0)
- Enabling condition > Repair information (Weight: 0)
- Negative mechanism > Increased material complexity (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

IGU and VIG retrofit. In some cases, it is possible to replace existing monolithic glass with a more efficient insulated glass unit (IGU) or vacuuminsulated glass (VIG), depending on the dimensions of the existing window system (Fig. 7). First, confirm that the glazing pocket has sufficient depth to accommodate the proposed unit. If the space is adequate, the next consideration is weight: IGUs can weigh two to three times more than monolithic glass, which may place significant stress on the supporting hinges of the sash. In casement windows, wider units relative to their height exert increased stress on operating hinges, potentially affecting long-term functionality and durability.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 8

Created: 23.6.2026 r. 9:20 by user, Modified: 23.6.2026 r. 9:20 by user

Other codes assigned to segment:

- Circular strategy > Upgrade (Weight: 0)
- Circular strategy > Refurbishment (Weight: 0)
- Obsolescence > Functional (Weight: 0)
- Enabling condition > Component accessibility (Weight: 0)
- Enabling condition > Repair information (Weight: 0)
- Negative mechanism > Increased material complexity (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)

IGU and VIG retrofit. In some cases, it is possible to replace existing monolithic glass with a more efficient insulated glass unit (IGU) or vacuuminsulated glass (VIG), depending on the dimensions of the existing window system (Fig. 7). First, confirm that the glazing pocket has sufficient depth to accommodate the proposed unit. If the space is adequate, the next consideration is weight: IGUs can weigh two to three times more than monolithic glass, which may place significant stress on the supporting hinges of the sash. In casement windows, wider units relative to their height exert increased stress on operating hinges, potentially affecting long-term functionality and durability.

Code: • Circular strategy > Upgrade Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 8

Created: 23.6.2026 r. 9:19 by user, Modified: 23.6.2026 r. 9:19 by user

Other codes assigned to segment:

- Circular strategy > Refurbishment (Weight: 0)
- Obsolescence > Functional (Weight: 0)
- Enabling condition > Component accessibility (Weight: 0)
- Enabling condition > Repair information (Weight: 0)
- Negative mechanism > Increased material complexity (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

IGU and VIG retrofit. In some cases, it is possible to replace existing monolithic glass with a more efficient insulated glass unit (IGU) or vacuuminsulated glass (VIG), depending on the dimensions of the existing window system (Fig. 7). First, confirm that the glazing pocket has sufficient depth to accommodate the proposed unit. If the space is adequate, the next consideration is weight: IGUs can weigh two to three times more than monolithic glass, which may place significant stress on the supporting hinges of the sash. In casement

windows, wider units relative to their height exert increased stress on operating hinges, potentially affecting long-term functionality and durability.

Code: • [Obsolescence > Functional](#) Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 8

Created: 23.6.2026 r. 9:19 by user, Modified: 23.6.2026 r. 9:19 by user

Other codes assigned to segment:

[Circular strategy > Upgrade](#) (Weight: 0)

[Circular strategy > Refurbishment](#) (Weight: 0)

[Enabling condition > Component accessibility](#) (Weight: 0)

[Enabling condition > Repair information](#) (Weight: 0)

[Negative mechanism > Increased material complexity](#) (Weight: 0)

[Negative mechanism > Burden shifting](#) (Weight: 0)

[Mitigation > Lifecycle verification](#) (Weight: 0)

IGU and VIG retrofit. In some cases, it is possible to replace existing monolithic glass with a more efficient insulated glass unit (IGU) or vacuuminsulated glass (VIG), depending on the dimensions of the existing window system (Fig. 7). First, confirm that the glazing pocket has sufficient depth to accommodate the proposed unit. If the space is adequate, the next consideration is weight: IGUs can weigh two to three times more than monolithic glass, which may place significant stress on the supporting hinges of the sash. In casement windows, wider units relative to their height exert increased stress on operating hinges, potentially affecting long-term functionality and durability.

Code: • [Negative mechanism > Burden shifting](#) Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 8

Created: 23.6.2026 r. 9:20 by user, Modified: 23.6.2026 r. 9:20 by user

Other codes assigned to segment:

[Circular strategy > Upgrade](#) (Weight: 0)

[Circular strategy > Refurbishment](#) (Weight: 0)

[Obsolescence > Functional](#) (Weight: 0)

[Enabling condition > Component accessibility](#) (Weight: 0)

[Enabling condition > Repair information](#) (Weight: 0)

[Negative mechanism > Increased material complexity](#) (Weight: 0)

[Mitigation > Lifecycle verification](#) (Weight: 0)

IGU and VIG retrofit. In some cases, it is possible to replace existing monolithic glass with a more efficient insulated glass unit (IGU) or vacuuminsulated glass (VIG), depending on the dimensions of the existing window system (Fig. 7). First, confirm that the glazing pocket has sufficient depth to accommodate the proposed unit. If the space is adequate, the next consideration is weight: IGUs can weigh two to three times more than monolithic glass, which may place significant stress on the supporting hinges of the sash. In casement windows, wider units relative to their height exert increased stress on operating hinges, potentially affecting long-term functionality and durability.

Code: • [Enabling condition > Repair information](#) Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 8

Created: 23.6.2026 r. 9:20 by user, Modified: 23.6.2026 r. 9:20 by user

Other codes assigned to segment:

[Circular strategy > Upgrade](#) (Weight: 0)

[Circular strategy > Refurbishment](#) (Weight: 0)

[Obsolescence > Functional](#) (Weight: 0)

[Enabling condition > Component accessibility](#) (Weight: 0)

[Negative mechanism > Increased material complexity](#) (Weight: 0)

[Negative mechanism > Burden shifting](#) (Weight: 0)

[Mitigation > Lifecycle verification](#) (Weight: 0)

Storm windows have been used for over a century to improve the performance of existing windows. For less than the cost of a full restoration, replication, or replacement, storm windows can reduce water infiltration, improve energy efficiency, and even lessen street noise. Installation may occur on either the interior or exterior of the window, and window operability should be considered when selecting placement and type. Exterior installation is generally preferred when water infiltration is the primary concern. Interior installation is generally favored when air infiltration and thermal comfort are the main issues. Additional factors should also be considered, including condensation potential, compatibility of dissimilar metals, glass types, and sound attenuation, among others

Code: • Circular strategy > Upgrade Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 8

Created: 23.6.2026 r. 9:21 by user, Modified: 23.6.2026 r. 9:21 by user

Other codes assigned to segment:

Circular strategy > Refurbishment (Weight: 0)

Obsolescence > Functional (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Negative mechanism > Increased material complexity (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Storm windows have been used for over a century to improve the performance of existing windows. For less than the cost of a full restoration, replication, or replacement, storm windows can reduce water infiltration, improve energy efficiency, and even lessen street noise. Installation may occur on either the interior or exterior of the window, and window operability should be considered when selecting placement and type. Exterior installation is generally preferred when water infiltration is the primary concern. Interior installation is generally favored when air infiltration and thermal comfort are the main issues. Additional factors should also be considered, including condensation potential, compatibility of dissimilar metals, glass types, and sound attenuation, among others

Code: • Obsolescence > Functional Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 8

Created: 23.6.2026 r. 9:22 by user, Modified: 23.6.2026 r. 9:22 by user

Other codes assigned to segment:

Circular strategy > Upgrade (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Negative mechanism > Increased material complexity (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Storm windows have been used for over a century to improve the performance of existing windows. For less than the cost of a full restoration, replication, or replacement, storm windows can reduce water infiltration, improve energy efficiency, and even lessen street noise. Installation may occur on either the interior or exterior of the window, and window operability should be considered when selecting placement and type. Exterior installation is generally preferred when water infiltration is the primary concern. Interior installation is generally favored when air infiltration and thermal comfort are the main issues. Additional factors should also be considered, including condensation potential, compatibility of dissimilar metals, glass types, and sound attenuation, among others

Code: • Negative mechanism > Burden shifting Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 8

Created: 23.6.2026 r. 9:22 by user, Modified: 23.6.2026 r. 9:22 by user

Other codes assigned to segment:

Circular strategy > Upgrade (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Negative mechanism > Increased material complexity (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Storm windows have been used for over a century to improve the performance of existing windows. For less than the cost of a full restoration, replication, or replacement, storm windows can reduce water infiltration, improve energy efficiency, and even lessen street noise. Installation may occur on either the interior or exterior of the window, and window operability should be considered when selecting placement and type. Exterior installation is generally preferred when water infiltration is the primary concern. Interior installation is generally favored when air infiltration and thermal comfort are the main issues. Additional factors should also be considered, including condensation potential, compatibility of dissimilar metals, glass types, and sound attenuation, among others

Code: • Circular strategy > Refurbishment Weight score: 0
 Search A - Window maintenance and component repair > A-GS-08, p. 8
 Created: 23.6.2026 r. 9:21 by user, Modified: 23.6.2026 r. 9:21 by user
 Other codes assigned to segment:
 Circular strategy > Upgrade (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Negative mechanism > Increased material complexity (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Storm windows have been used for over a century to improve the performance of existing windows. For less than the cost of a full restoration, replication, or replacement, storm windows can reduce water infiltration, improve energy efficiency, and even lessen street noise. Installation may occur on either the interior or exterior of the window, and window operability should be considered when selecting placement and type. Exterior installation is generally preferred when water infiltration is the primary concern. Interior installation is generally favored when air infiltration and thermal comfort are the main issues. Additional factors should also be considered, including condensation potential, compatibility of dissimilar metals, glass types, and sound attenuation, among others

Code: • Enabling condition > Repair information Weight score: 0
 Search A - Window maintenance and component repair > A-GS-08, p. 8
 Created: 23.6.2026 r. 9:22 by user, Modified: 23.6.2026 r. 9:22 by user
 Other codes assigned to segment:
 Circular strategy > Upgrade (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Negative mechanism > Increased material complexity (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Storm windows have been used for over a century to improve the performance of existing windows. For less than the cost of a full restoration, replication, or replacement, storm windows can reduce water infiltration, improve energy efficiency, and even lessen street noise. Installation may occur on either the interior or exterior of the window, and window operability should be considered when selecting placement and type. Exterior installation is generally preferred when water infiltration is the primary concern. Interior installation is generally favored when air infiltration and thermal comfort are the main

issues. Additional factors should also be considered, including condensation potential, compatibility of dissimilar metals, glass types, and sound attenuation, among others

Code: • Mitigation > Lifecycle verification Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 8

Created: 23.6.2026 r. 9:22 by user, Modified: 23.6.2026 r. 9:22 by user

Other codes assigned to segment:

Circular strategy > Upgrade (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)

Obsolescence > Functional (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Negative mechanism > Increased material complexity (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Storm windows have been used for

over a century to improve the performance of existing windows. For less than the cost of a full restoration, replication, or replacement, storm windows can reduce water infiltration, improve energy efficiency, and even lessen street noise. Installation may occur on either the interior or exterior of the window, and window operability should be considered when selecting placement and type. Exterior installation is generally preferred when water infiltration is the primary concern. Interior installation is generally favored when air infiltration and thermal comfort are the main issues. Additional factors should also be considered, including condensation potential, compatibility of dissimilar metals, glass types, and sound attenuation, among others

Code: • Negative mechanism > Increased material complexity Weight score: 0

Search A - Window maintenance and component repair > A-GS-08, p. 8

Created: 23.6.2026 r. 9:22 by user, Modified: 23.6.2026 r. 9:22 by user

Other codes assigned to segment:

Circular strategy > Upgrade (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)

Obsolescence > Functional (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The results indicate that the embodied environmental impact of the wood material alone remains relatively small while glazing and aluminium cladding dominate. Regarding the surface treatment, the varnish quantity as well as the presence of paint do not significantly influence the environmental impact. Differences between operable and fixed windows also reflect additional materials and hardware requirements, resulting in operable windows exhibiting higher environmental impacts across all assessed categories. The findings of this study highlight the important role of structural elements and additional components on the overall environmental impact regarding the complexity of a window

Code: • Obsolescence > Functional Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 1

Created: 23.6.2026 r. 10:14 by user, Modified: 23.6.2026 r. 10:14 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Negative mechanism > Increased material complexity (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The results indicate that the embodied environmental impact of the wood material alone remains relatively small while glazing and aluminium cladding dominate. Regarding the surface treatment, the

varnish quantity as well as the presence of paint do not significantly influence the environmental impact. Differences between operable and fixed windows also reflect additional materials and hardware requirements, resulting in operable windows exhibiting higher environmental impacts across all assessed categories. The findings of this study highlight the important role of structural elements and additional components on the overall environmental impact regarding the complexity of a window

Code: • Negative mechanism > Increased material complexity Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 1

Created: 23.6.2026 r. 10:14 by user, Modified: 23.6.2026 r. 10:14 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Functional (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The results indicate that the embodied environmental impact of the wood material alone remains relatively small while glazing and aluminium cladding dominate. Regarding the surface treatment, the varnish quantity as well as the presence of paint do not significantly influence the environmental impact. Differences between operable and fixed windows also reflect additional materials and hardware requirements, resulting in operable windows exhibiting higher environmental impacts across all assessed categories. The findings of this study highlight the important role of structural elements and additional components on the overall environmental impact regarding the complexity of a window

Code: • Negative mechanism > Burden shifting Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 1

Created: 23.6.2026 r. 10:14 by user, Modified: 23.6.2026 r. 10:14 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Functional (Weight: 0)

Negative mechanism > Increased material complexity (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The results indicate that the embodied environmental impact of the wood material alone remains relatively small while glazing and aluminium cladding dominate. Regarding the surface treatment, the varnish quantity as well as the presence of paint do not significantly influence the environmental impact. Differences between operable and fixed windows also reflect additional materials and hardware requirements, resulting in operable windows exhibiting higher environmental impacts across all assessed categories. The findings of this study highlight the important role of structural elements and additional components on the overall environmental impact regarding the complexity of a window

Code: • Mitigation > Lifecycle verification Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 1

Created: 23.6.2026 r. 10:14 by user, Modified: 23.6.2026 r. 10:14 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Functional (Weight: 0)

Negative mechanism > Increased material complexity (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

The results indicate that the embodied environmental impact of the wood material alone remains relatively small while glazing and aluminium cladding dominate. Regarding the surface treatment, the varnish quantity as well as the presence of paint do not significantly influence the environmental impact. Differences between operable and fixed windows also reflect additional materials and hardware requirements, resulting in operable windows exhibiting higher environmental impacts across all assessed categories. The findings of this study highlight the important role of structural elements and additional components on the overall environmental impact regarding the complexity of a window

Code: • Negative mechanism > Higher initial material use Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 1

Created: 23.6.2026 r. 10:14 by user, Modified: 23.6.2026 r. 10:14 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Functional (Weight: 0)

Negative mechanism > Increased material complexity (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The results indicate that the embodied environmental impact of the wood material alone remains relatively small while glazing and aluminium cladding dominate. Regarding the surface treatment, the varnish quantity as well as the presence of paint do not significantly influence the environmental impact. Differences between operable and fixed windows also reflect additional materials and hardware requirements, resulting in operable windows exhibiting higher environmental impacts across all assessed categories. The findings of this study highlight the important role of structural elements and additional components on the overall environmental impact regarding the complexity of a window.

Code: • Circular strategy > Upgrade Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 1

Created: 23.6.2026 r. 10:12 by user, Modified: 23.6.2026 r. 10:12 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Obsolescence > Functional (Weight: 0)

Negative mechanism > Increased material complexity (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The results indicate that the embodied environmental impact of the wood material alone remains relatively small while glazing and aluminium cladding dominate. Regarding the surface treatment, the varnish quantity as well as the presence of paint do not significantly influence the environmental impact. Differences between operable and fixed windows also reflect additional materials and hardware requirements, resulting in operable windows exhibiting higher environmental impacts across all assessed categories. The findings of this study highlight the important role of structural elements and additional components on the overall environmental impact regarding the complexity of a window.

Code: • Circular strategy > Maintenance Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 1

Created: 23.6.2026 r. 10:12 by user, Modified: 23.6.2026 r. 10:12 by user

Other codes assigned to segment:

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Functional (Weight: 0)

Negative mechanism > Increased material complexity (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)
Mitigation > Lifecycle verification (Weight: 0)

The addition of aluminium cladding presents an important feature that not only significantly extends the service life but also embodies the greatest environmental impact of all window components [29]. Deciding between a purely wooden window and a wood-aluminium solution, therefore, requires careful consideration of the life cycle and local maintenance conditions. Therefore, the study encompasses this aspect as a relevant component of the environmental evaluation.

Code: • Circular strategy > Maintenance Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 3

Created: 23.6.2026 r. 10:15 by user, Modified: 23.6.2026 r. 10:15 by user

Other codes assigned to segment:

Circular strategy > Upgrade (Weight: 0)
Obsolescence > Physical (Weight: 0)
Negative mechanism > Higher initial material use (Weight: 0)
Negative mechanism > Burden shifting (Weight: 0)
Mitigation > Lifecycle verification (Weight: 0)

The addition of aluminium cladding presents an important feature that not only significantly extends the service life but also embodies the greatest environmental impact of all window components [29]. Deciding between a purely wooden window and a wood-aluminium solution, therefore, requires careful consideration of the life cycle and local maintenance conditions. Therefore, the study encompasses this aspect as a relevant component of the environmental evaluation.

Code: • Circular strategy > Upgrade Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 3

Created: 23.6.2026 r. 10:15 by user, Modified: 23.6.2026 r. 10:15 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)
Obsolescence > Physical (Weight: 0)
Negative mechanism > Higher initial material use (Weight: 0)
Negative mechanism > Burden shifting (Weight: 0)
Mitigation > Lifecycle verification (Weight: 0)

The addition of aluminium cladding presents an important feature that not only significantly extends the service life but also embodies the greatest environmental impact of all window components [29]. Deciding between a purely wooden window and a wood-aluminium solution, therefore, requires careful consideration of the life cycle and local maintenance conditions. Therefore, the study encompasses this aspect as a relevant component of the environmental evaluation.

Code: • Negative mechanism > Higher initial material use Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 3

Created: 23.6.2026 r. 10:15 by user, Modified: 23.6.2026 r. 10:15 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)
Circular strategy > Upgrade (Weight: 0)
Obsolescence > Physical (Weight: 0)
Negative mechanism > Burden shifting (Weight: 0)
Mitigation > Lifecycle verification (Weight: 0)

The addition of aluminium cladding presents an important feature that not only significantly extends the service life but also embodies the greatest environmental impact of all window components [29]. Deciding between a purely wooden window and a wood-aluminium solution, therefore, requires careful consideration of the life cycle and local maintenance conditions. Therefore, the study encompasses this aspect as a relevant component of the environmental evaluation.

Code: • Negative mechanism > Burden shifting Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 3
 Created: 23.6.2026 r. 10:15 by user, Modified: 23.6.2026 r. 10:15 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The addition of aluminium cladding presents an important feature that not only significantly extends the service life but also embodies the greatest environmental impact of all window components [29]. Deciding between a purely wooden window and a wood-aluminium solution, therefore, requires careful consideration of the life cycle and local maintenance conditions. Therefore, the study encompasses this aspect as a relevant component of the environmental evaluation.

Code: • Obsolescence > Physical Weight score: 0
 Search A - Window maintenance and component repair > A-GS-11, p. 3
 Created: 23.6.2026 r. 10:15 by user, Modified: 23.6.2026 r. 10:15 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The addition of aluminium cladding presents an important feature that not only significantly extends the service life but also embodies the greatest environmental impact of all window components [29]. Deciding between a purely wooden window and a wood-aluminium solution, therefore, requires careful consideration of the life cycle and local maintenance conditions. Therefore, the study encompasses this aspect as a relevant component of the environmental evaluation.

Code: • Mitigation > Lifecycle verification Weight score: 0
 Search A - Window maintenance and component repair > A-GS-11, p. 3
 Created: 23.6.2026 r. 10:15 by user, Modified: 23.6.2026 r. 10:15 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)

Surprisingly, the maintenance stage has a substantially lower impact on the environment than the production stage. In the case of acrylic varnish, this is proven for each impact category and window type. Natural oil-based varnish reflects worse impacts than the water-based acrylic varnish. Particularly, the impacts of natural varnish application on TE and POC far exceed the impacts of the windows production stage.

Code: • Circular strategy > Maintenance Weight score: 0
 Search A - Window maintenance and component repair > A-GS-11, p. 8
 Created: 23.6.2026 r. 10:18 by user, Modified: 23.6.2026 r. 10:18 by user
 Other codes assigned to segment:
 Obsolescence > Physical (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Surprisingly, the maintenance stage has a substantially lower impact on the environment than the production stage. In the case of acrylic varnish, this is proven for each impact category and window type. Natural oil-based varnish reflects worse impacts than the water-based acrylic varnish. Particularly, the impacts of natural varnish application on TE and POC far exceed the impacts of the windows production stage.

Code: • Mitigation > Maintenance planning Weight score: 0
 Search A - Window maintenance and component repair > A-GS-11, p. 8
 Created: 23.6.2026 r. 10:18 by user, Modified: 23.6.2026 r. 10:18 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Surprisingly, the maintenance stage has a substantially lower impact on the environment than the production stage. In the case of acrylic varnish, this is proven for each impact category and window type. Natural oil-based varnish reflects worse impacts than the water-based acrylic varnish. Particularly, the impacts of natural varnish application on TE and POC far exceed the impacts of the windows production stage.

Code: • Mitigation > Lifecycle verification Weight score: 0
 Search A - Window maintenance and component repair > A-GS-11, p. 8
 Created: 23.6.2026 r. 10:18 by user, Modified: 23.6.2026 r. 10:18 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)

Surprisingly, the maintenance stage has a substantially lower impact on the environment than the production stage. In the case of acrylic varnish, this is proven for each impact category and window type. Natural oil-based varnish reflects worse impacts than the water-based acrylic varnish. Particularly, the impacts of natural varnish application on TE and POC far exceed the impacts of the windows production stage.

Code: • Obsolescence > Physical Weight score: 0
 Search A - Window maintenance and component repair > A-GS-11, p. 8
 Created: 23.6.2026 r. 10:18 by user, Modified: 23.6.2026 r. 10:18 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Surprisingly, the maintenance stage has a substantially lower impact on the environment than the production stage. In the case of acrylic varnish, this is proven for each impact category and window type. Natural oil-based varnish reflects worse impacts than the water-based acrylic varnish. Particularly, the impacts of natural varnish application on TE and POC far exceed the impacts of the windows production stage.

Code: • Negative mechanism > Burden shifting Weight score: 0
 Search A - Window maintenance and component repair > A-GS-11, p. 8
 Created: 23.6.2026 r. 10:18 by user, Modified: 23.6.2026 r. 10:18 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Obsolescence > Physical (Weight: 0)

Mitigation > Maintenance planning (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The glazing refers to the most dominant wood-based window component, shaping the environmental profile, particularly for fixed windows. As expected, larger glazing areas increase the glass's relative contribution to the total environmental impact. This highlights the strong dependence of window environmental performance on glazing properties and design across scenarios. Similar conclusions regarding the dominant role of glazing are also proven by Marinello Jorba et al. [2] and Agliata et al. [18]. For further comparison, the results reported by Rabbani et al. [6] indicate that electrochromic windows production emits 49.6 kg CO₂ eq/m² of which 9.79 kg CO₂ eq comes from the glass production. In our study, the highest impact scenario (S1, operable wooden window with aluminium cladding) shows a considerably higher GWP of 120.55 kg CO₂ eq/m², including 40.10 kg CO₂ eq for the triple glazing alone. Considering the S2 operable variant (frame without cladding and without paint), the GWP drops to 59.97 kg CO₂ eq/m², with 40.6 kg CO₂ eq attributed to the glazing. Although electrochromic windows have lower impacts related to the glass production, wood-based windows, particularly without aluminium cladding, can achieve comparable impacts per square metre in terms of GWP.

Code: • Obsolescence > Functional Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 11

Created: 23.6.2026 r. 10:19 by user, Modified: 23.6.2026 r. 10:19 by user

Other codes assigned to segment:

Circular strategy > Upgrade (Weight: 0)
 Negative mechanism > Increased material complexity (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The glazing refers to the most dominant wood-based window component, shaping the environmental profile, particularly for fixed windows. As expected, larger glazing areas increase the glass's relative contribution to the total environmental impact. This highlights the strong dependence of window environmental performance on glazing properties and design across scenarios. Similar conclusions regarding the dominant role of glazing are also proven by Marinello Jorba et al. [2] and Agliata et al. [18]. For further comparison, the results reported by Rabbani et al. [6] indicate that electrochromic windows production emits 49.6 kg CO₂ eq/m² of which 9.79 kg CO₂ eq comes from the glass production. In our study, the highest impact scenario (S1, operable wooden window with aluminium cladding) shows a considerably higher GWP of 120.55 kg CO₂ eq/m², including 40.10 kg CO₂ eq for the triple glazing alone. Considering the S2 operable variant (frame without cladding and without paint), the GWP drops to 59.97 kg CO₂ eq/m², with 40.6 kg CO₂ eq attributed to the glazing. Although electrochromic windows have lower impacts related to the glass production, wood-based windows, particularly without aluminium cladding, can achieve comparable impacts per square metre in terms of GWP.

Code: • Circular strategy > Upgrade Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 11

Created: 23.6.2026 r. 10:19 by user, Modified: 23.6.2026 r. 10:19 by user

Other codes assigned to segment:

Obsolescence > Functional (Weight: 0)
 Negative mechanism > Increased material complexity (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The glazing refers to the most dominant wood-based window component, shaping the environmental profile, particularly for fixed windows. As expected, larger glazing areas increase the glass's relative contribution to the total environmental impact. This highlights the strong dependence of window environmental performance on glazing properties and design across scenarios. Similar

conclusions regarding the dominant role of glazing are also proven by Marinello Jorba et al. [2] and Agliata et al. [18]. For further comparison, the results reported by Rabbani et al. [6] indicate that electrochromic windows production emits 49.6 kg CO₂ eq/m² of which 9.79 kg CO₂ eq comes from the glass production. In our study, the highest impact scenario (S1, operable wooden window with aluminium cladding) shows a considerably higher GWP of 120.55 kg CO₂ eq/m², including 40.10 kg CO₂ eq for the triple glazing alone. Considering the S2 operable variant (frame without cladding and without paint), the GWP drops to 59.97 kg CO₂ eq/m², with 40.6 kg CO₂ eq attributed to the glazing. Although electrochromic windows have lower impacts related to the glass production, wood-based windows, particularly without aluminium cladding, can achieve comparable impacts per square metre in terms of GWP.

Code: • Negative mechanism > Higher initial material use Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 11

Created: 23.6.2026 r. 10:19 by user, Modified: 23.6.2026 r. 10:19 by user

Other codes assigned to segment:

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Functional (Weight: 0)

Negative mechanism > Increased material complexity (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The glazing refers to the most dominant wood-based window component, shaping the environmental profile, particularly for fixed windows. As expected, larger glazing areas increase the glass's relative contribution to the total environmental impact. This highlights the strong dependence of window environmental performance on glazing properties and design across scenarios. Similar conclusions regarding the dominant role of glazing are also proven by Marinello Jorba et al. [2] and Agliata et al. [18]. For further comparison, the results reported by Rabbani et al. [6] indicate that electrochromic windows production emits 49.6 kg CO₂ eq/m² of which 9.79 kg CO₂ eq comes from the glass production. In our study, the highest impact scenario (S1, operable wooden window with aluminium cladding) shows a considerably higher GWP of 120.55 kg CO₂ eq/m², including 40.10 kg CO₂ eq for the triple glazing alone. Considering the S2 operable variant (frame without cladding and without paint), the GWP drops to 59.97 kg CO₂ eq/m², with 40.6 kg CO₂ eq attributed to the glazing. Although electrochromic windows have lower impacts related to the glass production, wood-based windows, particularly without aluminium cladding, can achieve comparable impacts per square metre in terms of GWP.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 11

Created: 23.6.2026 r. 10:20 by user, Modified: 23.6.2026 r. 10:20 by user

Other codes assigned to segment:

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Functional (Weight: 0)

Negative mechanism > Increased material complexity (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

The glazing refers to the most dominant wood-based window component, shaping the environmental profile, particularly for fixed windows. As expected, larger glazing areas increase the glass's relative contribution to the total environmental impact. This highlights the strong dependence of window environmental performance on glazing properties and design across scenarios. Similar conclusions regarding the dominant role of glazing are also proven by Marinello Jorba et al. [2] and Agliata et al. [18]. For further comparison, the results reported by Rabbani et al. [6] indicate that electrochromic windows production emits 49.6 kg CO₂ eq/m² of which 9.79 kg CO₂ eq comes from the glass production. In our study, the highest impact scenario (S1, operable wooden window with aluminium cladding) shows a considerably higher GWP of 120.55 kg CO₂ eq/m², including 40.10 kg CO₂ eq for the triple glazing alone. Considering the S2 operable variant (frame without cladding and without paint), the GWP drops to 59.97 kg CO₂ eq/m², with 40.6 kg CO₂ eq attributed to the glazing.

Although electrochromic windows have lower impacts related to the glass production, wood-based windows, particularly without aluminium cladding, can achieve comparable impacts per square metre in terms of GWP.

Code: • Negative mechanism > Increased material complexity Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 11

Created: 23.6.2026 r. 10:19 by user, Modified: 23.6.2026 r. 10:19 by user

Other codes assigned to segment:

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Functional (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Impact distribution changes in window components on overall environmental performance are particularly evident when external cladding is used (scenario S1). This increases the frame's impact, especially for operable windows, due to higher material use and structural complexity. Consequently, the operable window exhibits a higher embodied burdenBuildings 2025, 15, 4490 12 of 17 than the fixed variant. Comparison of the exterior window designs confirms that the presence of a cladding significantly increases the production stage environmental impact beyond that of maintaining a window without cladding for 50 years.

Code: • Circular strategy > Maintenance Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 11 - 12

Created: 23.6.2026 r. 10:20 by user, Modified: 23.6.2026 r. 10:20 by user

Other codes assigned to segment:

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Physical (Weight: 0)

Obsolescence > Functional (Weight: 0)

Negative mechanism > Increased material complexity (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Impact distribution changes in window components on overall environmental performance are particularly evident when external cladding is used (scenario S1). This increases the frame's impact, especially for operable windows, due to higher material use and structural complexity. Consequently, the operable window exhibits a higher embodied burdenBuildings 2025, 15, 4490 12 of 17 than the fixed variant. Comparison of the exterior window designs confirms that the presence of a cladding significantly increases the production stage environmental impact beyond that of maintaining a window without cladding for 50 years.

Code: • Obsolescence > Functional Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 11 - 12

Created: 23.6.2026 r. 10:20 by user, Modified: 23.6.2026 r. 10:20 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Physical (Weight: 0)

Negative mechanism > Increased material complexity (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Impact distribution changes in window components on overall environmental performance are particularly evident when external cladding is used (scenario S1). This increases the frame's impact, especially for operable windows, due to higher material use and structural complexity. Consequently, the operable window exhibits a higher embodied burdenBuildings 2025, 15, 4490 12 of 17

than the fixed variant. Comparison of the exterior window designs confirms that the presence of a cladding significantly increases the production stage environmental impact beyond that of maintaining a window without cladding for 50 years.

Code: • [Obsolescence > Physical](#) Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 11 - 12

Created: 23.6.2026 r. 10:20 by user, Modified: 23.6.2026 r. 10:20 by user

Other codes assigned to segment:

[Circular strategy > Maintenance](#) (Weight: 0)

[Circular strategy > Upgrade](#) (Weight: 0)

[Obsolescence > Functional](#) (Weight: 0)

[Negative mechanism > Increased material complexity](#) (Weight: 0)

[Negative mechanism > Higher initial material use](#) (Weight: 0)

[Negative mechanism > Burden shifting](#) (Weight: 0)

[Mitigation > Lifecycle verification](#) (Weight: 0)

Impact distribution changes in window components on overall environmental performance are particularly evident when external cladding is used (scenario S1). This increases the frame's impact, especially for operable windows, due to higher material use and structural complexity. Consequently, the operable window exhibits a higher embodied burdenBuildings 2025, 15, 4490 12 of 17 than the fixed variant. Comparison of the exterior window designs confirms that the presence of a cladding significantly increases the production stage environmental impact beyond that of maintaining a window without cladding for 50 years.

Code: • [Negative mechanism > Burden shifting](#) Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 11 - 12

Created: 23.6.2026 r. 10:21 by user, Modified: 23.6.2026 r. 10:21 by user

Other codes assigned to segment:

[Circular strategy > Maintenance](#) (Weight: 0)

[Circular strategy > Upgrade](#) (Weight: 0)

[Obsolescence > Physical](#) (Weight: 0)

[Obsolescence > Functional](#) (Weight: 0)

[Negative mechanism > Increased material complexity](#) (Weight: 0)

[Negative mechanism > Higher initial material use](#) (Weight: 0)

[Mitigation > Lifecycle verification](#) (Weight: 0)

Impact distribution changes in window components on overall environmental performance are particularly evident when external cladding is used (scenario S1). This increases the frame's impact, especially for operable windows, due to higher material use and structural complexity. Consequently, the operable window exhibits a higher embodied burdenBuildings 2025, 15, 4490 12 of 17 than the fixed variant. Comparison of the exterior window designs confirms that the presence of a cladding significantly increases the production stage environmental impact beyond that of maintaining a window without cladding for 50 years.

Code: • [Negative mechanism > Increased material complexity](#) Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 11 - 12

Created: 23.6.2026 r. 10:21 by user, Modified: 23.6.2026 r. 10:21 by user

Other codes assigned to segment:

[Circular strategy > Maintenance](#) (Weight: 0)

[Circular strategy > Upgrade](#) (Weight: 0)

[Obsolescence > Physical](#) (Weight: 0)

[Obsolescence > Functional](#) (Weight: 0)

[Negative mechanism > Higher initial material use](#) (Weight: 0)

[Negative mechanism > Burden shifting](#) (Weight: 0)

[Mitigation > Lifecycle verification](#) (Weight: 0)

Impact distribution changes in window components on overall environmental performance are particularly evident when external cladding is used (scenario S1). This increases the frame's impact,

especially for operable windows, due to higher material use and structural complexity. Consequently, the operable window exhibits a higher embodied burdenBuildings 2025, 15, 4490 12 of 17 than the fixed variant. Comparison of the exterior window designs confirms that the presence of a cladding significantly increases the production stage environmental impact beyond that of maintaining a window without cladding for 50 years.

Code: • Negative mechanism > Higher initial material use Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 11 - 12

Created: 23.6.2026 r. 10:21 by user, Modified: 23.6.2026 r. 10:21 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Physical (Weight: 0)

Obsolescence > Functional (Weight: 0)

Negative mechanism > Increased material complexity (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Impact distribution changes in window components on overall environmental performance are particularly evident when external cladding is used (scenario S1). This increases the frame's impact, especially for operable windows, due to higher material use and structural complexity. Consequently, the operable window exhibits a higher embodied burdenBuildings 2025, 15, 4490 12 of 17 than the fixed variant. Comparison of the exterior window designs confirms that the presence of a cladding significantly increases the production stage environmental impact beyond that of maintaining a window without cladding for 50 years.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 11 - 12

Created: 23.6.2026 r. 10:21 by user, Modified: 23.6.2026 r. 10:21 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Physical (Weight: 0)

Obsolescence > Functional (Weight: 0)

Negative mechanism > Increased material complexity (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Impact distribution changes in window components on overall environmental performance are particularly evident when external cladding is used (scenario S1). This increases the frame's impact, especially for operable windows, due to higher material use and structural complexity. Consequently, the operable window exhibits a higher embodied burdenBuildings 2025, 15, 4490 12 of 17 than the fixed variant. Comparison of the exterior window designs confirms that the presence of a cladding significantly increases the production stage environmental impact beyond that of maintaining a window without cladding for 50 years.

Code: • Circular strategy > Upgrade Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 11 - 12

Created: 23.6.2026 r. 10:20 by user, Modified: 23.6.2026 r. 10:20 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Obsolescence > Physical (Weight: 0)

Obsolescence > Functional (Weight: 0)

Negative mechanism > Increased material complexity (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Maintenance requirements follow the coating manufacturer's recommendations. Weather conditions influence the degradation of the wooden surface and the required maintenance frequency [39]. In this study, maintenance intervals are intentionally set shorter to ensure the windows remain in optimal condition throughout their service life, even though longer intervals may be sufficient [39,40]. The B2 stage contributes far less to the overall environmental impact than the production stage, particularly for water-based acrylic varnishes, which show relatively low burdens across all impact categories and window types. Natural oil-based varnishes show the highest impacts in TE and POC, mainly due to upstream environmental burden in the supply chain. The reliance on agricultural inputs and land use changes in the production of bio-based ingredients contributes significantly to their environmental impact [41], indicating that bio-based materials may not always offer improved sustainability [28].

Code: • Circular strategy > Maintenance Weight score: 0
 Search A - Window maintenance and component repair > A-GS-11, p. 12
 Created: 23.6.2026 r. 10:22 by user, Modified: 23.6.2026 r. 10:22 by user
 Other codes assigned to segment:
 Obsolescence > Physical (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Maintenance requirements follow the coating manufacturer's recommendations. Weather conditions influence the degradation of the wooden surface and the required maintenance frequency [39]. In this study, maintenance intervals are intentionally set shorter to ensure the windows remain in optimal condition throughout their service life, even though longer intervals may be sufficient [39,40]. The B2 stage contributes far less to the overall environmental impact than the production stage, particularly for water-based acrylic varnishes, which show relatively low burdens across all impact categories and window types. Natural oil-based varnishes show the highest impacts in TE and POC, mainly due to upstream environmental burden in the supply chain. The reliance on agricultural inputs and land use changes in the production of bio-based ingredients contributes significantly to their environmental impact [41], indicating that bio-based materials may not always offer improved sustainability [28].

Code: • Mitigation > Lifecycle verification Weight score: 0
 Search A - Window maintenance and component repair > A-GS-11, p. 12
 Created: 23.6.2026 r. 10:23 by user, Modified: 23.6.2026 r. 10:23 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)

Maintenance requirements follow the coating manufacturer's recommendations. Weather conditions influence the degradation of the wooden surface and the required maintenance frequency [39]. In this study, maintenance intervals are intentionally set shorter to ensure the windows remain in optimal condition throughout their service life, even though longer intervals may be sufficient [39,40]. The B2 stage contributes far less to the overall environmental impact than the production stage, particularly for water-based acrylic varnishes, which show relatively low burdens across all impact categories and window types. Natural oil-based varnishes show the highest impacts in TE and POC, mainly due to upstream environmental burden in the supply chain. The reliance on agricultural inputs and land use changes in the production of bio-based ingredients contributes significantly to their environmental impact [41], indicating that bio-based materials may not always offer improved sustainability [28].

Code: • Negative mechanism > Burden shifting Weight score: 0
 Search A - Window maintenance and component repair > A-GS-11, p. 12

Created: 23.6.2026 r. 10:23 by user, Modified: 23.6.2026 r. 10:23 by user

Other codes assigned to segment:

- Circular strategy > Maintenance (Weight: 0)
- Obsolescence > Physical (Weight: 0)
- Mitigation > Maintenance planning (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

Maintenance requirements follow the coating manufacturer's recommendations. Weather conditions influence the degradation of the wooden surface and the required maintenance frequency [39]. In this study, maintenance intervals are intentionally set shorter to ensure the windows remain in optimal condition throughout their service life, even though longer intervals may be sufficient [39,40]. The B2 stage contributes far less to the overall environmental impact than the production stage, particularly for water-based acrylic varnishes, which show relatively low burdens across all impact categories and window types. Natural oil-based varnishes show the highest impacts in TE and POC, mainly due to upstream environmental burden in the supply chain. The reliance on agricultural inputs and land use changes in the production of bio-based ingredients contributes significantly to their environmental impact [41], indicating that bio-based materials may not always offer improved sustainability [28].

Code: • Obsolescence > Physical Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 12

Created: 23.6.2026 r. 10:22 by user, Modified: 23.6.2026 r. 10:22 by user

Other codes assigned to segment:

- Circular strategy > Maintenance (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)
- Mitigation > Maintenance planning (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

Maintenance requirements follow the coating manufacturer's recommendations. Weather conditions influence the degradation of the wooden surface and the required maintenance frequency [39]. In this study, maintenance intervals are intentionally set shorter to ensure the windows remain in optimal condition throughout their service life, even though longer intervals may be sufficient [39,40]. The B2 stage contributes far less to the overall environmental impact than the production stage, particularly for water-based acrylic varnishes, which show relatively low burdens across all impact categories and window types. Natural oil-based varnishes show the highest impacts in TE and POC, mainly due to upstream environmental burden in the supply chain. The reliance on agricultural inputs and land use changes in the production of bio-based ingredients contributes significantly to their environmental impact [41], indicating that bio-based materials may not always offer improved sustainability [28].

Code: • Mitigation > Maintenance planning Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 12

Created: 23.6.2026 r. 10:23 by user, Modified: 23.6.2026 r. 10:23 by user

Other codes assigned to segment:

- Circular strategy > Maintenance (Weight: 0)
- Obsolescence > Physical (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

Optimising materials and construction according to window type and selected environmental indicators can significantly reduce the overall impact. Specifically, prioritising fixed windows, use of water-based acrylic varnish and avoidance of aluminium cladding where possible are recommended to lower the total environmental impacts associated with the wood-based windows production with regard to the maintenance stage.

Code: • Mitigation > Maintenance planning Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 14

Created: 23.6.2026 r. 10:25 by user, Modified: 23.6.2026 r. 10:25 by user

Other codes assigned to segment:

- Circular strategy > Maintenance (Weight: 0)
- Circular strategy > Upgrade (Weight: 0)
- Negative mechanism > Higher initial material use (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

Optimising materials and construction according to window type and selected environmental indicators can significantly reduce the overall impact. Specifically, prioritising fixed windows, use of water-based acrylic varnish and avoidance of aluminium cladding where possible are recommended to lower the total environmental impacts associated with the wood-based windows production with regard to the maintenance stage.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 14

Created: 23.6.2026 r. 10:25 by user, Modified: 23.6.2026 r. 10:25 by user

Other codes assigned to segment:

- Circular strategy > Maintenance (Weight: 0)
- Circular strategy > Upgrade (Weight: 0)
- Negative mechanism > Higher initial material use (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)
- Mitigation > Maintenance planning (Weight: 0)

Optimising materials and construction according to window type and selected environmental indicators can significantly reduce the overall impact. Specifically, prioritising fixed windows, use of water-based acrylic varnish and avoidance of aluminium cladding where possible are recommended to lower the total environmental impacts associated with the wood-based windows production with regard to the maintenance stage.

Code: • Negative mechanism > Burden shifting Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 14

Created: 23.6.2026 r. 10:25 by user, Modified: 23.6.2026 r. 10:25 by user

Other codes assigned to segment:

- Circular strategy > Maintenance (Weight: 0)
- Circular strategy > Upgrade (Weight: 0)
- Negative mechanism > Higher initial material use (Weight: 0)
- Mitigation > Maintenance planning (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

Optimising materials and construction according to window type and selected environmental indicators can significantly reduce the overall impact. Specifically, prioritising fixed windows, use of water-based acrylic varnish and avoidance of aluminium cladding where possible are recommended to lower the total environmental impacts associated with the wood-based windows production with regard to the maintenance stage.

Code: • Negative mechanism > Higher initial material use Weight score: 0

Search A - Window maintenance and component repair > A-GS-11, p. 14

Created: 23.6.2026 r. 10:25 by user, Modified: 23.6.2026 r. 10:25 by user

Other codes assigned to segment:

- Circular strategy > Maintenance (Weight: 0)
- Circular strategy > Upgrade (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)
- Mitigation > Maintenance planning (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

Optimising materials and construction according to window type and selected environmental indicators can significantly reduce the overall impact. Specifically, prioritising fixed windows, use of water-based acrylic varnish and avoidance of aluminium cladding where possible are recommended to lower the total environmental impacts associated with the wood-based windows production with regard to the maintenance stage.

Code: • Circular strategy > Upgrade Weight score: 0
 Search A - Window maintenance and component repair > A-GS-11, p. 14
 Created: 23.6.2026 r. 10:25 by user, Modified: 23.6.2026 r. 10:25 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Optimising materials and construction according to window type and selected environmental indicators can significantly reduce the overall impact. Specifically, prioritising fixed windows, use of water-based acrylic varnish and avoidance of aluminium cladding where possible are recommended to lower the total environmental impacts associated with the wood-based windows production with regard to the maintenance stage.

Code: • Circular strategy > Maintenance Weight score: 0
 Search A - Window maintenance and component repair > A-GS-11, p. 14
 Created: 23.6.2026 r. 10:25 by user, Modified: 23.6.2026 r. 10:25 by user
 Other codes assigned to segment:
 Circular strategy > Upgrade (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

This research investigates the integration of Design for Disassembly (DfD) principles into the lifecycle of high-performance facade systems, as complex architectural systems frequently utilize materials and methods that hinder circularity, such as fixed connections. Designing and advocating for future uses is challenging without explicit disassembly standards or supporting infrastructure. By analysing existing disassembly methodologies, a tailored approach was developed for high-performance façade design, leveraging Schüco KG's existing digital infrastructure to optimize disassembly workflows. This methodology focuses on the generation of practical disassembly outputs, for use cases across designing, manufacturing, and constructing façade systems. Coordinating with fabricators, a proof-of-concept for a Disassembly Application is developed to integrate with existing digital infrastructure. The case study on the AWS 75.SI+ window system highlights the feasibility and benefits of incorporating disassembly planning from the early design stage. This framework provides a scalable model for reducing construction waste, improving material reuse, and supporting circular economy goals for high-performance façade systems and across complex building systems.

Code: • Circular strategy > Direct reuse Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-02, p. 1
 Created: 23.6.2026 r. 10:37 by user, Modified: 23.6.2026 r. 10:37 by user
 Other codes assigned to segment:
 Circular strategy > Refurbishment (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

This research investigates the integration of Design for Disassembly (DfD) principles into the lifecycle of high-performance facade systems, as complex architectural systems frequently utilize materials and methods that hinder circularity, such as fixed connections. Designing and advocating for future uses is challenging without explicit disassembly standards or supporting infrastructure. By analysing existing disassembly methodologies, a tailored approach was developed for high-performance façade design, leveraging Schüco KG's existing digital infrastructure to optimize disassembly workflows. This methodology focuses on the generation of practical disassembly outputs, for use cases across designing, manufacturing, and constructing façade systems. Coordinating with fabricators, a proof-of-concept for a Disassembly Application is developed to integrate with existing digital infrastructure. The case study on the AWS 75.SI+ window system highlights the feasibility and benefits of incorporating disassembly planning from the early design stage. This framework provides a scalable model for reducing construction waste, improving material reuse, and supporting circular economy goals for high-performance façade systems and across complex building systems.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-02, p. 1

Created: 23.6.2026 r. 10:38 by user, Modified: 23.6.2026 r. 10:38 by user

Other codes assigned to segment:

Circular strategy > Refurbishment (Weight: 0)

Circular strategy > Direct reuse (Weight: 0)

Enabling condition > Disassembly (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Design intervention (Weight: 0)

This research investigates the integration of Design for Disassembly (DfD) principles into the lifecycle of high-performance facade systems, as complex architectural systems frequently utilize materials and methods that hinder circularity, such as fixed connections. Designing and advocating for future uses is challenging without explicit disassembly standards or supporting infrastructure. By analysing existing disassembly methodologies, a tailored approach was developed for high-performance façade design, leveraging Schüco KG's existing digital infrastructure to optimize disassembly workflows. This methodology focuses on the generation of practical disassembly outputs, for use cases across designing, manufacturing, and constructing façade systems. Coordinating with fabricators, a proof-of-concept for a Disassembly Application is developed to integrate with existing digital infrastructure. The case study on the AWS 75.SI+ window system highlights the feasibility and benefits of incorporating disassembly planning from the early design stage. This framework provides a scalable model for reducing construction waste, improving material reuse, and supporting circular economy goals for high-performance façade systems and across complex building systems.

Code: • Enabling condition > Service infrastructure Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-02, p. 1

Created: 23.6.2026 r. 10:38 by user, Modified: 23.6.2026 r. 10:38 by user

Other codes assigned to segment:

Circular strategy > Refurbishment (Weight: 0)

Circular strategy > Direct reuse (Weight: 0)

Enabling condition > Disassembly (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Design intervention (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

This research investigates the integration of Design for Disassembly (DfD) principles into the lifecycle of high-performance facade systems, as complex architectural systems frequently utilize materials and methods that hinder circularity, such as fixed connections. Designing and advocating for future

uses is challenging without explicit disassembly standards or supporting infrastructure. By analysing existing disassembly methodologies, a tailored approach was developed for high-performance façade design, leveraging Schüco KG's existing digital infrastructure to optimize disassembly workflows. This methodology focuses on the generation of practical disassembly outputs, for use cases across designing, manufacturing, and constructing façade systems. Coordinating with fabricators, a proof-of-concept for a Disassembly Application is developed to integrate with existing digital infrastructure. The case study on the AWS 75.SI+ window system highlights the feasibility and benefits of incorporating disassembly planning from the early design stage. This framework provides a scalable model for reducing construction waste, improving material reuse, and supporting circular economy goals for high-performance façade systems and across complex building systems.

Code: • Enabling condition > Disassembly Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-02, p. 1
 Created: 23.6.2026 r. 10:38 by user, Modified: 23.6.2026 r. 10:38 by user
 Other codes assigned to segment:
 Circular strategy > Refurbishment (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

This research investigates the integration of Design for Disassembly (DfD) principles into the lifecycle of high-performance facade systems, as complex architectural systems frequently utilize materials and methods that hinder circularity, such as fixed connections. Designing and advocating for future uses is challenging without explicit disassembly standards or supporting infrastructure. By analysing existing disassembly methodologies, a tailored approach was developed for high-performance façade design, leveraging Schüco KG's existing digital infrastructure to optimize disassembly workflows. This methodology focuses on the generation of practical disassembly outputs, for use cases across designing, manufacturing, and constructing façade systems. Coordinating with fabricators, a proof-of-concept for a Disassembly Application is developed to integrate with existing digital infrastructure. The case study on the AWS 75.SI+ window system highlights the feasibility and benefits of incorporating disassembly planning from the early design stage. This framework provides a scalable model for reducing construction waste, improving material reuse, and supporting circular economy goals for high-performance façade systems and across complex building systems.

Code: • Negative mechanism > Missing repair infrastructure Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-02, p. 1
 Created: 23.6.2026 r. 10:38 by user, Modified: 23.6.2026 r. 10:38 by user
 Other codes assigned to segment:
 Circular strategy > Refurbishment (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

This research investigates the integration of Design for Disassembly (DfD) principles into the lifecycle of high-performance facade systems, as complex architectural systems frequently utilize materials and methods that hinder circularity, such as fixed connections. Designing and advocating for future uses is challenging without explicit disassembly standards or supporting infrastructure. By analysing existing disassembly methodologies, a tailored approach was developed for high-performance façade design, leveraging Schüco KG's existing digital infrastructure to optimize disassembly workflows. This methodology focuses on the generation of practical disassembly outputs, for use cases across

designing, manufacturing, and constructing façade systems. Coordinating with fabricators, a proof-of-concept for a Disassembly Application is developed to integrate with existing digital infrastructure. The case study on the AWS 75.SI+ window system highlights the feasibility and benefits of incorporating disassembly planning from the early design stage. This framework provides a scalable model for reducing construction waste, improving material reuse, and supporting circular economy goals for high-performance façade systems and across complex building systems.

Code: • Circular strategy > Refurbishment Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-02, p. 1
 Created: 23.6.2026 r. 10:37 by user, Modified: 23.6.2026 r. 10:37 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

This research investigates the integration of Design for Disassembly (DfD) principles into the lifecycle of high-performance facade systems, as complex architectural systems frequently utilize materials and methods that hinder circularity, such as fixed connections. Designing and advocating for future uses is challenging without explicit disassembly standards or supporting infrastructure. By analysing existing disassembly methodologies, a tailored approach was developed for high-performance façade design, leveraging Schüco KG's existing digital infrastructure to optimize disassembly workflows. This methodology focuses on the generation of practical disassembly outputs, for use cases across designing, manufacturing, and constructing façade systems. Coordinating with fabricators, a proof-of-concept for a Disassembly Application is developed to integrate with existing digital infrastructure. The case study on the AWS 75.SI+ window system highlights the feasibility and benefits of incorporating disassembly planning from the early design stage. This framework provides a scalable model for reducing construction waste, improving material reuse, and supporting circular economy goals for high-performance façade systems and across complex building systems.

Code: • Mitigation > Design intervention Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-02, p. 1
 Created: 23.6.2026 r. 10:38 by user, Modified: 23.6.2026 r. 10:38 by user
 Other codes assigned to segment:
 Circular strategy > Refurbishment (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

This research investigates the integration of Design for Disassembly (DfD) principles into the lifecycle of high-performance facade systems, as complex architectural systems frequently utilize materials and methods that hinder circularity, such as fixed connections. Designing and advocating for future uses is challenging without explicit disassembly standards or supporting infrastructure. By analysing existing disassembly methodologies, a tailored approach was developed for high-performance façade design, leveraging Schüco KG's existing digital infrastructure to optimize disassembly workflows. This methodology focuses on the generation of practical disassembly outputs, for use cases across designing, manufacturing, and constructing façade systems. Coordinating with fabricators, a proof-of-concept for a Disassembly Application is developed to integrate with existing digital infrastructure. The case study on the AWS 75.SI+ window system highlights the feasibility and benefits of incorporating disassembly planning from the early design stage. This framework provides a scalable

model for reducing construction waste, improving material reuse, and supporting circular economy goals for high-performance façade systems and across complex building systems.

Code: • Enabling condition > Repair information Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-02, p. 1
 Created: 23.6.2026 r. 10:38 by user, Modified: 23.6.2026 r. 10:38 by user
 Other codes assigned to segment:
 Circular strategy > Refurbishment (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Frequent technological advancements increase the rate of technical obsolescence of façade systems. Instead of considering future technological upgrades as a design criterion, many façade systems are replaced in their entirety in favor of higher-performing systems.

Code: • Negative mechanism > Frequent upgrading Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-02, p. 1
 Created: 23.6.2026 r. 10:43 by user, Modified: 23.6.2026 r. 10:43 by user
 Other codes assigned to segment:
 Obsolescence > Functional (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

Frequent technological advancements increase the rate of technical obsolescence of façade systems. Instead of considering future technological upgrades as a design criterion, many façade systems are replaced in their entirety in favor of higher-performing systems.

Code: • Negative mechanism > Premature replacement Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-02, p. 1
 Created: 23.6.2026 r. 10:41 by user, Modified: 23.6.2026 r. 10:41 by user
 Other codes assigned to segment:
 Obsolescence > Functional (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Negative mechanism > Frequent upgrading (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

Frequent technological advancements increase the rate of technical obsolescence of façade systems. Instead of considering future technological upgrades as a design criterion, many façade systems are replaced in their entirety in favor of higher-performing systems.

Code: • Negative mechanism > Lock-in Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-02, p. 1
 Created: 23.6.2026 r. 10:43 by user, Modified: 23.6.2026 r. 10:43 by user
 Other codes assigned to segment:
 Obsolescence > Functional (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Frequent upgrading (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

Frequent technological advancements increase the rate of technical obsolescence of façade systems. Instead of considering future technological upgrades as a design criterion, many façade systems are replaced in their entirety in favor of higher-performing systems.

Code: ● Obsolescence > Functional Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-02, p. 1
 Created: 23.6.2026 r. 10:41 by user, Modified: 23.6.2026 r. 10:41 by user
 Other codes assigned to segment:
 Obsolescence > Technical (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Frequent upgrading (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

Frequent technological advancements increase the rate of technical obsolescence of façade systems. Instead of considering future technological upgrades as a design criterion, many façade systems are replaced in their entirety in favor of higher-performing systems.

Code: ● Mitigation > Design intervention Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-02, p. 1
 Created: 23.6.2026 r. 10:43 by user, Modified: 23.6.2026 r. 10:43 by user
 Other codes assigned to segment:
 Obsolescence > Functional (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Frequent upgrading (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)

Frequent technological advancements increase the rate of technical obsolescence of façade systems. Instead of considering future technological upgrades as a design criterion, many façade systems are replaced in their entirety in favor of higher-performing systems.

Code: ● Obsolescence > Technical Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-02, p. 1
 Created: 23.6.2026 r. 10:41 by user, Modified: 23.6.2026 r. 10:41 by user
 Other codes assigned to segment:
 Obsolescence > Functional (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Frequent upgrading (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

While façade systems often use critical and high-value materials such as aluminum and steel, high-performance systems rely on polymers and adhesives, that reach technical obsolescence sooner, to meet structural, thermal, and acoustic criteria. Although high thermal performance significantly lowers operational carbon, current practice uses fixed connections, permanently sealing non-circular materials to systems. Thus, although facade assemblies are historically among the most demountable, reconfigurable, and reusable building systems, the only End-of-Life options for these high-performance systems are recycling, incineration, and landfill.

Code: ● Negative mechanism > Increased material complexity Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-02, p. 2
 Created: 23.6.2026 r. 10:45 by user, Modified: 23.6.2026 r. 10:45 by user
 Other codes assigned to segment:
 Circular strategy > Recycling (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)

Negative mechanism > Lock-in (Weight: 0)
Mitigation > Design intervention (Weight: 0)

While façade systems often use critical and high-value materials such as aluminum and steel, high-performance systems rely on polymers and adhesives, that reach technical obsolescence sooner, to meet structural, thermal, and acoustic criteria. Although high thermal performance significantly lowers operational carbon, current practice uses fixed connections, permanently sealing non-circular materials to systems. Thus, although facade assemblies are historically among the most demountable, reconfigurable, and reusable building systems, the only End-of-Life options for these high-performance systems are recycling, incineration, and landfill.

Code: • Enabling condition > Disassembly Weight score: 0
Search B - Disassembly reuse and facade recovery > B-GS-02, p. 2
Created: 23.6.2026 r. 10:45 by user, Modified: 23.6.2026 r. 10:45 by user
Other codes assigned to segment:
Circular strategy > Recycling (Weight: 0)
Obsolescence > Technical (Weight: 0)
Negative mechanism > Increased material complexity (Weight: 0)
Negative mechanism > Burden shifting (Weight: 0)
Negative mechanism > Lock-in (Weight: 0)
Mitigation > Design intervention (Weight: 0)

While façade systems often use critical and high-value materials such as aluminum and steel, high-performance systems rely on polymers and adhesives, that reach technical obsolescence sooner, to meet structural, thermal, and acoustic criteria. Although high thermal performance significantly lowers operational carbon, current practice uses fixed connections, permanently sealing non-circular materials to systems. Thus, although facade assemblies are historically among the most demountable, reconfigurable, and reusable building systems, the only End-of-Life options for these high-performance systems are recycling, incineration, and landfill.

Code: • Circular strategy > Recycling Weight score: 0
Search B - Disassembly reuse and facade recovery > B-GS-02, p. 2
Created: 23.6.2026 r. 10:45 by user, Modified: 23.6.2026 r. 10:45 by user
Other codes assigned to segment:
Obsolescence > Technical (Weight: 0)
Enabling condition > Disassembly (Weight: 0)
Negative mechanism > Increased material complexity (Weight: 0)
Negative mechanism > Burden shifting (Weight: 0)
Negative mechanism > Lock-in (Weight: 0)
Mitigation > Design intervention (Weight: 0)

While façade systems often use critical and high-value materials such as aluminum and steel, high-performance systems rely on polymers and adhesives, that reach technical obsolescence sooner, to meet structural, thermal, and acoustic criteria. Although high thermal performance significantly lowers operational carbon, current practice uses fixed connections, permanently sealing non-circular materials to systems. Thus, although facade assemblies are historically among the most demountable, reconfigurable, and reusable building systems, the only End-of-Life options for these high-performance systems are recycling, incineration, and landfill.

Code: • Mitigation > Design intervention Weight score: 0
Search B - Disassembly reuse and facade recovery > B-GS-02, p. 2
Created: 23.6.2026 r. 10:46 by user, Modified: 23.6.2026 r. 10:46 by user
Other codes assigned to segment:
Circular strategy > Recycling (Weight: 0)
Obsolescence > Technical (Weight: 0)
Enabling condition > Disassembly (Weight: 0)
Negative mechanism > Increased material complexity (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)

While façade systems often use critical and high-value materials such as aluminum and steel, high-performance systems rely on polymers and adhesives, that reach technical obsolescence sooner, to meet structural, thermal, and acoustic criteria. Although high thermal performance significantly lowers operational carbon, current practice uses fixed connections, permanently sealing non-circular materials to systems. Thus, although facade assemblies are historically among the most demountable, reconfigurable, and reusable building systems, the only End-of-Life options for these high-performance systems are recycling, incineration, and landfill.

Code: • Negative mechanism > Burden shifting Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-02, p. 2
 Created: 23.6.2026 r. 10:45 by user, Modified: 23.6.2026 r. 10:45 by user
 Other codes assigned to segment:
 Circular strategy > Recycling (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Increased material complexity (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

While façade systems often use critical and high-value materials such as aluminum and steel, high-performance systems rely on polymers and adhesives, that reach technical obsolescence sooner, to meet structural, thermal, and acoustic criteria. Although high thermal performance significantly lowers operational carbon, current practice uses fixed connections, permanently sealing non-circular materials to systems. Thus, although facade assemblies are historically among the most demountable, reconfigurable, and reusable building systems, the only End-of-Life options for these high-performance systems are recycling, incineration, and landfill.

Code: • Negative mechanism > Lock-in Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-02, p. 2
 Created: 23.6.2026 r. 10:45 by user, Modified: 23.6.2026 r. 10:45 by user
 Other codes assigned to segment:
 Circular strategy > Recycling (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Increased material complexity (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

While façade systems often use critical and high-value materials such as aluminum and steel, high-performance systems rely on polymers and adhesives, that reach technical obsolescence sooner, to meet structural, thermal, and acoustic criteria. Although high thermal performance significantly lowers operational carbon, current practice uses fixed connections, permanently sealing non-circular materials to systems. Thus, although facade assemblies are historically among the most demountable, reconfigurable, and reusable building systems, the only End-of-Life options for these high-performance systems are recycling, incineration, and landfill.

Code: • Obsolescence > Technical Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-02, p. 2
 Created: 23.6.2026 r. 10:45 by user, Modified: 23.6.2026 r. 10:45 by user
 Other codes assigned to segment:
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Increased material complexity (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)

Negative mechanism > Lock-in (Weight: 0)
Mitigation > Design intervention (Weight: 0)

The system contains aluminum profiles that are thermally broken with polyamide (PA) or proprietary material Polythermid (PT) isolators, which are permanently rolled into the aluminum profiles with insulation in between each thermal break component. Gaskets are made of EPDM and PE foam is used for insulation. Corners and profile joints are fixed with aluminum cleats and sealed with epoxy in addition to steel screws. Locking bars are a combination of plastic and metal, and most fittings are metal and fully removable

Code: • Mitigation > Design intervention Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-02, p. 5

Created: 23.6.2026 r. 10:47 by user, Modified: 23.6.2026 r. 10:47 by user

Other codes assigned to segment:

Obsolescence > Technical (Weight: 0)
Enabling condition > Component accessibility (Weight: 0)
Enabling condition > Disassembly (Weight: 0)
Enabling condition > Repair information (Weight: 0)
Negative mechanism > Increased material complexity (Weight: 0)
Negative mechanism > Lock-in (Weight: 0)

The system contains aluminum profiles that are thermally broken with polyamide (PA) or proprietary material Polythermid (PT) isolators, which are permanently rolled into the aluminum profiles with insulation in between each thermal break component. Gaskets are made of EPDM and PE foam is used for insulation. Corners and profile joints are fixed with aluminum cleats and sealed with epoxy in addition to steel screws. Locking bars are a combination of plastic and metal, and most fittings are metal and fully removable

Code: • Negative mechanism > Increased material complexity Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-02, p. 5

Created: 23.6.2026 r. 10:47 by user, Modified: 23.6.2026 r. 10:47 by user

Other codes assigned to segment:

Obsolescence > Technical (Weight: 0)
Enabling condition > Component accessibility (Weight: 0)
Enabling condition > Disassembly (Weight: 0)
Enabling condition > Repair information (Weight: 0)
Negative mechanism > Lock-in (Weight: 0)
Mitigation > Design intervention (Weight: 0)

The system contains aluminum profiles that are thermally broken with polyamide (PA) or proprietary material Polythermid (PT) isolators, which are permanently rolled into the aluminum profiles with insulation in between each thermal break component. Gaskets are made of EPDM and PE foam is used for insulation. Corners and profile joints are fixed with aluminum cleats and sealed with epoxy in addition to steel screws. Locking bars are a combination of plastic and metal, and most fittings are metal and fully removable

Code: • Negative mechanism > Lock-in Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-02, p. 5

Created: 23.6.2026 r. 10:47 by user, Modified: 23.6.2026 r. 10:47 by user

Other codes assigned to segment:

Obsolescence > Technical (Weight: 0)
Enabling condition > Component accessibility (Weight: 0)
Enabling condition > Disassembly (Weight: 0)
Enabling condition > Repair information (Weight: 0)
Negative mechanism > Increased material complexity (Weight: 0)
Mitigation > Design intervention (Weight: 0)

The system contains aluminum profiles that are thermally broken with polyamide (PA) or proprietary material Polythermid (PT) isolators, which are permanently rolled into the aluminum profiles with insulation in between each thermal break component. Gaskets are made of EPDM and PE foam is used for insulation. Corners and profile joints are fixed with aluminum cleats and sealed with epoxy in addition to steel screws. Locking bars are a combination of plastic and metal, and most fittings are metal and fully removable

Code: • Enabling condition > Component accessibility Weight score: 0
Search B - Disassembly reuse and facade recovery > B-GS-02, p. 5
Created: 23.6.2026 r. 10:47 by user, Modified: 23.6.2026 r. 10:47 by user
Other codes assigned to segment:
Obsolescence > Technical (Weight: 0)
Enabling condition > Disassembly (Weight: 0)
Enabling condition > Repair information (Weight: 0)
Negative mechanism > Increased material complexity (Weight: 0)
Negative mechanism > Lock-in (Weight: 0)
Mitigation > Design intervention (Weight: 0)

The system contains aluminum profiles that are thermally broken with polyamide (PA) or proprietary material Polythermid (PT) isolators, which are permanently rolled into the aluminum profiles with insulation in between each thermal break component. Gaskets are made of EPDM and PE foam is used for insulation. Corners and profile joints are fixed with aluminum cleats and sealed with epoxy in addition to steel screws. Locking bars are a combination of plastic and metal, and most fittings are metal and fully removable

Code: • Enabling condition > Disassembly Weight score: 0
Search B - Disassembly reuse and facade recovery > B-GS-02, p. 5
Created: 23.6.2026 r. 10:47 by user, Modified: 23.6.2026 r. 10:47 by user
Other codes assigned to segment:
Obsolescence > Technical (Weight: 0)
Enabling condition > Component accessibility (Weight: 0)
Enabling condition > Repair information (Weight: 0)
Negative mechanism > Increased material complexity (Weight: 0)
Negative mechanism > Lock-in (Weight: 0)
Mitigation > Design intervention (Weight: 0)

The system contains aluminum profiles that are thermally broken with polyamide (PA) or proprietary material Polythermid (PT) isolators, which are permanently rolled into the aluminum profiles with insulation in between each thermal break component. Gaskets are made of EPDM and PE foam is used for insulation. Corners and profile joints are fixed with aluminum cleats and sealed with epoxy in addition to steel screws. Locking bars are a combination of plastic and metal, and most fittings are metal and fully removable

Code: • Enabling condition > Repair information Weight score: 0
Search B - Disassembly reuse and facade recovery > B-GS-02, p. 5
Created: 23.6.2026 r. 10:47 by user, Modified: 23.6.2026 r. 10:47 by user
Other codes assigned to segment:
Obsolescence > Technical (Weight: 0)
Enabling condition > Component accessibility (Weight: 0)
Enabling condition > Disassembly (Weight: 0)
Negative mechanism > Increased material complexity (Weight: 0)
Negative mechanism > Lock-in (Weight: 0)
Mitigation > Design intervention (Weight: 0)

The system contains aluminum profiles that are thermally broken with polyamide (PA) or proprietary material Polythermid (PT) isolators, which are permanently rolled into the aluminum profiles with insulation in between each thermal break component. Gaskets are made of EPDM and PE foam is

used for insulation. Corners and profile joints are fixed with aluminum cleats and sealed with epoxy in addition to steel screws. Locking bars are a combination of plastic and metal, and most fittings are metal and fully removable

Code: • [Obsolescence > Technical](#) Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-02, p. 5
 Created: 23.6.2026 r. 10:47 by user, Modified: 23.6.2026 r. 10:47 by user
 Other codes assigned to segment:
[Enabling condition > Component accessibility](#) (Weight: 0)
[Enabling condition > Disassembly](#) (Weight: 0)
[Enabling condition > Repair information](#) (Weight: 0)
[Negative mechanism > Increased material complexity](#) (Weight: 0)
[Negative mechanism > Lock-in](#) (Weight: 0)
[Mitigation > Design intervention](#) (Weight: 0)

Integrating disassembly to each system type starts with system network analysis. Following the process mapped in Durmisevic's Knowledge Model, all system components are plotted as nodes and all components and connections are labeled. This mapping highlights the hierarchies and dependencies and the order in which they can safely be separated from the total configuration. Network analysis also highlights which components are inaccessible for removal due to sequencing and permanent connections. This process establishes a disassembly sequence for each system type and provides the necessary framework for disassembly metrics via a component database.

Code: • [Enabling condition > Skilled labour](#) Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-02, p. 5
 Created: 23.6.2026 r. 10:48 by user, Modified: 23.6.2026 r. 10:48 by user
 Other codes assigned to segment:
[Circular strategy > Direct reuse](#) (Weight: 0)
[Enabling condition > Disassembly](#) (Weight: 0)
[Enabling condition > Repair information](#) (Weight: 0)
[Mitigation > Design intervention](#) (Weight: 0)
[Mitigation > Lifecycle verification](#) (Weight: 0)

Integrating disassembly to each system type starts with system network analysis. Following the process mapped in Durmisevic's Knowledge Model, all system components are plotted as nodes and all components and connections are labeled. This mapping highlights the hierarchies and dependencies and the order in which they can safely be separated from the total configuration. Network analysis also highlights which components are inaccessible for removal due to sequencing and permanent connections. This process establishes a disassembly sequence for each system type and provides the necessary framework for disassembly metrics via a component database.

Code: • [Mitigation > Lifecycle verification](#) Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-02, p. 5
 Created: 23.6.2026 r. 10:49 by user, Modified: 23.6.2026 r. 10:49 by user
 Other codes assigned to segment:
[Circular strategy > Direct reuse](#) (Weight: 0)
[Enabling condition > Disassembly](#) (Weight: 0)
[Enabling condition > Repair information](#) (Weight: 0)
[Enabling condition > Skilled labour](#) (Weight: 0)
[Mitigation > Design intervention](#) (Weight: 0)

Integrating disassembly to each system type starts with system network analysis. Following the process mapped in Durmisevic's Knowledge Model, all system components are plotted as nodes and all components and connections are labeled. This mapping highlights the hierarchies and dependencies and the order in which they can safely be separated from the total configuration. Network analysis also highlights which components are inaccessible for removal due to sequencing

and permanent connections. This process establishes a disassembly sequence for each system type and provides the necessary framework for disassembly metrics via a component database.

Code: • Enabling condition > Disassembly Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-02, p. 5
 Created: 23.6.2026 r. 10:48 by user, Modified: 23.6.2026 r. 10:48 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Skilled labour (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Integrating disassembly to each system type starts with system network analysis. Following the process mapped in Durmisevic's Knowledge Model, all system components are plotted as nodes and all components and connections are labeled. This mapping highlights the hierarchies and dependencies and the order in which they can safely be separated from the total configuration. Network analysis also highlights which components are inaccessible for removal due to sequencing and permanent connections. This process establishes a disassembly sequence for each system type and provides the necessary framework for disassembly metrics via a component database.

Code: • Mitigation > Design intervention Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-02, p. 5
 Created: 23.6.2026 r. 10:49 by user, Modified: 23.6.2026 r. 10:49 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Skilled labour (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Integrating disassembly to each system type starts with system network analysis. Following the process mapped in Durmisevic's Knowledge Model, all system components are plotted as nodes and all components and connections are labeled. This mapping highlights the hierarchies and dependencies and the order in which they can safely be separated from the total configuration. Network analysis also highlights which components are inaccessible for removal due to sequencing and permanent connections. This process establishes a disassembly sequence for each system type and provides the necessary framework for disassembly metrics via a component database.

Code: • Circular strategy > Direct reuse Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-02, p. 5
 Created: 23.6.2026 r. 10:49 by user, Modified: 23.6.2026 r. 10:49 by user
 Other codes assigned to segment:
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Skilled labour (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Integrating disassembly to each system type starts with system network analysis. Following the process mapped in Durmisevic's Knowledge Model, all system components are plotted as nodes and all components and connections are labeled. This mapping highlights the hierarchies and dependencies and the order in which they can safely be separated from the total configuration. Network analysis also highlights which components are inaccessible for removal due to sequencing and permanent connections. This process establishes a disassembly sequence for each system type and provides the necessary framework for disassembly metrics via a component database.

Code: • Enabling condition > Repair information Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-02, p. 5

Created: 23.6.2026 r. 10:48 by user, Modified: 23.6.2026 r. 10:48 by user

Other codes assigned to segment:

- Circular strategy > Direct reuse (Weight: 0)
- Enabling condition > Disassembly (Weight: 0)
- Enabling condition > Skilled labour (Weight: 0)
- Mitigation > Design intervention (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

The aluminum profiles are first rolled with thermal breaks in a permanent connection. After cutting and processing the profiles with end cuts, holes, notches, and slots, the corner cleats are nailed, screwed, or crimped into position, and then sealed with epoxy to secure mitered frame corners. In coordination with fabricators, these two permanent connections were highlighted as assembly challenges, as mistakes during these steps cannot be reversed.

Code: • Enabling condition > Disassembly Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-02, p. 5

Created: 23.6.2026 r. 10:51 by user, Modified: 23.6.2026 r. 10:51 by user

Other codes assigned to segment:

- Obsolescence > Technical (Weight: 0)
- Negative mechanism > Increased material complexity (Weight: 0)
- Negative mechanism > Lock-in (Weight: 0)
- Mitigation > Design intervention (Weight: 0)

The aluminum profiles are first rolled with thermal breaks in a permanent connection. After cutting and processing the profiles with end cuts, holes, notches, and slots, the corner cleats are nailed, screwed, or crimped into position, and then sealed with epoxy to secure mitered frame corners. In coordination with fabricators, these two permanent connections were highlighted as assembly challenges, as mistakes during these steps cannot be reversed.

Code: • Negative mechanism > Increased material complexity Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-02, p. 5

Created: 23.6.2026 r. 10:51 by user, Modified: 23.6.2026 r. 10:51 by user

Other codes assigned to segment:

- Obsolescence > Technical (Weight: 0)
- Enabling condition > Disassembly (Weight: 0)
- Negative mechanism > Lock-in (Weight: 0)
- Mitigation > Design intervention (Weight: 0)

The aluminum profiles are first rolled with thermal breaks in a permanent connection. After cutting and processing the profiles with end cuts, holes, notches, and slots, the corner cleats are nailed, screwed, or crimped into position, and then sealed with epoxy to secure mitered frame corners. In coordination with fabricators, these two permanent connections were highlighted as assembly challenges, as mistakes during these steps cannot be reversed.

Code: • Mitigation > Design intervention Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-02, p. 5

Created: 23.6.2026 r. 10:51 by user, Modified: 23.6.2026 r. 10:51 by user

Other codes assigned to segment:

- Obsolescence > Technical (Weight: 0)
- Enabling condition > Disassembly (Weight: 0)
- Negative mechanism > Increased material complexity (Weight: 0)
- Negative mechanism > Lock-in (Weight: 0)

The aluminum profiles are first rolled with thermal breaks in a permanent connection. After cutting and processing the profiles with end cuts, holes, notches, and slots, the corner cleats are nailed, screwed, or crimped into position, and then sealed with epoxy to secure mitered frame corners. In

coordination with fabricators, these two permanent connections were highlighted as assembly challenges, as mistakes during these steps cannot be reversed.

Code: • Negative mechanism > Lock-in Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-02, p. 5
 Created: 23.6.2026 r. 10:51 by user, Modified: 23.6.2026 r. 10:51 by user
 Other codes assigned to segment:
 Obsolescence > Technical (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Increased material complexity (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

The aluminum profiles are first rolled with thermal breaks in a permanent connection. After cutting and processing the profiles with end cuts, holes, notches, and slots, the corner cleats are nailed, screwed, or crimped into position, and then sealed with epoxy to secure mitered frame corners. In coordination with fabricators, these two permanent connections were highlighted as assembly challenges, as mistakes during these steps cannot be reversed.

Code: • Obsolescence > Technical Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-02, p. 5
 Created: 23.6.2026 r. 10:50 by user, Modified: 23.6.2026 r. 10:50 by user
 Other codes assigned to segment:
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Increased material complexity (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

Regular renovation of the built environment is essential for regulatory compliance and environmental impact reduction, with window upgrades significantly improving energy efficiency and reducing carbon footprints. Upgrading involves disassembling windows for repair or replacement and extending their life through reuse, repair, and remanufacturing. A Disassembly Map guides this process, showing different routes toward the end-of-life (EoL) for window components, ensuring efficient disassembly, repair, or replacement, thereby minimizing environmental impact. The 'Right to Repair' proposal in the EU promotes sustainable consumption, emphasizing the need for detailed disassembly planning. Robots can enhance disassembly efficiency, reduce costs, and decrease energy use. This study introduces a methodology for robotic window disassembly, focusing on an aluminum frame window to validate the method for upgrading window components. It includes a virtual simulation stage using RoboDk software and a demonstration stage with a UR10e robotic arm, emphasizing the critical role of Disassembly Maps in guiding efficient and sustainable upgrades.

Code: • Circular strategy > Repair Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 1
 Created: 23.6.2026 r. 11:05 by user, Modified: 23.6.2026 r. 11:05 by user
 Other codes assigned to segment:
 Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Regular renovation of the built environment is essential for regulatory compliance and environmental impact reduction, with window upgrades significantly improving energy efficiency and reducing carbon footprints. Upgrading involves disassembling windows for repair or replacement and extending their life through reuse, repair, and remanufacturing. A Disassembly Map guides this process, showing different routes toward the end-of-life (EoL) for window components, ensuring

efficient disassembly, repair, or replacement, thereby minimizing environmental impact. The 'Right to Repair' proposal in the EU promotes sustainable consumption, emphasizing the need for detailed disassembly planning. Robots can enhance disassembly efficiency, reduce costs, and decrease energy use. This study introduces a methodology for robotic window disassembly, focusing on an aluminum frame window to validate the method for upgrading window components. It includes a virtual simulation stage using RoboDk software and a demonstration stage with a UR10e robotic arm, emphasizing the critical role of Disassembly Maps in guiding efficient and sustainable upgrades.

Code: • Enabling condition > Disassembly Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 1
 Created: 23.6.2026 r. 11:06 by user, Modified: 23.6.2026 r. 11:06 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Regular renovation of the built environment is essential for regulatory compliance and environmental impact reduction, with window upgrades significantly improving energy efficiency and reducing carbon footprints. Upgrading involves disassembling windows for repair or replacement and extending their life through reuse, repair, and remanufacturing. A Disassembly Map guides this process, showing different routes toward the end-of-life (EoL) for window components, ensuring efficient disassembly, repair, or replacement, thereby minimizing environmental impact. The 'Right to Repair' proposal in the EU promotes sustainable consumption, emphasizing the need for detailed disassembly planning. Robots can enhance disassembly efficiency, reduce costs, and decrease energy use. This study introduces a methodology for robotic window disassembly, focusing on an aluminum frame window to validate the method for upgrading window components. It includes a virtual simulation stage using RoboDk software and a demonstration stage with a UR10e robotic arm, emphasizing the critical role of Disassembly Maps in guiding efficient and sustainable upgrades.

Code: • Mitigation > Design intervention Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 1
 Created: 23.6.2026 r. 11:06 by user, Modified: 23.6.2026 r. 11:06 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Regular renovation of the built environment is essential for regulatory compliance and environmental impact reduction, with window upgrades significantly improving energy efficiency and reducing carbon footprints. Upgrading involves disassembling windows for repair or replacement and extending their life through reuse, repair, and remanufacturing. A Disassembly Map guides this process, showing different routes toward the end-of-life (EoL) for window components, ensuring efficient disassembly, repair, or replacement, thereby minimizing environmental impact. The 'Right to Repair' proposal in the EU promotes sustainable consumption, emphasizing the need for detailed disassembly planning. Robots can enhance disassembly efficiency, reduce costs, and decrease energy use. This study introduces a methodology for robotic window disassembly, focusing on an aluminum frame window to validate the method for upgrading window components. It includes a virtual

simulation stage using RoboDk software and a demonstration stage with a UR10e robotic arm, emphasizing the critical role of Disassembly Maps in guiding efficient and sustainable upgrades.

Code: • Circular strategy > Component replacement Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 1
 Created: 23.6.2026 r. 11:05 by user, Modified: 23.6.2026 r. 11:05 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Regular renovation of the built environment is essential for regulatory compliance and environmental impact reduction, with window upgrades significantly improving energy efficiency and reducing carbon footprints. Upgrading involves disassembling windows for repair or replacement and extending their life through reuse, repair, and remanufacturing. A Disassembly Map guides this process, showing different routes toward the end-of-life (EoL) for window components, ensuring efficient disassembly, repair, or replacement, thereby minimizing environmental impact. The 'Right to Repair' proposal in the EU promotes sustainable consumption, emphasizing the need for detailed disassembly planning. Robots can enhance disassembly efficiency, reduce costs, and decrease energy use. This study introduces a methodology for robotic window disassembly, focusing on an aluminum frame window to validate the method for upgrading window components. It includes a virtual simulation stage using RoboDk software and a demonstration stage with a UR10e robotic arm, emphasizing the critical role of Disassembly Maps in guiding efficient and sustainable upgrades.

Code: • Circular strategy > Refurbishment Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 1
 Created: 23.6.2026 r. 11:06 by user, Modified: 23.6.2026 r. 11:06 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Regular renovation of the built environment is essential for regulatory compliance and environmental impact reduction, with window upgrades significantly improving energy efficiency and reducing carbon footprints. Upgrading involves disassembling windows for repair or replacement and extending their life through reuse, repair, and remanufacturing. A Disassembly Map guides this process, showing different routes toward the end-of-life (EoL) for window components, ensuring efficient disassembly, repair, or replacement, thereby minimizing environmental impact. The 'Right to Repair' proposal in the EU promotes sustainable consumption, emphasizing the need for detailed disassembly planning. Robots can enhance disassembly efficiency, reduce costs, and decrease energy use. This study introduces a methodology for robotic window disassembly, focusing on an aluminum frame window to validate the method for upgrading window components. It includes a virtual simulation stage using RoboDk software and a demonstration stage with a UR10e robotic arm, emphasizing the critical role of Disassembly Maps in guiding efficient and sustainable upgrades.

Code: • Enabling condition > Repair information Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 1
 Created: 23.6.2026 r. 11:06 by user, Modified: 23.6.2026 r. 11:06 by user

Other codes assigned to segment:

- Circular strategy > Repair (Weight: 0)
- Circular strategy > Component replacement (Weight: 0)
- Circular strategy > Upgrade (Weight: 0)
- Circular strategy > Refurbishment (Weight: 0)
- Enabling condition > Disassembly (Weight: 0)
- Mitigation > Design intervention (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

Regular renovation of the built environment is essential for regulatory compliance and environmental impact reduction, with window upgrades significantly improving energy efficiency and reducing carbon footprints. Upgrading involves disassembling windows for repair or replacement and extending their life through reuse, repair, and remanufacturing. A Disassembly Map guides this process, showing different routes toward the end-of-life (EoL) for window components, ensuring efficient disassembly, repair, or replacement, thereby minimizing environmental impact. The 'Right to Repair' proposal in the EU promotes sustainable consumption, emphasizing the need for detailed disassembly planning. Robots can enhance disassembly efficiency, reduce costs, and decrease energy use. This study introduces a methodology for robotic window disassembly, focusing on an aluminum frame window to validate the method for upgrading window components. It includes a virtual simulation stage using RoboDk software and a demonstration stage with a UR10e robotic arm, emphasizing the critical role of Disassembly Maps in guiding efficient and sustainable upgrades.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-05, p. 1

Created: 23.6.2026 r. 11:06 by user, Modified: 23.6.2026 r. 11:06 by user

Other codes assigned to segment:

- Circular strategy > Repair (Weight: 0)
- Circular strategy > Component replacement (Weight: 0)
- Circular strategy > Upgrade (Weight: 0)
- Circular strategy > Refurbishment (Weight: 0)
- Enabling condition > Disassembly (Weight: 0)
- Enabling condition > Repair information (Weight: 0)
- Mitigation > Design intervention (Weight: 0)

Regular renovation of the built environment is essential for regulatory compliance and environmental impact reduction, with window upgrades significantly improving energy efficiency and reducing carbon footprints. Upgrading involves disassembling windows for repair or replacement and extending their life through reuse, repair, and remanufacturing. A Disassembly Map guides this process, showing different routes toward the end-of-life (EoL) for window components, ensuring efficient disassembly, repair, or replacement, thereby minimizing environmental impact. The 'Right to Repair' proposal in the EU promotes sustainable consumption, emphasizing the need for detailed disassembly planning. Robots can enhance disassembly efficiency, reduce costs, and decrease energy use. This study introduces a methodology for robotic window disassembly, focusing on an aluminum frame window to validate the method for upgrading window components. It includes a virtual simulation stage using RoboDk software and a demonstration stage with a UR10e robotic arm, emphasizing the critical role of Disassembly Maps in guiding efficient and sustainable upgrades.

Code: • Circular strategy > Upgrade Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-05, p. 1

Created: 23.6.2026 r. 11:05 by user, Modified: 23.6.2026 r. 11:05 by user

Other codes assigned to segment:

- Circular strategy > Repair (Weight: 0)
- Circular strategy > Component replacement (Weight: 0)
- Circular strategy > Refurbishment (Weight: 0)
- Enabling condition > Disassembly (Weight: 0)
- Enabling condition > Repair information (Weight: 0)
- Mitigation > Design intervention (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

Upgrading windows involves disassembling their components for repair, replacement, or recycling, thus extending their lifespan and aligning within the New Consumer Agenda and the Circular Economy Action Plan, which aim to overcome barriers to repair and promote sustainability, contributing to the goal of making Europe the first climate-neutral continent by 2050.

Disassembly, therefore, is essential not only for extending the lifecycle of construction components but also for reducing waste through material recovery options such as reuse, remanufacturing, and

Aktas, B. and Charkhab, M. E.

35. Forum Bauinformatik, Hamburg, 2024 423recycling

Code: • Circular strategy > Direct reuse Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-05, p. 1 - 2

Created: 23.6.2026 r. 11:08 by user, Modified: 23.6.2026 r. 11:08 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)

Circular strategy > Component replacement (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Enabling condition > Disassembly (Weight: 0)

Mitigation > Design intervention (Weight: 0)

Upgrading windows involves disassembling their components for repair, replacement, or recycling, thus extending their lifespan and aligning within the New Consumer Agenda and the Circular Economy Action Plan, which aim to overcome barriers to repair and promote sustainability, contributing to the goal of making Europe the first climate-neutral continent by 2050.

Disassembly, therefore, is essential not only for extending the lifecycle of construction components but also for reducing waste through material recovery options such as reuse, remanufacturing, and

Aktas, B. and Charkhab, M. E.

35. Forum Bauinformatik, Hamburg, 2024 423recycling

Code: • Enabling condition > Disassembly Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-05, p. 1 - 2

Created: 23.6.2026 r. 11:08 by user, Modified: 23.6.2026 r. 11:08 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)

Circular strategy > Component replacement (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Circular strategy > Direct reuse (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Mitigation > Design intervention (Weight: 0)

Upgrading windows involves disassembling their components for repair, replacement, or recycling, thus extending their lifespan and aligning within the New Consumer Agenda and the Circular Economy Action Plan, which aim to overcome barriers to repair and promote sustainability, contributing to the goal of making Europe the first climate-neutral continent by 2050.

Disassembly, therefore, is essential not only for extending the lifecycle of construction components but also for reducing waste through material recovery options such as reuse, remanufacturing, and

Aktas, B. and Charkhab, M. E.

35. Forum Bauinformatik, Hamburg, 2024 423recycling

Code: • Circular strategy > Recycling Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-05, p. 1 - 2

Created: 23.6.2026 r. 11:08 by user, Modified: 23.6.2026 r. 11:08 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)

Circular strategy > Component replacement (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Circular strategy > Direct reuse (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

Upgrading windows involves disassembling their components for repair, replacement, or recycling, thus extending their lifespan and aligning within the New Consumer Agenda and the Circular Economy Action Plan, which aim to overcome barriers to repair and promote sustainability, contributing to the goal of making Europe the first climate-neutral continent by 2050.

Disassembly, therefore, is essential not only for extending the lifecycle of construction components but also for reducing waste through material recovery options such as reuse, remanufacturing, and Aktas, B. and Charkhab, M. E.

35. Forum Bauinformatik, Hamburg, 2024 423recycling

Code: • Mitigation > Design intervention Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-05, p. 1 - 2

Created: 23.6.2026 r. 11:09 by user, Modified: 23.6.2026 r. 11:09 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)
 Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)

Upgrading windows involves disassembling their components for repair, replacement, or recycling, thus extending their lifespan and aligning within the New Consumer Agenda and the Circular Economy Action Plan, which aim to overcome barriers to repair and promote sustainability, contributing to the goal of making Europe the first climate-neutral continent by 2050.

Disassembly, therefore, is essential not only for extending the lifecycle of construction components but also for reducing waste through material recovery options such as reuse, remanufacturing, and Aktas, B. and Charkhab, M. E.

35. Forum Bauinformatik, Hamburg, 2024 423recycling

Code: • Circular strategy > Upgrade Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-05, p. 1 - 2

Created: 23.6.2026 r. 11:08 by user, Modified: 23.6.2026 r. 11:08 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)
 Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

Upgrading windows involves disassembling their components for repair, replacement, or recycling, thus extending their lifespan and aligning within the New Consumer Agenda and the Circular Economy Action Plan, which aim to overcome barriers to repair and promote sustainability, contributing to the goal of making Europe the first climate-neutral continent by 2050.

Disassembly, therefore, is essential not only for extending the lifecycle of construction components but also for reducing waste through material recovery options such as reuse, remanufacturing, and Aktas, B. and Charkhab, M. E.

35. Forum Bauinformatik, Hamburg, 2024 423recycling

Code: • Circular strategy > Component replacement Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-05, p. 1 - 2

Created: 23.6.2026 r. 11:08 by user, Modified: 23.6.2026 r. 11:08 by user

Other codes assigned to segment:

- Circular strategy > Repair (Weight: 0)
- Circular strategy > Upgrade (Weight: 0)
- Circular strategy > Direct reuse (Weight: 0)
- Circular strategy > Recycling (Weight: 0)
- Enabling condition > Disassembly (Weight: 0)
- Mitigation > Design intervention (Weight: 0)

Upgrading windows involves disassembling their components for repair, replacement, or recycling, thus extending their lifespan and aligning within the New Consumer Agenda and the Circular Economy Action Plan, which aim to overcome barriers to repair and promote sustainability, contributing to the goal of making Europe the first climate-neutral continent by 2050.

Disassembly, therefore, is essential not only for extending the lifecycle of construction components but also for reducing waste through material recovery options such as reuse, remanufacturing, and

Aktas, B. and Charkhab, M. E.

35. Forum Bauinformatik, Hamburg, 2024 423recycling

Code: • Circular strategy > Repair Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-05, p. 1 - 2

Created: 23.6.2026 r. 11:08 by user, Modified: 23.6.2026 r. 11:08 by user

Other codes assigned to segment:

- Circular strategy > Component replacement (Weight: 0)
- Circular strategy > Upgrade (Weight: 0)
- Circular strategy > Direct reuse (Weight: 0)
- Circular strategy > Recycling (Weight: 0)
- Enabling condition > Disassembly (Weight: 0)
- Mitigation > Design intervention (Weight: 0)

However, most windows are designed and manufactured without considering EoL and disassembly, which significantly increases their environmental burden. Given the complexity of windows and other building components, detailed EoL scenarios must be developed to enhance sustainability.

Code: • Enabling condition > Disassembly Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-05, p. 2

Created: 23.6.2026 r. 11:11 by user, Modified: 23.6.2026 r. 11:11 by user

Other codes assigned to segment:

- Obsolescence > Technical (Weight: 0)
- Enabling condition > Repair information (Weight: 0)
- Negative mechanism > Lock-in (Weight: 0)
- Negative mechanism > Missing repair infrastructure (Weight: 0)
- Mitigation > Design intervention (Weight: 0)

However, most windows are designed and manufactured without considering EoL and disassembly, which significantly increases their environmental burden. Given the complexity of windows and other building components, detailed EoL scenarios must be developed to enhance sustainability.

Code: • Negative mechanism > Lock-in Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-05, p. 2

Created: 23.6.2026 r. 11:11 by user, Modified: 23.6.2026 r. 11:11 by user

Other codes assigned to segment:

- Obsolescence > Technical (Weight: 0)
- Enabling condition > Disassembly (Weight: 0)
- Enabling condition > Repair information (Weight: 0)
- Negative mechanism > Missing repair infrastructure (Weight: 0)
- Mitigation > Design intervention (Weight: 0)

However, most windows are designed and manufactured without considering EoL and disassembly, which significantly increases their environmental burden. Given the complexity of windows and other building components, detailed EoL scenarios must be developed to enhance sustainability.

Code: • Mitigation > Design intervention Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 2
 Created: 23.6.2026 r. 11:11 by user, Modified: 23.6.2026 r. 11:11 by user
 Other codes assigned to segment:
 Obsolescence > Technical (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)

However, most windows are designed and manufactured without considering EoL and disassembly, which significantly increases their environmental burden. Given the complexity of windows and other building components, detailed EoL scenarios must be developed to enhance sustainability.

Code: • Enabling condition > Repair information Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 2
 Created: 23.6.2026 r. 11:11 by user, Modified: 23.6.2026 r. 11:11 by user
 Other codes assigned to segment:
 Obsolescence > Technical (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

However, most windows are designed and manufactured without considering EoL and disassembly, which significantly increases their environmental burden. Given the complexity of windows and other building components, detailed EoL scenarios must be developed to enhance sustainability.

Code: • Obsolescence > Technical Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 2
 Created: 23.6.2026 r. 11:10 by user, Modified: 23.6.2026 r. 11:10 by user
 Other codes assigned to segment:
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

However, most windows are designed and manufactured without considering EoL and disassembly, which significantly increases their environmental burden. Given the complexity of windows and other building components, detailed EoL scenarios must be developed to enhance sustainability.

Code: • Negative mechanism > Missing repair infrastructure Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 2
 Created: 23.6.2026 r. 11:11 by user, Modified: 23.6.2026 r. 11:11 by user
 Other codes assigned to segment:
 Obsolescence > Technical (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

Effective disassembly planning requires a high level of detail regarding the physical interfaces among building components to identify physical constraints and sequence the necessary steps for dismantling [1]. This process defines and logically orders the deconstruction activities.

Code: • Mitigation > Design intervention Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 2
 Created: 23.6.2026 r. 11:13 by user, Modified: 23.6.2026 r. 11:13 by user
 Other codes assigned to segment:
 Enabling condition > Component accessibility (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Skilled labour (Weight: 0)

Effective disassembly planning requires a high level of detail regarding the physical interfaces among building components to identify physical constraints and sequence the necessary steps for dismantling [1]. This process defines and logically orders the deconstruction activities.

Code: • Enabling condition > Repair information Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 2
 Created: 23.6.2026 r. 11:12 by user, Modified: 23.6.2026 r. 11:12 by user
 Other codes assigned to segment:
 Enabling condition > Component accessibility (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Skilled labour (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

Effective disassembly planning requires a high level of detail regarding the physical interfaces among building components to identify physical constraints and sequence the necessary steps for dismantling [1]. This process defines and logically orders the deconstruction activities.

Code: • Enabling condition > Disassembly Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 2
 Created: 23.6.2026 r. 11:12 by user, Modified: 23.6.2026 r. 11:12 by user
 Other codes assigned to segment:
 Enabling condition > Component accessibility (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Skilled labour (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

Effective disassembly planning requires a high level of detail regarding the physical interfaces among building components to identify physical constraints and sequence the necessary steps for dismantling [1]. This process defines and logically orders the deconstruction activities.

Code: • Enabling condition > Component accessibility Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 2
 Created: 23.6.2026 r. 11:12 by user, Modified: 23.6.2026 r. 11:12 by user
 Other codes assigned to segment:
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Skilled labour (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

Effective disassembly planning requires a high level of detail regarding the physical interfaces among building components to identify physical constraints and sequence the necessary steps for dismantling [1]. This process defines and logically orders the deconstruction activities.

Code: • Enabling condition > Skilled labour Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 2
 Created: 23.6.2026 r. 11:12 by user, Modified: 23.6.2026 r. 11:12 by user
 Other codes assigned to segment:
 Enabling condition > Component accessibility (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)

Enabling condition > Repair information (Weight: 0)
Mitigation > Design intervention (Weight: 0)

construction documents such as drawings and specifications, which are highly detailed, are a necessity before constructing the building. On the other hand, construction documents are also challenging to get for many buildings and window types. Therefore, the first step of this case study is to search for the aluminum windows section drawing to model it and define the geometrical precedence and constraints before defining the disassembly map.

Code: • Mitigation > Design intervention Weight score: 0
Search B - Disassembly reuse and facade recovery > B-GS-05, p. 4
Created: 23.6.2026 r. 11:14 by user, Modified: 23.6.2026 r. 11:14 by user
Other codes assigned to segment:
Enabling condition > Component accessibility (Weight: 0)
Enabling condition > Disassembly (Weight: 0)
Enabling condition > Repair information (Weight: 0)
Negative mechanism > Missing repair infrastructure (Weight: 0)

construction documents such as drawings and specifications, which are highly detailed, are a necessity before constructing the building. On the other hand, construction documents are also challenging to get for many buildings and window types. Therefore, the first step of this case study is to search for the aluminum windows section drawing to model it and define the geometrical precedence and constraints before defining the disassembly map.

Code: • Enabling condition > Repair information Weight score: 0
Search B - Disassembly reuse and facade recovery > B-GS-05, p. 4
Created: 23.6.2026 r. 11:13 by user, Modified: 23.6.2026 r. 11:13 by user
Other codes assigned to segment:
Enabling condition > Component accessibility (Weight: 0)
Enabling condition > Disassembly (Weight: 0)
Negative mechanism > Missing repair infrastructure (Weight: 0)
Mitigation > Design intervention (Weight: 0)

construction documents such as drawings and specifications, which are highly detailed, are a necessity before constructing the building. On the other hand, construction documents are also challenging to get for many buildings and window types. Therefore, the first step of this case study is to search for the aluminum windows section drawing to model it and define the geometrical precedence and constraints before defining the disassembly map.

Code: • Negative mechanism > Missing repair infrastructure Weight score: 0
Search B - Disassembly reuse and facade recovery > B-GS-05, p. 4
Created: 23.6.2026 r. 11:14 by user, Modified: 23.6.2026 r. 11:14 by user
Other codes assigned to segment:
Enabling condition > Component accessibility (Weight: 0)
Enabling condition > Disassembly (Weight: 0)
Enabling condition > Repair information (Weight: 0)
Mitigation > Design intervention (Weight: 0)

construction documents such as drawings and specifications, which are highly detailed, are a necessity before constructing the building. On the other hand, construction documents are also challenging to get for many buildings and window types. Therefore, the first step of this case study is to search for the aluminum windows section drawing to model it and define the geometrical precedence and constraints before defining the disassembly map.

Code: • Enabling condition > Disassembly Weight score: 0
Search B - Disassembly reuse and facade recovery > B-GS-05, p. 4
Created: 23.6.2026 r. 11:14 by user, Modified: 23.6.2026 r. 11:14 by user

Other codes assigned to segment:

Enabling condition > Component accessibility (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

construction documents such as drawings and specifications, which are highly detailed, are a necessity before constructing the building. On the other hand, construction documents are also challenging to get for many buildings and window types. Therefore, the first step of this case study is to search for the aluminum windows section drawing to model it and define the geometrical precedence and constraints before defining the disassembly map.

Code: • Enabling condition > Component accessibility Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 4
 Created: 23.6.2026 r. 11:14 by user, Modified: 23.6.2026 r. 11:14 by user
 Other codes assigned to segment:
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

The disassembly map is formulated as a tree search problem, beginning from the assembled state and identifying a sequence of actions to reach the disassembled state for 0.4mx0.4m aluminum-framed single-glazed window units. (Figure 1). The 3D model of the window, which is also the assembled window model, is utilized for disassembly map development to investigate geometrical preferences, fastener locations, and details following a disassembly-by-assembly approach. This Aktas, B. and Charkhab, M. E.

35. Forum Bauinformatik, Hamburg, 2024 426process defines each step of robotic automation by disassembly motion sequences. The outer aluminum profiles were planned to be disassembled individually for the four edges. Subsequently, the gasket would be removed and recycled, and the single-glazed flat glass would be removed and replaced with double or triple-glazed ones.

Code: • Circular strategy > Component replacement Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 4 - 5
 Created: 23.6.2026 r. 11:15 by user, Modified: 23.6.2026 r. 11:15 by user
 Other codes assigned to segment:
 Circular strategy > Upgrade (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

The disassembly map is formulated as a tree search problem, beginning from the assembled state and identifying a sequence of actions to reach the disassembled state for 0.4mx0.4m aluminum-framed single-glazed window units. (Figure 1). The 3D model of the window, which is also the assembled window model, is utilized for disassembly map development to investigate geometrical preferences, fastener locations, and details following a disassembly-by-assembly approach. This Aktas, B. and Charkhab, M. E.

35. Forum Bauinformatik, Hamburg, 2024 426process defines each step of robotic automation by disassembly motion sequences. The outer aluminum profiles were planned to be disassembled individually for the four edges. Subsequently, the gasket would be removed and recycled, and the single-glazed flat glass would be removed and replaced with double or triple-glazed ones.

Code: • Circular strategy > Recycling Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 4 - 5
 Created: 23.6.2026 r. 11:15 by user, Modified: 23.6.2026 r. 11:15 by user

Other codes assigned to segment:

Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

The disassembly map is formulated as a tree search problem, beginning from the assembled state and identifying a sequence of actions to reach the disassembled state for 0.4mx0.4m aluminum-framed single-glazed window units. (Figure 1). The 3D model of the window, which is also the assembled window model, is utilized for disassembly map development to investigate geometrical preferences, fastener locations, and details following a disassembly-by-assembly approach. This Aktas, B. and Charkhab, M. E.

35. Forum Bauinformatik, Hamburg, 2024 426process defines each step of robotic automation by disassembly motion sequences. The outer aluminum profiles were planned to be disassembled individually for the four edges. Subsequently, the gasket would be removed and recycled, and the single-glazed flat glass would be removed and replaced with double or triple-glazed ones.

Code: • Mitigation > Design intervention Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-05, p. 4 - 5

Created: 23.6.2026 r. 11:16 by user, Modified: 23.6.2026 r. 11:16 by user

Other codes assigned to segment:

Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)

The disassembly map is formulated as a tree search problem, beginning from the assembled state and identifying a sequence of actions to reach the disassembled state for 0.4mx0.4m aluminum-framed single-glazed window units. (Figure 1). The 3D model of the window, which is also the assembled window model, is utilized for disassembly map development to investigate geometrical preferences, fastener locations, and details following a disassembly-by-assembly approach. This Aktas, B. and Charkhab, M. E.

35. Forum Bauinformatik, Hamburg, 2024 426process defines each step of robotic automation by disassembly motion sequences. The outer aluminum profiles were planned to be disassembled individually for the four edges. Subsequently, the gasket would be removed and recycled, and the single-glazed flat glass would be removed and replaced with double or triple-glazed ones.

Code: • Enabling condition > Repair information Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-05, p. 4 - 5

Created: 23.6.2026 r. 11:16 by user, Modified: 23.6.2026 r. 11:16 by user

Other codes assigned to segment:

Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

The disassembly map is formulated as a tree search problem, beginning from the assembled state and identifying a sequence of actions to reach the disassembled state for 0.4mx0.4m aluminum-framed single-glazed window units. (Figure 1). The 3D model of the window, which is also the assembled window model, is utilized for disassembly map development to investigate geometrical preferences, fastener locations, and details following a disassembly-by-assembly approach. This Aktas, B. and Charkhab, M. E.

35. Forum Bauinformatik, Hamburg, 2024 426process defines each step of robotic automation by

disassembly motion sequences. The outer aluminum profiles were planned to be disassembled individually for the four edges. Subsequently, the gasket would be removed and recycled, and the single-glazed flat glass would be removed and replaced with double or triple-glazed ones.

Code: • Enabling condition > Disassembly Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 4 - 5
 Created: 23.6.2026 r. 11:16 by user, Modified: 23.6.2026 r. 11:16 by user
 Other codes assigned to segment:
 Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

The disassembly map is formulated as a tree search problem, beginning from the assembled state and identifying a sequence of actions to reach the disassembled state for 0.4mx0.4m aluminum-framed single-glazed window units. (Figure 1). The 3D model of the window, which is also the assembled window model, is utilized for disassembly map development to investigate geometrical preferences, fastener locations, and details following a disassembly-by-assembly approach. This Aktas, B. and Charkhab, M. E.

35. Forum Bauinformatik, Hamburg, 2024 426process defines each step of robotic automation by disassembly motion sequences. The outer aluminum profiles were planned to be disassembled individually for the four edges. Subsequently, the gasket would be removed and recycled, and the single-glazed flat glass would be removed and replaced with double or triple-glazed ones.

Code: • Circular strategy > Upgrade Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 4 - 5
 Created: 23.6.2026 r. 11:15 by user, Modified: 23.6.2026 r. 11:15 by user
 Other codes assigned to segment:
 Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

The study highlighted the critical role of the Disassembly Map in defining disassembly path planning and motion sequences, ensuring an efficient and systematic process for manual and robotic disassembly through section drawings of the windows at the EoL stage. This research contributes to the field by developing a methodology for upgrading building components to meet regulatory requirements and reduce environmental impacts. Emphasizing the use of disassembly maps not only ensures the precision but also the effectiveness of the upgrading process, promoting sustainability in the built environment.

Code: • Mitigation > Lifecycle verification Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 7
 Created: 23.6.2026 r. 11:18 by user, Modified: 23.6.2026 r. 11:18 by user
 Other codes assigned to segment:
 Circular strategy > Upgrade (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

The study highlighted the critical role of the Disassembly Map in defining disassembly path planning and motion sequences, ensuring an efficient and systematic process for manual and robotic disassembly through section drawings of the windows at the EoL stage. This research contributes to the field by developing a methodology for upgrading building components to meet regulatory requirements and reduce environmental impacts. Emphasizing the use of disassembly maps not only

ensures the precision but also the effectiveness of the upgrading process, promoting sustainability in the built environment.

Code: • Enabling condition > Disassembly Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 7
 Created: 23.6.2026 r. 11:17 by user, Modified: 23.6.2026 r. 11:17 by user
 Other codes assigned to segment:
 Circular strategy > Upgrade (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The study highlighted the critical role of the Disassembly Map in defining disassembly path planning and motion sequences, ensuring an efficient and systematic process for manual and robotic disassembly through section drawings of the windows at the EoL stage. This research contributes to the field by developing a methodology for upgrading building components to meet regulatory requirements and reduce environmental impacts. Emphasizing the use of disassembly maps not only ensures the precision but also the effectiveness of the upgrading process, promoting sustainability in the built environment.

Code: • Enabling condition > Repair information Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 7
 Created: 23.6.2026 r. 11:17 by user, Modified: 23.6.2026 r. 11:17 by user
 Other codes assigned to segment:
 Circular strategy > Upgrade (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The study highlighted the critical role of the Disassembly Map in defining disassembly path planning and motion sequences, ensuring an efficient and systematic process for manual and robotic disassembly through section drawings of the windows at the EoL stage. This research contributes to the field by developing a methodology for upgrading building components to meet regulatory requirements and reduce environmental impacts. Emphasizing the use of disassembly maps not only ensures the precision but also the effectiveness of the upgrading process, promoting sustainability in the built environment.

Code: • Mitigation > Design intervention Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 7
 Created: 23.6.2026 r. 11:17 by user, Modified: 23.6.2026 r. 11:17 by user
 Other codes assigned to segment:
 Circular strategy > Upgrade (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The study highlighted the critical role of the Disassembly Map in defining disassembly path planning and motion sequences, ensuring an efficient and systematic process for manual and robotic disassembly through section drawings of the windows at the EoL stage. This research contributes to the field by developing a methodology for upgrading building components to meet regulatory requirements and reduce environmental impacts. Emphasizing the use of disassembly maps not only ensures the precision but also the effectiveness of the upgrading process, promoting sustainability in the built environment.

Code: • Circular strategy > Upgrade Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-05, p. 7
 Created: 23.6.2026 r. 11:17 by user, Modified: 23.6.2026 r. 11:17 by user

Other codes assigned to segment:

Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The main contributions of the thesis are:

Firstly, an exploration of the notion of 'component management'. This challenges the assumption that components removed from the building stock must either be: a) directly reused, which can often be impractical, and is rarely given due attention, or b) sent to waste management, which wastes embodied impacts. Instead, the role and implementation of repurposing and upcycling are described, alongside a procedure for comprehensively checking the practicality of direct reuse; Secondly, the development of an urban-level 'triage': a process to separate out components for reuse, repurposing and upcycling, from those for which downcycling or energy recovery are the best option. Key to the triage is an information system; the thesis reviews current means of understanding existing buildings as material banks and presents a new approach to gathering this information; Thirdly, a proposal for an innovative manufacturing enterprise using secondary timber in a new product: cross-laminated secondary timber. This provides an exemplar case study of the potential for industrial-scale upcycling. A proof of concept study is presented, with a preliminary examination of technical feasibility and leading the way for additional investigation of socioeconomic and environmental sustainability, and, ideally, future pilot- and commercial-scale implementation.

The implications of this product case study are synthesised with the other parts of the thesis in a discussion of areas for future research, policy and practical action to evolve a more nuanced and sustainable management of existing building components.

Code: • Enabling condition > Service infrastructure Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-10, p. 5
 Created: 23.6.2026 r. 11:21 by user, Modified: 23.6.2026 r. 11:21 by user
 Other codes assigned to segment:

Circular strategy > Refurbishment (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Business-model intervention (Weight: 0)
 Mitigation > Policy intervention (Weight: 0)

The main contributions of the thesis are:

Firstly, an exploration of the notion of 'component management'. This challenges the assumption that components removed from the building stock must either be: a) directly reused, which can often be impractical, and is rarely given due attention, or b) sent to waste management, which wastes embodied impacts. Instead, the role and implementation of repurposing and upcycling are described, alongside a procedure for comprehensively checking the practicality of direct reuse; Secondly, the development of an urban-level 'triage': a process to separate out components for reuse, repurposing and upcycling, from those for which downcycling or energy recovery are the best option. Key to the triage is an information system; the thesis reviews current means of understanding existing buildings as material banks and presents a new approach to gathering this information; Thirdly, a proposal for an innovative manufacturing enterprise using secondary timber in a new product: cross-laminated secondary timber. This provides an exemplar case study of the

potential for industrial-scale upcycling. A proof of concept study is presented, with a preliminary examination of technical feasibility and leading the way for additional investigation of socioeconomic and environmental sustainability, and, ideally, future pilot- and commercial-scale implementation.

The implications of this product case study are synthesised with the other parts of the thesis in a discussion of areas for future research, policy and practical action to evolve a more nuanced and sustainable management of existing building components.

Code: • Negative mechanism > Missing repair infrastructure Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-10, p. 5

Created: 23.6.2026 r. 11:21 by user, Modified: 23.6.2026 r. 11:21 by user

Other codes assigned to segment:

Circular strategy > Refurbishment (Weight: 0)

Circular strategy > Direct reuse (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Mitigation > Business-model intervention (Weight: 0)

Mitigation > Policy intervention (Weight: 0)

The main contributions of the thesis are:

Firstly, an exploration of the notion of ‘component management’. This challenges the assumption that components removed from the building stock must either be: a) directly reused, which can often be impractical, and is rarely given due attention, or b) sent to waste management, which wastes embodied impacts. Instead, the role and implementation of repurposing and upcycling are described, alongside a procedure for comprehensively checking the practicality of direct reuse; Secondly, the development of an urban-level ‘triage’: a process to separate out components for

reuse, repurposing and upcycling, from those for which downcycling or energy recovery are the best option. Key to the triage is an information system; the thesis reviews current means of understanding existing buildings as material banks and presents a new approach to gathering this information; Thirdly, a proposal for an innovative manufacturing enterprise using secondary timber in a new

product: cross-laminated secondary timber. This provides an exemplar case study of the potential for industrial-scale upcycling. A proof of concept study is presented, with a preliminary examination of technical feasibility and leading the way for additional investigation of socioeconomic and environmental sustainability, and, ideally, future pilot- and commercial-scale implementation.

The implications of this product case study are synthesised with the other parts of the thesis in a discussion of areas for future research, policy and practical action to evolve a more nuanced and sustainable management of existing building components.

Code: • Mitigation > Business-model intervention Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-10, p. 5

Created: 23.6.2026 r. 11:21 by user, Modified: 23.6.2026 r. 11:21 by user

Other codes assigned to segment:

Circular strategy > Refurbishment (Weight: 0)

Circular strategy > Direct reuse (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Policy intervention (Weight: 0)

The main contributions of the thesis are:

Firstly, an exploration of the notion of ‘component management’. This challenges the

assumption that components removed from the building stock must either be: a) directly reused, which can often be impractical, and is rarely given due attention, or b) sent to waste management, which wastes embodied impacts. Instead, the role and implementation of repurposing and upcycling are described, alongside a procedure for comprehensively checking the practicality of direct reuse; Secondly, the development of an urban-level 'triage': a process to separate out components for reuse, repurposing and upcycling, from those for which downcycling or energy recovery are the best option. Key to the triage is an information system; the thesis reviews current means of understanding existing buildings as material banks and presents a new approach to gathering this information; Thirdly, a proposal for an innovative manufacturing enterprise using secondary timber in a new product: cross-laminated secondary timber. This provides an exemplar case study of the potential for industrial-scale upcycling. A proof of concept study is presented, with a preliminary examination of technical feasibility and leading the way for additional investigation of socioeconomic and environmental sustainability, and, ideally, future pilot- and commercial-scale implementation.

The implications of this product case study are synthesised with the other parts of the thesis in a discussion of areas for future research, policy and practical action to evolve a more nuanced and sustainable management of existing building components.

Code: • Mitigation > Policy intervention Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-10, p. 5

Created: 23.6.2026 r. 11:21 by user, Modified: 23.6.2026 r. 11:21 by user

Other codes assigned to segment:

Circular strategy > Refurbishment (Weight: 0)

Circular strategy > Direct reuse (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Business-model intervention (Weight: 0)

The main contributions of the thesis are:

Firstly, an exploration of the notion of 'component management'. This challenges the assumption that components removed from the building stock must either be: a) directly reused, which can often be impractical, and is rarely given due attention, or b) sent to waste management, which wastes embodied impacts. Instead, the role and implementation of repurposing and upcycling are described, alongside a procedure for comprehensively checking the practicality of direct reuse; Secondly, the development of an urban-level 'triage': a process to separate out components for

reuse, repurposing and upcycling, from those for which downcycling or energy recovery are the best option. Key to the triage is an information system; the thesis reviews current means of understanding existing buildings as material banks and presents a new approach to gathering this information; Thirdly, a proposal for an innovative manufacturing enterprise using secondary timber in a new product: cross-laminated secondary timber. This provides an exemplar case study of the potential for industrial-scale upcycling. A proof of concept study is presented, with a preliminary examination of technical feasibility and leading the way for additional investigation of socioeconomic and environmental sustainability, and, ideally, future pilot- and commercial-scale implementation.

The implications of this product case study are synthesised with the other parts of the thesis in a discussion of areas for future research, policy and practical action to evolve a more nuanced and sustainable management of existing building components.

Code: • Circular strategy > Direct reuse Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-10, p. 5

Created: 23.6.2026 r. 11:20 by user, Modified: 23.6.2026 r. 11:20 by user

Other codes assigned to segment:

- Circular strategy > Refurbishment (Weight: 0)
- Circular strategy > Recycling (Weight: 0)
- Enabling condition > Repair information (Weight: 0)
- Enabling condition > Service infrastructure (Weight: 0)
- Negative mechanism > Missing repair infrastructure (Weight: 0)
- Mitigation > Business-model intervention (Weight: 0)
- Mitigation > Policy intervention (Weight: 0)

The main contributions of the thesis are:

Firstly, an exploration of the notion of 'component management'. This challenges the assumption that components removed from the building stock must either be: a) directly reused, which can often be impractical, and is rarely given due attention, or b) sent to waste management, which wastes embodied impacts. Instead, the role and implementation of repurposing and upcycling are described, alongside a procedure for comprehensively checking the practicality of direct reuse; Secondly, the development of an urban-level 'triage': a process to separate out components for reuse, repurposing and upcycling, from those for which downcycling or energy recovery are the best option. Key to the triage is an information system; the thesis reviews current means of understanding existing buildings as material banks and presents a new approach to gathering this information; Thirdly, a proposal for an innovative manufacturing enterprise using secondary timber in a new product: cross-laminated secondary timber. This provides an exemplar case study of the potential for industrial-scale upcycling. A proof of concept study is presented, with a preliminary examination of technical feasibility and leading the way for additional investigation of socioeconomic and environmental sustainability, and, ideally, future pilot- and commercial-scale implementation.

The implications of this product case study are synthesised with the other parts of the thesis in a discussion of areas for future research, policy and practical action to evolve a more nuanced and sustainable management of existing building components.

Code: • Circular strategy > Recycling Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-10, p. 5

Created: 23.6.2026 r. 11:20 by user, Modified: 23.6.2026 r. 11:20 by user

Other codes assigned to segment:

- Circular strategy > Refurbishment (Weight: 0)
- Circular strategy > Direct reuse (Weight: 0)
- Enabling condition > Repair information (Weight: 0)
- Enabling condition > Service infrastructure (Weight: 0)
- Negative mechanism > Missing repair infrastructure (Weight: 0)
- Mitigation > Business-model intervention (Weight: 0)
- Mitigation > Policy intervention (Weight: 0)

The main contributions of the thesis are:

Firstly, an exploration of the notion of 'component management'. This challenges the assumption that components removed from the building stock must either be: a) directly reused, which can often be impractical, and is rarely given due attention, or b) sent to waste management, which wastes embodied impacts. Instead, the role and implementation of repurposing and upcycling are described, alongside a procedure for comprehensively checking the practicality of direct reuse; Secondly, the development of an urban-level 'triage': a process to separate out components for reuse, repurposing and upcycling, from those for which downcycling or energy recovery are the best option. Key to the triage is an information system; the thesis reviews current means of

understanding existing buildings as material banks and presents a new approach to gathering this information; Thirdly, a proposal for an innovative manufacturing enterprise using secondary timber in a new

product: cross-laminated secondary timber. This provides an exemplar case study of the potential for industrial-scale upcycling. A proof of concept study is presented, with a preliminary examination of technical feasibility and leading the way for additional investigation of socioeconomic and environmental sustainability, and, ideally, future pilot- and commercial-scale implementation.

The implications of this product case study are synthesised with the other parts of the thesis in a discussion of areas for future research, policy and practical action to evolve a more nuanced and sustainable management of existing building components.

Code: • Circular strategy > Refurbishment Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-10, p. 5

Created: 23.6.2026 r. 11:20 by user, Modified: 23.6.2026 r. 11:20 by user

Other codes assigned to segment:

Circular strategy > Direct reuse (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Business-model intervention (Weight: 0)

Mitigation > Policy intervention (Weight: 0)

The main contributions of the thesis are:

Firstly, an exploration of the notion of 'component management'. This challenges the assumption that components removed from the building stock must either be: a) directly reused, which can often be impractical, and is rarely given due attention, or b) sent to waste management, which wastes embodied impacts. Instead, the role and implementation of repurposing and upcycling are described, alongside a procedure for comprehensively checking the practicality of direct reuse; Secondly, the development of an urban-level 'triage': a process to separate out components for

reuse, repurposing and upcycling, from those for which downcycling or energy recovery are the best option. Key to the triage is an information system; the thesis reviews current means of understanding existing buildings as material banks and presents a new approach to gathering this information; Thirdly, a proposal for an innovative manufacturing enterprise using secondary timber in a new product: cross-laminated secondary timber. This provides an exemplar case study of the potential for industrial-scale upcycling. A proof of concept study is presented, with a preliminary examination of technical feasibility and leading the way for additional investigation of socioeconomic and environmental sustainability, and, ideally, future pilot- and commercial-scale implementation.

The implications of this product case study are synthesised with the other parts of the thesis in a discussion of areas for future research, policy and practical action to evolve a more nuanced and sustainable management of existing building components.

Code: • Enabling condition > Repair information Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-10, p. 5

Created: 23.6.2026 r. 11:20 by user, Modified: 23.6.2026 r. 11:20 by user

Other codes assigned to segment:

Circular strategy > Refurbishment (Weight: 0)

Circular strategy > Direct reuse (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Business-model intervention (Weight: 0)
 Mitigation > Policy intervention (Weight: 0)

the direct transfer of components from one site to another leaves the demand project fully exposed to the risks of 'insufficient quantities of components', 'inconsistent quality' and 'lack of performance guarantees'. These uncertainties over quantity and quality are likely to be closely linked to 'risk in specifying reused components', concerns about 'additional construction complexity' and 'insurance constraints'. The development of reuse markets and supply chain infrastructure therefore has the potential to address several linked barriers to reuse by increasing confidence in component quantities and quality. The extent to which it can address them depends on the scope of activities undertaken by intermediaries in such an infrastructure; for instance, carrying out tests and providing warranties.

Barriers to reuse

Many of the publications note that existing codes and standards are composed based on the properties of new materials, and therefore encourage their use. An advocate for reuse (and effective spokesperson for the salvage industry), Thornton Kay, suggests this is not as significant a barrier as commonly thought: the UK Building Regulations, Regulation 7 (Materials and workmanship), allows the use of any material, as long as it is fit for purpose (BioRegional and Salvo, 2010; Salvo News, 2013). This is typically achieved by certification (Hurley and Hobbs, 2005), but that is not the only route to demonstrating fitness for purpose. Making specific reference to clauses in Regulation 7, Kay establishes that a) building control bodies should not impede the use of materials that are suitable for an intended purpose, and b) past experience, such as use in an existing building, can be used to demonstrate suitability (HM Government, 2013; Salvo News, 2013). Nonetheless, there is a burden on a member of the project team to make the claim of fitness for purpose, and if suppliers, specifiers and contractors show reluctance, as the research suggests, client confidence will remain low. Private homeowners purchasing reclaimed materials from salvage yards may be reassured, but for mainstream construction organisations and their insurers, some form of certification is the norm

Code: • Enabling condition > User acceptance Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-10, p. 49 - 50

Created: 23.6.2026 r. 12:12 by user, Modified: 23.6.2026 r. 12:12 by user

Other codes assigned to segment:

Circular strategy > Direct reuse (Weight: 0)

Obsolescence > Economic (Weight: 0)

Obsolescence > Regulatory (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Enabling condition > Certification (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Business-model intervention (Weight: 0)

Mitigation > Policy intervention (Weight: 0)

the direct transfer of components from one site to another leaves the demand project fully exposed to the risks of 'insufficient quantities of components', 'inconsistent quality' and 'lack of performance guarantees'. These uncertainties over quantity and quality are likely to be closely linked to 'risk in specifying reused components', concerns about 'additional construction complexity' and 'insurance constraints'. The development of reuse markets and supply chain infrastructure therefore has the potential to address several linked barriers to reuse by increasing confidence in component quantities and quality. The extent to which it can address them depends on the scope of activities undertaken by intermediaries in such an infrastructure; for instance, carrying out tests and providing warranties.

Barriers to reuse

Many of the publications note that existing codes and standards are composed based on the properties of new materials, and therefore encourage their use. An advocate for reuse (and effective spokesperson for the salvage industry), Thornton Kay, suggests this is not as significant a barrier as commonly thought: the UK Building Regulations, Regulation 7 (Materials and workmanship), allows the use of any material, as long as it is fit for purpose (BioRegional and Salvo, 2010; Salvo News, 2013). This is typically achieved by certification (Hurley and Hobbs, 2005), but that is not the only route to demonstrating fitness for purpose. Making specific reference to clauses in Regulation 7, Kay establishes that a) building control bodies should not impede the use of materials that are suitable for an intended purpose, and b) past experience, such as use in an existing building, can be used to demonstrate suitability (HM Government, 2013; Salvo News, 2013). Nonetheless, there is a burden on a member of the project team to make the claim of fitness for purpose, and if suppliers, specifiers and contractors show reluctance, as the research suggests, client confidence will remain low. Private homeowners purchasing reclaimed materials from salvage yards may be reassured, but for mainstream construction organisations and their insurers, some form of certification is the norm

Code: • Circular strategy > Direct reuse Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-10, p. 49 - 50

Created: 23.6.2026 r. 12:12 by user, Modified: 23.6.2026 r. 12:12 by user

Other codes assigned to segment:

Obsolescence > Economic (Weight: 0)

Obsolescence > Regulatory (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Enabling condition > Certification (Weight: 0)

Enabling condition > User acceptance (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Business-model intervention (Weight: 0)

Mitigation > Policy intervention (Weight: 0)

the direct transfer of

components from one site to another leaves the demand project fully exposed to the risks of 'insufficient quantities of components', 'inconsistent quality' and 'lack of performance guarantees'. These uncertainties over quantity and quality are likely to be closely linked to 'risk in specifying reused components', concerns about 'additional construction complexity' and 'insurance constraints'. The development of reuse markets and supply chain infrastructure therefore has the potential to address several linked barriers to reuse by increasing confidence in component quantities and quality. The extent to which it can address them depends on the scope of activities undertaken by intermediaries in such an infrastructure; for instance, carrying out tests and providing warranties.

Barriers to reuse

Many of the publications note that existing codes and standards are composed based on the properties of new materials, and therefore encourage their use. An advocate for reuse (and effective spokesperson for the salvage industry), Thornton Kay, suggests this is not as significant a barrier as commonly thought: the UK Building Regulations, Regulation 7 (Materials and workmanship), allows the use of any material, as long as it is fit for purpose (BioRegional and Salvo, 2010; Salvo News, 2013). This is typically achieved by certification (Hurley and Hobbs, 2005), but that is not the only route to demonstrating fitness for purpose. Making specific reference to clauses in Regulation 7, Kay establishes that a) building control bodies should not impede the use of materials that are suitable for an intended purpose, and b) past experience, such as use in an existing building, can be used to demonstrate suitability (HM Government, 2013; Salvo News, 2013). Nonetheless, there is a burden on a member of the project team to make the claim of fitness for purpose, and if suppliers, specifiers and contractors show reluctance, as the research suggests, client confidence will remain low. Private

homeowners purchasing reclaimed materials from salvage yards may be reassured, but for mainstream construction organisations and their insurers, some form of certification is the norm

Code: • [Obsolescence > Regulatory](#) Weight score: 0

[Search B - Disassembly reuse and facade recovery > B-GS-10, p. 49 - 50](#)

Created: 23.6.2026 r. 12:12 by user, Modified: 23.6.2026 r. 12:12 by user

Other codes assigned to segment:

[Circular strategy > Direct reuse](#) (Weight: 0)

[Obsolescence > Economic](#) (Weight: 0)

[Enabling condition > Service infrastructure](#) (Weight: 0)

[Enabling condition > Certification](#) (Weight: 0)

[Enabling condition > User acceptance](#) (Weight: 0)

[Negative mechanism > Missing repair infrastructure](#) (Weight: 0)

[Mitigation > Business-model intervention](#) (Weight: 0)

[Mitigation > Policy intervention](#) (Weight: 0)

the direct transfer of

components from one site to another leaves the demand project fully exposed to the risks of 'insufficient quantities of components', 'inconsistent quality' and 'lack of performance guarantees'. These uncertainties over quantity and quality are likely to be closely linked to 'risk in specifying reused components', concerns about 'additional construction complexity' and 'insurance constraints'. The development of reuse markets and supply chain infrastructure therefore has the potential to address several linked barriers to reuse by increasing confidence in component quantities and quality. The extent to which it can address them depends on the scope of activities undertaken by intermediaries in such an infrastructure; for instance, carrying out tests and providing warranties.

Barriers to reuse

Many of the publications note that existing codes and standards are composed based on the properties of new materials, and therefore encourage their use. An advocate for reuse (and effective spokesperson for the salvage industry), Thornton Kay, suggests this is not as significant a barrier as commonly thought: the UK Building Regulations, Regulation 7 (Materials and workmanship), allows the use of any material, as long as it is fit for purpose (BioRegional and Salvo, 2010; Salvo News, 2013). This is typically achieved by certification (Hurley and Hobbs, 2005), but that is not the only route to demonstrating fitness for purpose. Making specific reference to clauses in Regulation 7, Kay establishes that a) building control bodies should not impede the use of materials that are suitable for an intended purpose, and b) past experience, such as use in an existing building, can be used to demonstrate suitability (HM Government, 2013; Salvo News, 2013). Nonetheless, there is a burden on a member of the project team to make the claim of fitness for purpose, and if suppliers, specifiers and contractors show reluctance, as the research suggests, client confidence will remain low. Private homeowners purchasing reclaimed materials from salvage yards may be reassured, but for mainstream construction organisations and their insurers, some form of certification is the norm

Code: • [Negative mechanism > Missing repair infrastructure](#) Weight score: 0

[Search B - Disassembly reuse and facade recovery > B-GS-10, p. 49 - 50](#)

Created: 23.6.2026 r. 12:13 by user, Modified: 23.6.2026 r. 12:13 by user

Other codes assigned to segment:

[Circular strategy > Direct reuse](#) (Weight: 0)

[Obsolescence > Economic](#) (Weight: 0)

[Obsolescence > Regulatory](#) (Weight: 0)

[Enabling condition > Service infrastructure](#) (Weight: 0)

[Enabling condition > Certification](#) (Weight: 0)

[Enabling condition > User acceptance](#) (Weight: 0)

[Mitigation > Business-model intervention](#) (Weight: 0)

[Mitigation > Policy intervention](#) (Weight: 0)

the direct transfer of

components from one site to another leaves the demand project fully exposed to the risks of 'insufficient quantities of components', 'inconsistent quality' and 'lack of performance guarantees'. These uncertainties over quantity and quality are likely to be closely linked to 'risk in specifying reused components', concerns about 'additional construction complexity' and 'insurance constraints'. The development of reuse markets and supply chain infrastructure therefore has the potential to address several linked barriers to reuse by increasing confidence in component quantities and quality. The extent to which it can address them depends on the scope of activities undertaken by intermediaries in such an infrastructure; for instance, carrying out tests and providing warranties.

Barriers to reuse

Many of the publications note that existing codes and standards are composed based on the properties of new materials, and therefore encourage their use. An advocate for reuse (and effective spokesperson for the salvage industry), Thornton Kay, suggests this is not as significant a barrier as commonly thought: the UK Building Regulations, Regulation 7 (Materials and workmanship), allows the use of any material, as long as it is fit for purpose (BioRegional and Salvo, 2010; Salvo News, 2013). This is typically achieved by certification (Hurley and Hobbs, 2005), but that is not the only route to demonstrating fitness for purpose. Making specific reference to clauses in Regulation 7, Kay establishes that a) building control bodies should not impede the use of materials that are suitable for an intended purpose, and b) past experience, such as use in an existing building, can be used to demonstrate suitability (HM Government, 2013; Salvo News, 2013). Nonetheless, there is a burden on a member of the project team to make the claim of fitness for purpose, and if suppliers, specifiers and contractors show reluctance, as the research suggests, client confidence will remain low. Private homeowners purchasing reclaimed materials from salvage yards may be reassured, but for mainstream construction organisations and their insurers, some form of certification is the norm

Code: • Enabling condition > Certification Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-10, p. 49 - 50

Created: 23.6.2026 r. 12:12 by user, Modified: 23.6.2026 r. 12:12 by user

Other codes assigned to segment:

Circular strategy > Direct reuse (Weight: 0)

Obsolescence > Economic (Weight: 0)

Obsolescence > Regulatory (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Enabling condition > User acceptance (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Business-model intervention (Weight: 0)

Mitigation > Policy intervention (Weight: 0)

the direct transfer of

components from one site to another leaves the demand project fully exposed to the risks of 'insufficient quantities of components', 'inconsistent quality' and 'lack of performance guarantees'. These uncertainties over quantity and quality are likely to be closely linked to 'risk in specifying reused components', concerns about 'additional construction complexity' and 'insurance constraints'. The development of reuse markets and supply chain infrastructure therefore has the potential to address several linked barriers to reuse by increasing confidence in component quantities and quality. The extent to which it can address them depends on the scope of activities undertaken by intermediaries in such an infrastructure; for instance, carrying out tests and providing warranties.

Barriers to reuse

Many of the publications note that existing codes and standards are composed based on the properties of new materials, and therefore encourage their use. An advocate for reuse (and effective spokesperson for the salvage industry), Thornton Kay, suggests this is not as

significant a barrier as commonly thought: the UK Building Regulations, Regulation 7 (Materials and workmanship), allows the use of any material, as long as it is fit for purpose (BioRegional and Salvo, 2010; Salvo News, 2013). This is typically achieved by certification (Hurley and Hobbs, 2005), but that is not the only route to demonstrating fitness for purpose. Making specific reference to clauses in Regulation 7, Kay establishes that a) building control bodies should not impede the use of materials that are suitable for an intended purpose, and b) past experience, such as use in an existing building, can be used to demonstrate suitability (HM Government, 2013; Salvo News, 2013). Nonetheless, there is a burden on a member of the project team to make the claim of fitness for purpose, and if suppliers, specifiers and contractors show reluctance, as the research suggests, client confidence will remain low. Private homeowners purchasing reclaimed materials from salvage yards may be reassured, but for mainstream construction organisations and their insurers, some form of certification is the norm

Code: • [Obsolescence > Economic](#) Weight score: 0

Search B - [Disassembly reuse and facade recovery > B-GS-10](#), p. 49 - 50

Created: 23.6.2026 r. 12:12 by user, Modified: 23.6.2026 r. 12:12 by user

Other codes assigned to segment:

[Circular strategy > Direct reuse](#) (Weight: 0)

[Obsolescence > Regulatory](#) (Weight: 0)

[Enabling condition > Service infrastructure](#) (Weight: 0)

[Enabling condition > Certification](#) (Weight: 0)

[Enabling condition > User acceptance](#) (Weight: 0)

[Negative mechanism > Missing repair infrastructure](#) (Weight: 0)

[Mitigation > Business-model intervention](#) (Weight: 0)

[Mitigation > Policy intervention](#) (Weight: 0)

the direct transfer of components from one site to another leaves the demand project fully exposed to the risks of 'insufficient quantities of components', 'inconsistent quality' and 'lack of performance guarantees'. These uncertainties over quantity and quality are likely to be closely linked to 'risk in specifying reused components', concerns about 'additional construction complexity' and 'insurance constraints'. The development of reuse markets and supply chain infrastructure therefore has the potential to address several linked barriers to reuse by increasing confidence in component quantities and quality. The extent to which it can address them depends on the scope of activities undertaken by intermediaries in such an infrastructure; for instance, carrying out tests and providing warranties.

Barriers to reuse

Many of the publications note that existing codes and standards are composed based on the properties of new materials, and therefore encourage their use. An advocate for reuse (and effective spokesperson for the salvage industry), Thornton Kay, suggests this is not as significant a barrier as commonly thought: the UK Building Regulations, Regulation 7 (Materials and workmanship), allows the use of any material, as long as it is fit for purpose (BioRegional and Salvo, 2010; Salvo News, 2013). This is typically achieved by certification (Hurley and Hobbs, 2005), but that is not the only route to demonstrating fitness for purpose. Making specific reference to clauses in Regulation 7, Kay establishes that a) building control bodies should not impede the use of materials that are suitable for an intended purpose, and b) past experience, such as use in an existing building, can be used to demonstrate suitability (HM Government, 2013; Salvo News, 2013). Nonetheless, there is a burden on a member of the project team to make the claim of fitness for purpose, and if suppliers, specifiers and contractors show reluctance, as the research suggests, client confidence will remain low. Private homeowners purchasing reclaimed materials from salvage yards may be reassured, but for mainstream construction organisations and their insurers, some form of certification is the norm

Code: • [Enabling condition > Service infrastructure](#) Weight score: 0

Search B - [Disassembly reuse and facade recovery > B-GS-10](#), p. 49 - 50

Created: 23.6.2026 r. 12:13 by user, Modified: 23.6.2026 r. 12:13 by user

Other codes assigned to segment:

- Circular strategy > Direct reuse (Weight: 0)
- Obsolescence > Economic (Weight: 0)
- Obsolescence > Regulatory (Weight: 0)
- Enabling condition > Certification (Weight: 0)
- Enabling condition > User acceptance (Weight: 0)
- Negative mechanism > Missing repair infrastructure (Weight: 0)
- Mitigation > Business-model intervention (Weight: 0)
- Mitigation > Policy intervention (Weight: 0)

the direct transfer of

components from one site to another leaves the demand project fully exposed to the risks of 'insufficient quantities of components', 'inconsistent quality' and 'lack of performance guarantees'. These uncertainties over quantity and quality are likely to be closely linked to 'risk in specifying reused components', concerns about 'additional construction complexity' and 'insurance constraints'. The development of reuse markets and supply chain infrastructure therefore has the potential to address several linked barriers to reuse by increasing confidence in component quantities and quality. The extent to which it can address them depends on the scope of activities undertaken by intermediaries in such an infrastructure; for instance, carrying out tests and providing warranties.

Barriers to reuse

Many of the publications note that existing codes and standards are composed based on the properties of new materials, and therefore encourage their use. An advocate for reuse (and effective spokesperson for the salvage industry), Thornton Kay, suggests this is not as significant a barrier as commonly thought: the UK Building Regulations, Regulation 7 (Materials and workmanship), allows the use of any material, as long as it is fit for purpose (BioRegional and Salvo, 2010; Salvo News, 2013). This is typically achieved by certification (Hurley and Hobbs, 2005), but that is not the only route to demonstrating fitness for purpose. Making specific reference to clauses in Regulation 7, Kay establishes that a) building control bodies should not impede the use of materials that are suitable for an intended purpose, and b) past experience, such as use in an existing building, can be used to demonstrate suitability (HM Government, 2013; Salvo News, 2013). Nonetheless, there is a burden on a member of the project team to make the claim of fitness for purpose, and if suppliers, specifiers and contractors show reluctance, as the research suggests, client confidence will remain low. Private homeowners purchasing reclaimed materials from salvage yards may be reassured, but for mainstream construction organisations and their insurers, some form of certification is the norm

Code: • Mitigation > Business-model intervention Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-10, p. 49 - 50

Created: 23.6.2026 r. 12:14 by user, Modified: 23.6.2026 r. 12:14 by user

Other codes assigned to segment:

- Circular strategy > Direct reuse (Weight: 0)
- Obsolescence > Economic (Weight: 0)
- Obsolescence > Regulatory (Weight: 0)
- Enabling condition > Service infrastructure (Weight: 0)
- Enabling condition > Certification (Weight: 0)
- Enabling condition > User acceptance (Weight: 0)
- Negative mechanism > Missing repair infrastructure (Weight: 0)
- Mitigation > Policy intervention (Weight: 0)

the direct transfer of

components from one site to another leaves the demand project fully exposed to the risks of 'insufficient quantities of components', 'inconsistent quality' and 'lack of performance guarantees'. These uncertainties over quantity and quality are likely to be closely linked to 'risk in specifying reused components', concerns about 'additional construction complexity' and

'insurance constraints'. The development of reuse markets and supply chain infrastructure therefore has the potential to address several linked barriers to reuse by increasing confidence in component quantities and quality. The extent to which it can address them depends on the scope of activities undertaken by intermediaries in such an infrastructure; for instance, carrying out tests and providing warranties.

Barriers to reuse

Many of the publications note that existing codes and standards are composed based on the properties of new materials, and therefore encourage their use. An advocate for reuse (and effective spokesperson for the salvage industry), Thornton Kay, suggests this is not as significant a barrier as commonly thought: the UK Building Regulations, Regulation 7 (Materials and workmanship), allows the use of any material, as long as it is fit for purpose (BioRegional and Salvo, 2010; Salvo News, 2013). This is typically achieved by certification (Hurley and Hobbs, 2005), but that is not the only route to demonstrating fitness for purpose. Making specific reference to clauses in Regulation 7, Kay establishes that a) building control bodies should not impede the use of materials that are suitable for an intended purpose, and b) past experience, such as use in an existing building, can be used to demonstrate suitability (HM Government, 2013; Salvo News, 2013). Nonetheless, there is a burden on a member of the project team to make the claim of fitness for purpose, and if suppliers, specifiers and contractors show reluctance, as the research suggests, client confidence will remain low. Private homeowners purchasing reclaimed materials from salvage yards may be reassured, but for mainstream construction organisations and their insurers, some form of certification is the norm

Code: • Mitigation > Policy intervention Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-10, p. 49 - 50

Created: 23.6.2026 r. 12:14 by user, Modified: 23.6.2026 r. 12:14 by user

Other codes assigned to segment:

Circular strategy > Direct reuse (Weight: 0)

Obsolescence > Economic (Weight: 0)

Obsolescence > Regulatory (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Enabling condition > Certification (Weight: 0)

Enabling condition > User acceptance (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Business-model intervention (Weight: 0)

Discarding components to waste management is perceived as a safe, default position that was unchallenged in all case study projects (Figure 11a). This perception that components are unwanted and valueless sets in motion a series of steps from which it is difficult to recover any component performance. Components are treated as a liability rather than an asset, undergoing destructive demolition rather than careful dismantling. Contractors do not see value in creating the capacity to take a proactive role in sorting and redistributing these materials: instead, the skip service provides a simple release valve. Once discarded to a skip, they are 'waste', subject to further damage and mixed with other wastes. WTSs provide the interface with the rest of the waste management infrastructure, but are not set up as places for reclamation. By the time waste has reached the WTS, therefore, the management options are effectively limited to recycling, incineration with or without energy recovery, and landfill

Code: • Negative mechanism > Premature replacement Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-10, p. 94 - 95

Created: 23.6.2026 r. 11:26 by user, Modified: 23.6.2026 r. 11:26 by user

Other codes assigned to segment:

Circular strategy > Recycling (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)
Mitigation > Business-model intervention (Weight: 0)

Discarding components to waste management is perceived as a safe, default position that was unchallenged in all case study projects (Figure 11a). This perception that components are unwanted and valueless sets in motion a series of steps from which it is difficult to recover any component performance. Components are treated as a liability rather than an asset, undergoing destructive demolition rather than careful dismantling. Contractors do not see value in creating the capacity to take a proactive role in sorting and redistributing these materials: instead, the skip service provides a simple release valve. Once discarded to a skip, they are 'waste', subject to further damage and mixed with other wastes. WTSs provide the interface with the rest of the waste management infrastructure, but are not set up as places for reclamation. By the time waste has reached the WTS, therefore, the management options are effectively limited to recycling, incineration with or without energy recovery, and landfill

Code: • Mitigation > Business-model intervention Weight score: 0
Search B - Disassembly reuse and facade recovery > B-GS-10, p. 94 - 95
Created: 23.6.2026 r. 11:27 by user, Modified: 23.6.2026 r. 11:27 by user
Other codes assigned to segment:
Circular strategy > Recycling (Weight: 0)
Enabling condition > Service infrastructure (Weight: 0)
Negative mechanism > Premature replacement (Weight: 0)
Negative mechanism > Burden shifting (Weight: 0)
Negative mechanism > Missing repair infrastructure (Weight: 0)

Discarding components to waste management is perceived as a safe, default position that was unchallenged in all case study projects (Figure 11a). This perception that components are unwanted and valueless sets in motion a series of steps from which it is difficult to recover any component performance. Components are treated as a liability rather than an asset, undergoing destructive demolition rather than careful dismantling. Contractors do not see value in creating the capacity to take a proactive role in sorting and redistributing these materials: instead, the skip service provides a simple release valve. Once discarded to a skip, they are 'waste', subject to further damage and mixed with other wastes. WTSs provide the interface with the rest of the waste management infrastructure, but are not set up as places for reclamation. By the time waste has reached the WTS, therefore, the management options are effectively limited to recycling, incineration with or without energy recovery, and landfill

Code: • Circular strategy > Recycling Weight score: 0
Search B - Disassembly reuse and facade recovery > B-GS-10, p. 94 - 95
Created: 23.6.2026 r. 11:26 by user, Modified: 23.6.2026 r. 11:26 by user
Other codes assigned to segment:
Enabling condition > Service infrastructure (Weight: 0)
Negative mechanism > Premature replacement (Weight: 0)
Negative mechanism > Burden shifting (Weight: 0)
Negative mechanism > Missing repair infrastructure (Weight: 0)
Mitigation > Business-model intervention (Weight: 0)

Discarding components to waste management is perceived as a safe, default position that was unchallenged in all case study projects (Figure 11a). This perception that components are unwanted and valueless sets in motion a series of steps from which it is difficult to recover any component performance. Components are treated as a liability rather than an asset, undergoing destructive demolition rather than careful dismantling. Contractors do not see value in creating the capacity to take a proactive role in

sorting and redistributing these materials: instead, the skip service provides a simple release valve. Once discarded to a skip, they are 'waste', subject to further damage and mixed with other wastes. WTSs provide the interface with the rest of the waste management infrastructure, but are not set up as places for reclamation. By the time waste has reached the WTS, therefore, the management options are effectively limited to recycling, incineration with or without energy recovery, and landfill

Code: • Enabling condition > Service infrastructure Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-10, p. 94 - 95
 Created: 23.6.2026 r. 11:26 by user, Modified: 23.6.2026 r. 11:26 by user
 Other codes assigned to segment:
 Circular strategy > Recycling (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Business-model intervention (Weight: 0)

Discarding components to waste management is perceived as a safe, default position that was unchallenged in all case study projects (Figure 11a). This perception that components are unwanted and valueless sets in motion a series of steps from which it is difficult to recover any component performance. Components are treated as a liability rather than an asset, undergoing destructive demolition rather than careful dismantling. Contractors do not see value in creating the capacity to take a proactive role in sorting and redistributing these materials: instead, the skip service provides a simple release valve. Once discarded to a skip, they are 'waste', subject to further damage and mixed with other wastes. WTSs provide the interface with the rest of the waste management infrastructure, but are not set up as places for reclamation. By the time waste has reached the WTS, therefore, the management options are effectively limited to recycling, incineration with or without energy recovery, and landfill

Code: • Negative mechanism > Burden shifting Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-10, p. 94 - 95
 Created: 23.6.2026 r. 11:26 by user, Modified: 23.6.2026 r. 11:26 by user
 Other codes assigned to segment:
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Business-model intervention (Weight: 0)

Discarding components to waste management is perceived as a safe, default position that was unchallenged in all case study projects (Figure 11a). This perception that components are unwanted and valueless sets in motion a series of steps from which it is difficult to recover any component performance. Components are treated as a liability rather than an asset, undergoing destructive demolition rather than careful dismantling. Contractors do not see value in creating the capacity to take a proactive role in sorting and redistributing these materials: instead, the skip service provides a simple release valve. Once discarded to a skip, they are 'waste', subject to further damage and mixed with other wastes. WTSs provide the interface with the rest of the waste management infrastructure, but are not set up as places for reclamation. By the time waste has reached the WTS, therefore, the management options are effectively limited to recycling, incineration with or without energy recovery, and landfill

Code: • Negative mechanism > Missing repair infrastructure Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-10, p. 94 - 95
 Created: 23.6.2026 r. 11:26 by user, Modified: 23.6.2026 r. 11:26 by user

Other codes assigned to segment:

- Circular strategy > Recycling (Weight: 0)
- Enabling condition > Service infrastructure (Weight: 0)
- Negative mechanism > Premature replacement (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)
- Mitigation > Business-model intervention (Weight: 0)

If building components are to be reclaimed, demand must be recognised from the outset. To establish whether there is demand requires timely and appropriate information about existing building components. Nascent systems of component management are illustrated in Figure 11b,c, in which an audit provides the starting point. This could be in the form of a pre-redevelopment audit, or informal identification of reusable components, such as the practical knowledge that a demolition contractor may apply when tendering a job

Code: ● Enabling condition > Service infrastructure Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-10, p. 96
 Created: 23.6.2026 r. 11:28 by user, Modified: 23.6.2026 r. 11:28 by user

Other codes assigned to segment:

- Circular strategy > Direct reuse (Weight: 0)
- Enabling condition > Repair information (Weight: 0)
- Enabling condition > Business model (Weight: 0)
- Negative mechanism > Missing repair infrastructure (Weight: 0)
- Mitigation > Business-model intervention (Weight: 0)

If building components are to be reclaimed, demand must be recognised from the outset. To establish whether there is demand requires timely and appropriate information about existing building components. Nascent systems of component management are illustrated in Figure 11b,c, in which an audit provides the starting point. This could be in the form of a pre-redevelopment audit, or informal identification of reusable components, such as the practical knowledge that a demolition contractor may apply when tendering a job

Code: ● Enabling condition > Business model Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-10, p. 96
 Created: 23.6.2026 r. 11:28 by user, Modified: 23.6.2026 r. 11:28 by user

Other codes assigned to segment:

- Circular strategy > Direct reuse (Weight: 0)
- Enabling condition > Repair information (Weight: 0)
- Enabling condition > Service infrastructure (Weight: 0)
- Negative mechanism > Missing repair infrastructure (Weight: 0)
- Mitigation > Business-model intervention (Weight: 0)

If building components are to be reclaimed, demand must be recognised from the outset. To establish whether there is demand requires timely and appropriate information about existing building components. Nascent systems of component management are illustrated in Figure 11b,c, in which an audit provides the starting point. This could be in the form of a pre-redevelopment audit, or informal identification of reusable components, such as the practical knowledge that a demolition contractor may apply when tendering a job

Code: ● Enabling condition > Repair information Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-10, p. 96
 Created: 23.6.2026 r. 11:28 by user, Modified: 23.6.2026 r. 11:28 by user

Other codes assigned to segment:

- Circular strategy > Direct reuse (Weight: 0)
- Enabling condition > Service infrastructure (Weight: 0)
- Enabling condition > Business model (Weight: 0)
- Negative mechanism > Missing repair infrastructure (Weight: 0)
- Mitigation > Business-model intervention (Weight: 0)

If building components are to be reclaimed, demand must be recognised from the outset. To establish whether there is demand requires timely and appropriate information about existing building components. Nascent systems of component management are illustrated in Figure 11b,c, in which an audit provides the starting point. This could be in the form of a pre-redevelopment audit, or informal identification of reusable components, such as the practical knowledge that a demolition contractor may apply when tendering a job

Code: • Negative mechanism > Missing repair infrastructure Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-10, p. 96

Created: 23.6.2026 r. 11:28 by user, Modified: 23.6.2026 r. 11:28 by user

Other codes assigned to segment:

Circular strategy > Direct reuse (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Enabling condition > Business model (Weight: 0)

Mitigation > Business-model intervention (Weight: 0)

If building components are to be reclaimed, demand must be recognised from the outset. To establish whether there is demand requires timely and appropriate information about existing building components. Nascent systems of component management are illustrated in Figure 11b,c, in which an audit provides the starting point. This could be in the form of a pre-redevelopment audit, or informal identification of reusable components, such as the practical knowledge that a demolition contractor may apply when tendering a job

Code: • Mitigation > Business-model intervention Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-10, p. 96

Created: 23.6.2026 r. 11:28 by user, Modified: 23.6.2026 r. 11:28 by user

Other codes assigned to segment:

Circular strategy > Direct reuse (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Enabling condition > Business model (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

If building components are to be reclaimed, demand must be recognised from the outset. To establish whether there is demand requires timely and appropriate information about existing building components. Nascent systems of component management are illustrated in Figure 11b,c, in which an audit provides the starting point. This could be in the form of a pre-redevelopment audit, or informal identification of reusable components, such as the practical knowledge that a demolition contractor may apply when tendering a job

Code: • Circular strategy > Direct reuse Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-10, p. 96

Created: 23.6.2026 r. 11:27 by user, Modified: 23.6.2026 r. 11:27 by user

Other codes assigned to segment:

Enabling condition > Repair information (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Enabling condition > Business model (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Business-model intervention (Weight: 0)

The primary source of information about materials emerging from the case study projects was waste reporting. As well as being generated too late in the process, these reports do not provide a qualitative understanding of components that would be necessary in identifying demand. EWC codes or the categories of the European Waste Statistics Directive (EWC-Stat; European Commission, 2002) capture all types of waste, but do so in a way that is geared towards waste management, rather than component performance, despite the intention of

aiding waste prevention. Taking an example, there are only two EWC codes for timber construction waste: 17 02 01 (wood), and 17 02 04* (glass, plastic, and wood containing or contaminated with dangerous substances). The codes do not distinguish between a solid timber joist, a kitchen unit of particleboard and melamine, and a panelled door: distinctions that would need to be drawn in order to understand whether such components can serve others' needs. The point is demonstrated by Figure 12a: a 'soil and stone' pile with large blocks of cut granite in amongst soil and gravel. All are '17 05 04' as far as the EWC is concerned, and all attract the lower UK landfill tax rate for inert waste, but are very different in terms of embodied carbon, performance, and value. Measuring in terms of undifferentiated tonnage or volume of broad waste categories biases their disposal towards downcycling en masse.

Code: • Negative mechanism > Missing repair infrastructure Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-10, p. 97

Created: 23.6.2026 r. 11:30 by user, Modified: 23.6.2026 r. 11:30 by user

Other codes assigned to segment:

Circular strategy > Recycling (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Business-model intervention (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The primary source of information about materials emerging from the case study projects was waste reporting. As well as being generated too late in the process, these reports do not provide a qualitative understanding of components that would be necessary in identifying demand. EWC codes or the categories of the European Waste Statistics Directive (EWC-Stat; European Commission, 2002) capture all types of waste, but do so in a way that is geared towards waste management, rather than component performance, despite the intention of aiding waste prevention. Taking an example, there are only two EWC codes for timber construction waste: 17 02 01 (wood), and 17 02 04* (glass, plastic, and wood containing or contaminated with dangerous substances). The codes do not distinguish between a solid timber joist, a kitchen unit of particleboard and melamine, and a panelled door: distinctions that would need to be drawn in order to understand whether such components can serve others' needs. The point is demonstrated by Figure 12a: a 'soil and stone' pile with large blocks of cut granite in amongst soil and gravel. All are '17 05 04' as far as the EWC is concerned, and all attract the lower UK landfill tax rate for inert waste, but are very different in terms of embodied carbon, performance, and value. Measuring in terms of undifferentiated tonnage or volume of broad waste categories biases their disposal towards downcycling en masse.

Code: • Circular strategy > Recycling Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-10, p. 97

Created: 23.6.2026 r. 11:30 by user, Modified: 23.6.2026 r. 11:30 by user

Other codes assigned to segment:

Enabling condition > Repair information (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Business-model intervention (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The primary source of information about materials emerging from the case study projects was waste reporting. As well as being generated too late in the process, these reports do not provide a qualitative understanding of components that would be necessary in identifying demand. EWC codes or the categories of the European Waste Statistics Directive (EWC-Stat; European Commission, 2002) capture all types of waste, but do so in a way that is geared towards waste management, rather than component performance, despite the intention of aiding waste prevention. Taking an example, there are only two EWC codes for timber

construction waste: 17 02 01 (wood), and 17 02 04* (glass, plastic, and wood containing or contaminated with dangerous substances). The codes do not distinguish between a solid timber joist, a kitchen unit of particleboard and melamine, and a panelled door: distinctions that would need to be drawn in order to understand whether such components can serve others' needs. The point is demonstrated by Figure 12a: a 'soil and stone' pile with large blocks of cut granite in amongst soil and gravel. All are '17 05 04' as far as the EWC is concerned, and all attract the lower UK landfill tax rate for inert waste, but are very different in terms of embodied carbon, performance, and value. Measuring in terms of undifferentiated tonnage or volume of broad waste categories biases their disposal towards downcycling en masse.

Code: • Negative mechanism > Burden shifting Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-10, p. 97

Created: 23.6.2026 r. 11:29 by user, Modified: 23.6.2026 r. 11:29 by user

Other codes assigned to segment:

Circular strategy > Recycling (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Business-model intervention (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The primary source of information about materials emerging from the case study projects was waste reporting. As well as being generated too late in the process, these reports do not provide a qualitative understanding of components that would be necessary in identifying demand. EWC codes or the categories of the European Waste Statistics Directive (EWC-Stat; European Commission, 2002) capture all types of waste, but do so in a way that is geared towards waste management, rather than component performance, despite the intention of aiding waste prevention. Taking an example, there are only two EWC codes for timber construction waste: 17 02 01 (wood), and 17 02 04* (glass, plastic, and wood containing or contaminated with dangerous substances). The codes do not distinguish between a solid timber joist, a kitchen unit of particleboard and melamine, and a panelled door: distinctions that would need to be drawn in order to understand whether such components can serve others' needs. The point is demonstrated by Figure 12a: a 'soil and stone' pile with large blocks of cut granite in amongst soil and gravel. All are '17 05 04' as far as the EWC is concerned, and all attract the lower UK landfill tax rate for inert waste, but are very different in terms of embodied carbon, performance, and value. Measuring in terms of undifferentiated tonnage or volume of broad waste categories biases their disposal towards downcycling en masse.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-10, p. 97

Created: 23.6.2026 r. 11:30 by user, Modified: 23.6.2026 r. 11:30 by user

Other codes assigned to segment:

Circular strategy > Recycling (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Business-model intervention (Weight: 0)

The primary source of information about materials emerging from the case study projects was waste reporting. As well as being generated too late in the process, these reports do not provide a qualitative understanding of components that would be necessary in identifying demand. EWC codes or the categories of the European Waste Statistics Directive (EWC-Stat; European Commission, 2002) capture all types of waste, but do so in a way that is geared towards waste management, rather than component performance, despite the intention of aiding waste prevention. Taking an example, there are only two EWC codes for timber construction waste: 17 02 01 (wood), and 17 02 04* (glass, plastic, and wood containing or

contaminated with dangerous substances). The codes do not distinguish between a solid timber joist, a kitchen unit of particleboard and melamine, and a panelled door: distinctions that would need to be drawn in order to understand whether such components can serve others' needs. The point is demonstrated by Figure 12a: a 'soil and stone' pile with large blocks of cut granite in amongst soil and gravel. All are '17 05 04' as far as the EWC is concerned, and all attract the lower UK landfill tax rate for inert waste, but are very different in terms of embodied carbon, performance, and value. Measuring in terms of undifferentiated tonnage or volume of broad waste categories biases their disposal towards downcycling en masse.

Code: • Mitigation > Business-model intervention Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-10, p. 97

Created: 23.6.2026 r. 11:30 by user, Modified: 23.6.2026 r. 11:30 by user

Other codes assigned to segment:

Circular strategy > Recycling (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The primary source of information about materials emerging from the case study projects was waste reporting. As well as being generated too late in the process, these reports do not provide a qualitative understanding of components that would be necessary in identifying demand. EWC codes or the categories of the European Waste Statistics Directive (EWC-Stat; European Commission, 2002) capture all types of waste, but do so in a way that is geared towards waste management, rather than component performance, despite the intention of aiding waste prevention. Taking an example, there are only two EWC codes for timber construction waste: 17 02 01 (wood), and 17 02 04* (glass, plastic, and wood containing or contaminated with dangerous substances). The codes do not distinguish between a solid timber joist, a kitchen unit of particleboard and melamine, and a panelled door: distinctions that would need to be drawn in order to understand whether such components can serve others' needs. The point is demonstrated by Figure 12a: a 'soil and stone' pile with large blocks of cut granite in amongst soil and gravel. All are '17 05 04' as far as the EWC is concerned, and all attract the lower UK landfill tax rate for inert waste, but are very different in terms of embodied carbon, performance, and value. Measuring in terms of undifferentiated tonnage or volume of broad waste categories biases their disposal towards downcycling en masse.

Code: • Enabling condition > Repair information Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-10, p. 97

Created: 23.6.2026 r. 11:29 by user, Modified: 23.6.2026 r. 11:29 by user

Other codes assigned to segment:

Circular strategy > Recycling (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Business-model intervention (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

'Where we want to be': a triage process to support reuse, repurposing, and upcycling
Based on an analysis of C&D waste management processes, the previous section explained that for building components to be reclaimed, demand must exist and be identified from the outset; that to serve latent demand, specific components' qualities as well as quantities must be identified; and that to overcome the vantage point problem, this information must flow between supply and demand. It was noted, however, that the demand projects that can directly reuse components identified in this way are limited. For demolition practices to change (and thus increase the supply of reclaimed goods), there must be timely evidence of demand at scale. This section expands on the nascent model of component management illustrated in Figure 11

to address the identified needs and shortfalls. It describes a series of interlinked activities that are necessary to form a triage process

Code: • Circular strategy > Refurbishment Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-10, p. 98
 Created: 23.6.2026 r. 11:41 by user, Modified: 23.6.2026 r. 11:41 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Enabling condition > Business model (Weight: 0)
 Mitigation > Business-model intervention (Weight: 0)
 Mitigation > Policy intervention (Weight: 0)

‘Where we want to be’: a triage process to support reuse, repurposing, and upcycling
 Based on an analysis of C&D waste management processes, the previous section explained that for building components to be reclaimed, demand must exist and be identified from the outset; that to serve latent demand, specific components’ qualities as well as quantities must be identified; and that to overcome the vantage point problem, this information must flow between supply and demand. It was noted, however, that the demand projects that can directly reuse components identified in this way are limited. For demolition practices to change (and thus increase the supply of reclaimed goods), there must be timely evidence of demand at scale. This section expands on the nascent model of component management illustrated in Figure 11 to address the identified needs and shortfalls. It describes a series of interlinked activities that are necessary to form a triage process

Code: • Circular strategy > Recycling Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-10, p. 98
 Created: 23.6.2026 r. 11:42 by user, Modified: 23.6.2026 r. 11:42 by user
 Other codes assigned to segment:
 Circular strategy > Refurbishment (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Enabling condition > Business model (Weight: 0)
 Mitigation > Business-model intervention (Weight: 0)
 Mitigation > Policy intervention (Weight: 0)

‘Where we want to be’: a triage process to support reuse, repurposing, and upcycling
 Based on an analysis of C&D waste management processes, the previous section explained that for building components to be reclaimed, demand must exist and be identified from the outset; that to serve latent demand, specific components’ qualities as well as quantities must be identified; and that to overcome the vantage point problem, this information must flow between supply and demand. It was noted, however, that the demand projects that can directly reuse components identified in this way are limited. For demolition practices to change (and thus increase the supply of reclaimed goods), there must be timely evidence of demand at scale. This section expands on the nascent model of component management illustrated in Figure 11 to address the identified needs and shortfalls. It describes a series of interlinked activities that are necessary to form a triage process

Code: • Circular strategy > Direct reuse Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-10, p. 98
 Created: 23.6.2026 r. 11:41 by user, Modified: 23.6.2026 r. 11:41 by user
 Other codes assigned to segment:
 Circular strategy > Refurbishment (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > Repair information (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)
 Enabling condition > Business model (Weight: 0)
 Mitigation > Business-model intervention (Weight: 0)
 Mitigation > Policy intervention (Weight: 0)

‘Where we want to be’: a triage process to support reuse, repurposing, and upcycling
 Based on an analysis of C&D waste management processes, the previous section explained that for building components to be reclaimed, demand must exist and be identified from the outset; that to serve latent demand, specific components’ qualities as well as quantities must be identified; and that to overcome the vantage point problem, this information must flow between supply and demand. It was noted, however, that the demand projects that can directly reuse components identified in this way are limited. For demolition practices to change (and thus increase the supply of reclaimed goods), there must be timely evidence of demand at scale. This section expands on the nascent model of component management illustrated in Figure 11 to address the identified needs and shortfalls. It describes a series of interlinked activities that are necessary to form a triage process

Code: • Enabling condition > Business model Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-10, p. 98
 Created: 23.6.2026 r. 11:43 by user, Modified: 23.6.2026 r. 11:43 by user
 Other codes assigned to segment:
 Circular strategy > Refurbishment (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Mitigation > Business-model intervention (Weight: 0)
 Mitigation > Policy intervention (Weight: 0)

‘Where we want to be’: a triage process to support reuse, repurposing, and upcycling
 Based on an analysis of C&D waste management processes, the previous section explained that for building components to be reclaimed, demand must exist and be identified from the outset; that to serve latent demand, specific components’ qualities as well as quantities must be identified; and that to overcome the vantage point problem, this information must flow between supply and demand. It was noted, however, that the demand projects that can directly reuse components identified in this way are limited. For demolition practices to change (and thus increase the supply of reclaimed goods), there must be timely evidence of demand at scale. This section expands on the nascent model of component management illustrated in Figure 11 to address the identified needs and shortfalls. It describes a series of interlinked activities that are necessary to form a triage process

Code: • Mitigation > Business-model intervention Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-10, p. 98
 Created: 23.6.2026 r. 11:43 by user, Modified: 23.6.2026 r. 11:43 by user
 Other codes assigned to segment:
 Circular strategy > Refurbishment (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Enabling condition > Business model (Weight: 0)
 Mitigation > Policy intervention (Weight: 0)

‘Where we want to be’: a triage process to support reuse, repurposing, and upcycling
 Based on an analysis of C&D waste management processes, the previous section explained that for building components to be reclaimed, demand must exist and be identified from the

outset; that to serve latent demand, specific components' qualities as well as quantities must be identified; and that to overcome the vantage point problem, this information must flow between supply and demand. It was noted, however, that the demand projects that can directly reuse components identified in this way are limited. For demolition practices to change (and thus increase the supply of reclaimed goods), there must be timely evidence of demand at scale. This section expands on the nascent model of component management illustrated in Figure 11 to address the identified needs and shortfalls. It describes a series of interlinked activities that are necessary to form a triage process

Code: • Enabling condition > Service infrastructure Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-10, p. 98
 Created: 23.6.2026 r. 11:42 by user, Modified: 23.6.2026 r. 11:42 by user
 Other codes assigned to segment:
 Circular strategy > Refurbishment (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Business model (Weight: 0)
 Mitigation > Business-model intervention (Weight: 0)
 Mitigation > Policy intervention (Weight: 0)

'Where we want to be': a triage process to support reuse, repurposing, and upcycling
 Based on an analysis of C&D waste management processes, the previous section explained that for building components to be reclaimed, demand must exist and be identified from the outset; that to serve latent demand, specific components' qualities as well as quantities must be identified; and that to overcome the vantage point problem, this information must flow between supply and demand. It was noted, however, that the demand projects that can directly reuse components identified in this way are limited. For demolition practices to change (and thus increase the supply of reclaimed goods), there must be timely evidence of demand at scale. This section expands on the nascent model of component management illustrated in Figure 11 to address the identified needs and shortfalls. It describes a series of interlinked activities that are necessary to form a triage process

Code: • Mitigation > Policy intervention Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-10, p. 98
 Created: 23.6.2026 r. 11:43 by user, Modified: 23.6.2026 r. 11:43 by user
 Other codes assigned to segment:
 Circular strategy > Refurbishment (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Enabling condition > Business model (Weight: 0)
 Mitigation > Business-model intervention (Weight: 0)

'Where we want to be': a triage process to support reuse, repurposing, and upcycling
 Based on an analysis of C&D waste management processes, the previous section explained that for building components to be reclaimed, demand must exist and be identified from the outset; that to serve latent demand, specific components' qualities as well as quantities must be identified; and that to overcome the vantage point problem, this information must flow between supply and demand. It was noted, however, that the demand projects that can directly reuse components identified in this way are limited. For demolition practices to change (and thus increase the supply of reclaimed goods), there must be timely evidence of demand at scale. This section expands on the nascent model of component management illustrated in Figure 11 to address the identified needs and shortfalls. It describes a series of interlinked activities that are necessary to form a triage process

Code: • Enabling condition > Repair information Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-10, p. 98
 Created: 23.6.2026 r. 11:42 by user, Modified: 23.6.2026 r. 11:42 by user
 Other codes assigned to segment:
 Circular strategy > Refurbishment (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Enabling condition > Business model (Weight: 0)
 Mitigation > Business-model intervention (Weight: 0)
 Mitigation > Policy intervention (Weight: 0)

To enable exhaustive exploration of the potential for direct reuse, there is a need for effective information flows through a widely accessible database similar to an RMM. Figure 13a begins to address the identified shortfalls of RMMs by integrating an information gathering regime modelled on pre-redevelopment audits (NFDC/IDE, 2016). At present, the results of audits are used in isolation by the project team and only inform waste management decisions. If, instead, audits from many projects were collated in a database, with components that are to emerge as waste described systematically (and qualitatively as well as quantitatively), it would create a timely array of information. Potential demand actors could navigate the database online and agreements could be reached prior to soft strip, demolition, or deconstruction.

Code: • Mitigation > Policy intervention Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-10, p. 100
 Created: 23.6.2026 r. 12:01 by user, Modified: 23.6.2026 r. 12:01 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Enabling condition > Business model (Weight: 0)
 Mitigation > Business-model intervention (Weight: 0)

To enable exhaustive exploration of the potential for direct reuse, there is a need for effective information flows through a widely accessible database similar to an RMM. Figure 13a begins to address the identified shortfalls of RMMs by integrating an information gathering regime modelled on pre-redevelopment audits (NFDC/IDE, 2016). At present, the results of audits are used in isolation by the project team and only inform waste management decisions. If, instead, audits from many projects were collated in a database, with components that are to emerge as waste described systematically (and qualitatively as well as quantitatively), it would create a timely array of information. Potential demand actors could navigate the database online and agreements could be reached prior to soft strip, demolition, or deconstruction.

Code: • Mitigation > Business-model intervention Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-10, p. 100
 Created: 23.6.2026 r. 12:01 by user, Modified: 23.6.2026 r. 12:01 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Enabling condition > Business model (Weight: 0)
 Mitigation > Policy intervention (Weight: 0)

To enable exhaustive exploration of the potential for direct reuse, there is a need for effective information flows through a widely accessible database similar to an RMM. Figure 13a begins to address the identified shortfalls of RMMs by integrating an information gathering regime modelled on pre-redevelopment audits (NFDC/IDE, 2016). At present, the results of audits are

used in isolation by the project team and only inform waste management decisions. If, instead, audits from many projects were collated in a database, with components that are to emerge as waste described systematically (and qualitatively as well as quantitatively), it would create a timely array of information. Potential demand actors could navigate the database online and agreements could be reached prior to soft strip, demolition, or deconstruction.

Code: • Circular strategy > Direct reuse Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-10, p. 100
 Created: 23.6.2026 r. 12:00 by user, Modified: 23.6.2026 r. 12:00 by user
 Other codes assigned to segment:
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Enabling condition > Business model (Weight: 0)
 Mitigation > Business-model intervention (Weight: 0)
 Mitigation > Policy intervention (Weight: 0)

To enable exhaustive exploration of the potential for direct reuse, there is a need for effective information flows through a widely accessible database similar to an RMM. Figure 13a begins to address the identified shortfalls of RMMs by integrating an information gathering regime modelled on pre-redevelopment audits (NFDC/IDE, 2016). At present, the results of audits are used in isolation by the project team and only inform waste management decisions. If, instead, audits from many projects were collated in a database, with components that are to emerge as waste described systematically (and qualitatively as well as quantitatively), it would create a timely array of information. Potential demand actors could navigate the database online and agreements could be reached prior to soft strip, demolition, or deconstruction.

Code: • Enabling condition > Repair information Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-10, p. 100
 Created: 23.6.2026 r. 12:00 by user, Modified: 23.6.2026 r. 12:00 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Enabling condition > Business model (Weight: 0)
 Mitigation > Business-model intervention (Weight: 0)
 Mitigation > Policy intervention (Weight: 0)

To enable exhaustive exploration of the potential for direct reuse, there is a need for effective information flows through a widely accessible database similar to an RMM. Figure 13a begins to address the identified shortfalls of RMMs by integrating an information gathering regime modelled on pre-redevelopment audits (NFDC/IDE, 2016). At present, the results of audits are used in isolation by the project team and only inform waste management decisions. If, instead, audits from many projects were collated in a database, with components that are to emerge as waste described systematically (and qualitatively as well as quantitatively), it would create a timely array of information. Potential demand actors could navigate the database online and agreements could be reached prior to soft strip, demolition, or deconstruction.

Code: • Enabling condition > Business model Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-10, p. 100
 Created: 23.6.2026 r. 12:00 by user, Modified: 23.6.2026 r. 12:00 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Mitigation > Business-model intervention (Weight: 0)
 Mitigation > Policy intervention (Weight: 0)

To enable exhaustive exploration of the potential for direct reuse, there is a need for effective information flows through a widely accessible database similar to an RMM. Figure 13a begins to address the identified shortfalls of RMMs by integrating an information gathering regime modelled on pre-redevelopment audits (NFDC/IDE, 2016). At present, the results of audits are used in isolation by the project team and only inform waste management decisions. If, instead, audits from many projects were collated in a database, with components that are to emerge as waste described systematically (and qualitatively as well as quantitatively), it would create a timely array of information. Potential demand actors could navigate the database online and agreements could be reached prior to soft strip, demolition, or deconstruction.

Code: • Enabling condition > Service infrastructure Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-10, p. 100
 Created: 23.6.2026 r. 12:00 by user, Modified: 23.6.2026 r. 12:00 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Business model (Weight: 0)
 Mitigation > Business-model intervention (Weight: 0)
 Mitigation > Policy intervention (Weight: 0)

While conducting an LCA of reusable modular building components, it is essential to differentiate between the core components and subcomponents. This distinction is crucial because they may be subject to various value retention processes (VRPs), such as repair and replacement, yielding different environmental impacts.

Code: • Circular strategy > Direct reuse Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 3
 Created: 23.6.2026 r. 12:24 by user, Modified: 23.6.2026 r. 12:24 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Component replacement (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

While conducting an LCA of reusable modular building components, it is essential to differentiate between the core components and subcomponents. This distinction is crucial because they may be subject to various value retention processes (VRPs), such as repair and replacement, yielding different environmental impacts.

Code: • Negative mechanism > Higher initial material use Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 3
 Created: 23.6.2026 r. 12:24 by user, Modified: 23.6.2026 r. 12:24 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

While conducting an LCA of reusable modular building components, it is essential to differentiate between the core components and subcomponents. This distinction is crucial because they may be subject to various value retention processes (VRPs), such as repair and replacement, yielding different environmental impacts.

Code: • Circular strategy > Repair Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 3
 Created: 23.6.2026 r. 12:24 by user, Modified: 23.6.2026 r. 12:24 by user

Other codes assigned to segment:

Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

While conducting an LCA of reusable modular building components, it is essential to differentiate between the core components and subcomponents. This distinction is crucial because they may be subject to various value retention processes (VRPs), such as repair and replacement, yielding different environmental impacts.

Code: • Negative mechanism > Burden shifting Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-14, p. 3

Created: 23.6.2026 r. 12:24 by user, Modified: 23.6.2026 r. 12:24 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)
 Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

While conducting an LCA of reusable modular building components, it is essential to differentiate between the core components and subcomponents. This distinction is crucial because they may be subject to various value retention processes (VRPs), such as repair and replacement, yielding different environmental impacts.

Code: • Circular strategy > Component replacement Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-14, p. 3

Created: 23.6.2026 r. 12:24 by user, Modified: 23.6.2026 r. 12:24 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

While conducting an LCA of reusable modular building components, it is essential to differentiate between the core components and subcomponents. This distinction is crucial because they may be subject to various value retention processes (VRPs), such as repair and replacement, yielding different environmental impacts.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-14, p. 3

Created: 23.6.2026 r. 12:24 by user, Modified: 23.6.2026 r. 12:24 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)
 Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)

Such a “modularity” design achieves a DfD solution that allows the ease of assembly, disassembly, relocation, and reuse of the modular unit. The modular unit has dimensions of 12.2 m in length, 2.4 m in width, 1.9 m in height. The modular unit was manufactured using virgin materials without incorporating any recycled materials. The modular design therefore follows a “downstream reuse” concept, where virgin materials are used initially, but disassembly and reuse are prioritized after the building’s first life

Code: • Circular strategy > Direct reuse Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 5
 Created: 23.6.2026 r. 12:17 by user, Modified: 23.6.2026 r. 12:17 by user
 Other codes assigned to segment:
 Obsolescence > Technical (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Such a “modularity” design achieves a DfD solution that allows the ease of assembly, disassembly, relocation, and reuse of the modular unit. The modular unit has dimensions of 12.2 m in length, 2.4 m in width, 1.9 m in height. The modular unit was manufactured using virgin materials without incorporating any recycled materials. The modular design therefore follows a “downstream reuse” concept, where virgin materials are used initially, but disassembly and reuse are prioritized after the building’s first life

Code: • Enabling condition > Disassembly Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 5
 Created: 23.6.2026 r. 12:17 by user, Modified: 23.6.2026 r. 12:17 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Such a “modularity” design achieves a DfD solution that allows the ease of assembly, disassembly, relocation, and reuse of the modular unit. The modular unit has dimensions of 12.2 m in length, 2.4 m in width, 1.9 m in height. The modular unit was manufactured using virgin materials without incorporating any recycled materials. The modular design therefore follows a “downstream reuse” concept, where virgin materials are used initially, but disassembly and reuse are prioritized after the building’s first life

Code: • Obsolescence > Technical Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 5
 Created: 23.6.2026 r. 12:18 by user, Modified: 23.6.2026 r. 12:18 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Such a “modularity” design achieves a DfD solution that allows the ease of assembly, disassembly, relocation, and reuse of the modular unit. The modular unit has dimensions of 12.2 m in length, 2.4 m in width, 1.9 m in height. The modular unit was manufactured using virgin materials without incorporating any recycled materials. The modular design therefore follows a “downstream reuse” concept, where virgin materials are used initially, but disassembly and reuse are prioritized after the building’s first life

Code: • Mitigation > Lifecycle verification Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 5
 Created: 23.6.2026 r. 12:18 by user, Modified: 23.6.2026 r. 12:18 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

Such a “modularity” design achieves a DfD solution that allows the ease of assembly, disassembly, relocation, and reuse of the modular unit. The modular unit has dimensions of 12.2 m in length, 2.4 m in width, 1.9 m in height. The modular unit was manufactured using virgin materials without incorporating any recycled materials. The modular design therefore follows a “downstream reuse” concept, where virgin materials are used initially, but disassembly and reuse are prioritized after the building’s first life

Code: • Mitigation > Design intervention Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 5
 Created: 23.6.2026 r. 12:18 by user, Modified: 23.6.2026 r. 12:18 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Due to the lack of standardized modularity designs and connection systems in the market, reusing a specific modular unit in different modular systems is technically and economically challenging. Therefore, it is assumed that if the modular unit is functionally and technically reusable, it should be reused within the same building system in subsequent use cycle(s). If it is unsuitable for reuse, reaching the end of their designed lifespan, the modular unit should be demolished for recycling and disposal.

Code: • Circular strategy > Direct reuse Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 5
 Created: 23.6.2026 r. 12:19 by user, Modified: 23.6.2026 r. 12:19 by user
 Other codes assigned to segment:
 Circular strategy > Recycling (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Obsolescence > Economic (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

Due to the lack of standardized modularity designs and connection systems in the market, reusing a specific modular unit in different modular systems is technically and economically challenging. Therefore, it is assumed that if the modular unit is functionally and technically reusable, it should be reused within the same building system in subsequent use cycle(s). If it is unsuitable for reuse, reaching the end of their designed lifespan, the modular unit should be demolished for recycling and disposal.

Code: • Negative mechanism > Missing repair infrastructure Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 5
 Created: 23.6.2026 r. 12:20 by user, Modified: 23.6.2026 r. 12:20 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Obsolescence > Economic (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

Due to the lack of standardized modularity designs and connection systems in the market, reusing a specific modular unit in different modular systems is technically and economically challenging. Therefore, it is assumed that if the modular unit is functionally and technically reusable, it should be

reused within the same building system in subsequent use cycle(s). If it is unsuitable for reuse, reaching the end of their designed lifespan, the modular unit should be demolished for recycling and disposal.

Code: • Enabling condition > Disassembly Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 5
 Created: 23.6.2026 r. 12:20 by user, Modified: 23.6.2026 r. 12:20 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Obsolescence > Economic (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

Due to the lack of standardized modularity designs and connection systems in the market, reusing a specific modular unit in different modular systems is technically and economically challenging. Therefore, it is assumed that if the modular unit is functionally and technically reusable, it should be reused within the same building system in subsequent use cycle(s). If it is unsuitable for reuse, reaching the end of their designed lifespan, the modular unit should be demolished for recycling and disposal.

Code: • Negative mechanism > Lock-in Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 5
 Created: 23.6.2026 r. 12:20 by user, Modified: 23.6.2026 r. 12:20 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Obsolescence > Economic (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

Due to the lack of standardized modularity designs and connection systems in the market, reusing a specific modular unit in different modular systems is technically and economically challenging. Therefore, it is assumed that if the modular unit is functionally and technically reusable, it should be reused within the same building system in subsequent use cycle(s). If it is unsuitable for reuse, reaching the end of their designed lifespan, the modular unit should be demolished for recycling and disposal.

Code: • Circular strategy > Recycling Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 5
 Created: 23.6.2026 r. 12:19 by user, Modified: 23.6.2026 r. 12:19 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Obsolescence > Economic (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

Due to the lack of standardized modularity designs and connection systems in the market, reusing a specific modular unit in different modular systems is technically and economically challenging. Therefore, it is assumed that if the modular unit is functionally and technically reusable, it should be reused within the same building system in subsequent use cycle(s). If it is unsuitable for reuse,

reaching the end of their designed lifespan, the modular unit should be demolished for recycling and disposal.

Code: ● Mitigation > Design intervention Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 5
 Created: 23.6.2026 r. 12:20 by user, Modified: 23.6.2026 r. 12:20 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Obsolescence > Economic (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)

Due to the lack of standardized modularity designs and connection systems in the market, reusing a specific modular unit in different modular systems is technically and economically challenging. Therefore, it is assumed that if the modular unit is functionally and technically reusable, it should be reused within the same building system in subsequent use cycle(s). If it is unsuitable for reuse, reaching the end of their designed lifespan, the modular unit should be demolished for recycling and disposal.

Code: ● Obsolescence > Technical Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 5
 Created: 23.6.2026 r. 12:20 by user, Modified: 23.6.2026 r. 12:20 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Obsolescence > Economic (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

Due to the lack of standardized modularity designs and connection systems in the market, reusing a specific modular unit in different modular systems is technically and economically challenging. Therefore, it is assumed that if the modular unit is functionally and technically reusable, it should be reused within the same building system in subsequent use cycle(s). If it is unsuitable for reuse, reaching the end of their designed lifespan, the modular unit should be demolished for recycling and disposal.

Code: ● Obsolescence > Economic Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 5
 Created: 23.6.2026 r. 12:20 by user, Modified: 23.6.2026 r. 12:20 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

Through the use cycles, the inherent properties of the modular components are unchanged, and the materials are intended for direct reuse or use after repair or replacement within the same system. The assessment focused on the modular unit's core component (steel frame) and subcomponents (concrete slab and bolt-nut connector). The functional unit chosen in this LCA is the modular unit with a reference service period (i.e., the designed lifetime) of 50 years.

The LCA is conducted to evaluate the environmental burdens and credits associated with the reuse of these modular components across multiple cycles. Additionally, the impact of recycling non-reusable components is also evaluated.

Code: • Circular strategy > Direct reuse Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 5
 Created: 23.6.2026 r. 12:22 by user, Modified: 23.6.2026 r. 12:22 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Through the use cycles, the inherent properties of the modular components are unchanged, and the materials are intended for direct reuse or use after repair or replacement within the same system. The assessment focused on the modular unit's core component (steel frame) and subcomponents (concrete slab and bolt-nut connector). The functional unit chosen in this LCA is the modular unit with a reference service period (i.e., the designed lifetime) of 50 years.

The LCA is conducted to evaluate the environmental burdens and credits associated with the reuse of these modular components across multiple cycles. Additionally, the impact of recycling non-reusable components is also evaluated.

Code: • Circular strategy > Repair Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 5
 Created: 23.6.2026 r. 12:22 by user, Modified: 23.6.2026 r. 12:22 by user
 Other codes assigned to segment:
 Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Through the use cycles, the inherent properties of the modular components are unchanged, and the materials are intended for direct reuse or use after repair or replacement within the same system. The assessment focused on the modular unit's core component (steel frame) and subcomponents (concrete slab and bolt-nut connector). The functional unit chosen in this LCA is the modular unit with a reference service period (i.e., the designed lifetime) of 50 years.

The LCA is conducted to evaluate the environmental burdens and credits associated with the reuse of these modular components across multiple cycles. Additionally, the impact of recycling non-reusable components is also evaluated.

Code: • Negative mechanism > Burden shifting Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 5
 Created: 23.6.2026 r. 12:23 by user, Modified: 23.6.2026 r. 12:23 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Through the use cycles, the inherent properties of the modular components are unchanged, and the materials are intended for direct reuse or use after repair or replacement within the same system. The assessment focused on the modular unit's core component (steel frame) and subcomponents (concrete slab and bolt-nut connector). The functional unit chosen in this LCA is the modular unit with a reference service period (i.e., the designed lifetime) of 50 years.

The LCA is conducted to evaluate the environmental burdens and credits associated with the reuse of these modular components across multiple cycles. Additionally, the impact of recycling non-reusable components is also evaluated.

Code: • Circular strategy > Component replacement Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 5
 Created: 23.6.2026 r. 12:22 by user, Modified: 23.6.2026 r. 12:22 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Through the use cycles, the inherent properties of the modular components are unchanged, and the materials are intended for direct reuse or use after repair or replacement within the same system. The assessment focused on the modular unit's core component (steel frame) and subcomponents (concrete slab and bolt-nut connector). The functional unit chosen in this LCA is the modular unit with a reference service period (i.e., the designed lifetime) of 50 years.

The LCA is conducted to evaluate the environmental burdens and credits associated with the reuse of these modular components across multiple cycles. Additionally, the impact of recycling non-reusable components is also evaluated.

Code: • Circular strategy > Recycling Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 5
 Created: 23.6.2026 r. 12:22 by user, Modified: 23.6.2026 r. 12:22 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Through the use cycles, the inherent properties of the modular components are unchanged, and the materials are intended for direct reuse or use after repair or replacement within the same system. The assessment focused on the modular unit's core component (steel frame) and subcomponents (concrete slab and bolt-nut connector). The functional unit chosen in this LCA is the modular unit with a reference service period (i.e., the designed lifetime) of 50 years.

The LCA is conducted to evaluate the environmental burdens and credits associated with the reuse of these modular components across multiple cycles. Additionally, the impact of recycling non-reusable components is also evaluated.

Code: • Mitigation > Lifecycle verification Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 5
 Created: 23.6.2026 r. 12:22 by user, Modified: 23.6.2026 r. 12:22 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)

From an LCA perspective, assessing the potential reuse of modular components represents challenges due to the unknown future value retention processes. The reuse, recycling and disposal of modular components will take place later, and uncertainties surround variables such as the reusability rate, recyclability rate, and amount of waste disposed. To address this concern, a global sensitivity analysis has been conducted in this study to calculate probabilistic life cycle outcomes and

identify the key influential factors that significantly affect the LCA results. Therefore, overinterpreting the present results in deterministic or absolute terms should be avoided.

Code: • Circular strategy > Direct reuse Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 14
 Created: 23.6.2026 r. 12:26 by user, Modified: 23.6.2026 r. 12:26 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

From an LCA perspective, assessing the potential reuse of modular components represents challenges due to the unknown future value retention processes. The reuse, recycling and disposal of modular components will take place later, and uncertainties surround variables such as the reusability rate, recyclability rate, and amount of waste disposed. To address this concern, a global sensitivity analysis has been conducted in this study to calculate probabilistic life cycle outcomes and identify the key influential factors that significantly affect the LCA results. Therefore, overinterpreting the present results in deterministic or absolute terms should be avoided.

Code: • Circular strategy > Recycling Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 14
 Created: 23.6.2026 r. 12:26 by user, Modified: 23.6.2026 r. 12:26 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

From an LCA perspective, assessing the potential reuse of modular components represents challenges due to the unknown future value retention processes. The reuse, recycling and disposal of modular components will take place later, and uncertainties surround variables such as the reusability rate, recyclability rate, and amount of waste disposed. To address this concern, a global sensitivity analysis has been conducted in this study to calculate probabilistic life cycle outcomes and identify the key influential factors that significantly affect the LCA results. Therefore, overinterpreting the present results in deterministic or absolute terms should be avoided.

Code: • Circular strategy > Repair Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 14
 Created: 23.6.2026 r. 12:26 by user, Modified: 23.6.2026 r. 12:26 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

From an LCA perspective, assessing the potential reuse of modular components represents challenges due to the unknown future value retention processes. The reuse, recycling and disposal of modular components will take place later, and uncertainties surround variables such as the reusability rate, recyclability rate, and amount of waste disposed. To address this concern, a global sensitivity analysis has been conducted in this study to calculate probabilistic life cycle outcomes and identify the key influential factors that significantly affect the LCA results. Therefore, overinterpreting the present results in deterministic or absolute terms should be avoided.

Code: • Negative mechanism > Burden shifting Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 14
 Created: 23.6.2026 r. 12:26 by user, Modified: 23.6.2026 r. 12:26 by user

Other codes assigned to segment:

- Circular strategy > Repair (Weight: 0)
- Circular strategy > Direct reuse (Weight: 0)
- Circular strategy > Recycling (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

From an LCA perspective, assessing the potential reuse of modular components represents challenges due to the unknown future value retention processes. The reuse, recycling and disposal of modular components will take place later, and uncertainties surround variables such as the reusability rate, recyclability rate, and amount of waste disposed. To address this concern, a global sensitivity analysis has been conducted in this study to calculate probabilistic life cycle outcomes and identify the key influential factors that significantly affect the LCA results. Therefore, overinterpreting the present results in deterministic or absolute terms should be avoided.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-14, p. 14

Created: 23.6.2026 r. 12:26 by user, Modified: 23.6.2026 r. 12:26 by user

Other codes assigned to segment:

- Circular strategy > Repair (Weight: 0)
- Circular strategy > Direct reuse (Weight: 0)
- Circular strategy > Recycling (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)

These findings suggest that different levels of reusability and recyclability are warranted for core and non-core modular elements. Specifically, it is desirable to achieve a higher level of reusability (i.e., direct reuse) for the core steel frame to avoid the impact of repairs. The finding also suggests that promoting the recycling of the steel frame at its end-of-life can have a significant influence on GWP reductions. On the other hand, a relatively lower level of reusability (i. e., repairable) and recyclability for subcomponents is considered environmentally acceptable. However, the lowest level of reusability, which involves replacement of subcomponents, should be avoided to mitigate the impact of disposal of replaced materials and production of new materials. In view of this, it is crucial to consider appropriate design methods, disassembly procedures, and end-of-use-cycle treatments for individual modular elements and materials, with a view to enhancing their environmental performance of reuse and recycling. In other words, tailor-made design and deconstruction strategies should be developed for different modular components to maximize their reusability and recycling potential while minimizing the multi-use cycle environmental impacts.

Code: • Negative mechanism > Higher initial material use Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-14, p. 14

Created: 23.6.2026 r. 12:28 by user, Modified: 23.6.2026 r. 12:28 by user

Other codes assigned to segment:

- Circular strategy > Repair (Weight: 0)
- Circular strategy > Component replacement (Weight: 0)
- Circular strategy > Direct reuse (Weight: 0)
- Circular strategy > Recycling (Weight: 0)
- Negative mechanism > Premature replacement (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)
- Mitigation > Design intervention (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

These findings suggest that different levels of reusability and recyclability are warranted for core and non-core modular elements. Specifically, it is desirable to achieve a higher level of reusability (i.e., direct reuse) for the core steel frame to avoid the impact of repairs. The finding also suggests that promoting the recycling of the steel frame at its end-of-life can have a significant influence on GWP reductions. On the other hand, a relatively lower level of reusability (i. e., repairable) and recyclability for subcomponents is considered environmentally acceptable. However, the lowest level of

reusability, which involves replacement of subcomponents, should be avoided to mitigate the impact of disposal of replaced materials and production of new materials. In view of this, it is crucial to consider appropriate design methods, disassembly procedures, and end-of-use-cycle treatments for individual modular elements and materials, with a view to enhancing their environmental performance of reuse and recycling. In other words, tailor-made design and deconstruction strategies should be developed for different modular components to maximize their reusability and recycling potential while minimizing the multi-use cycle environmental impacts.

Code: • Circular strategy > Direct reuse Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-14, p. 14

Created: 23.6.2026 r. 12:27 by user, Modified: 23.6.2026 r. 12:27 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)

Circular strategy > Component replacement (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Design intervention (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

These findings suggest that different levels of reusability and recyclability are warranted for core and non-core modular elements. Specifically, it is desirable to achieve a higher level of reusability (i.e., direct reuse) for the core steel frame to avoid the impact of repairs. The finding also suggests that promoting the recycling of the steel frame at its end-of-life can have a significant influence on GWP reductions. On the other hand, a relatively lower level of reusability (i. e., repairable) and recyclability for subcomponents is considered environmentally acceptable. However, the lowest level of reusability, which involves replacement of subcomponents, should be avoided to mitigate the impact of disposal of replaced materials and production of new materials. In view of this, it is crucial to consider appropriate design methods, disassembly procedures, and end-of-use-cycle treatments for individual modular elements and materials, with a view to enhancing their environmental performance of reuse and recycling. In other words, tailor-made design and deconstruction strategies should be developed for different modular components to maximize their reusability and recycling potential while minimizing the multi-use cycle environmental impacts.

Code: • Circular strategy > Repair Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-14, p. 14

Created: 23.6.2026 r. 12:27 by user, Modified: 23.6.2026 r. 12:27 by user

Other codes assigned to segment:

Circular strategy > Component replacement (Weight: 0)

Circular strategy > Direct reuse (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Design intervention (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

These findings suggest that different levels of reusability and recyclability are warranted for core and non-core modular elements. Specifically, it is desirable to achieve a higher level of reusability (i.e., direct reuse) for the core steel frame to avoid the impact of repairs. The finding also suggests that promoting the recycling of the steel frame at its end-of-life can have a significant influence on GWP reductions. On the other hand, a relatively lower level of reusability (i. e., repairable) and recyclability for subcomponents is considered environmentally acceptable. However, the lowest level of reusability, which involves replacement of subcomponents, should be avoided to mitigate the impact of disposal of replaced materials and production of new materials. In view of this, it is crucial to

consider appropriate design methods, disassembly procedures, and end-of-use-cycle treatments for individual modular elements and materials, with a view to enhancing their environmental performance of reuse and recycling. In other words, tailor-made design and deconstruction strategies should be developed for different modular components to maximize their reusability and recycling potential while minimizing the multi-use cycle environmental impacts.

Code: • Negative mechanism > Premature replacement Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-14, p. 14

Created: 23.6.2026 r. 12:28 by user, Modified: 23.6.2026 r. 12:28 by user

Other codes assigned to segment:

- Circular strategy > Repair (Weight: 0)
- Circular strategy > Component replacement (Weight: 0)
- Circular strategy > Direct reuse (Weight: 0)
- Circular strategy > Recycling (Weight: 0)
- Negative mechanism > Higher initial material use (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)
- Mitigation > Design intervention (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

These findings suggest that different levels of reusability and recyclability are warranted for core and non-core modular elements. Specifically, it is desirable to achieve a higher level of reusability (i.e., direct reuse) for the core steel frame to avoid the impact of repairs. The finding also suggests that promoting the recycling of the steel frame at its end-of-life can have a significant influence on GWP reductions. On the other hand, a relatively lower level of reusability (i. e., repairable) and recyclability for subcomponents is considered environmentally acceptable. However, the lowest level of reusability, which involves replacement of subcomponents, should be avoided to mitigate the impact of disposal of replaced materials and production of new materials. In view of this, it is crucial to consider appropriate design methods, disassembly procedures, and end-of-use-cycle treatments for individual modular elements and materials, with a view to enhancing their environmental performance of reuse and recycling. In other words, tailor-made design and deconstruction strategies should be developed for different modular components to maximize their reusability and recycling potential while minimizing the multi-use cycle environmental impacts.

Code: • Circular strategy > Recycling Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-14, p. 14

Created: 23.6.2026 r. 12:27 by user, Modified: 23.6.2026 r. 12:27 by user

Other codes assigned to segment:

- Circular strategy > Repair (Weight: 0)
- Circular strategy > Component replacement (Weight: 0)
- Circular strategy > Direct reuse (Weight: 0)
- Negative mechanism > Premature replacement (Weight: 0)
- Negative mechanism > Higher initial material use (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)
- Mitigation > Design intervention (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

These findings suggest that different levels of reusability and recyclability are warranted for core and non-core modular elements. Specifically, it is desirable to achieve a higher level of reusability (i.e., direct reuse) for the core steel frame to avoid the impact of repairs. The finding also suggests that promoting the recycling of the steel frame at its end-of-life can have a significant influence on GWP reductions. On the other hand, a relatively lower level of reusability (i. e., repairable) and recyclability for subcomponents is considered environmentally acceptable. However, the lowest level of reusability, which involves replacement of subcomponents, should be avoided to mitigate the impact of disposal of replaced materials and production of new materials. In view of this, it is crucial to consider appropriate design methods, disassembly procedures, and end-of-use-cycle treatments for individual modular elements and materials, with a view to enhancing their environmental

performance of reuse and recycling. In other words, tailor-made design and deconstruction strategies should be developed for different modular components to maximize their reusability and recycling potential while minimizing the multi-use cycle environmental impacts.

Code: • Mitigation > Design intervention Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 14
 Created: 23.6.2026 r. 12:28 by user, Modified: 23.6.2026 r. 12:28 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

These findings suggest that different levels of reusability and recyclability are warranted for core and non-core modular elements. Specifically, it is desirable to achieve a higher level of reusability (i.e., direct reuse) for the core steel frame to avoid the impact of repairs. The finding also suggests that promoting the recycling of the steel frame at its end-of-life can have a significant influence on GWP reductions. On the other hand, a relatively lower level of reusability (i. e., repairable) and recyclability for subcomponents is considered environmentally acceptable. However, the lowest level of reusability, which involves replacement of subcomponents, should be avoided to mitigate the impact of disposal of replaced materials and production of new materials. In view of this, it is crucial to consider appropriate design methods, disassembly procedures, and end-of-use-cycle treatments for individual modular elements and materials, with a view to enhancing their environmental performance of reuse and recycling. In other words, tailor-made design and deconstruction strategies should be developed for different modular components to maximize their reusability and recycling potential while minimizing the multi-use cycle environmental impacts.

Code: • Circular strategy > Component replacement Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-14, p. 14
 Created: 23.6.2026 r. 12:27 by user, Modified: 23.6.2026 r. 12:27 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

These findings suggest that different levels of reusability and recyclability are warranted for core and non-core modular elements. Specifically, it is desirable to achieve a higher level of reusability (i.e., direct reuse) for the core steel frame to avoid the impact of repairs. The finding also suggests that promoting the recycling of the steel frame at its end-of-life can have a significant influence on GWP reductions. On the other hand, a relatively lower level of reusability (i. e., repairable) and recyclability for subcomponents is considered environmentally acceptable. However, the lowest level of reusability, which involves replacement of subcomponents, should be avoided to mitigate the impact of disposal of replaced materials and production of new materials. In view of this, it is crucial to consider appropriate design methods, disassembly procedures, and end-of-use-cycle treatments for individual modular elements and materials, with a view to enhancing their environmental performance of reuse and recycling. In other words, tailor-made design and deconstruction

strategies should be developed for different modular components to maximize their reusability and recycling potential while minimizing the multi-use cycle environmental impacts.

Code: • Negative mechanism > Burden shifting Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-14, p. 14

Created: 23.6.2026 r. 12:28 by user, Modified: 23.6.2026 r. 12:28 by user

Other codes assigned to segment:

- Circular strategy > Repair (Weight: 0)
- Circular strategy > Component replacement (Weight: 0)
- Circular strategy > Direct reuse (Weight: 0)
- Circular strategy > Recycling (Weight: 0)
- Negative mechanism > Premature replacement (Weight: 0)
- Negative mechanism > Higher initial material use (Weight: 0)
- Mitigation > Design intervention (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

These findings suggest that different levels of reusability and recyclability are warranted for core and non-core modular elements. Specifically, it is desirable to achieve a higher level of reusability (i.e., direct reuse) for the core steel frame to avoid the impact of repairs. The finding also suggests that promoting the recycling of the steel frame at its end-of-life can have a significant influence on GWP reductions. On the other hand, a relatively lower level of reusability (i. e., repairable) and recyclability for subcomponents is considered environmentally acceptable. However, the lowest level of reusability, which involves replacement of subcomponents, should be avoided to mitigate the impact of disposal of replaced materials and production of new materials. In view of this, it is crucial to consider appropriate design methods, disassembly procedures, and end-of-use-cycle treatments for individual modular elements and materials, with a view to enhancing their environmental performance of reuse and recycling. In other words, tailor-made design and deconstruction strategies should be developed for different modular components to maximize their reusability and recycling potential while minimizing the multi-use cycle environmental impacts.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-14, p. 14

Created: 23.6.2026 r. 12:28 by user, Modified: 23.6.2026 r. 12:28 by user

Other codes assigned to segment:

- Circular strategy > Repair (Weight: 0)
- Circular strategy > Component replacement (Weight: 0)
- Circular strategy > Direct reuse (Weight: 0)
- Circular strategy > Recycling (Weight: 0)
- Negative mechanism > Premature replacement (Weight: 0)
- Negative mechanism > Higher initial material use (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)
- Mitigation > Design intervention (Weight: 0)

This study underscores the importance of incorporating value retention processes into the life cycle assessment of building reuses, with a view to enhancing the rigor of the assessment outcomes. Value retention processes, such as repair and replacement, are the necessary activities undertaken to maintain or enhance the value of the modular component and enable continued reuse. By assessing the impacts of the value retention processes, this study offers valuable insights into the potential trade-offs between the environmental benefits and loads of

Y. Yang et al. Sustainable Production and Consumption 46 (2024) 543–558
557

reusing and recycling modular components. These findings can guide structural designers and contractors to implement appropriate design and deconstruction planning strategies, with the goal of avoiding value retention activities that have considerable environmental impacts.

Code: • Circular strategy > Repair Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-14, p. 14 - 15

Created: 23.6.2026 r. 12:30 by user, Modified: 23.6.2026 r. 12:30 by user

Other codes assigned to segment:

- Circular strategy > Component replacement (Weight: 0)
- Circular strategy > Direct reuse (Weight: 0)
- Negative mechanism > Higher initial material use (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)
- Mitigation > Design intervention (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

This study underscores the importance of incorporating value retention processes into the life cycle assessment of building reuses, with a view to enhancing the rigor of the assessment outcomes. Value retention processes, such as repair and replacement, are the necessary activities undertaken to maintain or enhance the value of the modular component and enable continued reuse. By assessing the impacts of the value retention processes, this study offers valuable insights into the potential trade-offs between the environmental benefits and loads of

Y. Yang et al. Sustainable Production and Consumption 46 (2024) 543–558
557

reusing and recycling modular components. These findings can guide structural designers and contractors to implement appropriate design and deconstruction planning strategies, with the goal of avoiding value retention activities that have considerable environmental impacts.

Code: • Negative mechanism > Burden shifting Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-14, p. 14 - 15

Created: 23.6.2026 r. 12:31 by user, Modified: 23.6.2026 r. 12:31 by user

Other codes assigned to segment:

- Circular strategy > Repair (Weight: 0)
- Circular strategy > Component replacement (Weight: 0)
- Circular strategy > Direct reuse (Weight: 0)
- Negative mechanism > Higher initial material use (Weight: 0)
- Mitigation > Design intervention (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

This study underscores the importance of incorporating value retention processes into the life cycle assessment of building reuses, with a view to enhancing the rigor of the assessment outcomes. Value retention processes, such as repair and replacement, are the necessary activities undertaken to maintain or enhance the value of the modular component and enable continued reuse. By assessing the impacts of the value retention processes, this study offers valuable insights into the potential trade-offs between the environmental benefits and loads of

Y. Yang et al. Sustainable Production and Consumption 46 (2024) 543–558
557

reusing and recycling modular components. These findings can guide structural designers and contractors to implement appropriate design and deconstruction planning strategies, with the goal of avoiding value retention activities that have considerable environmental impacts.

Code: • Circular strategy > Component replacement Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-14, p. 14 - 15

Created: 23.6.2026 r. 12:30 by user, Modified: 23.6.2026 r. 12:30 by user

Other codes assigned to segment:

- Circular strategy > Repair (Weight: 0)
- Circular strategy > Direct reuse (Weight: 0)
- Negative mechanism > Higher initial material use (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)
- Mitigation > Design intervention (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

This study underscores the importance of incorporating value retention processes into the life cycle assessment of building reuses, with a view to enhancing the rigor of the assessment outcomes. Value

retention processes, such as repair and replacement, are the necessary activities undertaken to maintain or enhance the value of the modular component and enable continued reuse. By assessing the impacts of the value retention processes, this study offers valuable insights into the potential trade-offs between the environmental benefits and loads of

Y. Yang et al. Sustainable Production and Consumption 46 (2024) 543–558

557

reusing and recycling modular components. These findings can guide structural designers and contractors to implement appropriate design and deconstruction planning strategies, with the goal of avoiding value retention activities that have considerable environmental impacts.

Code: • Mitigation > Design intervention Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-14, p. 14 - 15

Created: 23.6.2026 r. 12:31 by user, Modified: 23.6.2026 r. 12:31 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)

Circular strategy > Component replacement (Weight: 0)

Circular strategy > Direct reuse (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

This study underscores the importance of incorporating value retention processes into the life cycle assessment of building reuses, with a view to enhancing the rigor of the assessment outcomes. Value retention processes, such as repair and replacement, are the necessary activities undertaken to maintain or enhance the value of the modular component and enable continued reuse. By assessing the impacts of the value retention processes, this study offers valuable insights into the potential trade-offs between the environmental benefits and loads of

Y. Yang et al. Sustainable Production and Consumption 46 (2024) 543–558

557

reusing and recycling modular components. These findings can guide structural designers and contractors to implement appropriate design and deconstruction planning strategies, with the goal of avoiding value retention activities that have considerable environmental impacts.

Code: • Circular strategy > Direct reuse Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-14, p. 14 - 15

Created: 23.6.2026 r. 12:30 by user, Modified: 23.6.2026 r. 12:30 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)

Circular strategy > Component replacement (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Design intervention (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

This study underscores the importance of incorporating value retention processes into the life cycle assessment of building reuses, with a view to enhancing the rigor of the assessment outcomes. Value retention processes, such as repair and replacement, are the necessary activities undertaken to maintain or enhance the value of the modular component and enable continued reuse. By assessing the impacts of the value retention processes, this study offers valuable insights into the potential trade-offs between the environmental benefits and loads of

Y. Yang et al. Sustainable Production and Consumption 46 (2024) 543–558

557

reusing and recycling modular components. These findings can guide structural designers and contractors to implement appropriate design and deconstruction planning strategies, with the goal of avoiding value retention activities that have considerable environmental impacts.

Code: • Negative mechanism > Higher initial material use Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-14, p. 14 - 15

Created: 23.6.2026 r. 12:31 by user, Modified: 23.6.2026 r. 12:31 by user

Other codes assigned to segment:

- Circular strategy > Repair (Weight: 0)
- Circular strategy > Component replacement (Weight: 0)
- Circular strategy > Direct reuse (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)
- Mitigation > Design intervention (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

This study underscores the importance of incorporating value retention processes into the life cycle assessment of building reuses, with a view to enhancing the rigor of the assessment outcomes. Value retention processes, such as repair and replacement, are the necessary activities undertaken to maintain or enhance the value of the modular component and enable continued reuse. By assessing the impacts of the value retention processes, this study offers valuable insights into the potential trade-offs between the environmental benefits and loads of

Y. Yang et al. Sustainable Production and Consumption 46 (2024) 543–558

557

reusing and recycling modular components. These findings can guide structural designers and contractors to implement appropriate design and deconstruction planning strategies, with the goal of avoiding value retention activities that have considerable environmental impacts.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-14, p. 14 - 15

Created: 23.6.2026 r. 12:31 by user, Modified: 23.6.2026 r. 12:31 by user

Other codes assigned to segment:

- Circular strategy > Repair (Weight: 0)
- Circular strategy > Component replacement (Weight: 0)
- Circular strategy > Direct reuse (Weight: 0)
- Negative mechanism > Higher initial material use (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)
- Mitigation > Design intervention (Weight: 0)

ASSESSMENT METHOD

NUMBER OF STEPS TO COMPLETE ASSESSMENT

CRITERIA, TOTAL

FACTORS OF DISASSEMBLY CRITERIA

Durmisevic et al. (2003)

7:

- Functional decomposition
- Life-cycle coordination
- Relational pattern
- Systematisation
- Component specification
- Indicators of deconstruction
- Transformation capacity
- 48 • Geometry of product edge
- Standardisation of product edge
- Type of connection
- Accessibility to fixings and intermediary
- Tolerance
- Morphology of joint

Cottafava & Ritzen (2021)

4:

- Material circularity indicator (MCI)
- MCI through embedded factors
- System circularity indicator (SCI)
- Building circularity indicator (BCI)

10 • Building layer

- Connection type
- Connection accessibility
- Crossings
- Form containment

van Vliet

(2018)

4:

- MCI
- Product circularity indicator (PCI)
- SCI
- BCI

8 MCI

- Accessibility to connection
- Type of connection
- Assembly shape
- Independency
- Method of fabrication
- Assembly sequences
- Type of relational pattern
- SCI
- BCI

van Vliet et al.

(2021)

4:

- Assessment of products
- Disassembly potential (DP)
- Layers of brand and environmental cost indicator
- Building-level score 4 • Connection type
- Connection accessibility
- Independency
- Geometry of product edge

Code: • Enabling condition > Disassembly Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-19, p. 3

Created: 23.6.2026 r. 12:53 by user, Modified: 23.6.2026 r. 12:53 by user

Other codes assigned to segment:

Enabling condition > Component accessibility (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Mitigation > Design intervention (Weight: 0)

ASSESSMENT METHOD

NUMBER OF STEPS TO COMPLETE ASSESSMENT

CRITERIA, TOTAL

FACTORS OF DISASSEMBLY CRITERIA

Durmisevic et al. (2003)

7:

- Functional decomposition
- Life-cycle coordination
- Relational pattern
- Systematisation
- Component specification
- Indicators of deconstruction
- Transformation capacity
- 48 • Geometry of product edge
- Standardisation of product edge
- Type of connection
- Accessibility to fixings and intermediary
- Tolerance
- Morphology of joint

Cottafava & Ritzen (2021)

4:

- Material circularity indicator (MCI)
- MCI through embedded factors
- System circularity indicator (SCI)
- Building circularity indicator (BCI)

10 • Building layer

- Connection type
- Connection accessibility
- Crossings
- Form containment

van Vliet

(2018)

4:

- MCI
- Product circularity indicator (PCI)
- SCI
- BCI

8 MCI

- Accessibility to connection
- Type of connection
- Assembly shape
- Independency
- Method of fabrication
- Assembly sequences
- Type of relational pattern
- SCI
- BCI

van Vliet et al.

(2021)

4:

- Assessment of products
- Disassembly potential (DP)
- Layers of brand and environmental cost indicator
- Building-level score 4 • Connection type
- Connection accessibility

- Independency
- Geometry of product edge
 - Code: • Mitigation > Design intervention Weight score: 0
 - Search B - Disassembly reuse and facade recovery > B-GS-19, p. 3
 - Created: 23.6.2026 r. 12:54 by user, Modified: 23.6.2026 r. 12:54 by user
 - Other codes assigned to segment:
 - Enabling condition > Component accessibility (Weight: 0)
 - Enabling condition > Disassembly (Weight: 0)
 - Enabling condition > Repair information (Weight: 0)

ASSESSMENT METHOD

NUMBER OF STEPS TO COMPLETE ASSESSMENT

CRITERIA, TOTAL

FACTORS OF DISASSEMBLY CRITERIA

Durmisevic et al. (2003)

7:

- Functional decomposition
- Life-cycle coordination
- Relational pattern
- Systematisation
- Component specification
- Indicators of deconstruction
- Transformation capacity

48 • Geometry of product edge

- Standardisation of product edge
- Type of connection
- Accessibility to fixings and intermediary
- Tolerance
- Morphology of joint

Cottafava & Ritzen (2021)

4:

- Material circularity indicator (MCI)
- MCI through embedded factors
- System circularity indicator (SCI)
- Building circularity indicator (BCI)

10 • Building layer

- Connection type
- Connection accessibility
- Crossings
- Form containment

van Vliet

(2018)

4:

- MCI
- Product circularity indicator (PCI)
- SCI
- BCI

8 MCI

- Accessibility to connection
- Type of connection

- Assembly shape
- Independency
- Method of fabrication
- Assembly sequences
- Type of relational pattern
- SCI
- BCI

van Vliet et al.

(2021)

4:

- Assessment of products
- Disassembly potential (DP)
- Layers of brand and environmental cost indicator
- Building-level score 4 • Connection type
- Connection accessibility
- Independency
- Geometry of product edge

Code: • Enabling condition > Component accessibility Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-19, p. 3

Created: 23.6.2026 r. 12:53 by user, Modified: 23.6.2026 r. 12:53 by user

Other codes assigned to segment:

Enabling condition > Disassembly (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Mitigation > Design intervention (Weight: 0)

ASSESSMENT METHOD

NUMBER OF STEPS TO COMPLETE ASSESSMENT

CRITERIA, TOTAL

FACTORS OF DISASSEMBLY CRITERIA

Durmisevic et al. (2003)

7:

- Functional decomposition
- Life-cycle coordination
- Relational pattern
- Systematisation
- Component specification
- Indicators of deconstruction
- Transformation capacity
- 48 • Geometry of product edge
- Standardisation of product edge
- Type of connection
- Accessibility to fixings and intermediary
- Tolerance
- Morphology of joint

Cottafava & Ritzen (2021)

4:

- Material circularity indicator (MCI)
- MCI through embedded factors
- System circularity indicator (SCI)

- Building circularity indicator (BCI)

10 • Building layer

- Connection type
- Connection accessibility
- Crossings
- Form containment

van Vliet

(2018)

4:

- MCI
- Product circularity indicator (PCI)
- SCI
- BCI

8 MCI

- Accessibility to connection
- Type of connection
- Assembly shape
- Independency
- Method of fabrication
- Assembly sequences
- Type of relational pattern
- SCI
- BCI

van Vliet et al.

(2021)

4:

- Assessment of products
- Disassembly potential (DP)
- Layers of brand and environmental cost indicator
- Building-level score 4 • Connection type
- Connection accessibility
- Independency
- Geometry of product edge

Code: • Enabling condition > Repair information Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-19, p. 3

Created: 23.6.2026 r. 12:53 by user, Modified: 23.6.2026 r. 12:53 by user

Other codes assigned to segment:

Enabling condition > Component accessibility (Weight: 0)

Enabling condition > Disassembly (Weight: 0)

Mitigation > Design intervention (Weight: 0)

However, there is a lack of DP assessment methods suitable for supporting practical design decisions made during the early design phase.

The aim of this paper is to develop a robust, simple and reliable DP assessment method that connects (1) to the practical design process and (2) can be utilised with BIM data from the early phases to the detailed design phase. The research questions are as follows:

- How can a building's DP be evaluated during the early building design phases with the situationally available data used for LCA?
- How can quantitative values for building component DP be assessed in a sufficiently reliable and practical manner to improve existing LCA calculations, particularly for end-of-life (EoL) scenarios?

Code: • Enabling condition > Repair information Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-19, p. 4

Created: 23.6.2026 r. 12:45 by user, Modified: 23.6.2026 r. 12:45 by user

Other codes assigned to segment:

Enabling condition > Disassembly (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Design intervention (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

However, there is a lack of DP assessment methods suitable for supporting practical design decisions made during the early design phase.

The aim of this paper is to develop a robust, simple and reliable DP assessment method that connects (1) to the practical design process and (2) can be utilised with BIM data from the early phases to the detailed design phase. The research questions are as follows:

- How can a building's DP be evaluated during the early building design phases with the situationally available data used for LCA?
- How can quantitative values for building component DP be assessed in a sufficiently reliable and practical manner to improve existing LCA calculations, particularly for end-of-life (EoL) scenarios?

Code: • Mitigation > Design intervention Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-19, p. 4

Created: 23.6.2026 r. 12:47 by user, Modified: 23.6.2026 r. 12:47 by user

Other codes assigned to segment:

Enabling condition > Disassembly (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

However, there is a lack of DP assessment methods suitable for supporting practical design decisions made during the early design phase.

The aim of this paper is to develop a robust, simple and reliable DP assessment method that connects (1) to the practical design process and (2) can be utilised with BIM data from the early phases to the detailed design phase. The research questions are as follows:

- How can a building's DP be evaluated during the early building design phases with the situationally available data used for LCA?
- How can quantitative values for building component DP be assessed in a sufficiently reliable and practical manner to improve existing LCA calculations, particularly for end-of-life (EoL) scenarios?

Code: • Negative mechanism > Missing repair infrastructure Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-19, p. 4

Created: 23.6.2026 r. 12:48 by user, Modified: 23.6.2026 r. 12:48 by user

Other codes assigned to segment:

Enabling condition > Disassembly (Weight: 0)

Enabling condition > Repair information (Weight: 0)

Mitigation > Design intervention (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

However, there is a lack of DP assessment methods suitable for supporting practical design decisions made during the early design phase.

The aim of this paper is to develop a robust, simple and reliable DP assessment method that connects (1) to the practical design process and (2) can be utilised with BIM data from the early phases to the detailed design phase. The research questions are as follows:

- How can a building's DP be evaluated during the early building design phases with the situationally available data used for LCA?
- How can quantitative values for building component DP be assessed in a sufficiently reliable and practical manner to improve existing LCA calculations, particularly for end-of-life (EoL) scenarios?

Code: • Enabling condition > Disassembly Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-19, p. 4
 Created: 23.6.2026 r. 12:45 by user, Modified: 23.6.2026 r. 12:45 by user
 Other codes assigned to segment:
 Enabling condition > Repair information (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

However, there is a lack of DP assessment methods suitable for supporting practical design decisions made during the early design phase.

The aim of this paper is to develop a robust, simple and reliable DP assessment method that connects (1) to the practical design process and (2) can be utilised with BIM data from the early phases to the detailed design phase. The research questions are as follows:

- How can a building's DP be evaluated during the early building design phases with the situationally available data used for LCA?
- How can quantitative values for building component DP be assessed in a sufficiently reliable and practical manner to improve existing LCA calculations, particularly for end-of-life (EoL) scenarios?

Code: • Mitigation > Lifecycle verification Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-19, p. 4
 Created: 23.6.2026 r. 12:47 by user, Modified: 23.6.2026 r. 12:47 by user
 Other codes assigned to segment:
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

For all three projects studied, BIM data were insufficient to assess the DP of individual building products, apart from load-bearing steel beams, concrete columns and hollow-core slabs in Väre Module J (VAR) (Table 2). This limitation poses significant challenges for the applicability of all reviewed DP assessment methods (Table 1), which depend on precise product-based data, such as the independence of elements or the geometry of the product edge. Although complementary detail information was generally available through structural engineering drawings, evaluating this information across multiple sources—including element graphs, large-scale BIMs and detailed construction drawings—would require an extensive amount of additional work for DP assessment, even after final project materials have been completed. Furthermore, these documents are typically unavailable during early design phases, when preliminary BIM data are the only available source. Therefore, the provided BIM data pertain to material take-off granularity, necessitating a streamlined method for feasible DP assessment.

Code: • Mitigation > Design intervention Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-19, p. 6
 Created: 23.6.2026 r. 12:49 by user, Modified: 23.6.2026 r. 12:49 by user
 Other codes assigned to segment:
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

For all three projects studied, BIM data were insufficient to assess the DP of individual building products, apart from load-bearing steel beams, concrete columns and hollow-core slabs in Väre Module J (VAR) (Table 2). This limitation poses significant challenges for the applicability of all reviewed DP assessment methods (Table 1), which depend on precise product-based data, such as the independence of elements or the geometry of the product edge. Although complementary detail

information was generally available through structural engineering drawings, evaluating this information across multiple sources—including element graphs, large-scale BIMs and detailed construction drawings—would require an extensive amount of additional work for DP assessment, even after final project materials have been completed. Furthermore, these documents are typically unavailable during early design phases, when preliminary BIM data are the only available source. Therefore, the provided BIM data pertain to material take-off granularity, necessitating a streamlined method for feasible DP assessment.

Code: • Enabling condition > Disassembly Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-19, p. 6
 Created: 23.6.2026 r. 12:48 by user, Modified: 23.6.2026 r. 12:48 by user
 Other codes assigned to segment:
 Enabling condition > Repair information (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

For all three projects studied, BIM data were insufficient to assess the DP of individual building products, apart from load-bearing steel beams, concrete columns and hollow-core slabs in Väre Module J (VAR) (Table 2). This limitation poses significant challenges for the applicability of all reviewed DP assessment methods (Table 1), which depend on precise product-based data, such as the independence of elements or the geometry of the product edge. Although complementary detail information was generally available through structural engineering drawings, evaluating this information across multiple sources—including element graphs, large-scale BIMs and detailed construction drawings—would require an extensive amount of additional work for DP assessment, even after final project materials have been completed. Furthermore, these documents are typically unavailable during early design phases, when preliminary BIM data are the only available source. Therefore, the provided BIM data pertain to material take-off granularity, necessitating a streamlined method for feasible DP assessment.

Code: • Enabling condition > Repair information Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-19, p. 6
 Created: 23.6.2026 r. 12:48 by user, Modified: 23.6.2026 r. 12:48 by user
 Other codes assigned to segment:
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

For all three projects studied, BIM data were insufficient to assess the DP of individual building products, apart from load-bearing steel beams, concrete columns and hollow-core slabs in Väre Module J (VAR) (Table 2). This limitation poses significant challenges for the applicability of all reviewed DP assessment methods (Table 1), which depend on precise product-based data, such as the independence of elements or the geometry of the product edge. Although complementary detail information was generally available through structural engineering drawings, evaluating this information across multiple sources—including element graphs, large-scale BIMs and detailed construction drawings—would require an extensive amount of additional work for DP assessment, even after final project materials have been completed. Furthermore, these documents are typically unavailable during early design phases, when preliminary BIM data are the only available source. Therefore, the provided BIM data pertain to material take-off granularity, necessitating a streamlined method for feasible DP assessment.

Code: • Mitigation > Lifecycle verification Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-19, p. 6
 Created: 23.6.2026 r. 12:49 by user, Modified: 23.6.2026 r. 12:49 by user

Other codes assigned to segment:

- Enabling condition > Disassembly (Weight: 0)
- Enabling condition > Repair information (Weight: 0)
- Negative mechanism > Missing repair infrastructure (Weight: 0)
- Mitigation > Design intervention (Weight: 0)

For all three projects studied, BIM data were insufficient to assess the DP of individual building products, apart from load-bearing steel beams, concrete columns and hollow-core slabs in Väre Module J (VAR) (Table 2). This limitation poses significant challenges for the applicability of all reviewed DP assessment methods (Table 1), which depend on precise product-based data, such as the independence of elements or the geometry of the product edge. Although complementary detail information was generally available through structural engineering drawings, evaluating this information across multiple sources—including element graphs, large-scale BIMs and detailed construction drawings—would require an extensive amount of additional work for DP assessment, even after final project materials have been completed. Furthermore, these documents are typically unavailable during early design phases, when preliminary BIM data are the only available source. Therefore, the provided BIM data pertain to material take-off granularity, necessitating a streamlined method for feasible DP assessment.

Code: • Negative mechanism > Missing repair infrastructure Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-19, p. 6

Created: 23.6.2026 r. 12:49 by user, Modified: 23.6.2026 r. 12:49 by user

Other codes assigned to segment:

- Enabling condition > Disassembly (Weight: 0)
- Enabling condition > Repair information (Weight: 0)
- Mitigation > Design intervention (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

Durmisevic et al. (2017) identified six barriers to high reclamation of building elements and materials and the implications of Design for Disassembly:

1. Lack of data on the technical composition of buildings and element quality.
2. No certification tools for reusable elements.
3. Missing design, disassembly, and decision-making protocols.
4. Absence of reverse logistics and market strategies for reuse.
5. Uncertainty in managing long-term investment risks in reusable structures.
6. Buildings often get demolished due to the absence of supportive decision-making protocols rather than their low disassembly potential.

Code: • Mitigation > Policy intervention Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-20, p. 25

Created: 23.6.2026 r. 12:58 by user, Modified: 23.6.2026 r. 12:58 by user

Other codes assigned to segment:

- Enabling condition > Repair information (Weight: 0)
- Enabling condition > Service infrastructure (Weight: 0)
- Enabling condition > Business model (Weight: 0)
- Enabling condition > Certification (Weight: 0)
- Negative mechanism > Lock-in (Weight: 0)
- Negative mechanism > Missing repair infrastructure (Weight: 0)
- Mitigation > Business-model intervention (Weight: 0)

Durmisevic et al. (2017) identified six barriers to high reclamation of building elements and materials and the implications of Design for Disassembly:

1. Lack of data on the technical composition of buildings and element quality.
2. No certification tools for reusable elements.
3. Missing design, disassembly, and decision-making protocols.
4. Absence of reverse logistics and market strategies for reuse.
5. Uncertainty in managing long-term investment risks in reusable

structures.

6. Buildings often get demolished due to the absence of supportive decision-making protocols rather than their low disassembly potential.

Code: • Negative mechanism > Lock-in Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-20, p. 25

Created: 23.6.2026 r. 12:58 by user, Modified: 23.6.2026 r. 12:58 by user

Other codes assigned to segment:

Enabling condition > Repair information (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Enabling condition > Business model (Weight: 0)

Enabling condition > Certification (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Business-model intervention (Weight: 0)

Mitigation > Policy intervention (Weight: 0)

Durmisevic et al. (2017) identified six barriers to high reclamation of building elements and materials and the implications of Design for Disassembly:

1. Lack of data on the technical composition of buildings and element quality. 2. No certification tools for reusable elements.

3. Missing design, disassembly, and decision-making protocols. 4. Absence of reverse logistics and market strategies for reuse. 5. Uncertainty in managing long-term investment risks in reusable structures.

6. Buildings often get demolished due to the absence of supportive decision-making protocols rather than their low disassembly potential.

Code: • Enabling condition > Service infrastructure Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-20, p. 25

Created: 23.6.2026 r. 12:58 by user, Modified: 23.6.2026 r. 12:58 by user

Other codes assigned to segment:

Enabling condition > Repair information (Weight: 0)

Enabling condition > Business model (Weight: 0)

Enabling condition > Certification (Weight: 0)

Negative mechanism > Lock-in (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Business-model intervention (Weight: 0)

Mitigation > Policy intervention (Weight: 0)

Durmisevic et al. (2017) identified six barriers to high reclamation of building elements and materials and the implications of Design for Disassembly:

1. Lack of data on the technical composition of buildings and element quality. 2. No certification tools for reusable elements.

3. Missing design, disassembly, and decision-making protocols. 4. Absence of reverse logistics and market strategies for reuse. 5. Uncertainty in managing long-term investment risks in reusable structures.

6. Buildings often get demolished due to the absence of supportive decision-making protocols rather than their low disassembly potential.

Code: • Enabling condition > Repair information Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-20, p. 25

Created: 23.6.2026 r. 12:58 by user, Modified: 23.6.2026 r. 12:58 by user

Other codes assigned to segment:

Enabling condition > Service infrastructure (Weight: 0)

Enabling condition > Business model (Weight: 0)

Enabling condition > Certification (Weight: 0)

Negative mechanism > Lock-in (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Business-model intervention (Weight: 0)

Mitigation > Policy intervention (Weight: 0)

Durmisevic et al. (2017) identified six barriers to high reclamation of building elements and materials and the implications of Design for Disassembly:

1. Lack of data on the technical composition of buildings and element quality. 2. No certification tools for reusable elements.
3. Missing design, disassembly, and decision-making protocols. 4. Absence of reverse logistics and market strategies for reuse. 5. Uncertainty in managing long-term investment risks in reusable structures.
6. Buildings often get demolished due to the absence of supportive decision-making protocols rather than their low disassembly potential.

Code: • Negative mechanism > Missing repair infrastructure Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-20, p. 25

Created: 23.6.2026 r. 12:58 by user, Modified: 23.6.2026 r. 12:58 by user

Other codes assigned to segment:

Enabling condition > Repair information (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Enabling condition > Business model (Weight: 0)

Enabling condition > Certification (Weight: 0)

Negative mechanism > Lock-in (Weight: 0)

Mitigation > Business-model intervention (Weight: 0)

Mitigation > Policy intervention (Weight: 0)

Durmisevic et al. (2017) identified six barriers to high reclamation of building elements and materials and the implications of Design for Disassembly:

1. Lack of data on the technical composition of buildings and element quality. 2. No certification tools for reusable elements.
3. Missing design, disassembly, and decision-making protocols. 4. Absence of reverse logistics and market strategies for reuse. 5. Uncertainty in managing long-term investment risks in reusable structures.
6. Buildings often get demolished due to the absence of supportive decision-making protocols rather than their low disassembly potential.

Code: • Enabling condition > Business model Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-20, p. 25

Created: 23.6.2026 r. 12:58 by user, Modified: 23.6.2026 r. 12:58 by user

Other codes assigned to segment:

Enabling condition > Repair information (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Enabling condition > Certification (Weight: 0)

Negative mechanism > Lock-in (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Business-model intervention (Weight: 0)

Mitigation > Policy intervention (Weight: 0)

Durmisevic et al. (2017) identified six barriers to high reclamation of building elements and materials and the implications of Design for Disassembly:

1. Lack of data on the technical composition of buildings and element quality. 2. No certification tools for reusable elements.
3. Missing design, disassembly, and decision-making protocols. 4. Absence of reverse logistics and market strategies for reuse. 5. Uncertainty in managing long-term investment risks in reusable structures.
6. Buildings often get demolished due to the absence of supportive decision-making protocols rather than their low disassembly potential.

Code: • Enabling condition > Certification Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-20, p. 25

Created: 23.6.2026 r. 12:58 by user, Modified: 23.6.2026 r. 12:58 by user

Other codes assigned to segment:

- Enabling condition > Repair information (Weight: 0)
- Enabling condition > Service infrastructure (Weight: 0)
- Enabling condition > Business model (Weight: 0)
- Negative mechanism > Lock-in (Weight: 0)
- Negative mechanism > Missing repair infrastructure (Weight: 0)
- Mitigation > Business-model intervention (Weight: 0)
- Mitigation > Policy intervention (Weight: 0)

Durmisevic et al. (2017) identified six barriers to high reclamation of building elements and materials and the implications of Design for Disassembly:

1. Lack of data on the technical composition of buildings and element quality.
2. No certification tools for reusable elements.
3. Missing design, disassembly, and decision-making protocols.
4. Absence of reverse logistics and market strategies for reuse.
5. Uncertainty in managing long-term investment risks in reusable structures.
6. Buildings often get demolished due to the absence of supportive decision-making protocols rather than their low disassembly potential.

Code: • Mitigation > Business-model intervention Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-20, p. 25

Created: 23.6.2026 r. 12:58 by user, Modified: 23.6.2026 r. 12:58 by user

Other codes assigned to segment:

- Enabling condition > Repair information (Weight: 0)
- Enabling condition > Service infrastructure (Weight: 0)
- Enabling condition > Business model (Weight: 0)
- Enabling condition > Certification (Weight: 0)
- Negative mechanism > Lock-in (Weight: 0)
- Negative mechanism > Missing repair infrastructure (Weight: 0)
- Mitigation > Policy intervention (Weight: 0)

While disassembly typically occurs at the end of a product's life cycle, it should be considered from the outset (Marconi et al., 2019). It mainly occurs at component junctions, requiring an in-depth understanding of the connections.

Code: • Enabling condition > Component accessibility Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-20, p. 26

Created: 23.6.2026 r. 13:00 by user, Modified: 23.6.2026 r. 13:00 by user

Other codes assigned to segment:

- Enabling condition > Disassembly (Weight: 0)
- Enabling condition > Repair information (Weight: 0)
- Negative mechanism > Lock-in (Weight: 0)
- Mitigation > Design intervention (Weight: 0)

While disassembly typically occurs at the end of a product's life cycle, it should be considered from the outset (Marconi et al., 2019). It mainly occurs at component junctions, requiring an in-depth understanding of the connections.

Code: • Negative mechanism > Lock-in Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-20, p. 26

Created: 23.6.2026 r. 13:00 by user, Modified: 23.6.2026 r. 13:00 by user

Other codes assigned to segment:

- Enabling condition > Component accessibility (Weight: 0)
- Enabling condition > Disassembly (Weight: 0)
- Enabling condition > Repair information (Weight: 0)
- Mitigation > Design intervention (Weight: 0)

While disassembly typically occurs at the end of a product's life cycle, it should be considered from the outset (Marconi et al., 2019). It mainly occurs at component junctions, requiring an in-depth understanding of the connections.

Code: • Mitigation > Design intervention Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-20, p. 26
 Created: 23.6.2026 r. 13:00 by user, Modified: 23.6.2026 r. 13:00 by user
 Other codes assigned to segment:
 Enabling condition > Component accessibility (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)

While disassembly typically occurs at the end of a product's life cycle, it should be considered from the outset (Marconi et al., 2019). It mainly occurs at component junctions, requiring an in-depth understanding of the connections.

Code: • Enabling condition > Disassembly Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-20, p. 26
 Created: 23.6.2026 r. 13:00 by user, Modified: 23.6.2026 r. 13:00 by user
 Other codes assigned to segment:
 Enabling condition > Component accessibility (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

While disassembly typically occurs at the end of a product's life cycle, it should be considered from the outset (Marconi et al., 2019). It mainly occurs at component junctions, requiring an in-depth understanding of the connections.

Code: • Enabling condition > Repair information Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-20, p. 26
 Created: 23.6.2026 r. 13:00 by user, Modified: 23.6.2026 r. 13:00 by user
 Other codes assigned to segment:
 Enabling condition > Component accessibility (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

This particular façade, constructed in 2019 by Octatube, was designed considering eventual deconstruction. Every component and material is cataloged in a Madaster material passport for potential reuse

Code: • Circular strategy > Direct reuse Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-20, p. 94
 Created: 23.6.2026 r. 13:06 by user, Modified: 23.6.2026 r. 13:06 by user
 Other codes assigned to segment:
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Certification (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

This particular façade, constructed in 2019 by Octatube, was designed considering eventual deconstruction. Every component and material is cataloged in a Madaster material passport for potential reuse

Code: • Enabling condition > Disassembly Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-20, p. 94

Created: 23.6.2026 r. 13:07 by user, Modified: 23.6.2026 r. 13:07 by user

Other codes assigned to segment:

- Circular strategy > Direct reuse (Weight: 0)
- Enabling condition > Repair information (Weight: 0)
- Enabling condition > Certification (Weight: 0)
- Mitigation > Design intervention (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

This particular façade, constructed in 2019 by Octatube, was designed considering eventual deconstruction. Every component and material is cataloged in a Madaster material passport for potential reuse

Code: • Enabling condition > Repair information Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-20, p. 94

Created: 23.6.2026 r. 13:07 by user, Modified: 23.6.2026 r. 13:07 by user

Other codes assigned to segment:

- Circular strategy > Direct reuse (Weight: 0)
- Enabling condition > Disassembly (Weight: 0)
- Enabling condition > Certification (Weight: 0)
- Mitigation > Design intervention (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

This particular façade, constructed in 2019 by Octatube, was designed considering eventual deconstruction. Every component and material is cataloged in a Madaster material passport for potential reuse

Code: • Enabling condition > Certification Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-20, p. 94

Created: 23.6.2026 r. 13:07 by user, Modified: 23.6.2026 r. 13:07 by user

Other codes assigned to segment:

- Circular strategy > Direct reuse (Weight: 0)
- Enabling condition > Disassembly (Weight: 0)
- Enabling condition > Repair information (Weight: 0)
- Mitigation > Design intervention (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

This particular façade, constructed in 2019 by Octatube, was designed considering eventual deconstruction. Every component and material is cataloged in a Madaster material passport for potential reuse

Code: • Mitigation > Design intervention Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-20, p. 94

Created: 23.6.2026 r. 13:07 by user, Modified: 23.6.2026 r. 13:07 by user

Other codes assigned to segment:

- Circular strategy > Direct reuse (Weight: 0)
- Enabling condition > Disassembly (Weight: 0)
- Enabling condition > Repair information (Weight: 0)
- Enabling condition > Certification (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

This particular façade, constructed in 2019 by Octatube, was designed considering eventual deconstruction. Every component and material is cataloged in a Madaster material passport for potential reuse

Code: • Mitigation > Lifecycle verification Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-20, p. 94

Created: 23.6.2026 r. 13:08 by user, Modified: 23.6.2026 r. 13:08 by user

Other codes assigned to segment:

- Circular strategy > Direct reuse (Weight: 0)

Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Repair information (Weight: 0)
 Enabling condition > Certification (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

Taking into account aluminum recycling processes and material loss, as well as disassembly potential, the analysis predicts a 67% material reclamation rate for the aluminum profile. This is likely because the profile's design and connection methods, such as screws and connection geometry, may make disassembly more time-intensive, leading to a potentially destructive process. Examining the individual elements of the Insulated Glass Unit (IGU) reveals low material scores, primarily due to the use of adhesives in assembly, which necessitates a destructive disassembly process. However, when viewed as an entire system, the IGU's reclamation potential in terms of design is much higher, at 90%. Yet, it is important to recognize that these figures are subjective, as they stem from a semi-qualitative assessment of disassembly potential.

Code: • Circular strategy > Recycling Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-20, p. 105
 Created: 23.6.2026 r. 13:09 by user, Modified: 23.6.2026 r. 13:09 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Increased material complexity (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Taking into account aluminum recycling processes and material loss, as well as disassembly potential, the analysis predicts a 67% material reclamation rate for the aluminum profile. This is likely because the profile's design and connection methods, such as screws and connection geometry, may make disassembly more time-intensive, leading to a potentially destructive process. Examining the individual elements of the Insulated Glass Unit (IGU) reveals low material scores, primarily due to the use of adhesives in assembly, which necessitates a destructive disassembly process. However, when viewed as an entire system, the IGU's reclamation potential in terms of design is much higher, at 90%. Yet, it is important to recognize that these figures are subjective, as they stem from a semi-qualitative assessment of disassembly potential.

Code: • Negative mechanism > Burden shifting Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-20, p. 105
 Created: 23.6.2026 r. 13:09 by user, Modified: 23.6.2026 r. 13:09 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Increased material complexity (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Taking into account aluminum recycling processes and material loss, as well as disassembly potential, the analysis predicts a 67% material reclamation rate for the aluminum profile. This is likely because the profile's design and connection methods, such as screws and connection geometry, may make disassembly more time-intensive, leading to a potentially destructive process. Examining the individual elements of the Insulated Glass Unit (IGU) reveals low material scores, primarily due to the use of adhesives in assembly, which necessitates a destructive disassembly process. However, when viewed as an entire system, the IGU's reclamation potential in terms of design is much higher, at

90%. Yet, it is important to recognize that these figures are subjective, as they stem from a semi-qualitative assessment of disassembly potential.

Code: • Enabling condition > Disassembly Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-20, p. 105
 Created: 23.6.2026 r. 13:09 by user, Modified: 23.6.2026 r. 13:09 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Negative mechanism > Increased material complexity (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Taking into account aluminum recycling processes and material loss, as well as disassembly potential, the analysis predicts a 67% material reclamation rate for the aluminum profile. This is likely because the profile's design and connection methods, such as screws and connection geometry, may make disassembly more time-intensive, leading to a potentially destructive process. Examining the individual elements of the Insulated Glass Unit (IGU) reveals low material scores, primarily due to the use of adhesives in assembly, which necessitates a destructive disassembly process. However, when viewed as an entire system, the IGU's reclamation potential in terms of design is much higher, at 90%. Yet, it is important to recognize that these figures are subjective, as they stem from a semi-qualitative assessment of disassembly potential.

Code: • Negative mechanism > Lock-in Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-20, p. 105
 Created: 23.6.2026 r. 13:09 by user, Modified: 23.6.2026 r. 13:09 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Increased material complexity (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Taking into account aluminum recycling processes and material loss, as well as disassembly potential, the analysis predicts a 67% material reclamation rate for the aluminum profile. This is likely because the profile's design and connection methods, such as screws and connection geometry, may make disassembly more time-intensive, leading to a potentially destructive process. Examining the individual elements of the Insulated Glass Unit (IGU) reveals low material scores, primarily due to the use of adhesives in assembly, which necessitates a destructive disassembly process. However, when viewed as an entire system, the IGU's reclamation potential in terms of design is much higher, at 90%. Yet, it is important to recognize that these figures are subjective, as they stem from a semi-qualitative assessment of disassembly potential.

Code: • Circular strategy > Direct reuse Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-20, p. 105
 Created: 23.6.2026 r. 13:09 by user, Modified: 23.6.2026 r. 13:09 by user
 Other codes assigned to segment:
 Circular strategy > Recycling (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Increased material complexity (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Taking into account aluminum recycling processes and material loss, as well as disassembly potential, the analysis predicts a 67% material reclamation rate for the aluminum profile. This is likely because the profile's design and connection methods, such as screws and connection geometry, may make disassembly more time-intensive, leading to a potentially destructive process. Examining the individual elements of the Insulated Glass Unit (IGU) reveals low material scores, primarily due to the use of adhesives in assembly, which necessitates a destructive disassembly process. However, when viewed as an entire system, the IGU's reclamation potential in terms of design is much higher, at 90%. Yet, it is important to recognize that these figures are subjective, as they stem from a semi-qualitative assessment of disassembly potential.

Code: • [Obsolescence > Technical](#) Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-20, p. 105
 Created: 23.6.2026 r. 13:09 by user, Modified: 23.6.2026 r. 13:09 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Increased material complexity (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Taking into account aluminum recycling processes and material loss, as well as disassembly potential, the analysis predicts a 67% material reclamation rate for the aluminum profile. This is likely because the profile's design and connection methods, such as screws and connection geometry, may make disassembly more time-intensive, leading to a potentially destructive process. Examining the individual elements of the Insulated Glass Unit (IGU) reveals low material scores, primarily due to the use of adhesives in assembly, which necessitates a destructive disassembly process. However, when viewed as an entire system, the IGU's reclamation potential in terms of design is much higher, at 90%. Yet, it is important to recognize that these figures are subjective, as they stem from a semi-qualitative assessment of disassembly potential.

Code: • [Negative mechanism > Increased material complexity](#) Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-20, p. 105
 Created: 23.6.2026 r. 13:09 by user, Modified: 23.6.2026 r. 13:09 by user
 Other codes assigned to segment:
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Taking into account aluminum recycling processes and material loss, as well as disassembly potential, the analysis predicts a 67% material reclamation rate for the aluminum profile. This is likely because the profile's design and connection methods, such as screws and connection geometry, may make disassembly more time-intensive, leading to a potentially destructive process. Examining the individual elements of the Insulated Glass Unit (IGU) reveals low material scores, primarily due to the use of adhesives in assembly, which necessitates a destructive disassembly process. However, when viewed as an entire system, the IGU's reclamation potential in terms of design is much higher, at 90%. Yet, it is important to recognize that these figures are subjective, as they stem from a semi-qualitative assessment of disassembly potential.

Code: • [Mitigation > Lifecycle verification](#) Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-20, p. 105

Created: 23.6.2026 r. 13:10 by user, Modified: 23.6.2026 r. 13:10 by user

Other codes assigned to segment:

- Circular strategy > Direct reuse (Weight: 0)
- Circular strategy > Recycling (Weight: 0)
- Obsolescence > Technical (Weight: 0)
- Enabling condition > Disassembly (Weight: 0)
- Negative mechanism > Increased material complexity (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)
- Negative mechanism > Lock-in (Weight: 0)

A site visit to GSF Glasgroep (Figure 3.73) was conducted and highlights the importance of calculating disassembly times for enhanced material reclamation, implying financial considerations for reuse or other re-life scenario. GSF Glasgroep pioneered a circular production method for Insulated Glass Units (IGU), through an innovative process known as isoMAX Circu-therm (GSF Glasgroep, 2023). The times to disassemble de DGU, as per the site visit, cannot be compared, since the company processes the units in a distinct manner, using uncommon tools, which result in a material loss on both glazing and spacer. However, the process of removing the spacer from the glass was observed as time consuming, being the main barrier for disassembly

Code: • Mitigation > Design intervention Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-20, p. 106

Created: 23.6.2026 r. 13:12 by user, Modified: 23.6.2026 r. 13:12 by user

Other codes assigned to segment:

- Circular strategy > Component replacement (Weight: 0)
- Circular strategy > Recycling (Weight: 0)
- Enabling condition > Disassembly (Weight: 0)
- Enabling condition > Skilled labour (Weight: 0)
- Enabling condition > Service infrastructure (Weight: 0)
- Negative mechanism > Lock-in (Weight: 0)
- Negative mechanism > Missing repair infrastructure (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

A site visit to GSF Glasgroep (Figure 3.73) was conducted and highlights the importance of calculating disassembly times for enhanced material reclamation, implying financial considerations for reuse or other re-life scenario. GSF Glasgroep pioneered a circular production method for Insulated Glass Units (IGU), through an innovative process known as isoMAX Circu-therm (GSF Glasgroep, 2023). The times to disassemble de DGU, as per the site visit, cannot be compared, since the company processes the units in a distinct manner, using uncommon tools, which result in a material loss on both glazing and spacer. However, the process of removing the spacer from the glass was observed as time consuming, being the main barrier for disassembly

Code: • Circular strategy > Component replacement Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-20, p. 106

Created: 23.6.2026 r. 13:11 by user, Modified: 23.6.2026 r. 13:11 by user

Other codes assigned to segment:

- Circular strategy > Recycling (Weight: 0)
- Enabling condition > Disassembly (Weight: 0)
- Enabling condition > Skilled labour (Weight: 0)
- Enabling condition > Service infrastructure (Weight: 0)
- Negative mechanism > Lock-in (Weight: 0)
- Negative mechanism > Missing repair infrastructure (Weight: 0)
- Mitigation > Design intervention (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

A site visit to GSF Glasgroep (Figure 3.73) was conducted and highlights the importance of calculating disassembly times for enhanced material reclamation, implying financial considerations for reuse or other re-life scenario. GSF Glasgroep pioneered a circular production method for Insulated Glass Units (IGU), through an innovative process known as isoMAX Circu-therm (GSF Glasgroep, 2023). The

times to disassemble de DGU, as per the site visit, cannot be compared, since the company processes the units in a distinct manner, using uncommon tools, which result in a material loss on both glazing and spacer. However, the process of removing the spacer from the glass was observed as time consuming, being the main barrier for disassembly

Code: • Negative mechanism > Missing repair infrastructure Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-20, p. 106

Created: 23.6.2026 r. 13:11 by user, Modified: 23.6.2026 r. 13:11 by user

Other codes assigned to segment:

Circular strategy > Component replacement (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Enabling condition > Disassembly (Weight: 0)

Enabling condition > Skilled labour (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Negative mechanism > Lock-in (Weight: 0)

Mitigation > Design intervention (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

A site visit to GSF Glasgroep (Figure 3.73) was conducted and highlights the importance of calculating disassembly times for enhanced material reclamation, implying financial considerations for reuse or other re-life scenario. GSF Glasgroep pioneered a circular production method for Insulated Glass Units (IGU), through an innovative process known as isoMAX Circu-therm (GSF Glasgroep, 2023). The times to disassemble de DGU, as per the site visit, cannot be compared, since the company processes the units in a distinct manner, using uncommon tools, which result in a material loss on both glazing and spacer. However, the process of removing the spacer from the glass was observed as time consuming, being the main barrier for disassembly

Code: • Enabling condition > Service infrastructure Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-20, p. 106

Created: 23.6.2026 r. 13:11 by user, Modified: 23.6.2026 r. 13:11 by user

Other codes assigned to segment:

Circular strategy > Component replacement (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Enabling condition > Disassembly (Weight: 0)

Enabling condition > Skilled labour (Weight: 0)

Negative mechanism > Lock-in (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Design intervention (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

A site visit to GSF Glasgroep (Figure 3.73) was conducted and highlights the importance of calculating disassembly times for enhanced material reclamation, implying financial considerations for reuse or other re-life scenario. GSF Glasgroep pioneered a circular production method for Insulated Glass Units (IGU), through an innovative process known as isoMAX Circu-therm (GSF Glasgroep, 2023). The times to disassemble de DGU, as per the site visit, cannot be compared, since the company processes the units in a distinct manner, using uncommon tools, which result in a material loss on both glazing and spacer. However, the process of removing the spacer from the glass was observed as time consuming, being the main barrier for disassembly

Code: • Enabling condition > Skilled labour Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-20, p. 106

Created: 23.6.2026 r. 13:11 by user, Modified: 23.6.2026 r. 13:11 by user

Other codes assigned to segment:

Circular strategy > Component replacement (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Enabling condition > Disassembly (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Negative mechanism > Lock-in (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

A site visit to GSF Glasgroep (Figure 3.73) was conducted and highlights the importance of calculating disassembly times for enhanced material reclamation, implying financial considerations for reuse or other re-life scenario. GSF Glasgroep pioneered a circular production method for Insulated Glass Units (IGU), through an innovative process known as isoMAX Circu-therm (GSF Glasgroep, 2023). The times to disassemble de DGU, as per the site visit, cannot be compared, since the company processes the units in a distinct manner, using uncommon tools, which result in a material loss on both glazing and spacer. However, the process of removing the spacer from the glass was observed as time consuming, being the main barrier for disassembly

Code: • Negative mechanism > Lock-in Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-20, p. 106
 Created: 23.6.2026 r. 13:12 by user, Modified: 23.6.2026 r. 13:12 by user
 Other codes assigned to segment:
 Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Skilled labour (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

A site visit to GSF Glasgroep (Figure 3.73) was conducted and highlights the importance of calculating disassembly times for enhanced material reclamation, implying financial considerations for reuse or other re-life scenario. GSF Glasgroep pioneered a circular production method for Insulated Glass Units (IGU), through an innovative process known as isoMAX Circu-therm (GSF Glasgroep, 2023). The times to disassemble de DGU, as per the site visit, cannot be compared, since the company processes the units in a distinct manner, using uncommon tools, which result in a material loss on both glazing and spacer. However, the process of removing the spacer from the glass was observed as time consuming, being the main barrier for disassembly

Code: • Enabling condition > Disassembly Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-20, p. 106
 Created: 23.6.2026 r. 13:11 by user, Modified: 23.6.2026 r. 13:11 by user
 Other codes assigned to segment:
 Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > Skilled labour (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

A site visit to GSF Glasgroep (Figure 3.73) was conducted and highlights the importance of calculating disassembly times for enhanced material reclamation, implying financial considerations for reuse or other re-life scenario. GSF Glasgroep pioneered a circular production method for Insulated Glass Units (IGU), through an innovative process known as isoMAX Circu-therm (GSF Glasgroep, 2023). The times to disassemble de DGU, as per the site visit, cannot be compared, since the company processes the units in a distinct manner, using uncommon tools, which result in a material loss on both glazing and spacer. However, the process of removing the spacer from the glass was observed as time consuming, being the main barrier for disassembly

Code: • Circular strategy > Recycling Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-20, p. 106

Created: 23.6.2026 r. 13:11 by user, Modified: 23.6.2026 r. 13:11 by user

Other codes assigned to segment:

- Circular strategy > Component replacement (Weight: 0)
- Enabling condition > Disassembly (Weight: 0)
- Enabling condition > Skilled labour (Weight: 0)
- Enabling condition > Service infrastructure (Weight: 0)
- Negative mechanism > Lock-in (Weight: 0)
- Negative mechanism > Missing repair infrastructure (Weight: 0)
- Mitigation > Design intervention (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

A site visit to GSF Glasgroep (Figure 3.73) was conducted and highlights the importance of calculating disassembly times for enhanced material reclamation, implying financial considerations for reuse or other re-life scenario. GSF Glasgroep pioneered a circular production method for Insulated Glass Units (IGU), through an innovative process known as isoMAX Circu-therm (GSF Glasgroep, 2023). The times to disassemble de DGU, as per the site visit, cannot be compared, since the company processes the units in a distinct manner, using uncommon tools, which result in a material loss on both glazing and spacer. However, the process of removing the spacer from the glass was observed as time consuming, being the main barrier for disassembly

Code: • Mitigation > Lifecycle verification Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-20, p. 106

Created: 23.6.2026 r. 13:12 by user, Modified: 23.6.2026 r. 13:12 by user

Other codes assigned to segment:

- Circular strategy > Component replacement (Weight: 0)
- Circular strategy > Recycling (Weight: 0)
- Enabling condition > Disassembly (Weight: 0)
- Enabling condition > Skilled labour (Weight: 0)
- Enabling condition > Service infrastructure (Weight: 0)
- Negative mechanism > Lock-in (Weight: 0)
- Negative mechanism > Missing repair infrastructure (Weight: 0)
- Mitigation > Design intervention (Weight: 0)

Despite the rapid pace of product development, it might be beneficial to decrease the customization level in façade products, by standarizing connection types, thus reducing disassembly time and effort. This would ensure that façades can be effortlessly disassembled by any builder or deconstruction company, even in the absence of detailed information on disassembly sequences. A shift towards more standardized, simplified sections and connection systems, could eliminate the need for specialized assembly or disassembly tools.

7. Technological barriers regarding adhesive sealants could be overcome with the advancement of technology. For instance, laminated glazing is difficult to separate due to the adhesive interlayer. This results in difficult recycling and even reuse of elements and materials.

Code: • Obsolescence > Technical Weight score: 0

Search B - Disassembly reuse and facade recovery > B-GS-20, p. 118

Created: 23.6.2026 r. 13:13 by user, Modified: 23.6.2026 r. 13:13 by user

Other codes assigned to segment:

- Enabling condition > Component accessibility (Weight: 0)
- Enabling condition > Disassembly (Weight: 0)
- Negative mechanism > Increased material complexity (Weight: 0)
- Negative mechanism > Lock-in (Weight: 0)
- Mitigation > Design intervention (Weight: 0)
- Mitigation > Supplier commitment (Weight: 0)

Despite the rapid pace of product development, it might be beneficial to decrease the customization level in façade products, by standarizing connection types, thus reducing disassembly time and effort. This would ensure that façades can be effortlessly disassembled by any builder or

deconstruction company, even in the absence of detailed information on disassembly sequences. A shift towards more standardized, simplified sections and connection systems, could eliminate the need for specialized assembly or disassembly tools.

7. Technological barriers regarding adhesive sealants could be overcome with the advancement of technology. For instance, laminated glazing is difficult to separate due to the adhesive interlayer. This results in difficult recycling and even reuse of elements and materials.

Code: • Enabling condition > Component accessibility Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-20, p. 118
 Created: 23.6.2026 r. 13:13 by user, Modified: 23.6.2026 r. 13:13 by user
 Other codes assigned to segment:
 Obsolescence > Technical (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Increased material complexity (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Supplier commitment (Weight: 0)

Despite the rapid pace of product development, it might be beneficial to decrease the customization level in façade products, by standardizing connection types, thus reducing disassembly time and effort. This would ensure that façades can be effortlessly disassembled by any builder or deconstruction company, even in the absence of detailed information on disassembly sequences. A shift towards more standardized, simplified sections and connection systems, could eliminate the need for specialized assembly or disassembly tools.

7. Technological barriers regarding adhesive sealants could be overcome with the advancement of technology. For instance, laminated glazing is difficult to separate due to the adhesive interlayer. This results in difficult recycling and even reuse of elements and materials.

Code: • Mitigation > Design intervention Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-20, p. 118
 Created: 23.6.2026 r. 13:14 by user, Modified: 23.6.2026 r. 13:14 by user
 Other codes assigned to segment:
 Obsolescence > Technical (Weight: 0)
 Enabling condition > Component accessibility (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Increased material complexity (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Mitigation > Supplier commitment (Weight: 0)

Despite the rapid pace of product development, it might be beneficial to decrease the customization level in façade products, by standardizing connection types, thus reducing disassembly time and effort. This would ensure that façades can be effortlessly disassembled by any builder or deconstruction company, even in the absence of detailed information on disassembly sequences. A shift towards more standardized, simplified sections and connection systems, could eliminate the need for specialized assembly or disassembly tools.

7. Technological barriers regarding adhesive sealants could be overcome with the advancement of technology. For instance, laminated glazing is difficult to separate due to the adhesive interlayer. This results in difficult recycling and even reuse of elements and materials.

Code: • Negative mechanism > Increased material complexity Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-20, p. 118
 Created: 23.6.2026 r. 13:14 by user, Modified: 23.6.2026 r. 13:14 by user
 Other codes assigned to segment:
 Obsolescence > Technical (Weight: 0)
 Enabling condition > Component accessibility (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)

Mitigation > Design intervention (Weight: 0)
 Mitigation > Supplier commitment (Weight: 0)

Despite the rapid pace of product development, it might be beneficial to decrease the customization level in façade products, by standardizing connection types, thus reducing disassembly time and effort. This would ensure that façades can be effortlessly disassembled by any builder or deconstruction company, even in the absence of detailed information on disassembly sequences. A shift towards more standardized, simplified sections and connection systems, could eliminate the need for specialized assembly or disassembly tools.

7. Technological barriers regarding adhesive sealants could be overcome with the advancement of technology. For instance, laminated glazing is difficult to separate due to the adhesive interlayer. This results in difficult recycling and even reuse of elements and materials.

Code: • Negative mechanism > Lock-in Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-20, p. 118
 Created: 23.6.2026 r. 13:14 by user, Modified: 23.6.2026 r. 13:14 by user
 Other codes assigned to segment:
 Obsolescence > Technical (Weight: 0)
 Enabling condition > Component accessibility (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Increased material complexity (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Supplier commitment (Weight: 0)

Despite the rapid pace of product development, it might be beneficial to decrease the customization level in façade products, by standardizing connection types, thus reducing disassembly time and effort. This would ensure that façades can be effortlessly disassembled by any builder or deconstruction company, even in the absence of detailed information on disassembly sequences. A shift towards more standardized, simplified sections and connection systems, could eliminate the need for specialized assembly or disassembly tools.

7. Technological barriers regarding adhesive sealants could be overcome with the advancement of technology. For instance, laminated glazing is difficult to separate due to the adhesive interlayer. This results in difficult recycling and even reuse of elements and materials.

Code: • Mitigation > Supplier commitment Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-20, p. 118
 Created: 23.6.2026 r. 13:14 by user, Modified: 23.6.2026 r. 13:14 by user
 Other codes assigned to segment:
 Obsolescence > Technical (Weight: 0)
 Enabling condition > Component accessibility (Weight: 0)
 Enabling condition > Disassembly (Weight: 0)
 Negative mechanism > Increased material complexity (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Mitigation > Design intervention (Weight: 0)

Despite the rapid pace of product development, it might be beneficial to decrease the customization level in façade products, by standardizing connection types, thus reducing disassembly time and effort. This would ensure that façades can be effortlessly disassembled by any builder or deconstruction company, even in the absence of detailed information on disassembly sequences. A shift towards more standardized, simplified sections and connection systems, could eliminate the need for specialized assembly or disassembly tools.

7. Technological barriers regarding adhesive sealants could be overcome with the advancement of technology. For instance, laminated glazing is difficult to separate due to the adhesive interlayer. This results in difficult recycling and even reuse of elements and materials.

Code: • Enabling condition > Disassembly Weight score: 0
 Search B - Disassembly reuse and facade recovery > B-GS-20, p. 118

Created: 23.6.2026 r. 13:13 by user, Modified: 23.6.2026 r. 13:13 by user

Other codes assigned to segment:

- Obsolescence > Technical (Weight: 0)
- Enabling condition > Component accessibility (Weight: 0)
- Negative mechanism > Increased material complexity (Weight: 0)
- Negative mechanism > Lock-in (Weight: 0)
- Mitigation > Design intervention (Weight: 0)
- Mitigation > Supplier commitment (Weight: 0)

Although prolonging the lifetime of products, or their components, is an important strategy to achieve a more resource-efficient society [25], lifetime extension is only valid when the avoided impacts from production, e.g., energy use and the subsequent reduction in impact upstream such as raw material extraction and transportation, are lower than the impacts from producing more durable components or treating the returned product to extend its lifetime [20,26]. For many products, lifetime extension reduces environmental impacts per unit of function, but there are exceptions. For example, there can be environmental benefits to replacing active products, which consume energy and consumables during the use phase, with more energy-efficient ones rather than prolonging their lifetime [19,27]. Case-by-case decisions are necessary [28] for these types of products. Similarly, replacing single-use products, e.g., diapers and incontinence products, with more durable ones has also demonstrated both positive and negative results [18]. Passive durable products, however, hold a larger and more consistent potential for environmental improvement because their environmental impact is dominated by the production phase, usually the production of materials, while their physical lifetime is not fully exploited

Code: • Mitigation > Lifecycle verification Weight score: 0

Search C - Rebound and circularity theory > C-GS-01, p. 3

Created: 23.6.2026 r. 13:18 by user, Modified: 23.6.2026 r. 13:18 by user

Other codes assigned to segment:

- Circular strategy > Repair (Weight: 0)
- Circular strategy > Refurbishment (Weight: 0)
- Obsolescence > Physical (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)

Although prolonging the lifetime of products, or their components, is an important strategy to achieve a more resource-efficient society [25], lifetime extension is only valid when the avoided impacts from production, e.g., energy use and the subsequent reduction in impact upstream such as raw material extraction and transportation, are lower than the impacts from producing more durable components or treating the returned product to extend its lifetime [20,26]. For many products, lifetime extension reduces environmental impacts per unit of function, but there are exceptions. For example, there can be environmental benefits to replacing active products, which consume energy and consumables during the use phase, with more energy-efficient ones rather than prolonging their lifetime [19,27]. Case-by-case decisions are necessary [28] for these types of products. Similarly, replacing single-use products, e.g., diapers and incontinence products, with more durable ones has also demonstrated both positive and negative results [18]. Passive durable products, however, hold a larger and more consistent potential for environmental improvement because their environmental impact is dominated by the production phase, usually the production of materials, while their physical lifetime is not fully exploited

Code: • Negative mechanism > Burden shifting Weight score: 0

Search C - Rebound and circularity theory > C-GS-01, p. 3

Created: 23.6.2026 r. 13:18 by user, Modified: 23.6.2026 r. 13:18 by user

Other codes assigned to segment:

- Circular strategy > Repair (Weight: 0)
- Circular strategy > Refurbishment (Weight: 0)
- Obsolescence > Physical (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

Although prolonging the lifetime of products, or their components, is an important strategy to achieve a more resource-efficient society [25], lifetime extension is only valid when the avoided impacts from production, e.g., energy use and the subsequent reduction in impact upstream such as raw material extraction and transportation, are lower than the impacts from producing more durable components or treating the returned product to extend its lifetime [20,26]. For many products, lifetime extension reduces environmental impacts per unit of function, but there are exceptions. For example, there can be environmental benefits to replacing active products, which consume energy and consumables during the use phase, with more energy-efficient ones rather than prolonging their lifetime [19,27]. Case-by-case decisions are necessary [28] for these types of products. Similarly, replacing single-use products, e.g., diapers and incontinence products, with more durable ones has also demonstrated both positive and negative results [18]. Passive durable products, however, hold a larger and more consistent potential for environmental improvement because their environmental impact is dominated by the production phase, usually the production of materials, while their physical lifetime is not fully exploited

Code: • Circular strategy > Repair Weight score: 0

Search C - Rebound and circularity theory > C-GS-01, p. 3

Created: 23.6.2026 r. 13:17 by user, Modified: 23.6.2026 r. 13:17 by user

Other codes assigned to segment:

Circular strategy > Refurbishment (Weight: 0)

Obsolescence > Physical (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Although prolonging the lifetime of products, or their components, is an important strategy to achieve a more resource-efficient society [25], lifetime extension is only valid when the avoided impacts from production, e.g., energy use and the subsequent reduction in impact upstream such as raw material extraction and transportation, are lower than the impacts from producing more durable components or treating the returned product to extend its lifetime [20,26]. For many products, lifetime extension reduces environmental impacts per unit of function, but there are exceptions. For example, there can be environmental benefits to replacing active products, which consume energy and consumables during the use phase, with more energy-efficient ones rather than prolonging their lifetime [19,27]. Case-by-case decisions are necessary [28] for these types of products. Similarly, replacing single-use products, e.g., diapers and incontinence products, with more durable ones has also demonstrated both positive and negative results [18]. Passive durable products, however, hold a larger and more consistent potential for environmental improvement because their environmental impact is dominated by the production phase, usually the production of materials, while their physical lifetime is not fully exploited

Code: • Circular strategy > Refurbishment Weight score: 0

Search C - Rebound and circularity theory > C-GS-01, p. 3

Created: 23.6.2026 r. 13:17 by user, Modified: 23.6.2026 r. 13:17 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)

Obsolescence > Physical (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Although prolonging the lifetime of products, or their components, is an important strategy to achieve a more resource-efficient society [25], lifetime extension is only valid when the avoided impacts from production, e.g., energy use and the subsequent reduction in impact upstream such as raw material extraction and transportation, are lower than the impacts from producing more durable components or treating the returned product to extend its lifetime [20,26]. For many products,

lifetime extension reduces environmental impacts per unit of function, but there are exceptions. For example, there can be environmental benefits to replacing active products, which consume energy and consumables during the use phase, with more energy-efficient ones rather than prolonging their lifetime [19,27]. Case-by-case decisions are necessary [28] for these types of products. Similarly, replacing single-use products, e.g., diapers and incontinence products, with more durable ones has also demonstrated both positive and negative results [18]. Passive durable products, however, hold a larger and more consistent potential for environmental improvement because their environmental impact is dominated by the production phase, usually the production of materials, while their physical lifetime is not fully exploited

Code: • Obsolescence > Physical Weight score: 0

Search C - Rebound and circularity theory > C-GS-01, p. 3

Created: 23.6.2026 r. 13:18 by user, Modified: 23.6.2026 r. 13:18 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

For all circular offerings, product lifetime was extended through repair or refurbishment. Repair is defined as restoring the product “so it can be used in its original function,” while refurbishment is defined as “restoring an old product and bringing it up to date” [46]. In one case (Company C), a modular design that allowed for more efficient repair was implemented. The types of PSSs used as means to achieve the circular offerings were product-oriented services and use-oriented services

Code: • Mitigation > Business-model intervention Weight score: 0

Search C - Rebound and circularity theory > C-GS-01, p. 4

Created: 23.6.2026 r. 13:20 by user, Modified: 23.6.2026 r. 13:20 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Enabling condition > Business model (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

For all circular offerings, product lifetime was extended through repair or refurbishment. Repair is defined as restoring the product “so it can be used in its original function,” while refurbishment is defined as “restoring an old product and bringing it up to date” [46]. In one case (Company C), a modular design that allowed for more efficient repair was implemented. The types of PSSs used as means to achieve the circular offerings were product-oriented services and use-oriented services

Code: • Enabling condition > Business model Weight score: 0

Search C - Rebound and circularity theory > C-GS-01, p. 4

Created: 23.6.2026 r. 13:19 by user, Modified: 23.6.2026 r. 13:19 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Mitigation > Business-model intervention (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

For all circular offerings, product lifetime was extended through repair or refurbishment. Repair is defined as restoring the product “so it can be used in its original function,” while refurbishment is defined as “restoring an old product and bringing it up to date” [46]. In one case (Company C), a modular design that allowed for more efficient repair was implemented. The types of PSSs used as means to achieve the circular offerings were product-oriented services and use-oriented services

Code: • Circular strategy > Repair Weight score: 0

Search C - Rebound and circularity theory > C-GS-01, p. 4

Created: 23.6.2026 r. 13:19 by user, Modified: 23.6.2026 r. 13:19 by user

Other codes assigned to segment:

- Circular strategy > Refurbishment (Weight: 0)
- Enabling condition > Service infrastructure (Weight: 0)
- Enabling condition > Business model (Weight: 0)
- Mitigation > Business-model intervention (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

For all circular offerings, product lifetime was extended through repair or refurbishment. Repair is defined as restoring the product “so it can be used in its original function,” while refurbishment is defined as “restoring an old product and bringing it up to date” [46]. In one case (Company C), a modular design that allowed for more efficient repair was implemented. The types of PSSs used as means to achieve the circular offerings were product-oriented services and use-oriented services

Code: • Mitigation > Lifecycle verification Weight score: 0

Search C - Rebound and circularity theory > C-GS-01, p. 4

Created: 23.6.2026 r. 13:20 by user, Modified: 23.6.2026 r. 13:20 by user

Other codes assigned to segment:

- Circular strategy > Repair (Weight: 0)
- Circular strategy > Refurbishment (Weight: 0)
- Enabling condition > Service infrastructure (Weight: 0)
- Enabling condition > Business model (Weight: 0)
- Mitigation > Business-model intervention (Weight: 0)

For all circular offerings, product lifetime was extended through repair or refurbishment. Repair is defined as restoring the product “so it can be used in its original function,” while refurbishment is defined as “restoring an old product and bringing it up to date” [46]. In one case (Company C), a modular design that allowed for more efficient repair was implemented. The types of PSSs used as means to achieve the circular offerings were product-oriented services and use-oriented services

Code: • Circular strategy > Refurbishment Weight score: 0

Search C - Rebound and circularity theory > C-GS-01, p. 4

Created: 23.6.2026 r. 13:19 by user, Modified: 23.6.2026 r. 13:19 by user

Other codes assigned to segment:

- Circular strategy > Repair (Weight: 0)
- Enabling condition > Service infrastructure (Weight: 0)
- Enabling condition > Business model (Weight: 0)
- Mitigation > Business-model intervention (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

For all circular offerings, product lifetime was extended through repair or refurbishment. Repair is defined as restoring the product “so it can be used in its original function,” while refurbishment is defined as “restoring an old product and bringing it up to date” [46]. In one case (Company C), a modular design that allowed for more efficient repair was implemented. The types of PSSs used as means to achieve the circular offerings were product-oriented services and use-oriented services

Code: • Enabling condition > Service infrastructure Weight score: 0

Search C - Rebound and circularity theory > C-GS-01, p. 4

Created: 23.6.2026 r. 13:19 by user, Modified: 23.6.2026 r. 13:19 by user

Other codes assigned to segment:

- Circular strategy > Repair (Weight: 0)
- Circular strategy > Refurbishment (Weight: 0)
- Enabling condition > Business model (Weight: 0)
- Mitigation > Business-model intervention (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

The environmental impact was reduced much more, by 45–88%, with the largest reduction for the indicator “amount of iron ore” used over the life cycle. The main reasons for the difference in reduction between LCCman and LCA are due to: (i) costs not related to production, and (ii) costs and environmental impacts of the dominant phase.

Code: • Enabling condition > Business model Weight score: 0
 Search C - Rebound and circularity theory > C-GS-01, p. 7
 Created: 23.6.2026 r. 13:21 by user, Modified: 23.6.2026 r. 13:21 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The environmental impact was reduced much more, by 45–88%, with the largest reduction for the indicator “amount of iron ore” used over the life cycle. The main reasons for the difference in reduction between LCCman and LCA are due to: (i) costs not related to production, and (ii) costs and environmental impacts of the dominant phase.

Code: • Circular strategy > Refurbishment Weight score: 0
 Search C - Rebound and circularity theory > C-GS-01, p. 7
 Created: 23.6.2026 r. 13:21 by user, Modified: 23.6.2026 r. 13:21 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Enabling condition > Business model (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The environmental impact was reduced much more, by 45–88%, with the largest reduction for the indicator “amount of iron ore” used over the life cycle. The main reasons for the difference in reduction between LCCman and LCA are due to: (i) costs not related to production, and (ii) costs and environmental impacts of the dominant phase.

Code: • Mitigation > Lifecycle verification Weight score: 0
 Search C - Rebound and circularity theory > C-GS-01, p. 7
 Created: 23.6.2026 r. 13:21 by user, Modified: 23.6.2026 r. 13:21 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Enabling condition > Business model (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)

The environmental impact was reduced much more, by 45–88%, with the largest reduction for the indicator “amount of iron ore” used over the life cycle. The main reasons for the difference in reduction between LCCman and LCA are due to: (i) costs not related to production, and (ii) costs and environmental impacts of the dominant phase.

Code: • Circular strategy > Repair Weight score: 0
 Search C - Rebound and circularity theory > C-GS-01, p. 7
 Created: 23.6.2026 r. 13:21 by user, Modified: 23.6.2026 r. 13:21 by user
 Other codes assigned to segment:
 Circular strategy > Refurbishment (Weight: 0)
 Enabling condition > Business model (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The environmental impact was reduced much more, by 45–88%, with the largest reduction for the indicator “amount of iron ore” used over the life cycle. The main reasons for the difference in

reduction between LCCman and LCA are due to: (i) costs not related to production, and (ii) costs and environmental impacts of the dominant phase.

Code: • Negative mechanism > Burden shifting Weight score: 0

Search C - Rebound and circularity theory > C-GS-01, p. 7

Created: 23.6.2026 r. 13:21 by user, Modified: 23.6.2026 r. 13:21 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)

Enabling condition > Business model (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Figure 3 shows the results of the sensitivity analysis for Company A. The x-axis shows the number of uses (one use is the original BAU and 10 uses are the original circular offering), and the y-axis shows the normalized GWP and LCCman. The figure shows that the GWP and LCCman decrease as the number of uses increase. The decrease in the GWP for both the beach flag (Figure 3a) and the event tent (Figure 3c) drops sharply at the beginning, then flattens towards the end. Most of the environmental benefits of the circular offering were obtained after 4–5 uses. However, the improvements became less prominent if the product was reused 5 or 10 times.

Code: • Circular strategy > Direct reuse Weight score: 0

Search C - Rebound and circularity theory > C-GS-01, p. 9

Created: 23.6.2026 r. 13:22 by user, Modified: 23.6.2026 r. 13:22 by user

Other codes assigned to segment:

Circular strategy > Refurbishment (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Figure 3 shows the results of the sensitivity analysis for Company A. The x-axis shows the number of uses (one use is the original BAU and 10 uses are the original circular offering), and the y-axis shows the normalized GWP and LCCman. The figure shows that the GWP and LCCman decrease as the number of uses increase. The decrease in the GWP for both the beach flag (Figure 3a) and the event tent (Figure 3c) drops sharply at the beginning, then flattens towards the end. Most of the environmental benefits of the circular offering were obtained after 4–5 uses. However, the improvements became less prominent if the product was reused 5 or 10 times.

Code: • Negative mechanism > Burden shifting Weight score: 0

Search C - Rebound and circularity theory > C-GS-01, p. 9

Created: 23.6.2026 r. 13:23 by user, Modified: 23.6.2026 r. 13:23 by user

Other codes assigned to segment:

Circular strategy > Refurbishment (Weight: 0)

Circular strategy > Direct reuse (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Figure 3 shows the results of the sensitivity analysis for Company A. The x-axis shows the number of uses (one use is the original BAU and 10 uses are the original circular offering), and the y-axis shows the normalized GWP and LCCman. The figure shows that the GWP and LCCman decrease as the number of uses increase. The decrease in the GWP for both the beach flag (Figure 3a) and the event tent (Figure 3c) drops sharply at the beginning, then flattens towards the end. Most of the environmental benefits of the circular offering were obtained after 4–5 uses. However, the improvements became less prominent if the product was reused 5 or 10 times.

Code: • Circular strategy > Refurbishment Weight score: 0

Search C - Rebound and circularity theory > C-GS-01, p. 9

Created: 23.6.2026 r. 13:23 by user, Modified: 23.6.2026 r. 13:23 by user

Other codes assigned to segment:

Circular strategy > Direct reuse (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)
Mitigation > Lifecycle verification (Weight: 0)

Figure 3 shows the results of the sensitivity analysis for Company A. The x-axis shows the number of uses (one use is the original BAU and 10 uses are the original circular offering), and the y-axis shows the normalized GWP and LCCman. The figure shows that the GWP and LCCman decrease as the number of uses increase. The decrease in the GWP for both the beach flag (Figure 3a) and the event tent (Figure 3c) drops sharply at the beginning, then flattens towards the end. Most of the environmental benefits of the circular offering were obtained after 4–5 uses. However, the improvements became less prominent if the product was reused 5 or 10 times.

Code: • Mitigation > Lifecycle verification Weight score: 0
Search C - Rebound and circularity theory > C-GS-01, p. 9
Created: 23.6.2026 r. 13:23 by user, Modified: 23.6.2026 r. 13:23 by user
Other codes assigned to segment:
Circular strategy > Refurbishment (Weight: 0)
Circular strategy > Direct reuse (Weight: 0)
Negative mechanism > Burden shifting (Weight: 0)

In the circular offering, the door handle was redesigned for ease-of-repair, allowing for replacing the handle alone as opposed to the BAU scenario where the whole door is replaced. In this way, the lifetime for some of the door's components could be prolonged. Both options were done under a maintenance contract with the company, which inspects the inlets at prearranged intervals and guarantees the change of the door or door handle in the event of damage. Both the BAU and circular offerings extend the lifetime of the system through repair but differ with respect to how large a component is replaced at repair.

Code: • Mitigation > Business-model intervention Weight score: 0
Search C - Rebound and circularity theory > C-GS-01, p. 13
Created: 23.6.2026 r. 13:25 by user, Modified: 23.6.2026 r. 13:25 by user
Other codes assigned to segment:
Circular strategy > Repair (Weight: 0)
Circular strategy > Component replacement (Weight: 0)
Enabling condition > Component accessibility (Weight: 0)
Enabling condition > Service infrastructure (Weight: 0)
Mitigation > Design intervention (Weight: 0)

In the circular offering, the door handle was redesigned for ease-of-repair, allowing for replacing the handle alone as opposed to the BAU scenario where the whole door is replaced. In this way, the lifetime for some of the door's components could be prolonged. Both options were done under a maintenance contract with the company, which inspects the inlets at prearranged intervals and guarantees the change of the door or door handle in the event of damage. Both the BAU and circular offerings extend the lifetime of the system through repair but differ with respect to how large a component is replaced at repair.

Code: • Mitigation > Design intervention Weight score: 0
Search C - Rebound and circularity theory > C-GS-01, p. 13
Created: 23.6.2026 r. 13:25 by user, Modified: 23.6.2026 r. 13:25 by user
Other codes assigned to segment:
Circular strategy > Repair (Weight: 0)
Circular strategy > Component replacement (Weight: 0)
Enabling condition > Component accessibility (Weight: 0)
Enabling condition > Service infrastructure (Weight: 0)
Mitigation > Business-model intervention (Weight: 0)

In the circular offering, the door handle was redesigned for ease-of-repair, allowing for replacing the handle alone as opposed to the BAU scenario where the whole door is replaced. In this way, the lifetime for some of the door's components could be prolonged. Both options were done under a maintenance contract with the company, which inspects the inlets at prearranged intervals and guarantees the change of the door or door handle in the event of damage. Both the BAU and circular offerings extend the lifetime of the system through repair but differ with respect to how large a component is replaced at repair.

Code: • Enabling condition > Service infrastructure Weight score: 0
 Search C - Rebound and circularity theory > C-GS-01, p. 13
 Created: 23.6.2026 r. 13:24 by user, Modified: 23.6.2026 r. 13:24 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Component replacement (Weight: 0)
 Enabling condition > Component accessibility (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Business-model intervention (Weight: 0)

In the circular offering, the door handle was redesigned for ease-of-repair, allowing for replacing the handle alone as opposed to the BAU scenario where the whole door is replaced. In this way, the lifetime for some of the door's components could be prolonged. Both options were done under a maintenance contract with the company, which inspects the inlets at prearranged intervals and guarantees the change of the door or door handle in the event of damage. Both the BAU and circular offerings extend the lifetime of the system through repair but differ with respect to how large a component is replaced at repair.

Code: • Enabling condition > Component accessibility Weight score: 0
 Search C - Rebound and circularity theory > C-GS-01, p. 13
 Created: 23.6.2026 r. 13:24 by user, Modified: 23.6.2026 r. 13:24 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Component replacement (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Business-model intervention (Weight: 0)

In the circular offering, the door handle was redesigned for ease-of-repair, allowing for replacing the handle alone as opposed to the BAU scenario where the whole door is replaced. In this way, the lifetime for some of the door's components could be prolonged. Both options were done under a maintenance contract with the company, which inspects the inlets at prearranged intervals and guarantees the change of the door or door handle in the event of damage. Both the BAU and circular offerings extend the lifetime of the system through repair but differ with respect to how large a component is replaced at repair.

Code: • Circular strategy > Repair Weight score: 0
 Search C - Rebound and circularity theory > C-GS-01, p. 13
 Created: 23.6.2026 r. 13:24 by user, Modified: 23.6.2026 r. 13:24 by user
 Other codes assigned to segment:
 Circular strategy > Component replacement (Weight: 0)
 Enabling condition > Component accessibility (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Business-model intervention (Weight: 0)

In the circular offering, the door handle was redesigned for ease-of-repair, allowing for replacing the handle alone as opposed to the BAU scenario where the whole door is replaced. In this way, the lifetime for some of the door's components could be prolonged. Both options were done under a

maintenance contract with the company, which inspects the inlets at prearranged intervals and guarantees the change of the door or door handle in the event of damage. Both the BAU and circular offerings extend the lifetime of the system through repair but differ with respect to how large a component is replaced at repair.

Code: • Circular strategy > Component replacement Weight score: 0
 Search C - Rebound and circularity theory > C-GS-01, p. 13
 Created: 23.6.2026 r. 13:24 by user, Modified: 23.6.2026 r. 13:24 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Enabling condition > Component accessibility (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Business-model intervention (Weight: 0)

In the diagram without markets, a flow, say, from consumer to recycler to manufacturer implies 1:1 displacement—that every kg recycled prevents 1 kg of production from raw materials. The environmental benefit of recycling is then simply calculated as shown by equation (1):

$$E_{net} = (e_r - e_p) Q_r \quad (1)$$

with e_r and e_p being the environmental impact of producing one unit of secondary and primary material, respectively, and Q_r the amount of secondary material being produced. For ease of exposition, environmental impact is here measured using a one-dimensional impact indicator. In this engineering view, recycling generates environmental benefits as soon as $e_r / e_p < 1$, a condition met for many recycling activities. When the market is included between consumer and recycler, and between recycler and manufacturer, the outcome is more difficult to predict. The net environmental impact of an increase in circular economy activity is now (equation 2):

$$E_{net} = e_r Q_{Qr} + e_p Q_{Qp} \quad (2)$$

with Q_{Qr} being the change in secondary production and Q_{Qp} the resulting, market-mediated change in primary production. Recycling now reduces environmental impacts only if $Q_{Qp} < 0$ and $e_r / e_p < |Q_{Qp}| / Q_{Qr}$. In analogy to energy efficiency rebound, any circular economy activity with $P_{Prod} = Q_{Qr} + Q_{Qp} > 0$ is deemed to experience circular economy rebound. Comparing equation (1) with equation (2) shows that the engineering view implicitly assumes that $Q_{Qp} = -Q_{Qr}$.

Circular economy activities other than recycling that are likely to meet the condition $e_r / e_p < 1$ are product repair, for example, of garments, and product refurbishment, for example, of smartphones. In fact, there is a considerable body of research quantifying the environmental impacts e_r of many different recycling, refurbishment, and repair activities. In stark contrast, the impact of those activities on primary material and product production, Q_{Qp} , is largely unknown.

Code: • Circular strategy > Recycling Weight score: 0
 Search C - Rebound and circularity theory > C-GS-02, p. 4
 Created: 23.6.2026 r. 13:28 by user, Modified: 23.6.2026 r. 13:28 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

In the diagram without markets, a flow, say, from consumer to recycler to manufacturer implies 1:1 displacement—that every kg recycled prevents 1 kg of production from raw materials. The environmental benefit of recycling is then simply calculated as shown by equation (1):

$$E_{net} = (e_r - e_p) Q_r \quad (1)$$

with e_r and e_p being the environmental impact of producing one unit of secondary and primary material, respectively, and Q_r the amount of secondary material being produced. For ease of exposition, environmental impact is here measured using a one-dimensional impact indicator. In this engineering view, recycling generates environmental benefits as soon as $e_r / e_p < 1$, a condition met

for many recycling activities. When the market is included between consumer and recycler, and between recycler and manufacturer, the outcome is more difficult to predict. The net environmental impact of an increase in circular economy activity is now (equation 2):

$$E_{net} = e_r Q_{Qr} + e_p Q_{Qp} \quad (2)$$

with Q_{Qr} being the change in secondary production and Q_{Qp} the resulting, market-mediated change in primary production. Recycling now reduces environmental impacts only if $Q_{Qp} < 0$ and $e_r / e_p < |Q_{Qp}| / Q_{Qr}$. In analogy to energy efficiency rebound, any circular economy activity with $P_{Prod} = Q_{Qr} + Q_{Qp} > 0$ is deemed to experience circular economy rebound. Comparing equation (1) with equation (2) shows that the engineering view implicitly assumes that $Q_{Qp} = -Q_{Qr}$.

Circular economy activities other than recycling that are likely to meet the condition $e_r / e_p < 1$ are product repair, for example, of garments, and product refurbishment, for example, of smartphones. In fact, there is a considerable body of research quantifying the environmental impacts e_r of many different recycling, refurbishment, and repair activities. In stark contrast, the impact of those activities on primary material and product production, Q_{Qp} , is largely unknown.

Code: • Circular strategy > Refurbishment Weight score: 0

Search C - Rebound and circularity theory > C-GS-02, p. 4

Created: 23.6.2026 r. 13:28 by user, Modified: 23.6.2026 r. 13:28 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

In the diagram without markets, a flow, say, from consumer to recycler to manufacturer implies 1:1 displacement—that every kg recycled prevents 1 kg of production from raw materials. The environmental benefit of recycling is then simply calculated as shown by equation (1):

$$E_{net} = (e_r - e_p) Q_r \quad (1)$$

with e_r and e_p being the environmental impact of producing one unit of secondary and primary material, respectively, and Q_r the amount of secondary material being produced. For ease of exposition, environmental impact is here measured using a one-dimensional impact indicator. In this engineering view, recycling generates environmental benefits as soon as $e_r / e_p < 1$, a condition met for many recycling activities. When the market is included between consumer and recycler, and between recycler and manufacturer, the outcome is more difficult to predict. The net environmental impact of an increase in circular economy activity is now (equation 2):

$$E_{net} = e_r Q_{Qr} + e_p Q_{Qp} \quad (2)$$

with Q_{Qr} being the change in secondary production and Q_{Qp} the resulting, market-mediated change in primary production. Recycling now reduces environmental impacts only if $Q_{Qp} < 0$ and $e_r / e_p < |Q_{Qp}| / Q_{Qr}$. In analogy to energy efficiency rebound, any circular economy activity with $P_{Prod} = Q_{Qr} + Q_{Qp} > 0$ is deemed to experience circular economy rebound. Comparing equation (1) with equation (2) shows that the engineering view implicitly assumes that $Q_{Qp} = -Q_{Qr}$.

Circular economy activities other than recycling that are likely to meet the condition $e_r / e_p < 1$ are product repair, for example, of garments, and product refurbishment, for example, of smartphones. In fact, there is a considerable body of research quantifying the environmental impacts e_r of many different recycling, refurbishment, and repair activities. In stark contrast, the impact of those activities on primary material and product production, Q_{Qp} , is largely unknown.

Code: • Circular strategy > Repair Weight score: 0

Search C - Rebound and circularity theory > C-GS-02, p. 4

Created: 23.6.2026 r. 13:28 by user, Modified: 23.6.2026 r. 13:28 by user

Other codes assigned to segment:

Circular strategy > Refurbishment (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

In the diagram without markets, a flow, say, from consumer to recycler to manufacturer implies 1:1 displacement—that every kg recycled prevents 1 kg of production from raw materials. The environmental benefit of recycling is then simply calculated as shown by equation (1):

$$E_{net} = (e_r - e_p) Q_r \quad (1)$$

with e_r and e_p being the environmental impact of producing one unit of secondary and primary material, respectively, and Q_r the amount of secondary material being produced. For ease of exposition, environmental impact is here measured using a one-dimensional impact indicator. In this engineering view, recycling generates environmental benefits as soon as $e_r / e_p < 1$, a condition met for many recycling activities. When the market is included between consumer and recycler, and between recycler and manufacturer, the outcome is more difficult to predict. The net environmental impact of an increase in circular economy activity is now (equation 2):

$$E_{net} = e_r Q_{Qr} + e_p Q_{Qp} \quad (2)$$

with Q_{Qr} being the change in secondary production and Q_{Qp} the resulting, market-mediated change in primary production. Recycling now reduces environmental impacts only if $Q_{Qp} < 0$ and $e_r / e_p < |Q_{Qp}| / Q_{Qr}$. In analogy to energy efficiency rebound, any circular economy activity with $P_{Prod} = Q_{Qr} + Q_{Qp} > 0$ is deemed to experience circular economy rebound. Comparing equation (1) with equation (2) shows that the engineering view implicitly assumes that $Q_{Qp} = -Q_{Qr}$.

Circular economy activities other than recycling that are likely to meet the condition $e_r / e_p < 1$ are product repair, for example, of garments, and product refurbishment, for example, of smartphones. In fact, there is a considerable body of research quantifying the environmental impacts e_r of many different recycling, refurbishment, and repair activities. In stark contrast, the impact of those activities on primary material and product production, Q_{Qp} , is largely unknown.

Code: • Negative mechanism > Burden shifting Weight score: 0

Search C - Rebound and circularity theory > C-GS-02, p. 4

Created: 23.6.2026 r. 13:28 by user, Modified: 23.6.2026 r. 13:28 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

In the diagram without markets, a flow, say, from consumer to recycler to manufacturer implies 1:1 displacement—that every kg recycled prevents 1 kg of production from raw materials. The environmental benefit of recycling is then simply calculated as shown by equation (1):

$$E_{net} = (e_r - e_p) Q_r \quad (1)$$

with e_r and e_p being the environmental impact of producing one unit of secondary and primary material, respectively, and Q_r the amount of secondary material being produced. For ease of exposition, environmental impact is here measured using a one-dimensional impact indicator. In this engineering view, recycling generates environmental benefits as soon as $e_r / e_p < 1$, a condition met for many recycling activities. When the market is included between consumer and recycler, and between recycler and manufacturer, the outcome is more difficult to predict. The net environmental impact of an increase in circular economy activity is now (equation 2):

$$E_{net} = e_r Q_{Qr} + e_p Q_{Qp} \quad (2)$$

with Q_{Qr} being the change in secondary production and Q_{Qp} the resulting, market-mediated change in primary production. Recycling now reduces environmental impacts only if $Q_{Qp} < 0$ and $e_r / e_p < |Q_{Qp}| / Q_{Qr}$. In analogy to energy efficiency rebound, any circular economy activity with $P_{Prod} = Q_{Qr} + Q_{Qp} > 0$ is deemed to experience circular economy rebound. Comparing equation (1) with equation (2) shows that the engineering view implicitly assumes that $Q_{Qp} = -Q_{Qr}$.

Circular economy activities other than recycling that are likely to meet the condition $e_r / e_p < 1$ are product repair, for example, of garments, and product refurbishment, for example, of smartphones. In fact, there is a considerable body of research quantifying the environmental impacts e_r of many

different recycling, refurbishment, and repair activities. In stark contrast, the impact of those activities on primary material and product production, QQ_p , is largely unknown.

Code: • Mitigation > Lifecycle verification Weight score: 0
 Search C - Rebound and circularity theory > C-GS-02, p. 4
 Created: 23.6.2026 r. 13:28 by user, Modified: 23.6.2026 r. 13:28 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)

Activities in Q1 also have the potential to reduce overall environmental impact, but experience circular economy rebound, that is, $PProd > 0$. In the best case, this increase in total production and consumption only reduces the net environmental benefits, $Enet$, but does not reverse its sign (see figure 3). In the worst case, net environmental impact increases, resulting in circular economy backfire.

Code: • Mitigation > Lifecycle verification Weight score: 0
 Search C - Rebound and circularity theory > C-GS-02, p. 5
 Created: 23.6.2026 r. 13:29 by user, Modified: 23.6.2026 r. 13:29 by user
 Other codes assigned to segment:
 Negative mechanism > Increased comfort or use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)

Activities in Q1 also have the potential to reduce overall environmental impact, but experience circular economy rebound, that is, $PProd > 0$. In the best case, this increase in total production and consumption only reduces the net environmental benefits, $Enet$, but does not reverse its sign (see figure 3). In the worst case, net environmental impact increases, resulting in circular economy backfire.

Code: • Negative mechanism > Increased comfort or use Weight score: 0
 Search C - Rebound and circularity theory > C-GS-02, p. 5
 Created: 23.6.2026 r. 13:29 by user, Modified: 23.6.2026 r. 13:29 by user
 Other codes assigned to segment:
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Activities in Q1 also have the potential to reduce overall environmental impact, but experience circular economy rebound, that is, $PProd > 0$. In the best case, this increase in total production and consumption only reduces the net environmental benefits, $Enet$, but does not reverse its sign (see figure 3). In the worst case, net environmental impact increases, resulting in circular economy backfire.

Code: • Negative mechanism > Burden shifting Weight score: 0
 Search C - Rebound and circularity theory > C-GS-02, p. 5
 Created: 23.6.2026 r. 13:29 by user, Modified: 23.6.2026 r. 13:29 by user
 Other codes assigned to segment:
 Negative mechanism > Increased comfort or use (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Secondary goods may be insufficient substitutes for primary goods because they are of inferior quality or are otherwise less desirable to users. For example, recycled plastics and papers rarely compete directly with primary materials due to degradation in the quality of the polymer and shortening of fiber lengths during use, collection, and reprocessing (Allwood 2014). Recycled metals such as aluminum are contaminated with alloying materials, reducing their utility and value

(Nakajima et al. 2010). This means that recycled plastics, Zink and Geyer, Circular Economy Rebound
5RE SEARCH AND ANALYSIS

Production (# units)

Impacts

Primary production activity

Circular economy activity

0

1

Potential benefit

Realized benefit

2

Rebound

Figure 3 Circular economy rebound attributed to increased production. If no rebound occurs, the potential benefit is realized and impacts fall from E_0 to E_2 . However, $PP_{prod} > 0$ reduces the net benefit by $E_1 - E_2$, termed circular economy rebound. If rebound is sufficiently large, the benefit is eliminated entirely (i.e., $E_0 - E_2 = 0$), or backfire can occur (i.e., $E_0 - E_2 < 0$).

papers, and some metals are likely to be produced in addition to, rather than instead of, primary materials, and the potential benefits of recycling will be reduced.

Refurbished smartphones also rarely compete in the same market with primary smartphones (Geyer and Doctori Blass 2010; Skerlos et al. 2003). Instead, these secondhand phones are sold to consumers in developing countries who would not be able to afford a primary phone. Therefore, the appropriate comparison in this case is not between the impacts of refurbishing vs. primary production, but of refurbishing vs. no phone at all.³ Used phones are refurbished in addition to, rather than instead of, new phones, and the smartphone circular economy (how it is currently practiced) necessarily leads to rebound. The same is true for many used goods where technology changes too rapidly to support a secondhand market (Allwood 2014); in these cases, refurbishment or resale of used goods will necessarily increase net production and consumption.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search C - Rebound and circularity theory > C-GS-02, p. 5 - 6

Created: 23.6.2026 r. 13:31 by user, Modified: 23.6.2026 r. 13:31 by user

Other codes assigned to segment:

Circular strategy > Refurbishment (Weight: 0)

Circular strategy > Direct reuse (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Obsolescence > Functional (Weight: 0)

Obsolescence > Aesthetic (Weight: 0)

Obsolescence > User-driven (Weight: 0)

Negative mechanism > Increased comfort or use (Weight: 0)

Secondary goods may be insufficient substitutes for primary goods because they are of inferior quality or are otherwise less desirable to users. For example, recycled plastics and papers rarely compete directly with primary materials due to degradation in the quality of the polymer and shortening of fiber lengths during use, collection, and reprocessing (Allwood 2014). Recycled metals such as aluminum are contaminated with alloying materials, reducing their utility and value (Nakajima et al. 2010). This means that recycled plastics, Zink and Geyer, Circular Economy Rebound
5RE SEARCH AND ANALYSIS

Production (# units)

Impacts

Primary production activity

Circular economy activity

0

1

Potential benefit

Realized benefit

2

Rebound

Figure 3 Circular economy rebound attributed to increased production. If no rebound occurs, the potential benefit is realized and impacts fall from E_0 to E_2 . However, $PP_{rod} > 0$ reduces the net benefit by $E_1 - E_2$, termed circular economy rebound. If rebound is sufficiently large, the benefit is eliminated entirely (i.e., $E_0 - E_2 = 0$), or backfire can occur (i.e., $E_0 - E_2 < 0$).

papers, and some metals are likely to be produced in addition to, rather than instead of, primary materials, and the potential benefits of recycling will be reduced.

Refurbished smartphones also rarely compete in the same market with primary smartphones (Geyer and Doctori Blass 2010; Skerlos et al. 2003). Instead, these secondhand phones are sold to consumers in developing countries who would not be able to afford a primary phone. Therefore, the appropriate comparison in this case is not between the impacts of refurbishing vs. primary production, but of refurbishing vs. no phone at all.³ Used phones are refurbished in addition to, rather than instead of, new phones, and the smartphone circular economy (how it is currently practiced) necessarily leads to rebound. The same is true for many used goods where technology changes too rapidly to support a secondhand market (Allwood 2014); in these cases, refurbishment or resale of used goods will necessarily increase net production and consumption.

Code: • Negative mechanism > Increased comfort or use Weight score: 0

Search C - Rebound and circularity theory > C-GS-02, p. 5 - 6

Created: 23.6.2026 r. 13:31 by user, Modified: 23.6.2026 r. 13:31 by user

Other codes assigned to segment:

Circular strategy > Refurbishment (Weight: 0)

Circular strategy > Direct reuse (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Obsolescence > Functional (Weight: 0)

Obsolescence > Aesthetic (Weight: 0)

Obsolescence > User-driven (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Secondary goods may be insufficient substitutes for primary goods because they are of inferior quality or are otherwise less desirable to users. For example, recycled plastics and papers rarely compete directly with primary materials due to degradation in the quality of the polymer and shortening of fiber lengths during use, collection, and reprocessing (Allwood 2014). Recycled metals such as aluminum are contaminated with alloying materials, reducing their utility and value (Nakajima et al. 2010). This means that recycled plastics, Zink and Geyer, Circular Economy Rebound
5RE SEARCH AND ANALYSIS

Production (# units)

Impacts

Primary production activity

Circular economy activity

0

1

Potential benefit

Realized benefit

2

Rebound

Figure 3 Circular economy rebound attributed to increased production. If no rebound occurs, the potential benefit is realized and impacts fall from E_0 to E_2 . However, $PP_{rod} > 0$ reduces the net benefit by $E_1 - E_2$, termed circular economy rebound. If rebound is sufficiently large, the benefit is eliminated entirely (i.e., $E_0 - E_2 = 0$), or backfire can occur (i.e., $E_0 - E_2 < 0$).

papers, and some metals are likely to be produced in addition to, rather than instead of, primary materials, and the potential benefits of recycling will be reduced.

Refurbished smartphones also rarely compete in the same market with primary smartphones (Geyer and Doctori Blass 2010; Skerlos et al. 2003). Instead, these secondhand phones are sold to consumers in developing countries who would not be able to afford a primary phone. Therefore, the appropriate comparison in this case is not between the impacts of refurbishing vs. primary production, but of refurbishing vs. no phone at all.³ Used phones are refurbished in addition to, rather than instead of, new phones, and the smartphone circular economy (how it is currently practiced) necessarily leads to rebound. The same is true for many used goods where technology changes too rapidly to support a secondhand market (Allwood 2014); in these cases, refurbishment or resale of used goods will necessarily increase net production and consumption.

Code: • Circular strategy > Direct reuse Weight score: 0

Search C - Rebound and circularity theory > C-GS-02, p. 5 - 6

Created: 23.6.2026 r. 13:31 by user, Modified: 23.6.2026 r. 13:31 by user

Other codes assigned to segment:

Circular strategy > Refurbishment (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Obsolescence > Functional (Weight: 0)

Obsolescence > Aesthetic (Weight: 0)

Obsolescence > User-driven (Weight: 0)

Negative mechanism > Increased comfort or use (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Secondary goods may be insufficient substitutes for primary goods because they are of inferior quality or are otherwise less desirable to users. For example, recycled plastics and papers rarely compete directly with primary materials due to degradation in the quality of the polymer and shortening of fiber lengths during use, collection, and reprocessing (Allwood 2014). Recycled metals such as aluminum are contaminated with alloying materials, reducing their utility and value (Nakajima et al. 2010). This means that recycled plastics, Zink and Geyer, Circular Economy Rebound

5RE SEARCH AND ANALYSIS

Production (# units)

Impacts

Primary production activity

Circular economy activity

0

1

Potential benefit

Realized benefit

2

Rebound

Figure 3 Circular economy rebound attributed to increased production. If no rebound occurs, the potential benefit is realized and impacts fall from E_0 to E_2 . However, $PP_{prod} > 0$ reduces the net benefit by $E_1 - E_2$, termed circular economy rebound. If rebound is sufficiently large, the benefit is eliminated entirely (i.e., $E_0 - E_2 = 0$), or backfire can occur (i.e., $E_0 - E_2 < 0$).

papers, and some metals are likely to be produced in addition to, rather than instead of, primary materials, and the potential benefits of recycling will be reduced.

Refurbished smartphones also rarely compete in the same market with primary smartphones (Geyer and Doctori Blass 2010; Skerlos et al. 2003). Instead, these secondhand phones are sold to consumers in developing countries who would not be able to afford a primary phone. Therefore, the appropriate comparison in this case is not between the impacts of refurbishing vs. primary production, but of refurbishing vs. no phone at all.³ Used phones are refurbished in addition to, rather than instead of, new phones, and the smartphone circular economy (how it is currently practiced) necessarily leads to rebound. The same is true for many used goods where technology changes too rapidly to support a secondhand market (Allwood 2014); in these cases, refurbishment or resale of used goods will necessarily increase net production and consumption.

Code: ● Obsolescence > User-driven Weight score: 0
 Search C - Rebound and circularity theory > C-GS-02, p. 5 - 6
 Created: 23.6.2026 r. 13:31 by user, Modified: 23.6.2026 r. 13:31 by user
 Other codes assigned to segment:
 Circular strategy > Refurbishment (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Obsolescence > Aesthetic (Weight: 0)
 Negative mechanism > Increased comfort or use (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Secondary goods may be insufficient substitutes for primary goods because they are of inferior quality or are otherwise less desirable to users. For example, recycled plastics and papers rarely compete directly with primary materials due to degradation in the quality of the polymer and shortening of fiber lengths during use, collection, and reprocessing (Allwood 2014). Recycled metals such as aluminum are contaminated with alloying materials, reducing their utility and value (Nakajima et al. 2010). This means that recycled plastics, Zink and Geyer, Circular Economy Rebound

5RE SEARCH AND ANALYSIS

Production (# units)

Impacts

Primary production activity

Circular economy activity

0

1

Potential benefit

Realized benefit

2

Rebound

Figure 3 Circular economy rebound attributed to increased production. If no rebound occurs, the potential benefit is realized and impacts fall from E 0 to E 2. However, $PP_{rod} > 0$ reduces the net benefit by $E 1 - E 2$, termed circular economy rebound. If rebound is sufficiently large, the benefit is eliminated entirely (i.e., $E 0 - E 2 = 0$), or backfire can occur (i.e., $E 0 - E 2 < 0$).

papers, and some metals are likely to be produced in addition to, rather than instead of, primary materials, and the potential benefits of recycling will be reduced.

Refurbished smartphones also rarely compete in the same market with primary smartphones (Geyer and Doctori Blass 2010; Skerlos et al. 2003). Instead, these secondhand phones are sold to consumers in developing countries who would not be able to afford a primary phone. Therefore, the appropriate comparison in this case is not between the impacts of refurbishing vs. primary production, but of refurbishing vs. no phone at all.³ Used phones are refurbished in addition to, rather than instead of, new phones, and the smartphone circular economy (how it is currently practiced) necessarily leads to rebound. The same is true for many used goods where technology changes too rapidly to support a secondhand market (Allwood 2014); in these cases, refurbishment or resale of used goods will necessarily increase net production and consumption.

Code: ● Obsolescence > Aesthetic Weight score: 0
 Search C - Rebound and circularity theory > C-GS-02, p. 5 - 6
 Created: 23.6.2026 r. 13:31 by user, Modified: 23.6.2026 r. 13:31 by user
 Other codes assigned to segment:
 Circular strategy > Refurbishment (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Obsolescence > User-driven (Weight: 0)
 Negative mechanism > Increased comfort or use (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Secondary goods may be insufficient substitutes for primary goods because they are of inferior quality or are otherwise less desirable to users. For example, recycled plastics and papers rarely compete directly with primary materials due to degradation in the quality of the polymer and shortening of fiber lengths during use, collection, and reprocessing (Allwood 2014). Recycled metals such as aluminum are contaminated with alloying materials, reducing their utility and value (Nakajima et al. 2010). This means that recycled plastics, Zink and Geyer, Circular Economy Rebound

5RE SEARCH AND ANALYSIS

Production (# units)

Impacts

Primary production activity

Circular economy activity

0

1

Potential benefit

Realized benefit

2

Rebound

Figure 3 Circular economy rebound attributed to increased production. If no rebound occurs, the potential benefit is realized and impacts fall from E_0 to E_2 . However, $PP_{prod} > 0$ reduces the net benefit by $E_1 - E_2$, termed circular economy rebound. If rebound is sufficiently large, the benefit is eliminated entirely (i.e., $E_0 - E_2 = 0$), or backfire can occur (i.e., $E_0 - E_2 < 0$).

papers, and some metals are likely to be produced in addition to, rather than instead of, primary materials, and the potential benefits of recycling will be reduced.

Refurbished smartphones also rarely compete in the same market with primary smartphones (Geyer and Doctori Blass 2010; Skerlos et al. 2003). Instead, these secondhand phones are sold to consumers in developing countries who would not be able to afford a primary phone. Therefore, the appropriate comparison in this case is not between the impacts of refurbishing vs. primary production, but of refurbishing vs. no phone at all.³ Used phones are refurbished in addition to, rather than instead of, new phones, and the smartphone circular economy (how it is currently practiced) necessarily leads to rebound. The same is true for many used goods where technology changes too rapidly to support a secondhand market (Allwood 2014); in these cases, refurbishment or resale of used goods will necessarily increase net production and consumption.

Code: • Circular strategy > Refurbishment Weight score: 0

Search C - Rebound and circularity theory > C-GS-02, p. 5 - 6

Created: 23.6.2026 r. 13:31 by user, Modified: 23.6.2026 r. 13:31 by user

Other codes assigned to segment:

Circular strategy > Direct reuse (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Obsolescence > Functional (Weight: 0)

Obsolescence > Aesthetic (Weight: 0)

Obsolescence > User-driven (Weight: 0)

Negative mechanism > Increased comfort or use (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Secondary goods may be insufficient substitutes for primary goods because they are of inferior quality or are otherwise less desirable to users. For example, recycled plastics and papers rarely compete directly with primary materials due to degradation in the quality of the polymer and shortening of fiber lengths during use, collection, and reprocessing (Allwood 2014). Recycled metals such as aluminum are contaminated with alloying materials, reducing their utility and value (Nakajima et al. 2010). This means that recycled plastics, Zink and Geyer, Circular Economy Rebound

5RE SEARCH AND ANALYSIS

Production (# units)

Impacts

Primary production activity

Circular economy activity

0

1

Potential benefit

Realized benefit

2

Rebound

Figure 3 Circular economy rebound attributed to increased production. If no rebound occurs, the potential benefit is realized and impacts fall from E 0 to E 2. However, $PP_{rod} > 0$ reduces the net benefit by $E_1 - E_2$, termed circular economy rebound. If rebound is sufficiently large, the benefit is eliminated entirely (i.e., $E_0 - E_2 = 0$), or backfire can occur (i.e., $E_0 - E_2 < 0$).

papers, and some metals are likely to be produced in addition to, rather than instead of, primary materials, and the potential benefits of recycling will be reduced.

Refurbished smartphones also rarely compete in the same market with primary smartphones (Geyer and Doctori Blass 2010; Skerlos et al. 2003). Instead, these secondhand phones are sold to consumers in developing countries who would not be able to afford a primary phone. Therefore, the appropriate comparison in this case is not between the impacts of refurbishing vs. primary production, but of refurbishing vs. no phone at all.³ Used phones are refurbished in addition to, rather than instead of, new phones, and the smartphone circular economy (how it is currently practiced) necessarily leads to rebound. The same is true for many used goods where technology changes too rapidly to support a secondhand market (Allwood 2014); in these cases, refurbishment or resale of used goods will necessarily increase net production and consumption.

Code: • Obsolescence > Functional Weight score: 0

Search C - Rebound and circularity theory > C-GS-02, p. 5 - 6

Created: 23.6.2026 r. 13:31 by user, Modified: 23.6.2026 r. 13:31 by user

Other codes assigned to segment:

Circular strategy > Refurbishment (Weight: 0)

Circular strategy > Direct reuse (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Obsolescence > Aesthetic (Weight: 0)

Obsolescence > User-driven (Weight: 0)

Negative mechanism > Increased comfort or use (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Secondary goods may be insufficient substitutes for primary goods because they are of inferior quality or are otherwise less desirable to users. For example, recycled plastics and papers rarely compete directly with primary materials due to degradation in the quality of the polymer and shortening of fiber lengths during use, collection, and reprocessing (Allwood 2014). Recycled metals such as aluminum are contaminated with alloying materials, reducing their utility and value (Nakajima et al. 2010). This means that recycled plastics, Zink and Geyer, Circular Economy Rebound

5RE SEARCH AND ANALYSIS

Production (# units)

Impacts

Primary production activity

Circular economy activity

0

1

Potential benefit

Realized benefit

2

Rebound

Figure 3 Circular economy rebound attributed to increased production. If no rebound occurs, the potential benefit is realized and impacts fall from E 0 to E 2. However, $PP_{prod} > 0$ reduces the net benefit by $E_1 - E_2$, termed circular economy rebound. If rebound is sufficiently large, the benefit is eliminated entirely (i.e., $E_0 - E_2 = 0$), or backfire can occur (i.e., $E_0 - E_2 < 0$).

papers, and some metals are likely to be produced in addition to, rather than instead of, primary materials, and the potential benefits of recycling will be reduced.

Refurbished smartphones also rarely compete in the same market with primary smartphones (Geyer and Doctori Blass 2010; Skerlos et al. 2003). Instead, these secondhand phones are sold to consumers in developing countries who would not be able to afford a primary phone. Therefore, the appropriate comparison in this case is not between the impacts of refurbishing vs. primary production, but of refurbishing vs. no phone at all.³ Used phones are refurbished in addition to, rather than instead of, new phones, and the smartphone circular economy (how it is currently practiced) necessarily leads to rebound. The same is true for many used goods where technology changes too rapidly to support a secondhand market (Allwood 2014); in these cases, refurbishment or resale of used goods will necessarily increase net production and consumption.

Code: • Circular strategy > Recycling Weight score: 0

Search C - Rebound and circularity theory > C-GS-02, p. 5 - 6

Created: 23.6.2026 r. 13:31 by user, Modified: 23.6.2026 r. 13:31 by user

Other codes assigned to segment:

Circular strategy > Refurbishment (Weight: 0)

Circular strategy > Direct reuse (Weight: 0)

Obsolescence > Functional (Weight: 0)

Obsolescence > Aesthetic (Weight: 0)

Obsolescence > User-driven (Weight: 0)

Negative mechanism > Increased comfort or use (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

In order to entice buyers to purchase lowergrade materials, sellers offer them at a discount relative to primary materials. Downstream producers that choose to substitute with secondary material are now comparatively wealthier (the income effect) and can purchase more material and use it to make more products than they could before. It is likely that these products will also be of lower quality and will need to be sold to end users at a discount, multiplying the income effect. As a result of increased paper and plastic recycling, more goods are now produced, sold, and used.⁴

However, problem deepens because it is not necessary for the secondary good to be cheaper in order for it to impact prices and therefore lead to rebound. To see this, consider that increasing circular economy activities has the effect of increasing the supply of secondary goods (products, components, and materials) to markets. The secondary goods compete in these markets as substitutes for primary goods. Basic economic principles suggest that as supply of a good increases, both its price and the price of substitute goods will fall as suppliers compete for buyers and buyers of substitute goods switch to the cheaper good. This price decrease will cause demand for both the good and its substitutes to increase as consumers perceive themselves to be comparatively wealthier (the income effect) and make consumption decisions about all other goods in relation to the cheaper good (the substitution effect). Fringe buyers who were previously priced out of the good (or its substitutes) may also start to purchase the good. These demand increases will then increase prices, and these opposing forces will balance at a new equilibrium price and quantity. However, assuming a competitive market and normal goods, the new equilibrium price will necessarily be lower than the original price and the quantity consumed will be higher.

Code: • Enabling condition > User acceptance Weight score: 0

Search C - Rebound and circularity theory > C-GS-02, p. 6

Created: 23.6.2026 r. 13:34 by user, Modified: 23.6.2026 r. 13:34 by user

Other codes assigned to segment:

Obsolescence > Economic (Weight: 0)

Negative mechanism > Increased comfort or use (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

In order to entice buyers to purchase lowergrade materials, sellers offer them at a discount relative to primary materials. Downstream producers that choose to substitute with secondary material are now comparatively wealthier (the income effect) and can purchase more material and use it to make more products than they could before. It is likely that these products will also be of lower quality and will need to be sold to end users at a discount, multiplying the income effect. As a result of increased paper and plastic recycling, more goods are now produced, sold, and used.⁴

However, problem deepens because it is not necessary for the secondary good to be cheaper in order for it to impact prices and therefore lead to rebound. To see this, consider that increasing circular economy activities has the effect of increasing the supply of secondary goods (products, components, and materials) to markets. The secondary goods compete in these markets as substitutes for primary goods. Basic economic principles suggest that as supply of a good increases, both its price and the price of substitute goods will fall as suppliers compete for buyers and buyers of substitute goods switch to the cheaper good. This price decrease will cause demand for both the good and its substitutes to increase as consumers perceive themselves to be comparatively wealthier (the income effect) and make consumption decisions about all other goods in relation to the cheaper good (the substitution effect). Fringe buyers who were previously priced out of the good (or its substitutes) may also start to purchase the good. These demand increases will then increase prices, and these opposing forces will balance at a new equilibrium price and quantity. However, assuming a competitive market and normal goods, the new equilibrium price will necessarily be lower than the original price and the quantity consumed will be higher.

Code: • [Obsolescence > Economic](#) Weight score: 0

[Search C - Rebound and circularity theory > C-GS-02, p. 6](#)

Created: 23.6.2026 r. 13:34 by user, Modified: 23.6.2026 r. 13:34 by user

Other codes assigned to segment:

[Enabling condition > User acceptance](#) (Weight: 0)

[Negative mechanism > Increased comfort or use](#) (Weight: 0)

[Mitigation > Lifecycle verification](#) (Weight: 0)

In order to entice buyers to purchase lowergrade materials, sellers offer them at a discount relative to primary materials. Downstream producers that choose to substitute with secondary material are now comparatively wealthier (the income effect) and can purchase more material and use it to make more products than they could before. It is likely that these products will also be of lower quality and will need to be sold to end users at a discount, multiplying the income effect. As a result of increased paper and plastic recycling, more goods are now produced, sold, and used.⁴

However, problem deepens because it is not necessary for the secondary good to be cheaper in order for it to impact prices and therefore lead to rebound. To see this, consider that increasing circular economy activities has the effect of increasing the supply of secondary goods (products, components, and materials) to markets. The secondary goods compete in these markets as substitutes for primary goods. Basic economic principles suggest that as supply of a good increases, both its price and the price of substitute goods will fall as suppliers compete for buyers and buyers of substitute goods switch to the cheaper good. This price decrease will cause demand for both the good and its substitutes to increase as consumers perceive themselves to be comparatively wealthier (the income effect) and make consumption decisions about all other goods in relation to the cheaper good (the substitution effect). Fringe buyers who were previously priced out of the good (or its substitutes) may also start to purchase the good. These demand increases will then increase prices, and these opposing forces will balance at a new equilibrium price and quantity. However, assuming a competitive market and normal goods, the new equilibrium price will necessarily be lower than the original price and the quantity consumed will be higher.

Code: • [Negative mechanism > Increased comfort or use](#) Weight score: 0

[Search C - Rebound and circularity theory > C-GS-02, p. 6](#)

Created: 23.6.2026 r. 13:34 by user, Modified: 23.6.2026 r. 13:34 by user

Other codes assigned to segment:

Obsolescence > Economic (Weight: 0)
 Enabling condition > User acceptance (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

In order to entice buyers to purchase lowergrade materials, sellers offer them at a discount relative to primary materials. Downstream producers that choose to substitute with secondary material are now comparatively wealthier (the income effect) and can purchase more material and use it to make more products than they could before. It is likely that these products will also be of lower quality and will need to be sold to end users at a discount, multiplying the income effect. As a result of increased paper and plastic recycling, more goods are now produced, sold, and used.⁴

However, problem deepens because it is not necessary for the secondary good to be cheaper in order for it to impact prices and therefore lead to rebound. To see this, consider that increasing circular economy activities has the effect of increasing the supply of secondary goods (products, components, and materials) to markets. The secondary goods compete in these markets as substitutes for primary goods. Basic economic principles suggest that as supply of a good increases, both its price and the price of substitute goods will fall as suppliers compete for buyers and buyers of substitute goods switch to the cheaper good. This price decrease will cause demand for both the good and its substitutes to increase as consumers perceive themselves to be comparatively wealthier (the income effect) and make consumption decisions about all other goods in relation to the cheaper good (the substitution effect). Fringe buyers who were previously priced out of the good (or its substitutes) may also start to purchase the good. These demand increases will then increase prices, and these opposing forces will balance at a new equilibrium price and quantity. However, assuming a competitive market and normal goods, the new equilibrium price will necessarily be lower than the original price and the quantity consumed will be higher.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search C - Rebound and circularity theory > C-GS-02, p. 6

Created: 23.6.2026 r. 13:34 by user, Modified: 23.6.2026 r. 13:34 by user

Other codes assigned to segment:

Obsolescence > Economic (Weight: 0)
 Enabling condition > User acceptance (Weight: 0)
 Negative mechanism > Increased comfort or use (Weight: 0)

The result is that increasing the circular economy may prevent some primary production, but it may not prevent it on a one-to-one basis. Zink and colleagues (2016) demonstrated this in the context of recycling; the case is no different in the other circular economy activities of reuse, repair, or refurbishment—the only difference is the market being supplied. In the case of recycling, Zink and colleagues (2016) found that the primary factors determining displacement rate were the price responses of buyers and sellers of primary and recycled material and buyers' willingness to substitute between them.

Code: • Circular strategy > Recycling Weight score: 0

Search C - Rebound and circularity theory > C-GS-02, p. 7

Created: 23.6.2026 r. 13:36 by user, Modified: 23.6.2026 r. 13:36 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Enabling condition > User acceptance (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The result is that increasing the circular economy may prevent some primary production, but it may not prevent it on a one-to-one basis. Zink and colleagues (2016) demonstrated this in the context of

recycling; the case is no different in the other circular economy activities of reuse, repair, or refurbishment— the only difference is the market being supplied. In the case of recycling, Zink and colleagues (2016) found that the primary factors determining displacement rate were the price responses of buyers and sellers of primary and recycled material and buyers' willingness to substitute between them.

Code: • Enabling condition > User acceptance Weight score: 0
 Search C - Rebound and circularity theory > C-GS-02, p. 7
 Created: 23.6.2026 r. 13:36 by user, Modified: 23.6.2026 r. 13:36 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The result is that increasing the circular economy may prevent some primary production, but it may not prevent it on a one-to-one basis. Zink and colleagues (2016) demonstrated this in the context of recycling; the case is no different in the other circular economy activities of reuse, repair, or refurbishment— the only difference is the market being supplied. In the case of recycling, Zink and colleagues (2016) found that the primary factors determining displacement rate were the price responses of buyers and sellers of primary and recycled material and buyers' willingness to substitute between them.

Code: • Negative mechanism > Burden shifting Weight score: 0
 Search C - Rebound and circularity theory > C-GS-02, p. 7
 Created: 23.6.2026 r. 13:36 by user, Modified: 23.6.2026 r. 13:36 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > User acceptance (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The result is that increasing the circular economy may prevent some primary production, but it may not prevent it on a one-to-one basis. Zink and colleagues (2016) demonstrated this in the context of recycling; the case is no different in the other circular economy activities of reuse, repair, or refurbishment— the only difference is the market being supplied. In the case of recycling, Zink and colleagues (2016) found that the primary factors determining displacement rate were the price responses of buyers and sellers of primary and recycled material and buyers' willingness to substitute between them.

Code: • Circular strategy > Repair Weight score: 0
 Search C - Rebound and circularity theory > C-GS-02, p. 7
 Created: 23.6.2026 r. 13:35 by user, Modified: 23.6.2026 r. 13:35 by user
 Other codes assigned to segment:
 Circular strategy > Refurbishment (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > User acceptance (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The result is that increasing the circular economy may prevent some primary production, but it may not prevent it on a one-to-one basis. Zink and colleagues (2016) demonstrated this in the context of

recycling; the case is no different in the other circular economy activities of reuse, repair, or refurbishment— the only difference is the market being supplied. In the case of recycling, Zink and colleagues (2016) found that the primary factors determining displacement rate were the price responses of buyers and sellers of primary and recycled material and buyers' willingness to substitute between them.

Code: • Circular strategy > Refurbishment Weight score: 0
 Search C - Rebound and circularity theory > C-GS-02, p. 7
 Created: 23.6.2026 r. 13:35 by user, Modified: 23.6.2026 r. 13:35 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > User acceptance (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The result is that increasing the circular economy may prevent some primary production, but it may not prevent it on a one-to-one basis. Zink and colleagues (2016) demonstrated this in the context of recycling; the case is no different in the other circular economy activities of reuse, repair, or refurbishment— the only difference is the market being supplied. In the case of recycling, Zink and colleagues (2016) found that the primary factors determining displacement rate were the price responses of buyers and sellers of primary and recycled material and buyers' willingness to substitute between them.

Code: • Circular strategy > Direct reuse Weight score: 0
 Search C - Rebound and circularity theory > C-GS-02, p. 7
 Created: 23.6.2026 r. 13:35 by user, Modified: 23.6.2026 r. 13:35 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > User acceptance (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The result is that increasing the circular economy may prevent some primary production, but it may not prevent it on a one-to-one basis. Zink and colleagues (2016) demonstrated this in the context of recycling; the case is no different in the other circular economy activities of reuse, repair, or refurbishment— the only difference is the market being supplied. In the case of recycling, Zink and colleagues (2016) found that the primary factors determining displacement rate were the price responses of buyers and sellers of primary and recycled material and buyers' willingness to substitute between them.

Code: • Mitigation > Lifecycle verification Weight score: 0
 Search C - Rebound and circularity theory > C-GS-02, p. 7
 Created: 23.6.2026 r. 13:36 by user, Modified: 23.6.2026 r. 13:36 by user
 Other codes assigned to segment:
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > User acceptance (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)

CE solutions will generate multiple adverse effects (Font Vivanco, Mcdowall, et al., 2016), since it requires additional resources and infrastructure to be implemented, such the availability of spare

parts and different logistics infrastructure (Morseletto, 2020). Figge and Thorpe (2019) and Zink and Geyer (2017) demonstrate that the efficiency of the CE could be affected by how strategies are designed, which represents a theoretical limitation to achieve circularity.

Code: • Enabling condition > Service infrastructure Weight score: 0
 Search C - Rebound and circularity theory > C-GS-04, p. 2
 Created: 23.6.2026 r. 13:41 by user, Modified: 23.6.2026 r. 13:41 by user
 Other codes assigned to segment:
 Negative mechanism > Burden shifting (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

CE solutions will generate multiple adverse effects (Font Vivanco, Mcdowall, et al., 2016), since it requires additional resources and infrastructure to be implemented, such the availability of spare parts and different logistics infrastructure (Morseletto, 2020). Figge and Thorpe (2019) and Zink and Geyer (2017) demonstrate that the efficiency of the CE could be affected by how strategies are designed, which represents a theoretical limitation to achieve circularity.

Code: • Negative mechanism > Missing repair infrastructure Weight score: 0
 Search C - Rebound and circularity theory > C-GS-04, p. 2
 Created: 23.6.2026 r. 13:41 by user, Modified: 23.6.2026 r. 13:41 by user
 Other codes assigned to segment:
 Enabling condition > Service infrastructure (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

CE solutions will generate multiple adverse effects (Font Vivanco, Mcdowall, et al., 2016), since it requires additional resources and infrastructure to be implemented, such the availability of spare parts and different logistics infrastructure (Morseletto, 2020). Figge and Thorpe (2019) and Zink and Geyer (2017) demonstrate that the efficiency of the CE could be affected by how strategies are designed, which represents a theoretical limitation to achieve circularity.

Code: • Negative mechanism > Burden shifting Weight score: 0
 Search C - Rebound and circularity theory > C-GS-04, p. 2
 Created: 23.6.2026 r. 13:41 by user, Modified: 23.6.2026 r. 13:41 by user
 Other codes assigned to segment:
 Enabling condition > Service infrastructure (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

CE solutions will generate multiple adverse effects (Font Vivanco, Mcdowall, et al., 2016), since it requires additional resources and infrastructure to be implemented, such the availability of spare parts and different logistics infrastructure (Morseletto, 2020). Figge and Thorpe (2019) and Zink and Geyer (2017) demonstrate that the efficiency of the CE could be affected by how strategies are designed, which represents a theoretical limitation to achieve circularity.

Code: • Mitigation > Lifecycle verification Weight score: 0
 Search C - Rebound and circularity theory > C-GS-04, p. 2
 Created: 23.6.2026 r. 13:41 by user, Modified: 23.6.2026 r. 13:41 by user
 Other codes assigned to segment:
 Enabling condition > Service infrastructure (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)

The term circular economy rebound (CER) is used when the eco-efficiency of a productive system is offset by an increase in production or consumption (Zink and Geyer, 2017). The adoption and recognition of the importance of CER is justified due CE is primarily interested in material flows, not

energy efficiency, so it is fundamental to evaluate the suitability of borrowing RE definitions, classifications, and mechanisms from energy economics. To illustrate the difference between perspectives, a common example of CER is the increase in material usage per product when recycled material is used so, at end-of-life, it generates a greater amount of waste per unit of product. Zink and Geyer (2017) highlight the importance to recognize two mechanisms (1) the product substitution effect and (2) the price effect of CE strategies, corresponding to what Greening et al. (2000), in their seminal work on energy economics, point out as direct and secondary effects.

Code: • Negative mechanism > Burden shifting Weight score: 0

Search C - Rebound and circularity theory > C-GS-04, p. 2

Created: 23.6.2026 r. 13:42 by user, Modified: 23.6.2026 r. 13:42 by user

Other codes assigned to segment:

Negative mechanism > Increased comfort or use (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The term circular economy rebound (CER) is used when the eco-efficiency of a productive system is offset by an increase in production or consumption (Zink and Geyer, 2017). The adoption and recognition of the importance of CER is justified due CE is primarily interested in material flows, not energy efficiency, so it is fundamental to evaluate the suitability of borrowing RE definitions, classifications, and mechanisms from energy economics. To illustrate the difference between perspectives, a common example of CER is the increase in material usage per product when recycled material is used so, at end-of-life, it generates a greater amount of waste per unit of product. Zink and Geyer (2017) highlight the importance to recognize two mechanisms (1) the product substitution effect and (2) the price effect of CE strategies, corresponding to what Greening et al. (2000), in their seminal work on energy economics, point out as direct and secondary effects.

Code: • Negative mechanism > Increased comfort or use Weight score: 0

Search C - Rebound and circularity theory > C-GS-04, p. 2

Created: 23.6.2026 r. 13:42 by user, Modified: 23.6.2026 r. 13:42 by user

Other codes assigned to segment:

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The term circular economy rebound (CER) is used when the eco-efficiency of a productive system is offset by an increase in production or consumption (Zink and Geyer, 2017). The adoption and recognition of the importance of CER is justified due CE is primarily interested in material flows, not energy efficiency, so it is fundamental to evaluate the suitability of borrowing RE definitions, classifications, and mechanisms from energy economics. To illustrate the difference between perspectives, a common example of CER is the increase in material usage per product when recycled material is used so, at end-of-life, it generates a greater amount of waste per unit of product. Zink and Geyer (2017) highlight the importance to recognize two mechanisms (1) the product substitution effect and (2) the price effect of CE strategies, corresponding to what Greening et al. (2000), in their seminal work on energy economics, point out as direct and secondary effects.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search C - Rebound and circularity theory > C-GS-04, p. 2

Created: 23.6.2026 r. 13:43 by user, Modified: 23.6.2026 r. 13:43 by user

Other codes assigned to segment:

Negative mechanism > Increased comfort or use (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

the circular economy strategies are implemented to improve the system's environmental efficiency, reduce resource use, improve the middle-of-life and the end-of-life, and better integrate the network actors. But, as one change is implemented, it can cause RE. A high number of papers in our sample (24) presents some evidence of CE strategies triggering RE and making it difficult for the CE to reach

its potential. Horvath et al. (2019) directly emphasize this contradiction, as the developments to improve waste treatment efficiency will encourage business as usual and generate more waste than before. Dace et al. (2014) also conclude that a recycling strategy for plastic packaging will cause landfills to grow quickly, as we need a large amount of secondary materials to substitute virgin materials and obtain the same properties.

Another effect of recycling is a social one, raised by Sinha et al. (2016), where the valorization of electronics waste can induce informal recycling and material losses. According to the authors, the recovery rate of informal recycling is lower than the regular recycling of the CE system. The easy access to CE strategies needs to be considered for the success of CE. Dace et al. (2014) discovered that there has to be a container nearby to sort the waste for people to start considering recycling.

The same logic is applied to reuse or repair, Niero et al. (2021)

exemplify this with the consumers' adoption of reuse of shampoo bottles dependence on the proximity of a store to refill it. van Loon et al. (2021)

argue that if the product does not require energy use during use, the reuse and remanufacturing strategy will have a positive impact. On the other hand, if the product consumes energy during use, replacing it with new ones can avoid the negative effects of old and polluting products,

Code: • Circular strategy > Recycling Weight score: 0

Search C - Rebound and circularity theory > C-GS-04, p. 4

Created: 23.6.2026 r. 13:44 by user, Modified: 23.6.2026 r. 13:44 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)

Circular strategy > Direct reuse (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Negative mechanism > Increased comfort or use (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

the circular economy strategies are implemented to improve the system's environmental efficiency, reduce resource use, improve the middle-of-life and the end-of-life, and better integrate the network actors. But, as one change is implemented, it can cause RE. A high number of papers in our sample (24) presents some evidence of CE strategies triggering RE and making it difficult for the CE to reach its potential. Horvath et al. (2019) directly emphasize this contradiction, as the developments to improve waste treatment efficiency will encourage business as usual and generate more waste than before. Dace et al. (2014) also conclude that a recycling strategy for plastic packaging will cause landfills to grow quickly, as we need a large amount of secondary materials to substitute virgin materials and obtain the same properties.

Another effect of recycling is a social one, raised by Sinha et al. (2016), where the valorization of electronics waste can induce informal recycling and material losses. According to the authors, the recovery rate of informal recycling is lower than the regular recycling of the CE system. The easy access to CE strategies needs to be considered for the success of CE. Dace et al. (2014) discovered that there has to be a container nearby to sort the waste for people to start considering recycling.

The same logic is applied to reuse or repair, Niero et al. (2021)

exemplify this with the consumers' adoption of reuse of shampoo bottles dependence on the proximity of a store to refill it. van Loon et al. (2021)

argue that if the product does not require energy use during use, the reuse and remanufacturing strategy will have a positive impact. On the other hand, if the product consumes energy during use, replacing it with new ones can avoid the negative effects of old and polluting products,

Code: • Enabling condition > Service infrastructure Weight score: 0

Search C - Rebound and circularity theory > C-GS-04, p. 4

Created: 23.6.2026 r. 13:45 by user, Modified: 23.6.2026 r. 13:45 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)

Circular strategy > Direct reuse (Weight: 0)

Circular strategy > Recycling (Weight: 0)
 Negative mechanism > Increased comfort or use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

the circular economy strategies are implemented to improve the system's environmental efficiency, reduce resource use, improve the middle-of-life and the end-of-life, and better integrate the network actors. But, as one change is implemented, it can cause RE. A high number of papers in our sample (24) presents some evidence of CE strategies triggering RE and making it difficult for the CE to reach its potential. Horvath et al. (2019) directly emphasize this contradiction, as the developments to improve waste treatment efficiency will encourage business as usual and generate more waste than before. Dace et al. (2014) also conclude that a recycling strategy for plastic packaging will cause landfills to grow quickly, as we need a large amount of secondary materials to substitute virgin materials and obtain the same properties.

Another effect of recycling is a social one, raised by Sinha et al. (2016), where the valorization of electronics waste can induce informal recycling and material losses. According to the authors, the recovery rate of informal recycling is lower than the regular recycling of the CE system. The easy access to CE strategies needs to be considered for the success of CE. Dace et al. (2014) discovered that there has to be a container nearby to sort the waste for people to start considering recycling. The same logic is applied to reuse or repair, Niero et al. (2021)

exemplify this with the consumers' adoption of reuse of shampoo bottles dependence on the proximity of a store to refill it. van Loon et al. (2021)

argue that if the product does not require energy use during use, the reuse and remanufacturing strategy will have a positive impact. On the other hand, if the product consumes energy during use, replacing it with new ones can avoid the negative effects of old and polluting products,

Code: • Negative mechanism > Increased comfort or use Weight score: 0

Search C - Rebound and circularity theory > C-GS-04, p. 4

Created: 23.6.2026 r. 13:45 by user, Modified: 23.6.2026 r. 13:45 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)
 Circular strategy > Direct reuse (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

the circular economy strategies are implemented to improve the system's environmental efficiency, reduce resource use, improve the middle-of-life and the end-of-life, and better integrate the network actors. But, as one change is implemented, it can cause RE. A high number of papers in our sample (24) presents some evidence of CE strategies triggering RE and making it difficult for the CE to reach its potential. Horvath et al. (2019) directly emphasize this contradiction, as the developments to improve waste treatment efficiency will encourage business as usual and generate more waste than before. Dace et al. (2014) also conclude that a recycling strategy for plastic packaging will cause landfills to grow quickly, as we need a large amount of secondary materials to substitute virgin materials and obtain the same properties.

Another effect of recycling is a social one, raised by Sinha et al. (2016), where the valorization of electronics waste can induce informal recycling and material losses. According to the authors, the recovery rate of informal recycling is lower than the regular recycling of the CE system. The easy access to CE strategies needs to be considered for the success of CE. Dace et al. (2014) discovered that there has to be a container nearby to sort the waste for people to start considering recycling. The same logic is applied to reuse or repair, Niero et al. (2021)

exemplify this with the consumers' adoption of reuse of shampoo bottles dependence on the proximity of a store to refill it. van Loon et al. (2021)

argue that if the product does not require energy use during use, the reuse and remanufacturing strategy will have a positive impact. On the other hand, if the product consumes energy during use, replacing it with new ones can avoid the negative effects of old and polluting products,

Code: • Circular strategy > Direct reuse Weight score: 0

Search C - Rebound and circularity theory > C-GS-04, p. 4

Created: 23.6.2026 r. 13:44 by user, Modified: 23.6.2026 r. 13:44 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Negative mechanism > Increased comfort or use (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

the circular economy strategies are implemented to improve the system's environmental efficiency, reduce resource use, improve the middle-of-life and the end-of-life, and better integrate the network actors. But, as one change is implemented, it can cause RE. A high number of papers in our sample (24) presents some evidence of CE strategies triggering RE and making it difficult for the CE to reach its potential. Horvath et al. (2019) directly emphasize this contradiction, as the developments to improve waste treatment efficiency will encourage business as usual and generate more waste than before. Dace et al. (2014) also conclude that a recycling strategy for plastic packaging will cause landfills to grow quickly, as we need a large amount of secondary materials to substitute virgin materials and obtain the same properties.

Another effect of recycling is a social one, raised by Sinha et al. (2016), where the valorization of electronics waste can induce informal recycling and material losses. According to the authors, the recovery rate of informal recycling is lower than the regular recycling of the CE system. The easy access to CE strategies needs to be considered for the success of CE. Dace et al. (2014) discovered that there has to be a container nearby to sort the waste for people to start considering recycling. The same logic is applied to reuse or repair, Niero et al. (2021)

exemplify this with the consumers' adoption of reuse of shampoo bottles dependence on the proximity of a store to refill it. van Loon et al. (2021)

argue that if the product does not require energy use during use, the reuse and remanufacturing strategy will have a positive impact. On the other hand, if the product consumes energy during use, replacing it with new ones can avoid the negative effects of old and polluting products,

Code: • Circular strategy > Repair Weight score: 0

Search C - Rebound and circularity theory > C-GS-04, p. 4

Created: 23.6.2026 r. 13:44 by user, Modified: 23.6.2026 r. 13:44 by user

Other codes assigned to segment:

Circular strategy > Direct reuse (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Negative mechanism > Increased comfort or use (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

the circular economy strategies are implemented to improve the system's environmental efficiency, reduce resource use, improve the middle-of-life and the end-of-life, and better integrate the network actors. But, as one change is implemented, it can cause RE. A high number of papers in our sample (24) presents some evidence of CE strategies triggering RE and making it difficult for the CE to reach its potential. Horvath et al. (2019) directly emphasize this contradiction, as the developments to improve waste treatment efficiency will encourage business as usual and generate more waste than before. Dace et al. (2014) also conclude that a recycling strategy for plastic packaging will cause landfills to grow quickly, as we need a large amount of secondary materials to substitute virgin materials and obtain the same properties.

Another effect of recycling is a social one, raised by Sinha et al. (2016), where the valorization of electronics waste can induce informal recycling and material losses. According to the authors, the recovery rate of informal recycling is lower than the regular recycling of the CE system. The easy access to CE strategies needs to be considered for the success of CE. Dace et al. (2014) discovered that there has to be a container nearby to sort the waste for people to start considering recycling. The same logic is applied to reuse or repair, Niero et al. (2021) exemplify this with the consumers' adoption of reuse of shampoo bottles dependence on the proximity of a store to refill it. van Loon et al. (2021) argue that if the product does not require energy use during use, the reuse and remanufacturing strategy will have a positive impact. On the other hand, if the product consumes energy during use, replacing it with new ones can avoid the negative effects of old and polluting products,

Code: • Negative mechanism > Burden shifting Weight score: 0

Search C - Rebound and circularity theory > C-GS-04, p. 4

Created: 23.6.2026 r. 13:44 by user, Modified: 23.6.2026 r. 13:44 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)

Circular strategy > Direct reuse (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Negative mechanism > Increased comfort or use (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

the circular economy strategies are implemented to improve the system's environmental efficiency, reduce resource use, improve the middle-of-life and the end-of-life, and better integrate the network actors. But, as one change is implemented, it can cause RE. A high number of papers in our sample (24) presents some evidence of CE strategies triggering RE and making it difficult for the CE to reach its potential. Horvath et al. (2019) directly emphasize this contradiction, as the developments to improve waste treatment efficiency will encourage business as usual and generate more waste than before. Dace et al. (2014) also conclude that a recycling strategy for plastic packaging will cause landfills to grow quickly, as we need a large amount of secondary materials to substitute virgin materials and obtain the same properties.

Another effect of recycling is a social one, raised by Sinha et al. (2016), where the valorization of electronics waste can induce informal recycling and material losses. According to the authors, the recovery rate of informal recycling is lower than the regular recycling of the CE system. The easy access to CE strategies needs to be considered for the success of CE. Dace et al. (2014) discovered that there has to be a container nearby to sort the waste for people to start considering recycling. The same logic is applied to reuse or repair, Niero et al. (2021) exemplify this with the consumers' adoption of reuse of shampoo bottles dependence on the proximity of a store to refill it. van Loon et al. (2021) argue that if the product does not require energy use during use, the reuse and remanufacturing strategy will have a positive impact. On the other hand, if the product consumes energy during use, replacing it with new ones can avoid the negative effects of old and polluting products,

Code: • Mitigation > Lifecycle verification Weight score: 0

Search C - Rebound and circularity theory > C-GS-04, p. 4

Created: 23.6.2026 r. 13:45 by user, Modified: 23.6.2026 r. 13:45 by user

Other codes assigned to segment:

Circular strategy > Repair (Weight: 0)

Circular strategy > Direct reuse (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Negative mechanism > Increased comfort or use (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Some of the CE strategies to be implemented may require more transportation, spare parts, additional industrial processes, and new sales or trade points. This process rarely occurs in the same geographical location (Ottelin et al., 2020). Siderius and Poldner (2021) are emphatic that some rebound effects' causes come directly from the transition. "This includes the need for infrastructure and transportation between organizations that can now benefit from each other's waste streams and the increased necessity for (dis-) assembly practices due to modular design. This type of rebound is inherent to the transition, as it embodies the energy required to move from one system to the next" (Siderius and Poldner, 2021, p. 6). The authors argue that a separation between transitional CER and the design, economic and behavioral CER is useful to propose better mitigation strategies.

Code: • Negative mechanism > Burden shifting Weight score: 0

Search C - Rebound and circularity theory > C-GS-04, p. 5

Created: 23.6.2026 r. 13:46 by user, Modified: 23.6.2026 r. 13:46 by user

Other codes assigned to segment:

Enabling condition > Disassembly (Weight: 0)

Enabling condition > Spare parts (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Negative mechanism > Additional transport (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Some of the CE strategies to be implemented may require more transportation, spare parts, additional industrial processes, and new sales or trade points. This process rarely occurs in the same geographical location (Ottelin et al., 2020). Siderius and Poldner (2021) are emphatic that some rebound effects' causes come directly from the transition. "This includes the need for infrastructure and transportation between organizations that can now benefit from each other's waste streams and the increased necessity for (dis-) assembly practices due to modular design. This type of rebound is inherent to the transition, as it embodies the energy required to move from one system to the next" (Siderius and Poldner, 2021, p. 6). The authors argue that a separation between transitional CER and the design, economic and behavioral CER is useful to propose better mitigation strategies.

Code: • Enabling condition > Service infrastructure Weight score: 0

Search C - Rebound and circularity theory > C-GS-04, p. 5

Created: 23.6.2026 r. 13:46 by user, Modified: 23.6.2026 r. 13:46 by user

Other codes assigned to segment:

Enabling condition > Disassembly (Weight: 0)

Enabling condition > Spare parts (Weight: 0)

Negative mechanism > Additional transport (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Some of the CE strategies to be implemented may require more transportation, spare parts, additional industrial processes, and new sales or trade points. This process rarely occurs in the same geographical location (Ottelin et al., 2020). Siderius and Poldner (2021) are emphatic that some rebound effects' causes come directly from the transition. "This includes the need for infrastructure and transportation between organizations that can now benefit from each other's waste streams and the increased necessity for (dis-) assembly practices due to modular design. This type of rebound is inherent to the transition, as it embodies the energy required to move from one system to the next" (Siderius and Poldner, 2021, p. 6). The authors argue that a separation between transitional CER and the design, economic and behavioral CER is useful to propose better mitigation strategies.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search C - Rebound and circularity theory > C-GS-04, p. 5

Created: 23.6.2026 r. 13:47 by user, Modified: 23.6.2026 r. 13:47 by user

Other codes assigned to segment:

Enabling condition > Disassembly (Weight: 0)

Enabling condition > Spare parts (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Negative mechanism > Additional transport (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)

Some of the CE strategies to be implemented may require more transportation, spare parts, additional industrial processes, and new sales or trade points. This process rarely occurs in the same geographical location (Ottelin et al., 2020). Siderius and Poldner (2021) are emphatic that some rebound effects' causes come directly from the transition. "This includes the need for infrastructure and transportation between organizations that can now benefit from each other's waste streams and the increased necessity for (dis-) assembly practices due to modular design. This type of rebound is inherent to the transition, as it embodies the energy required to move from one system to the next" (Siderius and Poldner, 2021, p. 6). The authors argue that a separation between transitional CER and the design, economic and behavioral CER is useful to propose better mitigation strategies.

Code: • Negative mechanism > Additional transport Weight score: 0
 Search C - Rebound and circularity theory > C-GS-04, p. 5
 Created: 23.6.2026 r. 13:46 by user, Modified: 23.6.2026 r. 13:46 by user
 Other codes assigned to segment:
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Spare parts (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Some of the CE strategies to be implemented may require more transportation, spare parts, additional industrial processes, and new sales or trade points. This process rarely occurs in the same geographical location (Ottelin et al., 2020). Siderius and Poldner (2021) are emphatic that some rebound effects' causes come directly from the transition. "This includes the need for infrastructure and transportation between organizations that can now benefit from each other's waste streams and the increased necessity for (dis-) assembly practices due to modular design. This type of rebound is inherent to the transition, as it embodies the energy required to move from one system to the next" (Siderius and Poldner, 2021, p. 6). The authors argue that a separation between transitional CER and the design, economic and behavioral CER is useful to propose better mitigation strategies.

Code: • Enabling condition > Spare parts Weight score: 0
 Search C - Rebound and circularity theory > C-GS-04, p. 5
 Created: 23.6.2026 r. 13:46 by user, Modified: 23.6.2026 r. 13:46 by user
 Other codes assigned to segment:
 Enabling condition > Disassembly (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Negative mechanism > Additional transport (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Negative mechanism > Missing repair infrastructure (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Some of the CE strategies to be implemented may require more transportation, spare parts, additional industrial processes, and new sales or trade points. This process rarely occurs in the same geographical location (Ottelin et al., 2020). Siderius and Poldner (2021) are emphatic that some rebound effects' causes come directly from the transition. "This includes the need for infrastructure and transportation between organizations that can now benefit from each other's waste streams and the increased necessity for (dis-) assembly practices due to modular design. This type of rebound is inherent to the transition, as it embodies the energy required to move from one system to the next"

(Siderius and Poldner, 2021, p. 6). The authors argue that a separation between transitional CER and the design, economic and behavioral CER is useful to propose better mitigation strategies.

Code: • Negative mechanism > Missing repair infrastructure Weight score: 0

Search C - Rebound and circularity theory > C-GS-04, p. 5

Created: 23.6.2026 r. 13:46 by user, Modified: 23.6.2026 r. 13:46 by user

Other codes assigned to segment:

Enabling condition > Disassembly (Weight: 0)

Enabling condition > Spare parts (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Negative mechanism > Additional transport (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Some of the CE strategies to be implemented may require more transportation, spare parts, additional industrial processes, and new sales or trade points. This process rarely occurs in the same geographical location (Ottelin et al., 2020). Siderius and Poldner (2021) are emphatic that some rebound effects' causes come directly from the transition. "This includes the need for infrastructure and transportation between organizations that can now benefit from each other's waste streams and the increased necessity for (dis-) assembly practices due to modular design. This type of rebound is inherent to the transition, as it embodies the energy required to move from one system to the next" (Siderius and Poldner, 2021, p. 6). The authors argue that a separation between transitional CER and the design, economic and behavioral CER is useful to propose better mitigation strategies.

Code: • Enabling condition > Disassembly Weight score: 0

Search C - Rebound and circularity theory > C-GS-04, p. 5

Created: 23.6.2026 r. 13:47 by user, Modified: 23.6.2026 r. 13:47 by user

Other codes assigned to segment:

Enabling condition > Spare parts (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Negative mechanism > Additional transport (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Negative mechanism > Missing repair infrastructure (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

new technology brings a problem of obsolescence. Incremental innovation leads to a short product lifecycle and more consumption (Laurenti et al., 2016a), and new technologies do not always completely replace the old ones. Dzombak et al. (2019) state that some products are replaced by newer ones (LED lamps) before the end of life, with the justification of being more efficient, which increases the waste generated

Code: • Negative mechanism > Premature replacement Weight score: 0

Search C - Rebound and circularity theory > C-GS-04, p. 6

Created: 23.6.2026 r. 13:48 by user, Modified: 23.6.2026 r. 13:48 by user

Other codes assigned to segment:

Obsolescence > Functional (Weight: 0)

Obsolescence > Technical (Weight: 0)

Negative mechanism > Frequent upgrading (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

new technology brings a problem of obsolescence. Incremental innovation leads to a short product lifecycle and more consumption (Laurenti et al., 2016a), and new technologies do not always completely replace the old ones. Dzombak et al. (2019) state that some products are replaced by newer ones (LED lamps) before the end of life, with the justification of being more efficient, which increases the waste generated

Code: • Obsolescence > Functional Weight score: 0

Search C - Rebound and circularity theory > C-GS-04, p. 6

Created: 23.6.2026 r. 13:48 by user, Modified: 23.6.2026 r. 13:48 by user

Other codes assigned to segment:

- Obsolescence > Technical (Weight: 0)
- Negative mechanism > Premature replacement (Weight: 0)
- Negative mechanism > Frequent upgrading (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

new technology brings a problem of obsolescence. Incremental innovation leads to a short product lifecycle and more consumption (Laurenti et al., 2016a), and new technologies do not always completely replace the old ones. Dzombak et al. (2019) state that some products are replaced by newer ones (LED lamps) before the end of life, with the justification of being more efficient, which increases the waste generated

Code: • Negative mechanism > Burden shifting Weight score: 0

Search C - Rebound and circularity theory > C-GS-04, p. 6

Created: 23.6.2026 r. 13:48 by user, Modified: 23.6.2026 r. 13:48 by user

Other codes assigned to segment:

- Obsolescence > Functional (Weight: 0)
- Obsolescence > Technical (Weight: 0)
- Negative mechanism > Premature replacement (Weight: 0)
- Negative mechanism > Frequent upgrading (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

new technology brings a problem of obsolescence. Incremental innovation leads to a short product lifecycle and more consumption (Laurenti et al., 2016a), and new technologies do not always completely replace the old ones. Dzombak et al. (2019) state that some products are replaced by newer ones (LED lamps) before the end of life, with the justification of being more efficient, which increases the waste generated

Code: • Obsolescence > Technical Weight score: 0

Search C - Rebound and circularity theory > C-GS-04, p. 6

Created: 23.6.2026 r. 13:48 by user, Modified: 23.6.2026 r. 13:48 by user

Other codes assigned to segment:

- Obsolescence > Functional (Weight: 0)
- Negative mechanism > Premature replacement (Weight: 0)
- Negative mechanism > Frequent upgrading (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

new technology brings a problem of obsolescence. Incremental innovation leads to a short product lifecycle and more consumption (Laurenti et al., 2016a), and new technologies do not always completely replace the old ones. Dzombak et al. (2019) state that some products are replaced by newer ones (LED lamps) before the end of life, with the justification of being more efficient, which increases the waste generated

Code: • Negative mechanism > Frequent upgrading Weight score: 0

Search C - Rebound and circularity theory > C-GS-04, p. 6

Created: 23.6.2026 r. 13:48 by user, Modified: 23.6.2026 r. 13:48 by user

Other codes assigned to segment:

- Obsolescence > Functional (Weight: 0)
- Obsolescence > Technical (Weight: 0)
- Negative mechanism > Premature replacement (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

new technology brings a problem of obsolescence. Incremental innovation leads to a short product lifecycle and more consumption (Laurenti et al., 2016a), and new technologies do not always completely replace the old ones. Dzombak et al. (2019) state that some products are replaced by newer ones (LED lamps) before the end of life, with the justification of being more efficient, which increases the waste generated

Code: • Mitigation > Lifecycle verification Weight score: 0
 Search C - Rebound and circularity theory > C-GS-04, p. 6
 Created: 23.6.2026 r. 13:48 by user, Modified: 23.6.2026 r. 13:48 by user
 Other codes assigned to segment:
 Obsolescence > Functional (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Frequent upgrading (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)

The new business models create a variety of demands for secondary goods and trades. Makov and Font Vivanco et al. (2018) suggest that the existence of a secondary market leads to an increase in consumption, i.e., in the case of smartphones, users can purchase a second-hand unit in addition to the ones they already have. The ecodesign choices also create other needs of new connections within and between firms, and these new connections can initiate a RE within the same system, or as pointed out by Zink and Geyer (2017), they can create a new market system for the end of life of final goods. However, it also creates more interactions, and the results for society are more challenging to anticipate. It is possible to see that the new business models can enable CE efficiency by bringing circular strategies closer to consumers. But Freeman (2018) goes further and points out that the interaction between mechanisms to improve efficiency is correlated – at the very least – to the creation of entirely new industries and new rebound effects.

Code: • Mitigation > Business-model intervention Weight score: 0
 Search C - Rebound and circularity theory > C-GS-04, p. 6
 Created: 23.6.2026 r. 13:50 by user, Modified: 23.6.2026 r. 13:50 by user
 Other codes assigned to segment:
 Enabling condition > Business model (Weight: 0)
 Negative mechanism > Increased comfort or use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The new business models create a variety of demands for secondary goods and trades. Makov and Font Vivanco et al. (2018) suggest that the existence of a secondary market leads to an increase in consumption, i.e., in the case of smartphones, users can purchase a second-hand unit in addition to the ones they already have. The ecodesign choices also create other needs of new connections within and between firms, and these new connections can initiate a RE within the same system, or as pointed out by Zink and Geyer (2017), they can create a new market system for the end of life of final goods. However, it also creates more interactions, and the results for society are more challenging to anticipate. It is possible to see that the new business models can enable CE efficiency by bringing circular strategies closer to consumers. But Freeman (2018) goes further and points out that the interaction between mechanisms to improve efficiency is correlated – at the very least – to the creation of entirely new industries and new rebound effects.

Code: • Negative mechanism > Increased comfort or use Weight score: 0
 Search C - Rebound and circularity theory > C-GS-04, p. 6
 Created: 23.6.2026 r. 13:49 by user, Modified: 23.6.2026 r. 13:49 by user
 Other codes assigned to segment:
 Enabling condition > Business model (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Business-model intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The new business models create a variety of demands for secondary goods and trades. Makov and Font Vivanco et al. (2018) suggest that the existence of a secondary market leads to an increase in consumption, i.e., in the case of smartphones, users can purchase a second-hand unit in addition to the ones they already have. The ecodesign choices also create other needs of new connections within and between firms, and these new connections can initiate a RE within the same system, or as pointed out by Zink and Geyer (2017), they can create a new market system for the end of life of final goods. However, it also creates more interactions, and the results for society are more challenging to anticipate. It is possible to see that the new business models can enable CE efficiency by bringing circular strategies closer to consumers. But Freeman (2018) goes further and points out that the interaction between mechanisms to improve efficiency is correlated – at the very least – to the creation of entirely new industries and new rebound effects.

Code: • Enabling condition > Business model Weight score: 0

Search C - Rebound and circularity theory > C-GS-04, p. 6

Created: 23.6.2026 r. 13:49 by user, Modified: 23.6.2026 r. 13:49 by user

Other codes assigned to segment:

Negative mechanism > Increased comfort or use (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Business-model intervention (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The new business models create a variety of demands for secondary goods and trades. Makov and Font Vivanco et al. (2018) suggest that the existence of a secondary market leads to an increase in consumption, i.e., in the case of smartphones, users can purchase a second-hand unit in addition to the ones they already have. The ecodesign choices also create other needs of new connections within and between firms, and these new connections can initiate a RE within the same system, or as pointed out by Zink and Geyer (2017), they can create a new market system for the end of life of final goods. However, it also creates more interactions, and the results for society are more challenging to anticipate. It is possible to see that the new business models can enable CE efficiency by bringing circular strategies closer to consumers. But Freeman (2018) goes further and points out that the interaction between mechanisms to improve efficiency is correlated – at the very least – to the creation of entirely new industries and new rebound effects.

Code: • Negative mechanism > Burden shifting Weight score: 0

Search C - Rebound and circularity theory > C-GS-04, p. 6

Created: 23.6.2026 r. 13:49 by user, Modified: 23.6.2026 r. 13:49 by user

Other codes assigned to segment:

Enabling condition > Business model (Weight: 0)

Negative mechanism > Increased comfort or use (Weight: 0)

Mitigation > Business-model intervention (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The new business models create a variety of demands for secondary goods and trades. Makov and Font Vivanco et al. (2018) suggest that the existence of a secondary market leads to an increase in consumption, i.e., in the case of smartphones, users can purchase a second-hand unit in addition to the ones they already have. The ecodesign choices also create other needs of new connections within and between firms, and these new connections can initiate a RE within the same system, or as pointed out by Zink and Geyer (2017), they can create a new market system for the end of life of final goods. However, it also creates more interactions, and the results for society are more challenging to anticipate. It is possible to see that the new business models can enable CE efficiency by bringing circular strategies closer to consumers. But Freeman (2018) goes further and points out that the interaction between mechanisms to improve efficiency is correlated – at the very least – to the creation of entirely new industries and new rebound effects.

Code: • Mitigation > Lifecycle verification Weight score: 0
 Search C - Rebound and circularity theory > C-GS-04, p. 6
 Created: 23.6.2026 r. 13:50 by user, Modified: 23.6.2026 r. 13:50 by user
 Other codes assigned to segment:
 Enabling condition > Business model (Weight: 0)
 Negative mechanism > Increased comfort or use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Business-model intervention (Weight: 0)

none of the current circularity metrics are addressing all the predefined requirements. Most of the CE measurement indices and single assessment indicators are measuring progress in only a few of the defined goals, and none of them fulfil the requirements regarding emissions levels, employment, and socio-economic improvements measurement (except for the LCA-derived assessment indicators). This narrow scope of the circularity metrics is a shortcoming, since metrics that exclusively measure progress in a few CE goals could lead to burden shifting. For instance, a circular strategy may increase resource recirculation at the expense of additional environmental burden elsewhere, as reported by several case studies, e.g. on organic waste treatment (Cobo et al. (2018) and concrete recycling (Cullen, 2017). Circular strategies may also increase circularity by exporting materials for recycling in places with low environmental and health standards, increasing social impacts somewhere else

Code: • Circular strategy > Recycling Weight score: 0
 Search C - Rebound and circularity theory > C-GS-10, p. 9
 Created: 23.6.2026 r. 14:04 by user, Modified: 23.6.2026 r. 14:04 by user
 Other codes assigned to segment:
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

none of the current circularity metrics are addressing all the predefined requirements. Most of the CE measurement indices and single assessment indicators are measuring progress in only a few of the defined goals, and none of them fulfil the requirements regarding emissions levels, employment, and socio-economic improvements measurement (except for the LCA-derived assessment indicators). This narrow scope of the circularity metrics is a shortcoming, since metrics that exclusively measure progress in a few CE goals could lead to burden shifting. For instance, a circular strategy may increase resource recirculation at the expense of additional environmental burden elsewhere, as reported by several case studies, e.g. on organic waste treatment (Cobo et al. (2018) and concrete recycling (Cullen, 2017). Circular strategies may also increase circularity by exporting materials for recycling in places with low environmental and health standards, increasing social impacts somewhere else

Code: • Negative mechanism > Burden shifting Weight score: 0
 Search C - Rebound and circularity theory > C-GS-10, p. 9
 Created: 23.6.2026 r. 14:03 by user, Modified: 23.6.2026 r. 14:03 by user
 Other codes assigned to segment:
 Circular strategy > Recycling (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

none of the current circularity metrics are addressing all the predefined requirements. Most of the CE measurement indices and single assessment indicators are measuring progress in only a few of the defined goals, and none of them fulfil the requirements regarding emissions levels, employment, and socio-economic improvements measurement (except for the LCA-derived assessment indicators). This narrow scope of the circularity metrics is a shortcoming, since metrics that exclusively measure progress in a few CE goals could lead to burden shifting. For instance, a circular strategy may increase resource recirculation at the expense of additional environmental burden

elsewhere, as reported by several case studies, e.g. on organic waste treatment (Cobo et al. (2018) and concrete recycling (Cullen, 2017). Circular strategies may also increase circularity by exporting materials for recycling in places with low environmental and health standards, increasing social impacts somewhere else

Code: • Mitigation > Lifecycle verification Weight score: 0
 Search C - Rebound and circularity theory > C-GS-10, p. 9
 Created: 23.6.2026 r. 14:03 by user, Modified: 23.6.2026 r. 14:03 by user
 Other codes assigned to segment:
 Circular strategy > Recycling (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)

One of the main challenges in measuring the circularity of products and services is related to the allocation of impacts between the initial cycles of products (or materials) and their subsequent recycled or recovered cycles. This issue links with the “open-loop recycling” problem described within the LCA methodology in Section 3.2.1. In cases where the materials are recycled into the same products/functions, these credits are translated into reduced virgin material use (i.e. in a closed loop). However, when materials are recycled into new products/functions (e.g. open-loop recycling), the reviewed metrics have different approaches to allocate the credits between the recycler and the user of recycled materials, and to account for the decrease of quality (downcycling).

Code: • Negative mechanism > Burden shifting Weight score: 0
 Search C - Rebound and circularity theory > C-GS-10, p. 11
 Created: 23.6.2026 r. 14:04 by user, Modified: 23.6.2026 r. 14:04 by user
 Other codes assigned to segment:
 Circular strategy > Recycling (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

One of the main challenges in measuring the circularity of products and services is related to the allocation of impacts between the initial cycles of products (or materials) and their subsequent recycled or recovered cycles. This issue links with the “open-loop recycling” problem described within the LCA methodology in Section 3.2.1. In cases where the materials are recycled into the same products/functions, these credits are translated into reduced virgin material use (i.e. in a closed loop). However, when materials are recycled into new products/functions (e.g. open-loop recycling), the reviewed metrics have different approaches to allocate the credits between the recycler and the user of recycled materials, and to account for the decrease of quality (downcycling).

Code: • Circular strategy > Recycling Weight score: 0
 Search C - Rebound and circularity theory > C-GS-10, p. 11
 Created: 23.6.2026 r. 14:04 by user, Modified: 23.6.2026 r. 14:04 by user
 Other codes assigned to segment:
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

One of the main challenges in measuring the circularity of products and services is related to the allocation of impacts between the initial cycles of products (or materials) and their subsequent recycled or recovered cycles. This issue links with the “open-loop recycling” problem described within the LCA methodology in Section 3.2.1. In cases where the materials are recycled into the same products/functions, these credits are translated into reduced virgin material use (i.e. in a closed loop). However, when materials are recycled into new products/functions (e.g. open-loop recycling), the reviewed metrics have different approaches to allocate the credits between the recycler and the user of recycled materials, and to account for the decrease of quality (downcycling).

Code: • Mitigation > Lifecycle verification Weight score: 0
 Search C - Rebound and circularity theory > C-GS-10, p. 11
 Created: 23.6.2026 r. 14:04 by user, Modified: 23.6.2026 r. 14:04 by user

Other codes assigned to segment:

Circular strategy > Recycling (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

The general tendency in the reviewed metrics is to give credits to the products that are being recycled, i.e. the 0:100 approach with substitution. However, some metrics give credits to the products using the recycled material (i.e. the 100:00 or cut-off approach), and in many cases, the approach is not clear or specified.

Code: • Negative mechanism > Burden shifting Weight score: 0

Search C - Rebound and circularity theory > C-GS-10, p. 11

Created: 23.6.2026 r. 14:05 by user, Modified: 23.6.2026 r. 14:05 by user

Other codes assigned to segment:

Circular strategy > Recycling (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The general tendency in the reviewed metrics is to give credits to the products that are being recycled, i.e. the 0:100 approach with substitution. However, some metrics give credits to the products using the recycled material (i.e. the 100:00 or cut-off approach), and in many cases, the approach is not clear or specified.

Code: • Circular strategy > Recycling Weight score: 0

Search C - Rebound and circularity theory > C-GS-10, p. 11

Created: 23.6.2026 r. 14:05 by user, Modified: 23.6.2026 r. 14:05 by user

Other codes assigned to segment:

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The general tendency in the reviewed metrics is to give credits to the products that are being recycled, i.e. the 0:100 approach with substitution. However, some metrics give credits to the products using the recycled material (i.e. the 100:00 or cut-off approach), and in many cases, the approach is not clear or specified.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search C - Rebound and circularity theory > C-GS-10, p. 11

Created: 23.6.2026 r. 14:05 by user, Modified: 23.6.2026 r. 14:05 by user

Other codes assigned to segment:

Circular strategy > Recycling (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

the major challenges of current circularity metrics relate to (1) difficulties in measuring the CE goals in all the sustainability dimensions, (2) evaluating the scarcity of used materials, and (2) underrepresenting the complexities of multiple cycles (multifunctionality) and the consequences of material downcycling.

Code: • Circular strategy > Direct reuse Weight score: 0

Search C - Rebound and circularity theory > C-GS-10, p. 13

Created: 23.6.2026 r. 14:07 by user, Modified: 23.6.2026 r. 14:07 by user

Other codes assigned to segment:

Circular strategy > Recycling (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

the major challenges of current circularity metrics relate to (1) difficulties in measuring the CE goals in all the sustainability dimensions, (2) evaluating the scarcity of used materials, and (2)

underrepresenting the complexities of multiple cycles (multifunctionality) and the consequences of material downcycling.

Code: • Negative mechanism > Burden shifting Weight score: 0

Search C - Rebound and circularity theory > C-GS-10, p. 13

Created: 23.6.2026 r. 14:07 by user, Modified: 23.6.2026 r. 14:07 by user

Other codes assigned to segment:

Circular strategy > Direct reuse (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

the major challenges of current circularity metrics relate to (1) difficulties in measuring the CE goals in all the sustainability dimensions, (2) evaluating the scarcity of used materials, and (2) underrepresenting the complexities of multiple cycles (multifunctionality) and the consequences of material downcycling.

Code: • Circular strategy > Recycling Weight score: 0

Search C - Rebound and circularity theory > C-GS-10, p. 13

Created: 23.6.2026 r. 14:07 by user, Modified: 23.6.2026 r. 14:07 by user

Other codes assigned to segment:

Circular strategy > Direct reuse (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

the major challenges of current circularity metrics relate to (1) difficulties in measuring the CE goals in all the sustainability dimensions, (2) evaluating the scarcity of used materials, and (2) underrepresenting the complexities of multiple cycles (multifunctionality) and the consequences of material downcycling.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search C - Rebound and circularity theory > C-GS-10, p. 13

Created: 23.6.2026 r. 14:07 by user, Modified: 23.6.2026 r. 14:07 by user

Other codes assigned to segment:

Circular strategy > Direct reuse (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

a good circularity metric, aiming to measure the contribution of circular strategies to sustainable development, should be comprehensive enough to avoid burden shifting from reduced material consumption to increased environmental, economic or social impacts.

Code: • Negative mechanism > Burden shifting Weight score: 0

Search C - Rebound and circularity theory > C-GS-10, p. 13

Created: 23.6.2026 r. 14:08 by user, Modified: 23.6.2026 r. 14:08 by user

Other codes assigned to segment:

Mitigation > Lifecycle verification (Weight: 0)

a good circularity metric, aiming to measure the contribution of circular strategies to sustainable development, should be comprehensive enough to avoid burden shifting from reduced material consumption to increased environmental, economic or social impacts.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search C - Rebound and circularity theory > C-GS-10, p. 13

Created: 23.6.2026 r. 14:08 by user, Modified: 23.6.2026 r. 14:08 by user

Other codes assigned to segment:

Negative mechanism > Burden shifting (Weight: 0)

Additionally, most EPDs are cradle-to-gate assessments, which leave out the question of use phase impacts entirely. This trend can, in part, be explained by the difficulty of approximating an accurate service life for window frames—a long-life product whose replacement can hinge on a number of factors, from esthetics to performance to user needs. Although a product manufacturer may wish to provide a set of default scenarios in an EPD from which the building assessor may choose, the number of potential influencing factors on the use phase makes this impractical, due to the large number of combinations of factors related to climate, wear, and maintenance practices. However, when examining material use on buildings that must achieve high-performance standards for 80–100 years, the assumed service life of a component such as window frames does matter, as a single full replacement of an assembly will effectively double the product's life cycle impacts.

Code: • Circular strategy > Maintenance Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-01, p. 2
 Created: 23.6.2026 r. 14:12 by user, Modified: 23.6.2026 r. 14:12 by user
 Other codes assigned to segment:
 Circular strategy > Refurbishment (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Additionally, most EPDs are cradle-to-gate assessments, which leave out the question of use phase impacts entirely. This trend can, in part, be explained by the difficulty of approximating an accurate service life for window frames—a long-life product whose replacement can hinge on a number of factors, from esthetics to performance to user needs. Although a product manufacturer may wish to provide a set of default scenarios in an EPD from which the building assessor may choose, the number of potential influencing factors on the use phase makes this impractical, due to the large number of combinations of factors related to climate, wear, and maintenance practices. However, when examining material use on buildings that must achieve high-performance standards for 80–100 years, the assumed service life of a component such as window frames does matter, as a single full replacement of an assembly will effectively double the product's life cycle impacts.

Code: • Negative mechanism > Premature replacement Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-01, p. 2
 Created: 23.6.2026 r. 14:12 by user, Modified: 23.6.2026 r. 14:12 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Additionally, most EPDs are cradle-to-gate assessments, which leave out the question of use phase impacts entirely. This trend can, in part, be explained by the difficulty of approximating an accurate service life for window frames—a long-life product whose replacement can hinge on a number of factors, from esthetics to performance to user needs. Although a product manufacturer may wish to provide a set of default scenarios in an EPD from which the building assessor may choose, the number of potential influencing factors on the use phase makes this impractical, due to the large number of combinations of factors related to climate, wear, and maintenance practices. However, when examining material use on buildings that must achieve high-performance standards for 80–100 years, the assumed service life of a component such as window frames does matter, as a single full replacement of an assembly will effectively double the product's life cycle impacts.

Code: • Obsolescence > Physical Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-01, p. 2
 Created: 23.6.2026 r. 14:12 by user, Modified: 23.6.2026 r. 14:12 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Additionally, most EPDs are cradle-to-gate assessments, which leave out the question of use phase impacts entirely. This trend can, in part, be explained by the difficulty of approximating an accurate service life for window frames—a long-life product whose replacement can hinge on a number of factors, from esthetics to performance to user needs. Although a product manufacturer may wish to provide a set of default scenarios in an EPD from which the building assessor may choose, the number of potential influencing factors on the use phase makes this impractical, due to the large number of combinations of factors related to climate, wear, and maintenance practices. However, when examining material use on buildings that must achieve high-performance standards for 80–100 years, the assumed service life of a component such as window frames does matter, as a single full replacement of an assembly will effectively double the product's life cycle impacts.

Code: • Circular strategy > Refurbishment Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-01, p. 2

Created: 23.6.2026 r. 14:12 by user, Modified: 23.6.2026 r. 14:12 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Additionally, most EPDs are cradle-to-gate assessments, which leave out the question of use phase impacts entirely. This trend can, in part, be explained by the difficulty of approximating an accurate service life for window frames—a long-life product whose replacement can hinge on a number of factors, from esthetics to performance to user needs. Although a product manufacturer may wish to provide a set of default scenarios in an EPD from which the building assessor may choose, the number of potential influencing factors on the use phase makes this impractical, due to the large number of combinations of factors related to climate, wear, and maintenance practices. However, when examining material use on buildings that must achieve high-performance standards for 80–100 years, the assumed service life of a component such as window frames does matter, as a single full replacement of an assembly will effectively double the product's life cycle impacts.

Code: • Negative mechanism > Higher initial material use Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-01, p. 2

Created: 23.6.2026 r. 14:12 by user, Modified: 23.6.2026 r. 14:12 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Additionally, most EPDs are cradle-to-gate assessments, which leave out the question of use phase impacts entirely. This trend can, in part, be explained by the difficulty of approximating an accurate service life for window frames—a long-life product whose replacement can hinge on a number of factors, from esthetics to performance to user needs. Although a product manufacturer may wish to provide a set of default scenarios in an EPD from which the building assessor may choose, the number of potential influencing factors on the use phase makes this impractical, due to the large number of combinations of factors related to climate, wear, and maintenance practices. However,

when examining material use on buildings that must achieve high-performance standards for 80–100 years, the assumed service life of a component such as window frames does matter, as a single full replacement of an assembly will effectively double the product's life cycle impacts.

Code: • Mitigation > Lifecycle verification Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-01, p. 2
 Created: 23.6.2026 r. 14:12 by user, Modified: 23.6.2026 r. 14:12 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)

The model uses an incremental allocation methodology to account for periodic component replacements associated with functional failure and wear rather than a pre-emptive, timed replacement of whole frames across a building. This methodology is consistent with an approach to maintenance scenarios that maintains functional equivalence across assembly types and practices based on consistent window performance throughout the full building life cycle, rather than assuming a full replacement of every window in a building on a timed cycle.

Code: • Mitigation > Maintenance planning Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-01, p. 6
 Created: 23.6.2026 r. 14:14 by user, Modified: 23.6.2026 r. 14:14 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Component replacement (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The model uses an incremental allocation methodology to account for periodic component replacements associated with functional failure and wear rather than a pre-emptive, timed replacement of whole frames across a building. This methodology is consistent with an approach to maintenance scenarios that maintains functional equivalence across assembly types and practices based on consistent window performance throughout the full building life cycle, rather than assuming a full replacement of every window in a building on a timed cycle.

Code: • Obsolescence > Physical Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-01, p. 6
 Created: 23.6.2026 r. 14:14 by user, Modified: 23.6.2026 r. 14:14 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Repair (Weight: 0)
 Circular strategy > Component replacement (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The model uses an incremental allocation methodology to account for periodic component replacements associated with functional failure and wear rather than a pre-emptive, timed replacement of whole frames across a building. This methodology is consistent with an approach to maintenance scenarios that maintains functional equivalence across assembly types and practices based on consistent window performance throughout the full building life cycle, rather than assuming a full replacement of every window in a building on a timed cycle.

Code: • Circular strategy > Maintenance Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-01, p. 6
 Created: 23.6.2026 r. 14:13 by user, Modified: 23.6.2026 r. 14:13 by user

Other codes assigned to segment:

- Circular strategy > Repair (Weight: 0)
- Circular strategy > Component replacement (Weight: 0)
- Obsolescence > Physical (Weight: 0)
- Mitigation > Maintenance planning (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

The model uses an incremental allocation methodology to account for periodic component replacements associated with functional failure and wear rather than a pre-emptive, timed replacement of whole frames across a building. This methodology is consistent with an approach to maintenance scenarios that maintains functional equivalence across assembly types and practices based on consistent window performance throughout the full building life cycle, rather than assuming a full replacement of every window in a building on a timed cycle.

Code: • Circular strategy > Repair Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-01, p. 6

Created: 23.6.2026 r. 14:13 by user, Modified: 23.6.2026 r. 14:13 by user

Other codes assigned to segment:

- Circular strategy > Maintenance (Weight: 0)
- Circular strategy > Component replacement (Weight: 0)
- Obsolescence > Physical (Weight: 0)
- Mitigation > Maintenance planning (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

The model uses an incremental allocation methodology to account for periodic component replacements associated with functional failure and wear rather than a pre-emptive, timed replacement of whole frames across a building. This methodology is consistent with an approach to maintenance scenarios that maintains functional equivalence across assembly types and practices based on consistent window performance throughout the full building life cycle, rather than assuming a full replacement of every window in a building on a timed cycle.

Code: • Circular strategy > Component replacement Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-01, p. 6

Created: 23.6.2026 r. 14:13 by user, Modified: 23.6.2026 r. 14:13 by user

Other codes assigned to segment:

- Circular strategy > Maintenance (Weight: 0)
- Circular strategy > Repair (Weight: 0)
- Obsolescence > Physical (Weight: 0)
- Mitigation > Maintenance planning (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

The model uses an incremental allocation methodology to account for periodic component replacements associated with functional failure and wear rather than a pre-emptive, timed replacement of whole frames across a building. This methodology is consistent with an approach to maintenance scenarios that maintains functional equivalence across assembly types and practices based on consistent window performance throughout the full building life cycle, rather than assuming a full replacement of every window in a building on a timed cycle.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-01, p. 6

Created: 23.6.2026 r. 14:14 by user, Modified: 23.6.2026 r. 14:14 by user

Other codes assigned to segment:

- Circular strategy > Maintenance (Weight: 0)
- Circular strategy > Repair (Weight: 0)
- Circular strategy > Component replacement (Weight: 0)
- Obsolescence > Physical (Weight: 0)
- Mitigation > Maintenance planning (Weight: 0)

As aluminum-clad wood and PVCu window frames are intended to be maintenance-free products with factory-finished surfaces, the protective

Table 1 Maintenance activities for aluminum, wood, aluminum-clad wood, and PVCu window frames

Scenario 1: warranty period, little to no maintenance

Scenario 2: low maintenance, shorter life

Scenario 3: high maintenance, longer life

Anodized aluminum 25 years: full frame replacement No cleaning of frames

20 years: replace weather sealing 2 %/year: replace hardware 60 years: full frame replacement

Annual cleaning of frames 20 years: replace weather sealing 2 %/year: replace hardware +80 years:

full frame replacement Painted wood 20 years: full frame replacement 12 years: repaint when coatings have failed 15 years: replace weather sealing 2 %/year: replace hardware 30 years: full frame replacement

8 years: repaint and treat wood on regular basis to prevent failure

15 years: replace weather sealing 2 %/year: replace hardware +80 years: full frame replacement

Aluminum-clad wood

20 years: full frame replacement No refinishing 15 years: replace weather sealing 2 %/year: replace hardware 30 years: full frame replacement

Not viable due to inability to reinforce

factory-sealed joint between materials with additional maintenance

PVCu 20 years: full frame replacement No refinishing

15 years: replace weather sealing 2 %/year: replace hardware 30 years: full frame replacement

Not viable due to inability to reapply or prolong

lifespan of UV-inhibiting coating that protects primary material

1650 Int J Life Cycle Assess (2016) 21:1645–1657 finishes are expected to degrade over time and cannot be renewed through additional maintenance, leading to base material failure. In the case of aluminum-clad windows, this failure occurs as joints in the aluminum facings open, allowing moisture penetration to the underlying wood that leads to rot. In PVCu windows, the UV-inhibiting coating breaks down, causing the PVC to become brittle and crack.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-01, p. 6 - 7

Created: 23.6.2026 r. 14:15 by user, Modified: 23.6.2026 r. 14:15 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Obsolescence > Physical (Weight: 0)

Obsolescence > Technical (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Negative mechanism > Lock-in (Weight: 0)

As aluminum-clad wood and PVCu window frames are intended to be maintenance-free products with factory-finished surfaces, the protective

Table 1 Maintenance activities for aluminum, wood, aluminum-clad wood, and PVCu window frames

Scenario 1: warranty period, little to no maintenance

Scenario 2: low maintenance, shorter life

Scenario 3: high maintenance, longer life

Anodized aluminum 25 years: full frame replacement No cleaning of frames

20 years: replace weather sealing 2 %/year: replace hardware 60 years: full frame replacement

Annual cleaning of frames 20 years: replace weather sealing 2 %/year: replace hardware +80 years:

full frame replacement Painted wood 20 years: full frame replacement 12 years: repaint when coatings have failed 15 years: replace weather sealing 2 %/year: replace hardware 30 years: full frame replacement

8 years: repaint and treat wood on regular basis to prevent failure

15 years: replace weather sealing 2 %/year: replace hardware +80 years: full frame replacement

Aluminum-clad wood

20 years: full frame replacement No refinishing 15 years: replace weather sealing 2 %/year: replace hardware 30 years: full frame replacement

Not viable due to inability to reinforce

factory-sealed joint between materials with additional maintenance

PVCu 20 years: full frame replacement No refinishing

15 years: replace weather sealing 2 %/year: replace hardware 30 years: full frame replacement

Not viable due to inability to reapply or prolong

lifespan of UV-inhibiting coating that protects primary material

1650 Int J Life Cycle Assess (2016) 21:1645–1657 finishes are expected to degrade over time and cannot be renewed through additional maintenance, leading to base material failure. In the case of aluminum-clad windows, this failure occurs as joints in the aluminum facings open, allowing moisture penetration to the underlying wood that leads to rot. In PVCu windows, the UV-inhibiting coating breaks down, causing the PVC to become brittle and crack.

Code: • [Obsolescence](#) > [Technical](#) Weight score: 0

[Search D - Window LCA and replacement decisions](#) > [D-GS-01](#), p. 6 - 7

Created: 23.6.2026 r. 14:14 by user, Modified: 23.6.2026 r. 14:14 by user

Other codes assigned to segment:

[Circular strategy](#) > [Maintenance](#) (Weight: 0)

[Obsolescence](#) > [Physical](#) (Weight: 0)

[Negative mechanism](#) > [Premature replacement](#) (Weight: 0)

[Negative mechanism](#) > [Lock-in](#) (Weight: 0)

[Mitigation](#) > [Lifecycle verification](#) (Weight: 0)

As aluminum-clad wood and PVCu window frames are intended to be maintenance-free products with factory-finished surfaces, the protective

Table 1 Maintenance activities for aluminum, wood, aluminum-clad wood, and PVCu window frames

Scenario 1: warranty period, little to no maintenance

Scenario 2: low maintenance, shorter life

Scenario 3: high maintenance, longer life

Anodized aluminum 25 years: full frame replacement No cleaning of frames

20 years: replace weather sealing 2 %/year: replace hardware 60 years: full frame replacement

Annual cleaning of frames 20 years: replace weather sealing 2 %/year: replace hardware +80 years: full frame replacement Painted wood 20 years: full frame replacement 12 years: repaint when coatings have failed 15 years: replace weather sealing 2 %/year: replace hardware 30 years: full frame replacement

8 years: repaint and treat wood on regular basis to prevent failure

15 years: replace weather sealing 2 %/year: replace hardware +80 years: full frame replacement

Aluminum-clad wood

20 years: full frame replacement No refinishing 15 years: replace weather sealing 2 %/year: replace hardware 30 years: full frame replacement

Not viable due to inability to reinforce

factory-sealed joint between materials with additional maintenance

PVCu 20 years: full frame replacement No refinishing

15 years: replace weather sealing 2 %/year: replace hardware 30 years: full frame replacement

Not viable due to inability to reapply or prolong

lifespan of UV-inhibiting coating that protects primary material

1650 Int J Life Cycle Assess (2016) 21:1645–1657 finishes are expected to degrade over time and cannot be renewed through additional maintenance, leading to base material failure. In the case of aluminum-clad windows, this failure occurs as joints in the aluminum facings open, allowing moisture penetration to the underlying wood that leads to rot. In PVCu windows, the UV-inhibiting coating breaks down, causing the PVC to become brittle and crack.

Code: • Negative mechanism > Lock-in Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-01, p. 6 - 7
 Created: 23.6.2026 r. 14:15 by user, Modified: 23.6.2026 r. 14:15 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

As aluminum-clad wood and PVCu window frames are intended to be maintenance-free products with factory-finished surfaces, the protective

Table 1 Maintenance activities for aluminum, wood, aluminum-clad wood, and PVCu window frames

Scenario 1: warranty period, little to no maintenance

Scenario 2: low maintenance, shorter life

Scenario 3: high maintenance, longer life

Anodized aluminum 25 years: full frame replacement No cleaning of frames

20 years: replace weather sealing 2 %/year: replace hardware 60 years: full frame replacement

Annual cleaning of frames 20 years: replace weather sealing 2 %/year: replace hardware +80 years:

full frame replacement Painted wood 20 years: full frame replacement 12 years: repaint when coatings have failed 15 years: replace weather sealing 2 %/year: replace hardware 30 years: full frame replacement

8 years: repaint and treat wood on regular basis to prevent failure

15 years: replace weather sealing 2 %/year: replace hardware +80 years: full frame replacement

Aluminum-clad wood

20 years: full frame replacement No refinishing 15 years: replace weather sealing 2 %/year: replace hardware 30 years: full frame replacement

Not viable due to inability to reinforce

factory-sealed joint between materials with additional maintenance

PVCu 20 years: full frame replacement No refinishing

15 years: replace weather sealing 2 %/year: replace hardware 30 years: full frame replacement

Not viable due to inability to reapply or prolong

lifespan of UV-inhibiting coating that protects primary material

1650 Int J Life Cycle Assess (2016) 21:1645–1657 finishes are expected to degrade over time and cannot be renewed through additional maintenance, leading to base material failure. In the case of aluminum-clad windows, this failure occurs as joints in the aluminum facings open, allowing moisture penetration to the underlying wood that leads to rot. In PVCu windows, the UV-inhibiting coating breaks down, causing the PVC to become brittle and crack.

Code: • Circular strategy > Maintenance Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-01, p. 6 - 7
 Created: 23.6.2026 r. 14:15 by user, Modified: 23.6.2026 r. 14:15 by user
 Other codes assigned to segment:
 Obsolescence > Physical (Weight: 0)
 Obsolescence > Technical (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Lock-in (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

As aluminum-clad wood and PVCu window frames are intended to be maintenance-free products with factory-finished surfaces, the protective

Table 1 Maintenance activities for aluminum, wood, aluminum-clad wood, and PVCu window frames

Scenario 1: warranty period, little to no maintenance

Scenario 2: low maintenance, shorter life

Scenario 3: high maintenance, longer life

Anodized aluminum 25 years: full frame replacement No cleaning of frames

20 years: replace weather sealing 2 %/year: replace hardware 60 years: full frame replacement

Annual cleaning of frames 20 years: replace weather sealing 2 %/year: replace hardware +80 years:

full frame replacement Painted wood 20 years: full frame replacement 12 years: repaint when coatings have failed 15 years: replace weather sealing 2 %/year: replace hardware 30 years: full frame replacement

8 years: repaint and treat wood on regular basis to prevent failure

15 years: replace weather sealing 2 %/year: replace hardware +80 years: full frame replacement

Aluminum-clad wood

20 years: full frame replacement No refinishing 15 years: replace weather sealing 2 %/year: replace hardware 30 years: full frame replacement

Not viable due to inability to reinforce

factory-sealed joint between materials with additional maintenance

PVCu 20 years: full frame replacement No refinishing

15 years: replace weather sealing 2 %/year: replace hardware 30 years: full frame replacement

Not viable due to inability to reapply or prolong

lifespan of UV-inhibiting coating that protects primary material

1650 Int J Life Cycle Assess (2016) 21:1645–1657 finishes are expected to degrade over time and cannot be renewed through additional maintenance, leading to base material failure. In the case of aluminum-clad windows, this failure occurs as joints in the aluminum facings open, allowing moisture penetration to the underlying wood that leads to rot. In PVCu windows, the UV-inhibiting coating breaks down, causing the PVC to become brittle and crack.

Code: • Negative mechanism > Premature replacement Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-01, p. 6 - 7

Created: 23.6.2026 r. 14:15 by user, Modified: 23.6.2026 r. 14:15 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Obsolescence > Physical (Weight: 0)

Obsolescence > Technical (Weight: 0)

Negative mechanism > Lock-in (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

As aluminum-clad wood and PVCu window frames are intended to be maintenance-free products with factory-finished surfaces, the protective

Table 1 Maintenance activities for aluminum, wood, aluminum-clad wood, and PVCu window frames

Scenario 1: warranty period, little to no maintenance

Scenario 2: low maintenance, shorter life

Scenario 3: high maintenance, longer life

Anodized aluminum 25 years: full frame replacement No cleaning of frames

20 years: replace weather sealing 2 %/year: replace hardware 60 years: full frame replacement

Annual cleaning of frames 20 years: replace weather sealing 2 %/year: replace hardware +80 years:

full frame replacement Painted wood 20 years: full frame replacement 12 years: repaint when coatings have failed 15 years: replace weather sealing 2 %/year: replace hardware 30 years: full frame replacement

8 years: repaint and treat wood on regular basis to prevent failure

15 years: replace weather sealing 2 %/year: replace hardware +80 years: full frame replacement

Aluminum-clad wood

20 years: full frame replacement No refinishing 15 years: replace weather sealing 2 %/year: replace hardware 30 years: full frame replacement

Not viable due to inability to reinforce

factory-sealed joint between materials with additional maintenance

PVCu 20 years: full frame replacement No refinishing

15 years: replace weather sealing 2 %/year: replace hardware 30 years: full frame replacement

Not viable due to inability to reapply or prolong

lifespan of UV-inhibiting coating that protects primary material

1650 Int J Life Cycle Assess (2016) 21:1645–1657 finishes are expected to degrade over time and cannot be renewed through additional maintenance, leading to base material failure. In the case of aluminum-clad windows, this failure occurs as joints in the aluminum facings open, allowing moisture penetration to the underlying wood that leads to rot. In PVCu windows, the UV-inhibiting coating breaks down, causing the PVC to become brittle and crack.

Code: • [Obsolescence > Physical](#) Weight score: 0

[Search D - Window LCA and replacement decisions > D-GS-01](#), p. 6 - 7

Created: 23.6.2026 r. 14:15 by user, Modified: 23.6.2026 r. 14:15 by user

Other codes assigned to segment:

[Circular strategy > Maintenance](#) (Weight: 0)

[Obsolescence > Technical](#) (Weight: 0)

[Negative mechanism > Premature replacement](#) (Weight: 0)

[Negative mechanism > Lock-in](#) (Weight: 0)

[Mitigation > Lifecycle verification](#) (Weight: 0)

Each modeling practice is internally consistent, but represents a different view of the relationship between primary and recycled material by focusing either on the recycled content going into the product stream or the recovered material coming out (Frischknecht 2010), as described in Table 2. This methodological focus affects the environmental impact results of a study, influencing recommendations for policy and decision-making.

Code: • [Negative mechanism > Burden shifting](#) Weight score: 0

[Search D - Window LCA and replacement decisions > D-GS-01](#), p. 7

Created: 23.6.2026 r. 14:16 by user, Modified: 23.6.2026 r. 14:16 by user

Other codes assigned to segment:

[Circular strategy > Recycling](#) (Weight: 0)

[Mitigation > Lifecycle verification](#) (Weight: 0)

Each modeling practice is internally consistent, but represents a different view of the relationship between primary and recycled material by focusing either on the recycled content going into the product stream or the recovered material coming out (Frischknecht 2010), as described in Table 2. This methodological focus affects the environmental impact results of a study, influencing recommendations for policy and decision-making.

Code: • [Circular strategy > Recycling](#) Weight score: 0

[Search D - Window LCA and replacement decisions > D-GS-01](#), p. 7

Created: 23.6.2026 r. 14:16 by user, Modified: 23.6.2026 r. 14:16 by user

Other codes assigned to segment:

[Negative mechanism > Burden shifting](#) (Weight: 0)

[Mitigation > Lifecycle verification](#) (Weight: 0)

Each modeling practice is internally consistent, but represents a different view of the relationship between primary and recycled material by focusing either on the recycled content going into the product stream or the recovered material coming out (Frischknecht 2010), as described in Table 2. This methodological focus affects the environmental impact results of a study, influencing recommendations for policy and decision-making.

Code: • [Mitigation > Lifecycle verification](#) Weight score: 0

[Search D - Window LCA and replacement decisions > D-GS-01](#), p. 7

Created: 23.6.2026 r. 14:16 by user, Modified: 23.6.2026 r. 14:16 by user

Other codes assigned to segment:

Circular strategy > Recycling (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Differences between use scenarios within each material indicate that assumptions regarding durability and length of useable life assumed for window frame assemblies are significant for every material and should be the subject of further research. For aluminum frames, there is a 52 % decrease in the calculated global warming potential and a 45 % decrease in calculated fossil fuel depletion between Scenario 1 and Scenario 3. Studies of aluminum window frames using warranty periods as a means of estimating service life may therefore be approximately doubling their impact estimates when compared to the expectations for building products that are well-maintained

Code: • Negative mechanism > Higher initial material use Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-01, p. 7

Created: 23.6.2026 r. 14:17 by user, Modified: 23.6.2026 r. 14:17 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Obsolescence > Physical (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Mitigation > Maintenance planning (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Differences between use scenarios within each material indicate that assumptions regarding durability and length of useable life assumed for window frame assemblies are significant for every material and should be the subject of further research. For aluminum frames, there is a 52 % decrease in the calculated global warming potential and a 45 % decrease in calculated fossil fuel depletion between Scenario 1 and Scenario 3. Studies of aluminum window frames using warranty periods as a means of estimating service life may therefore be approximately doubling their impact estimates when compared to the expectations for building products that are well-maintained

Code: • Circular strategy > Maintenance Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-01, p. 7

Created: 23.6.2026 r. 14:17 by user, Modified: 23.6.2026 r. 14:17 by user

Other codes assigned to segment:

Obsolescence > Physical (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Mitigation > Maintenance planning (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Differences between use scenarios within each material indicate that assumptions regarding durability and length of useable life assumed for window frame assemblies are significant for every material and should be the subject of further research. For aluminum frames, there is a 52 % decrease in the calculated global warming potential and a 45 % decrease in calculated fossil fuel depletion between Scenario 1 and Scenario 3. Studies of aluminum window frames using warranty periods as a means of estimating service life may therefore be approximately doubling their impact estimates when compared to the expectations for building products that are well-maintained

Code: • Negative mechanism > Premature replacement Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-01, p. 7

Created: 23.6.2026 r. 14:17 by user, Modified: 23.6.2026 r. 14:17 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Obsolescence > Physical (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Mitigation > Maintenance planning (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Differences between use scenarios within each material indicate that assumptions regarding durability and length of useable life assumed for window frame assemblies are significant for every material and should be the subject of further research. For aluminum frames, there is a 52 % decrease in the calculated global warming potential and a 45 % decrease in calculated fossil fuel depletion between Scenario 1 and Scenario 3. Studies of aluminum window frames using warranty periods as a means of estimating service life may therefore be approximately doubling their impact estimates when compared to the expectations for building products that are well-maintained

Code: • Mitigation > Maintenance planning Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-01, p. 7

Created: 23.6.2026 r. 14:17 by user, Modified: 23.6.2026 r. 14:17 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Obsolescence > Physical (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Differences between use scenarios within each material indicate that assumptions regarding durability and length of useable life assumed for window frame assemblies are significant for every material and should be the subject of further research. For aluminum frames, there is a 52 % decrease in the calculated global warming potential and a 45 % decrease in calculated fossil fuel depletion between Scenario 1 and Scenario 3. Studies of aluminum window frames using warranty periods as a means of estimating service life may therefore be approximately doubling their impact estimates when compared to the expectations for building products that are well-maintained

Code: • Obsolescence > Physical Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-01, p. 7

Created: 23.6.2026 r. 14:17 by user, Modified: 23.6.2026 r. 14:17 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Mitigation > Maintenance planning (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Differences between use scenarios within each material indicate that assumptions regarding durability and length of useable life assumed for window frame assemblies are significant for every material and should be the subject of further research. For aluminum frames, there is a 52 % decrease in the calculated global warming potential and a 45 % decrease in calculated fossil fuel depletion between Scenario 1 and Scenario 3. Studies of aluminum window frames using warranty periods as a means of estimating service life may therefore be approximately doubling their impact estimates when compared to the expectations for building products that are well-maintained

Code: • Mitigation > Lifecycle verification Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-01, p. 7

Created: 23.6.2026 r. 14:17 by user, Modified: 23.6.2026 r. 14:17 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Obsolescence > Physical (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Mitigation > Maintenance planning (Weight: 0)

While this model was built to measure the importance of durability and maintenance in the use stage of the life cycle, it also makes clear that material reclamation and recycling at end of life is the most

significant contributor to reducing the embodied environmental burdens of window frame products. Aluminum frames in particular are adversely affected when life cycle stages are simplified, excluding the benefits associated with durability, low maintenance requirements, or end of life (EAA 2013). The very attributes cited in several existing studies (Sinha and Kutnar 2012; Asif et al. 2007) as contributing to the high impacts of initial frame production contribute to decreased impacts during the product life cycle, especially when end of life recycling is considered.

Code: • Circular strategy > Recycling Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-01, p. 8
 Created: 23.6.2026 r. 14:18 by user, Modified: 23.6.2026 r. 14:18 by user
 Other codes assigned to segment:
 Enabling condition > Service infrastructure (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

While this model was built to measure the importance of durability and maintenance in the use stage of the life cycle, it also makes clear that material reclamation and recycling at end of life is the most significant contributor to reducing the embodied environmental burdens of window frame products. Aluminum frames in particular are adversely affected when life cycle stages are simplified, excluding the benefits associated with durability, low maintenance requirements, or end of life (EAA 2013). The very attributes cited in several existing studies (Sinha and Kutnar 2012; Asif et al. 2007) as contributing to the high impacts of initial frame production contribute to decreased impacts during the product life cycle, especially when end of life recycling is considered.

Code: • Negative mechanism > Burden shifting Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-01, p. 8
 Created: 23.6.2026 r. 14:18 by user, Modified: 23.6.2026 r. 14:18 by user
 Other codes assigned to segment:
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

While this model was built to measure the importance of durability and maintenance in the use stage of the life cycle, it also makes clear that material reclamation and recycling at end of life is the most significant contributor to reducing the embodied environmental burdens of window frame products. Aluminum frames in particular are adversely affected when life cycle stages are simplified, excluding the benefits associated with durability, low maintenance requirements, or end of life (EAA 2013). The very attributes cited in several existing studies (Sinha and Kutnar 2012; Asif et al. 2007) as contributing to the high impacts of initial frame production contribute to decreased impacts during the product life cycle, especially when end of life recycling is considered.

Code: • Enabling condition > Service infrastructure Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-01, p. 8
 Created: 23.6.2026 r. 14:18 by user, Modified: 23.6.2026 r. 14:18 by user
 Other codes assigned to segment:
 Circular strategy > Recycling (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

While this model was built to measure the importance of durability and maintenance in the use stage of the life cycle, it also makes clear that material reclamation and recycling at end of life is the most significant contributor to reducing the embodied environmental burdens of window frame products. Aluminum frames in particular are adversely affected when life cycle stages are simplified, excluding the benefits associated with durability, low maintenance requirements, or end of life (EAA 2013). The very attributes cited in several existing studies (Sinha and Kutnar 2012; Asif et al. 2007) as

contributing to the high impacts of initial frame production contribute to decreased impacts during the product life cycle, especially when end of life recycling is considered.

Code: • Mitigation > Lifecycle verification Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-01, p. 8
 Created: 23.6.2026 r. 14:18 by user, Modified: 23.6.2026 r. 14:18 by user
 Other codes assigned to segment:
 Circular strategy > Recycling (Weight: 0)
 Enabling condition > Service infrastructure (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)

Swedish property owners commonly prefer replacing windows, despite high investment costs and long payback periods, due to a perceived certainty of energy and cost savings compared to maintaining them (BeBo, 2022). However, according to the literature review by Souviron et al. (2019), the climate benefit of window replacements is uncertain, considering the potential trade-off effect between the reduced operational GWP (O-GWP) and the E-GWP of energy-efficient windows. Although the E-GWP of windows is generally smaller than that of components accounting for a larger proportion of a building's mass (e.g. walls and slabs), their contributing impact percentage is high considering their relatively low material weight (Feng et al., 2023). Further uncertainty arises when considering window types with various SL and maintenance needs. For instance, aluminum-clad (AW) and polyvinyl chloride (PVC: PW) windows might need relatively little maintenance (Litti et al., 2018; Asif, 2019), but more frequent replacement (Litti et al., 2018). In addition, replacement and maintenance rates influence windows' life cycle (LC) impact (Eberhardt et al., 2018; Grant, 2010). Shifting cost and climate impacts across the LC stages could alter the ranking of preferable intervention measures. In addition, the uniqueness and complexity of renovation projects cause uncertainties (Noori et al., 2016) originating from (inter alia) sparse information and unforeseen conditions (Uotila et al., 2020). In LC studies, transparency about uncertainties and sensitivities is essential to increase the generalizability of LCA results (Souviron et al., 2019). Therefore, transparency and sensitivity analyses may be especially relevant for the generalizability of LC studies on renovation measures.

Code: • Circular strategy > Upgrade Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-02, p. 2
 Created: 24.6.2026 r. 12:11 by user, Modified: 24.6.2026 r. 12:11 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Obsolescence > Economic (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Swedish property owners commonly prefer replacing windows, despite high investment costs and long payback periods, due to a perceived certainty of energy and cost savings compared to maintaining them (BeBo, 2022). However, according to the literature review by Souviron et al. (2019), the climate benefit of window replacements is uncertain, considering the potential trade-off effect between the reduced operational GWP (O-GWP) and the E-GWP of energy-efficient windows. Although the E-GWP of windows is generally smaller than that of components accounting for a larger proportion of a building's mass (e.g. walls and slabs), their contributing impact percentage is high considering their relatively low material weight (Feng et al., 2023). Further uncertainty arises when considering window types with various SL and maintenance needs. For instance, aluminum-clad (AW) and polyvinyl chloride (PVC: PW) windows might need relatively little maintenance (Litti et al., 2018; Asif, 2019), but more frequent replacement (Litti et al., 2018). In addition, replacement and maintenance rates influence windows' life cycle (LC) impact (Eberhardt et al., 2018; Grant, 2010). Shifting cost and climate impacts across the LC stages could alter the ranking of preferable

intervention measures. In addition, the uniqueness and complexity of renovation projects cause uncertainties (Noori et al., 2016) originating from (inter alia) sparse information and unforeseen conditions (Uotila et al., 2020). In LC studies, transparency about uncertainties and sensitivities is essential to increase the generalizability of LCA results (Souviron et al., 2019). Therefore, transparency and sensitivity analyses may be especially relevant for the generalizability of LC studies on renovation measures.

Code: • Negative mechanism > Premature replacement Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-02, p. 2

Created: 24.6.2026 r. 12:11 by user, Modified: 24.6.2026 r. 12:11 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Functional (Weight: 0)

Obsolescence > Economic (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Swedish property owners commonly prefer replacing windows, despite high investment costs and long payback periods, due to a perceived certainty of energy and cost savings compared to maintaining them (BeBo, 2022). However, according to the literature review by Souviron et al. (2019), the climate benefit of window replacements is uncertain, considering the potential trade-off effect between the reduced operational GWP (O-GWP) and the E-GWP of energy-efficient windows. Although the E-GWP of windows is generally smaller than that of components accounting for a larger proportion of a building's mass (e.g. walls and slabs), their contributing impact percentage is high considering their relatively low material weight (Feng et al., 2023). Further uncertainty arises when considering window types with various SL and maintenance needs. For instance, aluminum-clad (AW) and polyvinyl chloride (PVC: PW) windows might need relatively little maintenance (Litti et al., 2018; Asif, 2019), but more frequent replacement (Litti et al., 2018). In addition, replacement and maintenance rates influence windows' life cycle (LC) impact (Eberhardt et al., 2018; Grant, 2010). Shifting cost and climate impacts across the LC stages could alter the ranking of preferable intervention measures. In addition, the uniqueness and complexity of renovation projects cause uncertainties (Noori et al., 2016) originating from (inter alia) sparse information and unforeseen conditions (Uotila et al., 2020). In LC studies, transparency about uncertainties and sensitivities is essential to increase the generalizability of LCA results (Souviron et al., 2019). Therefore, transparency and sensitivity analyses may be especially relevant for the generalizability of LC studies on renovation measures.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-02, p. 2

Created: 24.6.2026 r. 12:11 by user, Modified: 24.6.2026 r. 12:11 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Functional (Weight: 0)

Obsolescence > Economic (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Swedish property owners commonly prefer replacing windows, despite high investment costs and long payback periods, due to a perceived certainty of energy and cost savings compared to maintaining them (BeBo, 2022). However, according to the literature review by Souviron et al. (2019), the climate benefit of window replacements is uncertain, considering the potential trade-off effect between the reduced operational GWP (O-GWP) and the E-GWP of energy-efficient windows. Although the E-GWP of windows is generally smaller than that of components accounting for a larger proportion of a building's mass (e.g. walls and slabs), their contributing impact percentage is high

considering their relatively low material weight (Feng et al., 2023). Further uncertainty arises when considering window types with various SL and maintenance needs. For instance, aluminum-clad (AW) and polyvinyl chloride (PVC: PW) windows might need relatively little maintenance (Litti et al., 2018; Asif, 2019), but more frequent replacement (Litti et al., 2018). In addition, replacement and maintenance rates influence windows' life cycle (LC) impact (Eberhardt et al., 2018; Grant, 2010). Shifting cost and climate impacts across the LC stages could alter the ranking of preferable intervention measures. In addition, the uniqueness and complexity of renovation projects cause uncertainties (Noori et al., 2016) originating from (inter alia) sparse information and unforeseen conditions (Uotila et al., 2020). In LC studies, transparency about uncertainties and sensitivities is essential to increase the generalizability of LCA results (Souviron et al., 2019). Therefore, transparency and sensitivity analyses may be especially relevant for the generalizability of LC studies on renovation measures.

Code: • [Obsolescence > Functional](#) Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-02, p. 2
 Created: 24.6.2026 r. 12:11 by user, Modified: 24.6.2026 r. 12:11 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Obsolescence > Economic (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Swedish property owners commonly prefer replacing windows, despite high investment costs and long payback periods, due to a perceived certainty of energy and cost savings compared to maintaining them (BeBo, 2022). However, according to the literature review by Souviron et al. (2019), the climate benefit of window replacements is uncertain, considering the potential trade-off effect between the reduced operational GWP (O-GWP) and the E-GWP of energy-efficient windows. Although the E-GWP of windows is generally smaller than that of components accounting for a larger proportion of a building's mass (e.g. walls and slabs), their contributing impact percentage is high considering their relatively low material weight (Feng et al., 2023). Further uncertainty arises when considering window types with various SL and maintenance needs. For instance, aluminum-clad (AW) and polyvinyl chloride (PVC: PW) windows might need relatively little maintenance (Litti et al., 2018; Asif, 2019), but more frequent replacement (Litti et al., 2018). In addition, replacement and maintenance rates influence windows' life cycle (LC) impact (Eberhardt et al., 2018; Grant, 2010). Shifting cost and climate impacts across the LC stages could alter the ranking of preferable intervention measures. In addition, the uniqueness and complexity of renovation projects cause uncertainties (Noori et al., 2016) originating from (inter alia) sparse information and unforeseen conditions (Uotila et al., 2020). In LC studies, transparency about uncertainties and sensitivities is essential to increase the generalizability of LCA results (Souviron et al., 2019). Therefore, transparency and sensitivity analyses may be especially relevant for the generalizability of LC studies on renovation measures.

Code: • [Negative mechanism > Burden shifting](#) Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-02, p. 2
 Created: 24.6.2026 r. 12:11 by user, Modified: 24.6.2026 r. 12:11 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Obsolescence > Economic (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Swedish property owners commonly prefer replacing windows, despite high investment costs and long payback periods, due to a perceived certainty of energy and cost savings compared to maintaining them (BeBo, 2022). However, according to the literature review by Souviron et al. (2019), the climate benefit of window replacements is uncertain, considering the potential trade-off effect between the reduced operational GWP (O-GWP) and the E-GWP of energy-efficient windows. Although the E-GWP of windows is generally smaller than that of components accounting for a larger proportion of a building's mass (e.g. walls and slabs), their contributing impact percentage is high considering their relatively low material weight (Feng et al., 2023). Further uncertainty arises when considering window types with various SL and maintenance needs. For instance, aluminum-clad (AW) and polyvinyl chloride (PVC: PW) windows might need relatively little maintenance (Litti et al., 2018; Asif, 2019), but more frequent replacement (Litti et al., 2018). In addition, replacement and maintenance rates influence windows' life cycle (LC) impact (Eberhardt et al., 2018; Grant, 2010). Shifting cost and climate impacts across the LC stages could alter the ranking of preferable intervention measures. In addition, the uniqueness and complexity of renovation projects cause uncertainties (Noori et al., 2016) originating from (inter alia) sparse information and unforeseen conditions (Uotila et al., 2020). In LC studies, transparency about uncertainties and sensitivities is essential to increase the generalizability of LCA results (Souviron et al., 2019). Therefore, transparency and sensitivity analyses may be especially relevant for the generalizability of LC studies on renovation measures.

Code: • [Obsolescence > Economic](#) Weight score: 0

[Search D - Window LCA and replacement decisions > D-GS-02, p. 2](#)

Created: 24.6.2026 r. 12:11 by user, Modified: 24.6.2026 r. 12:11 by user

Other codes assigned to segment:

[Circular strategy > Maintenance](#) (Weight: 0)

[Circular strategy > Upgrade](#) (Weight: 0)

[Obsolescence > Functional](#) (Weight: 0)

[Negative mechanism > Premature replacement](#) (Weight: 0)

[Negative mechanism > Burden shifting](#) (Weight: 0)

[Mitigation > Lifecycle verification](#) (Weight: 0)

Swedish property owners commonly prefer replacing windows, despite high investment costs and long payback periods, due to a perceived certainty of energy and cost savings compared to maintaining them (BeBo, 2022). However, according to the literature review by Souviron et al. (2019), the climate benefit of window replacements is uncertain, considering the potential trade-off effect between the reduced operational GWP (O-GWP) and the E-GWP of energy-efficient windows. Although the E-GWP of windows is generally smaller than that of components accounting for a larger proportion of a building's mass (e.g. walls and slabs), their contributing impact percentage is high considering their relatively low material weight (Feng et al., 2023). Further uncertainty arises when considering window types with various SL and maintenance needs. For instance, aluminum-clad (AW) and polyvinyl chloride (PVC: PW) windows might need relatively little maintenance (Litti et al., 2018; Asif, 2019), but more frequent replacement (Litti et al., 2018). In addition, replacement and maintenance rates influence windows' life cycle (LC) impact (Eberhardt et al., 2018; Grant, 2010). Shifting cost and climate impacts across the LC stages could alter the ranking of preferable intervention measures. In addition, the uniqueness and complexity of renovation projects cause uncertainties (Noori et al., 2016) originating from (inter alia) sparse information and unforeseen conditions (Uotila et al., 2020). In LC studies, transparency about uncertainties and sensitivities is essential to increase the generalizability of LCA results (Souviron et al., 2019). Therefore, transparency and sensitivity analyses may be especially relevant for the generalizability of LC studies on renovation measures.

Code: • [Circular strategy > Maintenance](#) Weight score: 0

[Search D - Window LCA and replacement decisions > D-GS-02, p. 2](#)

Created: 24.6.2026 r. 12:11 by user, Modified: 24.6.2026 r. 12:11 by user

Other codes assigned to segment:

- Circular strategy > Upgrade (Weight: 0)
- Obsolescence > Functional (Weight: 0)
- Obsolescence > Economic (Weight: 0)
- Negative mechanism > Premature replacement (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

Option 1: Periodic maintenance (the base scenario) implies recurring maintenance activities to restore and maintain the function and appearance of existing windows. This involves cleaning, sanding, priming, painting and partly re-puttying at a contractor's estimated 10-year maintenance interval throughout the RSP.

Option 2: Replacement p Periodic maintenance assumes the degeneration of existing windows, ending their SL and replacement with new WW that are periodically maintained throughout the remaining RSP, as in Option 1.

Option 3: Replacement comprises three sub-options involving new windows with different frame materials (3.1 Wooden, 3.2 Aluminum-clad, or 3.3 PVC) left without maintenance during the RSP. This aligns with common conceptions of AW and PW being "maintenance free" (Asif et al., 2005), while WWs are left unmaintained, in contrast with Options 1 and 2. Additional window replacements were accounted for when the estimated window SL fell below the RSP.

Code: • Circular strategy > Component replacement Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-02, p. 6

Created: 24.6.2026 r. 12:13 by user, Modified: 24.6.2026 r. 12:13 by user

Other codes assigned to segment:

- Circular strategy > Maintenance (Weight: 0)
- Circular strategy > Upgrade (Weight: 0)
- Obsolescence > Physical (Weight: 0)
- Obsolescence > Functional (Weight: 0)
- Mitigation > Maintenance planning (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

Option 1: Periodic maintenance (the base scenario) implies recurring maintenance activities to restore and maintain the function and appearance of existing windows. This involves cleaning, sanding, priming, painting and partly re-puttying at a contractor's estimated 10-year maintenance interval throughout the RSP.

Option 2: Replacement p Periodic maintenance assumes the degeneration of existing windows, ending their SL and replacement with new WW that are periodically maintained throughout the remaining RSP, as in Option 1.

Option 3: Replacement comprises three sub-options involving new windows with different frame materials (3.1 Wooden, 3.2 Aluminum-clad, or 3.3 PVC) left without maintenance during the RSP. This aligns with common conceptions of AW and PW being "maintenance free" (Asif et al., 2005), while WWs are left unmaintained, in contrast with Options 1 and 2. Additional window replacements were accounted for when the estimated window SL fell below the RSP.

Code: • Circular strategy > Maintenance Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-02, p. 6

Created: 24.6.2026 r. 12:13 by user, Modified: 24.6.2026 r. 12:13 by user

Other codes assigned to segment:

- Circular strategy > Component replacement (Weight: 0)
- Circular strategy > Upgrade (Weight: 0)
- Obsolescence > Physical (Weight: 0)
- Obsolescence > Functional (Weight: 0)
- Mitigation > Maintenance planning (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

Option 1: Periodic maintenance (the base scenario) implies recurring maintenance activities to restore and maintain the function and appearance of existing windows. This involves cleaning, sanding, priming, painting and partly re-puttying at a contractor's estimated 10-year maintenance interval throughout the RSP.

Option 2: Replacement þ Periodic maintenance assumes the degeneration of existing windows, ending their SL and replacement with new WW that are periodically maintained throughout the remaining RSP, as in Option 1.

Option 3: Replacement comprises three sub-options involving new windows with different frame materials (3.1 Wooden, 3.2 Aluminum-clad, or 3.3 PVC) left without maintenance during the RSP. This aligns with common conceptions of AW and PW being "maintenance free" (Asif et al., 2005), while WWs are left unmaintained, in contrast with Options 1 and 2. Additional window replacements were accounted for when the estimated window SL fell below the RSP.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-02, p. 6

Created: 24.6.2026 r. 12:13 by user, Modified: 24.6.2026 r. 12:13 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Component replacement (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Physical (Weight: 0)

Obsolescence > Functional (Weight: 0)

Mitigation > Maintenance planning (Weight: 0)

Option 1: Periodic maintenance (the base scenario) implies recurring maintenance activities to restore and maintain the function and appearance of existing windows. This involves cleaning, sanding, priming, painting and partly re-puttying at a contractor's estimated 10-year maintenance interval throughout the RSP.

Option 2: Replacement þ Periodic maintenance assumes the degeneration of existing windows, ending their SL and replacement with new WW that are periodically maintained throughout the remaining RSP, as in Option 1.

Option 3: Replacement comprises three sub-options involving new windows with different frame materials (3.1 Wooden, 3.2 Aluminum-clad, or 3.3 PVC) left without maintenance during the RSP. This aligns with common conceptions of AW and PW being "maintenance free" (Asif et al., 2005), while WWs are left unmaintained, in contrast with Options 1 and 2. Additional window replacements were accounted for when the estimated window SL fell below the RSP.

Code: • Obsolescence > Physical Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-02, p. 6

Created: 24.6.2026 r. 12:13 by user, Modified: 24.6.2026 r. 12:13 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Component replacement (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Functional (Weight: 0)

Mitigation > Maintenance planning (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Option 1: Periodic maintenance (the base scenario) implies recurring maintenance activities to restore and maintain the function and appearance of existing windows. This involves cleaning, sanding, priming, painting and partly re-puttying at a contractor's estimated 10-year maintenance interval throughout the RSP.

Option 2: Replacement þ Periodic maintenance assumes the degeneration of existing windows, ending their SL and replacement with new WW that are periodically maintained throughout the remaining RSP, as in Option 1.

Option 3: Replacement comprises three sub-options involving new windows with different frame

materials (3.1 Wooden, 3.2 Aluminum-clad, or 3.3 PVC) left without maintenance during the RSP. This aligns with common conceptions of AW and PW being “maintenance free” (Asif et al., 2005), while WWs are left unmaintained, in contrast with Options 1 and 2. Additional window replacements were accounted for when the estimated window SL fell below the RSP.

Code: • Mitigation > Maintenance planning Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-02, p. 6
 Created: 24.6.2026 r. 12:13 by user, Modified: 24.6.2026 r. 12:13 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Option 1: Periodic maintenance (the base scenario) implies recurring maintenance activities to restore and maintain the function and appearance of existing windows. This involves cleaning, sanding, priming, painting and partly re-puttying at a contractor’s estimated 10-year maintenance interval throughout the RSP.

Option 2: Replacement þ Periodic maintenance assumes the degeneration of existing windows, ending their SL and replacement with new WW that are periodically maintained throughout the remaining RSP, as in Option 1.

Option 3: Replacement comprises three sub-options involving new windows with different frame materials (3.1 Wooden, 3.2 Aluminum-clad, or 3.3 PVC) left without maintenance during the RSP. This aligns with common conceptions of AW and PW being “maintenance free” (Asif et al., 2005), while WWs are left unmaintained, in contrast with Options 1 and 2. Additional window replacements were accounted for when the estimated window SL fell below the RSP.

Code: • Circular strategy > Upgrade Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-02, p. 6
 Created: 24.6.2026 r. 12:13 by user, Modified: 24.6.2026 r. 12:13 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Component replacement (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Option 1: Periodic maintenance (the base scenario) implies recurring maintenance activities to restore and maintain the function and appearance of existing windows. This involves cleaning, sanding, priming, painting and partly re-puttying at a contractor’s estimated 10-year maintenance interval throughout the RSP.

Option 2: Replacement þ Periodic maintenance assumes the degeneration of existing windows, ending their SL and replacement with new WW that are periodically maintained throughout the remaining RSP, as in Option 1.

Option 3: Replacement comprises three sub-options involving new windows with different frame materials (3.1 Wooden, 3.2 Aluminum-clad, or 3.3 PVC) left without maintenance during the RSP. This aligns with common conceptions of AW and PW being “maintenance free” (Asif et al., 2005), while WWs are left unmaintained, in contrast with Options 1 and 2. Additional window replacements were accounted for when the estimated window SL fell below the RSP.

Code: • Obsolescence > Functional Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-02, p. 6
 Created: 24.6.2026 r. 12:13 by user, Modified: 24.6.2026 r. 12:13 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Component replacement (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The results show that Option 1, maintaining existing windows, has the lowest LC Parameter Initial input Alternative Unit

Reference study period, RSP 60 100 years District heating cost 31 143 EUR/MWh

MID HIGH LOW

Emission factor 25 495 0 gCO₂-eq/kWh U-value, windows 1.8 2.3 1.3 W/m²

K Discount rate 4.4 10 0 %

Intervention rates

Maintenance rate 10 4 15 years

Window replacement rate:

- Wooden 40 30 50 years - Aluminum-clad 50 40 60 years - PVC 30 20 40 years Source(s): Table created by authors

Table 1. Initial data input used

for the "MID 60" scenario and the alternative inputs for the sensitivity analysis

Maintaining or replacing windows

773

Downloaded from <http://www.emerald.com/ijbpa/article-pdf/43/4/766/9795652/ijbpa-11-2023-0179.pdf> by guest on 18 June 2026GWP and costs among the studied options. Comparing the replacement scenarios suggests that PW (3.3) is the cheapest option but has the highest climate impact.

Code: • Negative mechanism > Higher initial material use Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-02, p. 8 - 9

Created: 24.6.2026 r. 12:15 by user, Modified: 24.6.2026 r. 12:15 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The results show that Option 1, maintaining existing windows, has the lowest LC Parameter Initial input Alternative Unit

Reference study period, RSP 60 100 years District heating cost 31 143 EUR/MWh

MID HIGH LOW

Emission factor 25 495 0 gCO₂-eq/kWh U-value, windows 1.8 2.3 1.3 W/m²

K Discount rate 4.4 10 0 %

Intervention rates

Maintenance rate 10 4 15 years

Window replacement rate:

- Wooden 40 30 50 years - Aluminum-clad 50 40 60 years - PVC 30 20 40 years Source(s): Table created by authors

Table 1. Initial data input used

for the "MID 60" scenario and the alternative inputs for the sensitivity analysis

Maintaining or replacing windows

773

Downloaded from <http://www.emerald.com/ijbpa/article-pdf/43/4/766/9795652/ijbpa-11-2023-0179.pdf>

0179.pdf by guest on 18 June 2026GWP and costs among the studied options. Comparing the replacement scenarios suggests that PW (3.3) is the cheapest option but has the highest climate impact.

Code: • Negative mechanism > Premature replacement Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-02, p. 8 - 9

Created: 24.6.2026 r. 12:15 by user, Modified: 24.6.2026 r. 12:15 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Obsolescence > Physical (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Mitigation > Maintenance planning (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The results show that Option 1, maintaining existing windows, has the lowest LC Parameter Initial input Alternative Unit

Reference study period, RSP 60 100 years District heating cost 31 143 EUR/MWh

MID HIGH LOW

Emission factor 25 495 0 gCO₂-eq/kWh U-value, windows 1.8 2.3 1.3 W/m²

K Discount rate 4.4 10 0 %

Intervention rates

Maintenance rate 10 4 15 years

Window replacement rate:

- Wooden 40 30 50 years - Aluminum-clad 50 40 60 years - PVC 30 20 40 years Source(s): Table created by authors

Table 1. Initial data input used

for the "MID 60" scenario and the alternative inputs for the sensitivity analysis

Maintaining or replacing windows

773

Downloaded from <http://www.emerald.com/ijbpa/article-pdf/43/4/766/9795652/ijbpa-11-2023-0179.pdf> by guest on 18 June 2026GWP and costs among the studied options. Comparing the replacement scenarios suggests that PW (3.3) is the cheapest option but has the highest climate impact.

Code: • Mitigation > Maintenance planning Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-02, p. 8 - 9

Created: 24.6.2026 r. 12:15 by user, Modified: 24.6.2026 r. 12:15 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Obsolescence > Physical (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The results show that Option 1, maintaining existing windows, has the lowest LC Parameter Initial input Alternative Unit

Reference study period, RSP 60 100 years District heating cost 31 143 EUR/MWh

MID HIGH LOW

Emission factor 25 495 0 gCO₂-eq/kWh U-value, windows 1.8 2.3 1.3 W/m²

K Discount rate 4.4 10 0 %

Intervention rates

Maintenance rate 10 4 15 years

Window replacement rate:

- Wooden 40 30 50 years - Aluminum-clad 50 40 60 years - PVC 30 20 40 years Source(s): Table created by authors

Table 1. Initial data input used

for the "MID 60" scenario and the alternative inputs for the sensitivity analysis

Maintaining or replacing windows

773

Downloaded from <http://www.emerald.com/ijbpa/article-pdf/43/4/766/9795652/ijbpa-11-2023-0179.pdf> by guest on 18 June 2026GWP and costs among the studied options. Comparing the replacement scenarios suggests that PW (3.3) is the cheapest option but has the highest climate impact.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-02, p. 8 - 9

Created: 24.6.2026 r. 12:15 by user, Modified: 24.6.2026 r. 12:15 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Obsolescence > Physical (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Mitigation > Maintenance planning (Weight: 0)

The results show that Option 1, maintaining existing windows, has the lowest LC Parameter Initial input Alternative Unit

Reference study period, RSP 60 100 years District heating cost 31 143 EUR/MWh

MID HIGH LOW

Emission factor 25 495 0 gCO₂-eq/kWh U-value, windows 1.8 2.3 1.3 W/m²

K Discount rate 4.4 10 0 %

Intervention rates

Maintenance rate 10 4 15 years

Window replacement rate:

- Wooden 40 30 50 years - Aluminum-clad 50 40 60 years - PVC 30 20 40 years Source(s): Table created by authors

Table 1. Initial data input used

for the "MID 60" scenario and the alternative inputs for the sensitivity analysis

Maintaining or replacing windows

773

Downloaded from <http://www.emerald.com/ijbpa/article-pdf/43/4/766/9795652/ijbpa-11-2023-0179.pdf> by guest on 18 June 2026GWP and costs among the studied options. Comparing the replacement scenarios suggests that PW (3.3) is the cheapest option but has the highest climate impact.

Code: • Obsolescence > Physical Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-02, p. 8 - 9

Created: 24.6.2026 r. 12:15 by user, Modified: 24.6.2026 r. 12:15 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Mitigation > Maintenance planning (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The results show that Option 1, maintaining existing windows, has the lowest LC Parameter Initial input Alternative Unit

Reference study period, RSP 60 100 years District heating cost 31 143 EUR/MWh

MID HIGH LOW

Emission factor 25 495 0 gCO₂-eq/kWh U-value, windows 1.8 2.3 1.3 W/m²

K Discount rate 4.4 10 0 %

Intervention rates

Maintenance rate 10 4 15 years

Window replacement rate:

- Wooden 40 30 50 years - Aluminum-clad 50 40 60 years - PVC 30 20 40 years Source(s): Table created by authors

Table 1. Initial data input used

for the "MID 60" scenario and the alternative inputs for the sensitivity analysis

Maintaining or replacing windows

773

Downloaded from <http://www.emerald.com/ijbpa/article-pdf/43/4/766/9795652/ijbpa-11-2023-0179.pdf> by guest on 18 June 2026 GWP and costs among the studied options. Comparing the replacement scenarios suggests that PW (3.3) is the cheapest option but has the highest climate impact.

Code: • Circular strategy > Maintenance Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-02, p. 8 - 9

Created: 24.6.2026 r. 12:14 by user, Modified: 24.6.2026 r. 12:14 by user

Other codes assigned to segment:

Obsolescence > Physical (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Mitigation > Maintenance planning (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The energy simulations showed a reduced OHE of 3.3 kWh/m² GFA per year when replacing the windows with new ones. Comparing the LCA-GWP and LCC results, this reduction (green columns below 0 in Figure 2) impacts the LC GWP more than the costs. Over a 60-year RSP, the saved heating energy from a window replacement translates into a reduced GWP of 4.8 kg CO₂-eq/m² GFA, which can be compared to the new windows' E-GWP of 7.3–43.8 kg CO₂-eq/m² GFA. On the other hand, the saved operational costs from reduced OHE do not result in net-zero LC costs; the investment costs can be recouped by, at best, 4% for the cheapest scenario (3.3 PW).

Code: • Mitigation > Lifecycle verification Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-02, p. 10

Created: 24.6.2026 r. 12:16 by user, Modified: 24.6.2026 r. 12:16 by user

Other codes assigned to segment:

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Functional (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

The energy simulations showed a reduced OHE of 3.3 kWh/m² GFA per year when replacing the windows with new ones. Comparing the LCA-GWP and LCC results, this reduction (green columns below 0 in Figure 2) impacts the LC GWP more than the costs. Over a 60-year RSP, the saved heating energy from a window replacement translates into a reduced GWP of 4.8 kg CO₂-eq/m² GFA, which can be compared to the new windows' E-GWP of 7.3–43.8 kg CO₂-eq/m² GFA. On the other hand, the saved operational costs from reduced OHE do not result in net-zero LC costs; the investment costs can be recouped by, at best, 4% for the cheapest scenario (3.3 PW).

Code: • Negative mechanism > Higher initial material use Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-02, p. 10

Created: 24.6.2026 r. 12:16 by user, Modified: 24.6.2026 r. 12:16 by user

Other codes assigned to segment:

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Functional (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The energy simulations showed a reduced OHE of 3.3 kWh/m² GFA per year when replacing the windows with new ones. Comparing the LCA-GWP and LCC results, this reduction (green columns below 0 in Figure 2) impacts the LC GWP more than the costs. Over a 60-year RSP, the saved heating energy from a window replacement translates into a reduced GWP of 4.8 kg CO₂-eq/m² GFA, which can be compared to the new windows' E-GWP of 7.3–43.8 kg CO₂-eq/m² GFA. On the other hand, the saved operational costs from reduced OHE do not result in net-zero LC costs; the investment costs can be recouped by, at best, 4% for the cheapest scenario (3.3 PW).

Code: • Negative mechanism > Burden shifting Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-02, p. 10

Created: 24.6.2026 r. 12:16 by user, Modified: 24.6.2026 r. 12:16 by user

Other codes assigned to segment:

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Functional (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The energy simulations showed a reduced OHE of 3.3 kWh/m² GFA per year when replacing the windows with new ones. Comparing the LCA-GWP and LCC results, this reduction (green columns below 0 in Figure 2) impacts the LC GWP more than the costs. Over a 60-year RSP, the saved heating energy from a window replacement translates into a reduced GWP of 4.8 kg CO₂-eq/m² GFA, which can be compared to the new windows' E-GWP of 7.3–43.8 kg CO₂-eq/m² GFA. On the other hand, the saved operational costs from reduced OHE do not result in net-zero LC costs; the investment costs can be recouped by, at best, 4% for the cheapest scenario (3.3 PW).

Code: • Circular strategy > Upgrade Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-02, p. 10

Created: 24.6.2026 r. 12:16 by user, Modified: 24.6.2026 r. 12:16 by user

Other codes assigned to segment:

Obsolescence > Functional (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The energy simulations showed a reduced OHE of 3.3 kWh/m² GFA per year when replacing the windows with new ones. Comparing the LCA-GWP and LCC results, this reduction (green columns below 0 in Figure 2) impacts the LC GWP more than the costs. Over a 60-year RSP, the saved heating energy from a window replacement translates into a reduced GWP of 4.8 kg CO₂-eq/m² GFA, which can be compared to the new windows' E-GWP of 7.3–43.8 kg CO₂-eq/m² GFA. On the other hand, the saved operational costs from reduced OHE do not result in net-zero LC costs; the investment costs can be recouped by, at best, 4% for the cheapest scenario (3.3 PW).

Code: • Obsolescence > Functional Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-02, p. 10

Created: 24.6.2026 r. 12:16 by user, Modified: 24.6.2026 r. 12:16 by user

Other codes assigned to segment:

Circular strategy > Upgrade (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Adjusting the DH emission factor reveals that window replacement is more beneficial the higher the emission factor, especially if the building operates for a longer period of time (see Figure 4). For a high emission factor, a replacement is always better to minimize the LC GWP, regardless of the RSP or intervention rate. For PW with a high replacement rate over 100 years, an emission factor of 125

gCO₂-eq/kWh would be enough to reach net-zero GWP. In contrast, operational cost savings are slight and cannot compensate for the high investment costs, even with Sweden's most expensive DH. To pay back the cheapest scenario, PW with a low replacement rate, the DH price must be at least 702 EUR/MWh over 60 years of operation. Maintaining windows remains economically preferable, except at a high intervention rate when AW and PW are slightly cheaper.

Code: • [Obsolescence > Functional](#) Weight score: 0

[Search D - Window LCA and replacement decisions > D-GS-02, p. 10](#)

Created: 24.6.2026 r. 12:18 by user, Modified: 24.6.2026 r. 12:18 by user

Other codes assigned to segment:

[Circular strategy > Maintenance](#) (Weight: 0)

[Circular strategy > Upgrade](#) (Weight: 0)

[Negative mechanism > Burden shifting](#) (Weight: 0)

[Mitigation > Lifecycle verification](#) (Weight: 0)

Adjusting the DH emission factor reveals that window replacement is more beneficial the higher the emission factor, especially if the building operates for a longer period of time (see Figure 4). For a high emission factor, a replacement is always better to minimize the LC GWP, regardless of the RSP or intervention rate. For PW with a high replacement rate over 100 years, an emission factor of 125 gCO₂-eq/kWh would be enough to reach net-zero GWP. In contrast, operational cost savings are slight and cannot compensate for the high investment costs, even with Sweden's most expensive DH. To pay back the cheapest scenario, PW with a low replacement rate, the DH price must be at least 702 EUR/MWh over 60 years of operation. Maintaining windows remains economically preferable, except at a high intervention rate when AW and PW are slightly cheaper.

Code: • [Negative mechanism > Burden shifting](#) Weight score: 0

[Search D - Window LCA and replacement decisions > D-GS-02, p. 10](#)

Created: 24.6.2026 r. 12:18 by user, Modified: 24.6.2026 r. 12:18 by user

Other codes assigned to segment:

[Circular strategy > Maintenance](#) (Weight: 0)

[Circular strategy > Upgrade](#) (Weight: 0)

[Obsolescence > Functional](#) (Weight: 0)

[Mitigation > Lifecycle verification](#) (Weight: 0)

Adjusting the DH emission factor reveals that window replacement is more beneficial the higher the emission factor, especially if the building operates for a longer period of time (see Figure 4). For a high emission factor, a replacement is always better to minimize the LC GWP, regardless of the RSP or intervention rate. For PW with a high replacement rate over 100 years, an emission factor of 125 gCO₂-eq/kWh would be enough to reach net-zero GWP. In contrast, operational cost savings are slight and cannot compensate for the high investment costs, even with Sweden's most expensive DH. To pay back the cheapest scenario, PW with a low replacement rate, the DH price must be at least 702 EUR/MWh over 60 years of operation. Maintaining windows remains economically preferable, except at a high intervention rate when AW and PW are slightly cheaper.

Code: • [Circular strategy > Maintenance](#) Weight score: 0

[Search D - Window LCA and replacement decisions > D-GS-02, p. 10](#)

Created: 24.6.2026 r. 12:18 by user, Modified: 24.6.2026 r. 12:18 by user

Other codes assigned to segment:

[Circular strategy > Upgrade](#) (Weight: 0)

[Obsolescence > Functional](#) (Weight: 0)

[Negative mechanism > Burden shifting](#) (Weight: 0)

[Mitigation > Lifecycle verification](#) (Weight: 0)

Adjusting the DH emission factor reveals that window replacement is more beneficial the higher the emission factor, especially if the building operates for a longer period of time (see Figure 4). For a high emission factor, a replacement is always better to minimize the LC GWP, regardless of the RSP

or intervention rate. For PW with a high replacement rate over 100 years, an emission factor of 125 gCO₂-eq/kWh would be enough to reach net-zero GWP. In contrast, operational cost savings are slight and cannot compensate for the high investment costs, even with Sweden's most expensive DH. To pay back the cheapest scenario, PW with a low replacement rate, the DH price must be at least 702 EUR/MWh over 60 years of operation. Maintaining windows remains economically preferable, except at a high intervention rate when AW and PW are slightly cheaper.

Code: • Circular strategy > Upgrade Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-02, p. 10

Created: 24.6.2026 r. 12:18 by user, Modified: 24.6.2026 r. 12:18 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Obsolescence > Functional (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Adjusting the DH emission factor reveals that window replacement is more beneficial the higher the emission factor, especially if the building operates for a longer period of time (see Figure 4). For a high emission factor, a replacement is always better to minimize the LC GWP, regardless of the RSP or intervention rate. For PW with a high replacement rate over 100 years, an emission factor of 125 gCO₂-eq/kWh would be enough to reach net-zero GWP. In contrast, operational cost savings are slight and cannot compensate for the high investment costs, even with Sweden's most expensive DH. To pay back the cheapest scenario, PW with a low replacement rate, the DH price must be at least 702 EUR/MWh over 60 years of operation. Maintaining windows remains economically preferable, except at a high intervention rate when AW and PW are slightly cheaper.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-02, p. 10

Created: 24.6.2026 r. 12:18 by user, Modified: 24.6.2026 r. 12:18 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Functional (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

the LCA-GWP was more sensitive to parameter changes compared to the LCC, indicating a higher level of uncertainty when choosing the most climateconscious option. Among the parameters, the DH's emission factor and the original windows' U-value were the most critical, with high values favoring replacement over maintenance.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-02, p. 16

Created: 24.6.2026 r. 12:20 by user, Modified: 24.6.2026 r. 12:20 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

the LCA-GWP was more sensitive to parameter changes compared to the LCC, indicating a higher level of uncertainty when choosing the most climateconscious option. Among the parameters, the DH's emission factor and the original windows' U-value were the most critical, with high values favoring replacement over maintenance.

Code: • Circular strategy > Maintenance Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-02, p. 16

Created: 24.6.2026 r. 12:19 by user, Modified: 24.6.2026 r. 12:19 by user

Other codes assigned to segment:

Circular strategy > Upgrade (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

the LCA-GWP was more sensitive to parameter changes compared to the LCC, indicating a higher level of uncertainty when choosing the most climateconscious option. Among the parameters, the DH's emission factor and the original windows' U-value were the most critical, with high values favoring replacement over maintenance.

Code: • Negative mechanism > Burden shifting Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-02, p. 16

Created: 24.6.2026 r. 12:19 by user, Modified: 24.6.2026 r. 12:19 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

the LCA-GWP was more sensitive to parameter changes compared to the LCC, indicating a higher level of uncertainty when choosing the most climateconscious option. Among the parameters, the DH's emission factor and the original windows' U-value were the most critical, with high values favoring replacement over maintenance.

Code: • Circular strategy > Upgrade Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-02, p. 16

Created: 24.6.2026 r. 12:19 by user, Modified: 24.6.2026 r. 12:19 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Results indicate that building characteristics are important, as the two buildings show contrary LC results, even within the same local context. Swapping the buildings' location affects initial LC performance, and which sensitivity parameters appear critical. The parameters influence the LC results, but only a few alter the window interventions' rankings, with heating emission factor having most impact. Reducing LC climate impact and costs by replacing windows is not guaranteed, as it depends on building characteristics and local context. Instead, window maintenance may be preferred for its lower material use. The study's integrated LC approach provides insights for harmonizing renovation budgets with climate targets. Adapting renovation measures to each specific case is key for choosing the best option, balancing operational and embodied impacts.

Code: • Circular strategy > Maintenance Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-03, p. 1

Created: 24.6.2026 r. 12:31 by user, Modified: 24.6.2026 r. 12:31 by user

Other codes assigned to segment:

Circular strategy > Upgrade (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Results indicate that building characteristics are important, as the two buildings show contrary LC results, even within the same local context. Swapping the buildings' location affects initial LC performance, and which sensitivity parameters appear critical. The parameters influence the LC results, but only a few alter the window interventions' rankings, with heating emission factor having most impact. Reducing LC climate impact and costs by replacing windows is not guaranteed, as it

depends on building characteristics and local context. Instead, window maintenance may be preferred for its lower material use. The study's integrated LC approach provides insights for harmonizing renovation budgets with climate targets. Adapting renovation measures to each specific case is key for choosing the best option, balancing operational and embodied impacts.

Code: • [Obsolescence > Functional](#) Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-03, p. 1
 Created: 24.6.2026 r. 12:31 by user, Modified: 24.6.2026 r. 12:31 by user
 Other codes assigned to segment:
[Circular strategy > Maintenance](#) (Weight: 0)
[Circular strategy > Upgrade](#) (Weight: 0)
[Negative mechanism > Higher initial material use](#) (Weight: 0)
[Negative mechanism > Burden shifting](#) (Weight: 0)
[Mitigation > Lifecycle verification](#) (Weight: 0)

Results indicate that building characteristics are important, as the two buildings show contrary LC results, even within the same local context. Swapping the buildings' location affects initial LC performance, and which sensitivity parameters appear critical. The parameters influence the LC results, but only a few alter the window interventions' rankings, with heating emission factor having most impact. Reducing LC climate impact and costs by replacing windows is not guaranteed, as it depends on building characteristics and local context. Instead, window maintenance may be preferred for its lower material use. The study's integrated LC approach provides insights for harmonizing renovation budgets with climate targets. Adapting renovation measures to each specific case is key for choosing the best option, balancing operational and embodied impacts.

Code: • [Negative mechanism > Burden shifting](#) Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-03, p. 1
 Created: 24.6.2026 r. 12:31 by user, Modified: 24.6.2026 r. 12:31 by user
 Other codes assigned to segment:
[Circular strategy > Maintenance](#) (Weight: 0)
[Circular strategy > Upgrade](#) (Weight: 0)
[Obsolescence > Functional](#) (Weight: 0)
[Negative mechanism > Higher initial material use](#) (Weight: 0)
[Mitigation > Lifecycle verification](#) (Weight: 0)

Results indicate that building characteristics are important, as the two buildings show contrary LC results, even within the same local context. Swapping the buildings' location affects initial LC performance, and which sensitivity parameters appear critical. The parameters influence the LC results, but only a few alter the window interventions' rankings, with heating emission factor having most impact. Reducing LC climate impact and costs by replacing windows is not guaranteed, as it depends on building characteristics and local context. Instead, window maintenance may be preferred for its lower material use. The study's integrated LC approach provides insights for harmonizing renovation budgets with climate targets. Adapting renovation measures to each specific case is key for choosing the best option, balancing operational and embodied impacts.

Code: • [Circular strategy > Upgrade](#) Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-03, p. 1
 Created: 24.6.2026 r. 12:31 by user, Modified: 24.6.2026 r. 12:31 by user
 Other codes assigned to segment:
[Circular strategy > Maintenance](#) (Weight: 0)
[Obsolescence > Functional](#) (Weight: 0)
[Negative mechanism > Higher initial material use](#) (Weight: 0)
[Negative mechanism > Burden shifting](#) (Weight: 0)
[Mitigation > Lifecycle verification](#) (Weight: 0)

Results indicate that building characteristics are important, as the two buildings show contrary LC results, even within the same local context. Swapping the buildings' location affects initial LC

performance, and which sensitivity parameters appear critical. The parameters influence the LC results, but only a few alter the window interventions' rankings, with heating emission factor having most impact. Reducing LC climate impact and costs by replacing windows is not guaranteed, as it depends on building characteristics and local context. Instead, window maintenance may be preferred for its lower material use. The study's integrated LC approach provides insights for harmonizing renovation budgets with climate targets. Adapting renovation measures to each specific case is key for choosing the best option, balancing operational and embodied impacts.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-03, p. 1

Created: 24.6.2026 r. 12:31 by user, Modified: 24.6.2026 r. 12:31 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Functional (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Results indicate that building characteristics are important, as the two buildings show contrary LC results, even within the same local context. Swapping the buildings' location affects initial LC performance, and which sensitivity parameters appear critical. The parameters influence the LC results, but only a few alter the window interventions' rankings, with heating emission factor having most impact. Reducing LC climate impact and costs by replacing windows is not guaranteed, as it depends on building characteristics and local context. Instead, window maintenance may be preferred for its lower material use. The study's integrated LC approach provides insights for harmonizing renovation budgets with climate targets. Adapting renovation measures to each specific case is key for choosing the best option, balancing operational and embodied impacts.

Code: • Negative mechanism > Higher initial material use Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-03, p. 1

Created: 24.6.2026 r. 12:31 by user, Modified: 24.6.2026 r. 12:31 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Functional (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

It is evident that there is no one-size-fits-all solution. Instead, the appropriateness of a renovation measure hinges on the specific conditions of each building [1,12], such as their characteristics and location. The building characteristics, such as initial thermal properties, play a key role in the success rate of EEMs [23] and their potential for operational heating energy (OHE) savings [21]. The diversity of buildings, with characteristics depending on the construction techniques, building standards and preferences prevailing during specific time periods, can be modeled by using archetypes representing groups of buildings with similar characteristics [24], which may enhance the representativeness of case studies exploring the LC impact of optional renovation measures.

Geographical location directly impacts a building's local context, including variations in climate conditions, energy production mixes and resource availability. The climate influences the energy needed for space heating [4]. For example, Northern Sweden's high heating demand results from the subarctic climate with long winter seasons and limited solar heat gain. The OHE use in Swedish multi-residential buildings is effectively reduced by improving the building envelope with energy-efficient windows [9,10]. In this context, property owners often choose to replace windows instead of maintaining them, believing that new windows offer more reliable operational energy and cost savings

Code: • Obsolescence > Functional Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-03, p. 2
 Created: 24.6.2026 r. 12:32 by user, Modified: 24.6.2026 r. 12:32 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Obsolescence > Economic (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

It is evident that there is no one-size-fits-all solution. Instead, the appropriateness of a renovation measure hinges on the specific conditions of each building [1,12], such as their characteristics and location. The building characteristics, such as initial thermal properties, play a key role in the success rate of EEMs [23] and their potential for operational heating energy (OHE) savings [21]. The diversity of buildings, with characteristics depending on the construction techniques, building standards and preferences prevailing during specific time periods, can be modeled by using archetypes representing groups of buildings with similar characteristics [24], which may enhance the representativeness of case studies exploring the LC impact of optional renovation measures.

Geographical location directly impacts a building's local context, including variations in climate conditions, energy production mixes and resource availability. The climate influences the energy needed for space heating [4]. For example, Northern Sweden's high heating demand results from the subarctic climate with long winter seasons and limited solar heat gain. The OHE use in Swedish multi-residential buildings is effectively reduced by improving the building envelope with energy-efficient windows [9,10]. In this context, property owners often choose to replace windows instead of maintaining them, believing that new windows offer more reliable operational energy and cost savings

Code: • Circular strategy > Upgrade Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-03, p. 2
 Created: 24.6.2026 r. 12:32 by user, Modified: 24.6.2026 r. 12:32 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Obsolescence > Economic (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

It is evident that there is no one-size-fits-all solution. Instead, the appropriateness of a renovation measure hinges on the specific conditions of each building [1,12], such as their characteristics and location. The building characteristics, such as initial thermal properties, play a key role in the success rate of EEMs [23] and their potential for operational heating energy (OHE) savings [21]. The diversity of buildings, with characteristics depending on the construction techniques, building standards and preferences prevailing during specific time periods, can be modeled by using archetypes representing groups of buildings with similar characteristics [24], which may enhance the representativeness of case studies exploring the LC impact of optional renovation measures.

Geographical location directly impacts a building's local context, including variations in climate conditions, energy production mixes and resource availability. The climate influences the energy needed for space heating [4]. For example, Northern Sweden's high heating demand results from the subarctic climate with long winter seasons and limited solar heat gain. The OHE use in Swedish multi-residential buildings is effectively reduced by improving the building envelope with energy-efficient windows [9,10]. In this context, property owners often choose to replace windows instead of maintaining them, believing that new windows offer more reliable operational energy and cost savings

Code: • Negative mechanism > Premature replacement Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-03, p. 2

Created: 24.6.2026 r. 12:33 by user, Modified: 24.6.2026 r. 12:33 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Obsolescence > Economic (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

It is evident that there is no one-size-fits-all solution. Instead, the appropriateness of a renovation measure hinges on the specific conditions of each building [1,12], such as their characteristics and location. The building characteristics, such as initial thermal properties, play a key role in the success rate of EEMs [23] and their potential for operational heating energy (OHE) savings [21]. The diversity of buildings, with characteristics depending on the construction techniques, building standards and preferences prevailing during specific time periods, can be modeled by using archetypes representing groups of buildings with similar characteristics [24], which may enhance the representativeness of case studies exploring the LC impact of optional renovation measures.

Geographical location directly impacts a building's local context, including variations in climate conditions, energy production mixes and resource availability. The climate influences the energy needed for space heating [4]. For example, Northern Sweden's high heating demand results from the subarctic climate with long winter seasons and limited solar heat gain. The OHE use in Swedish multi-residential buildings is effectively reduced by improving the building envelope with energy-efficient windows [9,10]. In this context, property owners often choose to replace windows instead of maintaining them, believing that new windows offer more reliable operational energy and cost savings

Code: • Circular strategy > Maintenance Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-03, p. 2

Created: 24.6.2026 r. 12:32 by user, Modified: 24.6.2026 r. 12:32 by user

Other codes assigned to segment:

Circular strategy > Upgrade (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Obsolescence > Economic (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

It is evident that there is no one-size-fits-all solution. Instead, the appropriateness of a renovation measure hinges on the specific conditions of each building [1,12], such as their characteristics and location. The building characteristics, such as initial thermal properties, play a key role in the success rate of EEMs [23] and their potential for operational heating energy (OHE) savings [21]. The diversity of buildings, with characteristics depending on the construction techniques, building standards and preferences prevailing during specific time periods, can be modeled by using archetypes representing groups of buildings with similar characteristics [24], which may enhance the representativeness of case studies exploring the LC impact of optional renovation measures.

Geographical location directly impacts a building's local context, including variations in climate conditions, energy production mixes and resource availability. The climate influences the energy needed for space heating [4]. For example, Northern Sweden's high heating demand results from the subarctic climate with long winter seasons and limited solar heat gain. The OHE use in Swedish multi-residential buildings is effectively reduced by improving the building envelope with energy-efficient windows [9,10]. In this context, property owners often choose to replace windows instead of maintaining them, believing that new windows offer more reliable operational energy and cost savings

Code: • Mitigation > Lifecycle verification Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-03, p. 2

Created: 24.6.2026 r. 12:33 by user, Modified: 24.6.2026 r. 12:33 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Obsolescence > Economic (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)

It is evident that there is no one-size-fits-all solution. Instead, the appropriateness of a renovation measure hinges on the specific conditions of each building [1,12], such as their characteristics and location. The building characteristics, such as initial thermal properties, play a key role in the success rate of EEMs [23] and their potential for operational heating energy (OHE) savings [21]. The diversity of buildings, with characteristics depending on the construction techniques, building standards and preferences prevailing during specific time periods, can be modeled by using archetypes representing groups of buildings with similar characteristics [24], which may enhance the representativeness of case studies exploring the LC impact of optional renovation measures.

Geographical location directly impacts a building's local context, including variations in climate conditions, energy production mixes and resource availability. The climate influences the energy needed for space heating [4]. For example, Northern Sweden's high heating demand results from the subarctic climate with long winter seasons and limited solar heat gain. The OHE use in Swedish multi-residential buildings is effectively reduced by improving the building envelope with energy-efficient windows [9,10]. In this context, property owners often choose to replace windows instead of maintaining them, believing that new windows offer more reliable operational energy and cost savings

Code: • Obsolescence > Economic Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-03, p. 2

Created: 24.6.2026 r. 12:33 by user, Modified: 24.6.2026 r. 12:33 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

EEMs can aid in decarbonizing the building sector and managing the ageing building stock's maintenance needs, but there can be trade-offs between the product and operational stages. Achieving the lowest environmental impact requires consideration of both energy performance and embodied GWP (E-GWP) from renovation [20]. For instance, Mathew et al. [18] studied the trade-offs between energy efficiency, LC GWP and conserving heritage values for an Irish 1920s' cottage. While new windows reduced energy use, reducing operational CO₂ emissions could lead to higher LC GWP due to the E-GWP, potentially offsetting the benefits of reduced O-GWP [18,27]. A survey study by Lindstrom [7] on the potential for transitioning the window sector to a circular economy showed that concerned stakeholders advocated for maintenance to move from the current prevailing linear economy of Swedish windows. However, Swedish property owners generally prefer window replacement despite long payback periods and high investment costs

Code: • Negative mechanism > Premature replacement Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-03, p. 3

Created: 24.6.2026 r. 12:35 by user, Modified: 24.6.2026 r. 12:35 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

EEMs can aid in decarbonizing the building sector and managing the ageing building stock's maintenance needs, but there can be trade-offs between the product and operational stages. Achieving the lowest environmental impact requires consideration of both energy performance and embodied GWP (E-GWP) from renovation [20]. For instance, Mathew et al. [18] studied the trade-offs between energy efficiency, LC GWP and conserving heritage values for an Irish 1920s' cottage. While new windows reduced energy use, reducing operational CO2 emissions could lead to higher LC GWP due to the E-GWP, potentially offsetting the benefits of reduced O-GWP [18,27]. A survey study by Lindstrom [7] on the potential for transitioning the window sector to a circular economy showed that concerned stakeholders advocated for maintenance to move from the current prevailing linear economy of Swedish windows. However, Swedish property owners generally prefer window replacement despite long payback periods and high investment costs

Code: • Circular strategy > Maintenance Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-03, p. 3
 Created: 24.6.2026 r. 12:34 by user, Modified: 24.6.2026 r. 12:34 by user
 Other codes assigned to segment:
 Circular strategy > Upgrade (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

EEMs can aid in decarbonizing the building sector and managing the ageing building stock's maintenance needs, but there can be trade-offs between the product and operational stages. Achieving the lowest environmental impact requires consideration of both energy performance and embodied GWP (E-GWP) from renovation [20]. For instance, Mathew et al. [18] studied the trade-offs between energy efficiency, LC GWP and conserving heritage values for an Irish 1920s' cottage. While new windows reduced energy use, reducing operational CO2 emissions could lead to higher LC GWP due to the E-GWP, potentially offsetting the benefits of reduced O-GWP [18,27]. A survey study by Lindstrom [7] on the potential for transitioning the window sector to a circular economy showed that concerned stakeholders advocated for maintenance to move from the current prevailing linear economy of Swedish windows. However, Swedish property owners generally prefer window replacement despite long payback periods and high investment costs

Code: • Negative mechanism > Higher initial material use Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-03, p. 3
 Created: 24.6.2026 r. 12:35 by user, Modified: 24.6.2026 r. 12:35 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

EEMs can aid in decarbonizing the building sector and managing the ageing building stock's maintenance needs, but there can be trade-offs between the product and operational stages. Achieving the lowest environmental impact requires consideration of both energy performance and embodied GWP (E-GWP) from renovation [20]. For instance, Mathew et al. [18] studied the trade-offs between energy efficiency, LC GWP and conserving heritage values for an Irish 1920s' cottage. While new windows reduced energy use, reducing operational CO2 emissions could lead to higher LC GWP due to the E-GWP, potentially offsetting the benefits of reduced O-GWP [18,27]. A survey study by Lindstrom [7] on the potential for transitioning the window sector to a circular economy showed that concerned stakeholders advocated for maintenance to move from the current prevailing linear economy of Swedish windows. However, Swedish property owners generally prefer window replacement despite long payback periods and high investment costs

Code: • Circular strategy > Upgrade Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-03, p. 3
 Created: 24.6.2026 r. 12:34 by user, Modified: 24.6.2026 r. 12:34 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

EEMs can aid in decarbonizing the building sector and managing the ageing building stock's maintenance needs, but there can be trade-offs between the product and operational stages. Achieving the lowest environmental impact requires consideration of both energy performance and embodied GWP (E-GWP) from renovation [20]. For instance, Mathew et al. [18] studied the trade-offs between energy efficiency, LC GWP and conserving heritage values for an Irish 1920s' cottage. While new windows reduced energy use, reducing operational CO2 emissions could lead to higher LC GWP due to the E-GWP, potentially offsetting the benefits of reduced O-GWP [18,27]. A survey study by Lindstrom [7] on the potential for transitioning the window sector to a circular economy showed that concerned stakeholders advocated for maintenance to move from the current prevailing linear economy of Swedish windows. However, Swedish property owners generally prefer window replacement despite long payback periods and high investment costs

Code: • Negative mechanism > Burden shifting Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-03, p. 3
 Created: 24.6.2026 r. 12:35 by user, Modified: 24.6.2026 r. 12:35 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

EEMs can aid in decarbonizing the building sector and managing the ageing building stock's maintenance needs, but there can be trade-offs between the product and operational stages. Achieving the lowest environmental impact requires consideration of both energy performance and embodied GWP (E-GWP) from renovation [20]. For instance, Mathew et al. [18] studied the trade-offs between energy efficiency, LC GWP and conserving heritage values for an Irish 1920s' cottage. While new windows reduced energy use, reducing operational CO2 emissions could lead to higher LC GWP due to the E-GWP, potentially offsetting the benefits of reduced O-GWP [18,27]. A survey study by Lindstrom [7] on the potential for transitioning the window sector to a circular economy showed that concerned stakeholders advocated for maintenance to move from the current prevailing linear economy of Swedish windows. However, Swedish property owners generally prefer window replacement despite long payback periods and high investment costs

Code: • Mitigation > Lifecycle verification Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-03, p. 3
 Created: 24.6.2026 r. 12:35 by user, Modified: 24.6.2026 r. 12:35 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)

Furthermore, the studied replacement options necessitate the management of existing windows, either through disposal or reuse in other buildings. The related impacts would be allocated to the

windows' end-of-life, Modules C and D. However, previous research has found that Modules C and D have limited contribution to both the LC climate impact of renovation [33,39] and the associated costs [40] in cold climates. Therefore, since these modules are unlikely to be decisive in comparing the window interventions, which is the primary focus of this study, they were excluded. Also, knowledge of windows' end-of-life (C) and reuse/recycle potential (D) is limited [27]. Estimating these modules' impact involves uncertain predictions as the related activities extend far into the future due to the window's relatively long SL (see section 4.5). Consequently, potential biogenic carbon fixing for wooden products was excluded from A1–A5, and so were the residual values or deconstruction costs at the end of the windows' SL. Nevertheless, it is important to note that options involving window replacement are at a disadvantage due to additional waste and costs related to dismantlement of windows—whereas maintaining existing windows avoids these extra burdens.

Code: • Negative mechanism > Burden shifting Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-03, p. 5

Created: 24.6.2026 r. 12:40 by user, Modified: 24.6.2026 r. 12:40 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Furthermore, the studied replacement options necessitate the management of existing windows, either through disposal or reuse in other buildings. The related impacts would be allocated to the windows' end-of-life, Modules C and D. However, previous research has found that Modules C and D have limited contribution to both the LC climate impact of renovation [33,39] and the associated costs [40] in cold climates. Therefore, since these modules are unlikely to be decisive in comparing the window interventions, which is the primary focus of this study, they were excluded. Also, knowledge of windows' end-of-life (C) and reuse/recycle potential (D) is limited [27]. Estimating these modules' impact involves uncertain predictions as the related activities extend far into the future due to the window's relatively long SL (see section 4.5). Consequently, potential biogenic carbon fixing for wooden products was excluded from A1–A5, and so were the residual values or deconstruction costs at the end of the windows' SL. Nevertheless, it is important to note that options involving window replacement are at a disadvantage due to additional waste and costs related to dismantlement of windows—whereas maintaining existing windows avoids these extra burdens.

Code: • Negative mechanism > Higher initial material use Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-03, p. 5

Created: 24.6.2026 r. 12:40 by user, Modified: 24.6.2026 r. 12:40 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Recycling (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Furthermore, the studied replacement options necessitate the management of existing windows, either through disposal or reuse in other buildings. The related impacts would be allocated to the windows' end-of-life, Modules C and D. However, previous research has found that Modules C and D have limited contribution to both the LC climate impact of renovation [33,39] and the associated costs [40] in cold climates. Therefore, since these modules are unlikely to be decisive in comparing the window interventions, which is the primary focus of this study, they were excluded. Also, knowledge of windows' end-of-life (C) and reuse/recycle potential (D) is limited [27]. Estimating these modules' impact involves uncertain predictions as the related activities extend far into the future due to the window's relatively long SL (see section 4.5). Consequently, potential biogenic

carbon fixing for wooden products was excluded from A1–A5, and so were the residual values or deconstruction costs at the end of the windows' SL. Nevertheless, it is important to note that options involving window replacement are at a disadvantage due to additional waste and costs related to dismantlement of windows—whereas maintaining existing windows avoids these extra burdens.

Code: • Circular strategy > Recycling Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-03, p. 5
 Created: 24.6.2026 r. 12:39 by user, Modified: 24.6.2026 r. 12:39 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Furthermore, the studied replacement options necessitate the management of existing windows, either through disposal or reuse in other buildings. The related impacts would be allocated to the windows' end-of-life, Modules C and D. However, previous research has found that Modules C and D have limited contribution to both the LC climate impact of renovation [33,39] and the associated costs [40] in cold climates. Therefore, since these modules are unlikely to be decisive in comparing the window interventions, which is the primary focus of this study, they were excluded. Also, knowledge of windows' end-of-life (C) and reuse/recycle potential (D) is limited [27]. Estimating these modules' impact involves uncertain predictions as the related activities extend far into the future due to the window's relatively long SL (see section 4.5). Consequently, potential biogenic carbon fixing for wooden products was excluded from A1–A5, and so were the residual values or deconstruction costs at the end of the windows' SL. Nevertheless, it is important to note that options involving window replacement are at a disadvantage due to additional waste and costs related to dismantlement of windows—whereas maintaining existing windows avoids these extra burdens.

Code: • Circular strategy > Maintenance Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-03, p. 5
 Created: 24.6.2026 r. 12:39 by user, Modified: 24.6.2026 r. 12:39 by user
 Other codes assigned to segment:
 Circular strategy > Recycling (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Furthermore, the studied replacement options necessitate the management of existing windows, either through disposal or reuse in other buildings. The related impacts would be allocated to the windows' end-of-life, Modules C and D. However, previous research has found that Modules C and D have limited contribution to both the LC climate impact of renovation [33,39] and the associated costs [40] in cold climates. Therefore, since these modules are unlikely to be decisive in comparing the window interventions, which is the primary focus of this study, they were excluded. Also, knowledge of windows' end-of-life (C) and reuse/recycle potential (D) is limited [27]. Estimating these modules' impact involves uncertain predictions as the related activities extend far into the future due to the window's relatively long SL (see section 4.5). Consequently, potential biogenic carbon fixing for wooden products was excluded from A1–A5, and so were the residual values or deconstruction costs at the end of the windows' SL. Nevertheless, it is important to note that options involving window replacement are at a disadvantage due to additional waste and costs related to dismantlement of windows—whereas maintaining existing windows avoids these extra burdens.

Code: • Negative mechanism > Premature replacement Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-03, p. 5
 Created: 24.6.2026 r. 12:39 by user, Modified: 24.6.2026 r. 12:39 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Furthermore, the studied replacement options necessitate the management of existing windows, either through disposal or reuse in other buildings. The related impacts would be allocated to the windows' end-of-life, Modules C and D. However, previous research has found that Modules C and D have limited contribution to both the LC climate impact of renovation [33,39] and the associated costs [40] in cold climates. Therefore, since these modules are unlikely to be decisive in comparing the window interventions, which is the primary focus of this study, they were excluded. Also, knowledge of windows' end-of-life (C) and reuse/recycle potential (D) is limited [27]. Estimating these modules' impact involves uncertain predictions as the related activities extend far into the future due to the window's relatively long SL (see section 4.5). Consequently, potential biogenic carbon fixing for wooden products was excluded from A1–A5, and so were the residual values or deconstruction costs at the end of the windows' SL. Nevertheless, it is important to note that options involving window replacement are at a disadvantage due to additional waste and costs related to dismantlement of windows—whereas maintaining existing windows avoids these extra burdens.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-03, p. 5

Created: 24.6.2026 r. 12:40 by user, Modified: 24.6.2026 r. 12:40 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Recycling (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)

Building G holds historical value [41] and was preserved when mine-induced ground deformations required moving the building from its original location. Although not legally protected, building renovations need to be carried out carefully to preserve Building G's features [42]. Therefore, wooden windows are the only type considered for replacement. The following options were investigated for Building G:

1) Periodic maintenance represents Building L's first option of maintaining the original windows every ten years. The main difference is an initial comprehensive window restoration, due to the windows' poor condition, followed by minor periodic maintenance. Restoration involves dismantling the historical windows, transporting them to a workshop by truck, and removing and applying putty composed of linseed oil and chalk—a traditional sealant used for bedding in glass in wooden window frames. In contrast, Building L's windows are maintained in place using a modern silicon sealant. It is assumed that the restoration will improve the windows' thermal transmittance by approximately 0.3 W/m²

K, based on Fredlund's [43] measurements of a similar window restoration.

2) Replacement + Periodic maintenance aligns with Building L, involving installation of new wooden triple-glazed windows, maintained every ten years.

3) Replacement involves installing new windows as Option 2, but without maintenance, requiring additional replacements when the windows exceed their SL.

4) Secondary glazing is exclusive to Building G's double-glazed windows. An extra windowpane, with an aluminum profile and finishing wooden lists, and a 40-year SL, is added from the inside to the restored and maintained windows in Option 1. The contractor offering the unique solution must travel across the country for the installation. In this study, all transportation-related impacts were allocated to Building G. However, in practice, the contractor can avoid long-distance commuting for

individual projects by coordinating multiple projects, meaning that transportation impacts could be allocated to several projects.

Code: • Negative mechanism > Burden shifting Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-03, p. 7

Created: 24.6.2026 r. 12:38 by user, Modified: 24.6.2026 r. 12:38 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Component replacement (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Functional (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Negative mechanism > Additional transport (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Building G holds historical value [41] and was preserved when mine-induced ground deformations required moving the building from its original location. Although not legally protected, building renovations need to be carried out carefully to preserve Building G's features [42]. Therefore, wooden windows are the only type considered for replacement. The following options were investigated for Building G:

1) Periodic maintenance represents Building L's first option of maintaining the original windows every ten years. The main difference is an initial comprehensive window restoration, due to the windows' poor condition, followed by minor periodic maintenance. Restoration involves dismantling the historical windows, transporting them to a workshop by truck, and removing and applying putty composed of linseed oil and chalk—a traditional sealant used for bedding in glass in wooden window frames. In contrast, Building L's windows are maintained in place using a modern silicon sealant. It is assumed that the restoration will improve the windows' thermal transmittance by approximately 0.3 W/m²

K, based on Fredlund's [43] measurements of a similar window restoration.

2) Replacement + Periodic maintenance aligns with Building L, involving installation of new wooden triple-glazed windows, maintained every ten years.

3) Replacement involves installing new windows as Option 2, but without maintenance, requiring additional replacements when the windows exceed their SL.

4) Secondary glazing is exclusive to Building G's double-glazed windows. An extra windowpane, with an aluminum profile and finishing wooden lists, and a 40-year SL, is added from the inside to the restored and maintained windows in Option 1. The contractor offering the unique solution must travel across the country for the installation. In this study, all transportation-related impacts were allocated to Building G. However, in practice, the contractor can avoid long-distance commuting for individual projects by coordinating multiple projects, meaning that transportation impacts could be allocated to several projects.

Code: • Circular strategy > Component replacement Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-03, p. 7

Created: 24.6.2026 r. 12:37 by user, Modified: 24.6.2026 r. 12:37 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Functional (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Negative mechanism > Additional transport (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Building G holds historical value [41] and was preserved when mine-induced ground deformations required moving the building from its original location. Although not legally protected, building renovations need to be carried out carefully to preserve Building G's features [42]. Therefore,

wooden windows are the only type considered for replacement. The following options were investigated for Building G:

1) Periodic maintenance represents Building L's first option of maintaining the original windows every ten years. The main difference is an initial comprehensive window restoration, due to the windows' poor condition, followed by minor periodic maintenance. Restoration involves dismantling the historical windows, transporting them to a workshop by truck, and removing and applying putty composed of linseed oil and chalk—a traditional sealant used for bedding in glass in wooden window frames. In contrast, Building L's windows are maintained in place using a modern silicon sealant. It is assumed that the restoration will improve the windows' thermal transmittance by approximately 0.3 W/m²

K, based on Fredlund's [43] measurements of a similar window restoration.

2) Replacement + Periodic maintenance aligns with Building L, involving installation of new wooden triple-glazed windows, maintained every ten years.

3) Replacement involves installing new windows as Option 2, but without maintenance, requiring additional replacements when the windows exceed their SL.

4) Secondary glazing is exclusive to Building G's double-glazed windows. An extra windowpane, with an aluminum profile and finishing wooden lists, and a 40-year SL, is added from the inside to the restored and maintained windows in Option 1. The contractor offering the unique solution must travel across the country for the installation. In this study, all transportation-related impacts were allocated to Building G. However, in practice, the contractor can avoid long-distance commuting for individual projects by coordinating multiple projects, meaning that transportation impacts could be allocated to several projects.

Code: • Negative mechanism > Additional transport Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-03, p. 7

Created: 24.6.2026 r. 12:38 by user, Modified: 24.6.2026 r. 12:38 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Component replacement (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Functional (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Building G holds historical value [41] and was preserved when mine-induced ground deformations required moving the building from its original location. Although not legally protected, building renovations need to be carried out carefully to preserve Building G's features [42]. Therefore, wooden windows are the only type considered for replacement. The following options were investigated for Building G:

1) Periodic maintenance represents Building L's first option of maintaining the original windows every ten years. The main difference is an initial comprehensive window restoration, due to the windows' poor condition, followed by minor periodic maintenance. Restoration involves dismantling the historical windows, transporting them to a workshop by truck, and removing and applying putty composed of linseed oil and chalk—a traditional sealant used for bedding in glass in wooden window frames. In contrast, Building L's windows are maintained in place using a modern silicon sealant. It is assumed that the restoration will improve the windows' thermal transmittance by approximately 0.3 W/m²

K, based on Fredlund's [43] measurements of a similar window restoration.

2) Replacement + Periodic maintenance aligns with Building L, involving installation of new wooden triple-glazed windows, maintained every ten years.

3) Replacement involves installing new windows as Option 2, but without maintenance, requiring additional replacements when the windows exceed their SL.

4) Secondary glazing is exclusive to Building G's double-glazed windows. An extra windowpane, with

an aluminum profile and finishing wooden lists, and a 40-year SL, is added from the inside to the restored and maintained windows in Option 1. The contractor offering the unique solution must travel across the country for the installation. In this study, all transportation-related impacts were allocated to Building G. However, in practice, the contractor can avoid long-distance commuting for individual projects by coordinating multiple projects, meaning that transportation impacts could be allocated to several projects.

Code: • [Obsolescence > Functional](#) Weight score: 0

[Search D - Window LCA and replacement decisions > D-GS-03, p. 7](#)

Created: 24.6.2026 r. 12:37 by user, Modified: 24.6.2026 r. 12:37 by user

Other codes assigned to segment:

[Circular strategy > Maintenance](#) (Weight: 0)

[Circular strategy > Component replacement](#) (Weight: 0)

[Circular strategy > Upgrade](#) (Weight: 0)

[Enabling condition > Service infrastructure](#) (Weight: 0)

[Negative mechanism > Additional transport](#) (Weight: 0)

[Negative mechanism > Burden shifting](#) (Weight: 0)

[Mitigation > Lifecycle verification](#) (Weight: 0)

Building G holds historical value [41] and was preserved when mine-induced ground deformations required moving the building from its original location. Although not legally protected, building renovations need to be carried out carefully to preserve Building G's features [42]. Therefore, wooden windows are the only type considered for replacement. The following options were investigated for Building G:

1) Periodic maintenance represents Building L's first option of maintaining the original windows every ten years. The main difference is an initial comprehensive window restoration, due to the windows' poor condition, followed by minor periodic maintenance. Restoration involves dismantling the historical windows, transporting them to a workshop by truck, and removing and applying putty composed of linseed oil and chalk—a traditional sealant used for bedding in glass in wooden window frames. In contrast, Building L's windows are maintained in place using a modern silicon sealant. It is assumed that the restoration will improve the windows' thermal transmittance by approximately 0.3 W/m²

K, based on Fredlund's [43] measurements of a similar window restoration.

2) Replacement + Periodic maintenance aligns with Building L, involving installation of new wooden triple-glazed windows, maintained every ten years.

3) Replacement involves installing new windows as Option 2, but without maintenance, requiring additional replacements when the windows exceed their SL.

4) Secondary glazing is exclusive to Building G's double-glazed windows. An extra windowpane, with an aluminum profile and finishing wooden lists, and a 40-year SL, is added from the inside to the restored and maintained windows in Option 1. The contractor offering the unique solution must travel across the country for the installation. In this study, all transportation-related impacts were allocated to Building G. However, in practice, the contractor can avoid long-distance commuting for individual projects by coordinating multiple projects, meaning that transportation impacts could be allocated to several projects.

Code: • [Circular strategy > Upgrade](#) Weight score: 0

[Search D - Window LCA and replacement decisions > D-GS-03, p. 7](#)

Created: 24.6.2026 r. 12:37 by user, Modified: 24.6.2026 r. 12:37 by user

Other codes assigned to segment:

[Circular strategy > Maintenance](#) (Weight: 0)

[Circular strategy > Component replacement](#) (Weight: 0)

[Obsolescence > Functional](#) (Weight: 0)

[Enabling condition > Service infrastructure](#) (Weight: 0)

[Negative mechanism > Additional transport](#) (Weight: 0)

[Negative mechanism > Burden shifting](#) (Weight: 0)

[Mitigation > Lifecycle verification](#) (Weight: 0)

Building G holds historical value [41] and was preserved when mine-induced ground deformations required moving the building from its original location. Although not legally protected, building renovations need to be carried out carefully to preserve Building G's features [42]. Therefore, wooden windows are the only type considered for replacement. The following options were investigated for Building G:

1) Periodic maintenance represents Building L's first option of maintaining the original windows every ten years. The main difference is an initial comprehensive window restoration, due to the windows' poor condition, followed by minor periodic maintenance. Restoration involves dismantling the historical windows, transporting them to a workshop by truck, and removing and applying putty composed of linseed oil and chalk—a traditional sealant used for bedding in glass in wooden window frames. In contrast, Building L's windows are maintained in place using a modern silicon sealant. It is assumed that the restoration will improve the windows' thermal transmittance by approximately 0.3 W/m²

K, based on Fredlund's [43] measurements of a similar window restoration.

2) Replacement + Periodic maintenance aligns with Building L, involving installation of new wooden triple-glazed windows, maintained every ten years.

3) Replacement involves installing new windows as Option 2, but without maintenance, requiring additional replacements when the windows exceed their SL.

4) Secondary glazing is exclusive to Building G's double-glazed windows. An extra windowpane, with an aluminum profile and finishing wooden lists, and a 40-year SL, is added from the inside to the restored and maintained windows in Option 1. The contractor offering the unique solution must travel across the country for the installation. In this study, all transportation-related impacts were allocated to Building G. However, in practice, the contractor can avoid long-distance commuting for individual projects by coordinating multiple projects, meaning that transportation impacts could be allocated to several projects.

Code: • Circular strategy > Maintenance Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-03, p. 7

Created: 24.6.2026 r. 12:37 by user, Modified: 24.6.2026 r. 12:37 by user

Other codes assigned to segment:

Circular strategy > Component replacement (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Functional (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Negative mechanism > Additional transport (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

Building G holds historical value [41] and was preserved when mine-induced ground deformations required moving the building from its original location. Although not legally protected, building renovations need to be carried out carefully to preserve Building G's features [42]. Therefore, wooden windows are the only type considered for replacement. The following options were investigated for Building G:

1) Periodic maintenance represents Building L's first option of maintaining the original windows every ten years. The main difference is an initial comprehensive window restoration, due to the windows' poor condition, followed by minor periodic maintenance. Restoration involves dismantling the historical windows, transporting them to a workshop by truck, and removing and applying putty composed of linseed oil and chalk—a traditional sealant used for bedding in glass in wooden window frames. In contrast, Building L's windows are maintained in place using a modern silicon sealant. It is assumed that the restoration will improve the windows' thermal transmittance by approximately 0.3 W/m²

K, based on Fredlund's [43] measurements of a similar window restoration.

2) Replacement + Periodic maintenance aligns with Building L, involving installation of new wooden

triple-glazed windows, maintained every ten years.

3) Replacement involves installing new windows as Option 2, but without maintenance, requiring additional replacements when the windows exceed their SL.

4) Secondary glazing is exclusive to Building G's double-glazed windows. An extra windowpane, with an aluminum profile and finishing wooden lists, and a 40-year SL, is added from the inside to the restored and maintained windows in Option 1. The contractor offering the unique solution must travel across the country for the installation. In this study, all transportation-related impacts were allocated to Building G. However, in practice, the contractor can avoid long-distance commuting for individual projects by coordinating multiple projects, meaning that transportation impacts could be allocated to several projects.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-03, p. 7

Created: 24.6.2026 r. 12:38 by user, Modified: 24.6.2026 r. 12:38 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Component replacement (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Functional (Weight: 0)

Enabling condition > Service infrastructure (Weight: 0)

Negative mechanism > Additional transport (Weight: 0)

Negative mechanism > Burden shifting (Weight: 0)

Building G holds historical value [41] and was preserved when mine-induced ground deformations required moving the building from its original location. Although not legally protected, building renovations need to be carried out carefully to preserve Building G's features [42]. Therefore, wooden windows are the only type considered for replacement. The following options were investigated for Building G:

1) Periodic maintenance represents Building L's first option of maintaining the original windows every ten years. The main difference is an initial comprehensive window restoration, due to the windows' poor condition, followed by minor periodic maintenance. Restoration involves dismantling the historical windows, transporting them to a workshop by truck, and removing and applying putty composed of linseed oil and chalk—a traditional sealant used for bedding in glass in wooden window frames. In contrast, Building L's windows are maintained in place using a modern silicon sealant. It is assumed that the restoration will improve the windows' thermal transmittance by approximately 0.3 W/m²

K, based on Fredlund's [43] measurements of a similar window restoration.

2) Replacement + Periodic maintenance aligns with Building L, involving installation of new wooden triple-glazed windows, maintained every ten years.

3) Replacement involves installing new windows as Option 2, but without maintenance, requiring additional replacements when the windows exceed their SL.

4) Secondary glazing is exclusive to Building G's double-glazed windows. An extra windowpane, with an aluminum profile and finishing wooden lists, and a 40-year SL, is added from the inside to the restored and maintained windows in Option 1. The contractor offering the unique solution must travel across the country for the installation. In this study, all transportation-related impacts were allocated to Building G. However, in practice, the contractor can avoid long-distance commuting for individual projects by coordinating multiple projects, meaning that transportation impacts could be allocated to several projects.

Code: • Enabling condition > Service infrastructure Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-03, p. 7

Created: 24.6.2026 r. 12:38 by user, Modified: 24.6.2026 r. 12:38 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Component replacement (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Obsolescence > Functional (Weight: 0)
 Negative mechanism > Additional transport (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Non-energy-related renovations, such as maintaining original windows, can be the most climate-friendly option, especially for buildings with good initial thermal performance supplied with low-carbon DH. On the other hand, increasing windows' energy-efficiency can help buildings with poor thermal properties in certain scenarios—such as colder climate zones and higher DH emission factors—reach net-zero LC GWP before the RSP ends.

Code: • Obsolescence > Functional Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-03, p. 20
 Created: 24.6.2026 r. 12:41 by user, Modified: 24.6.2026 r. 12:41 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Non-energy-related renovations, such as maintaining original windows, can be the most climate-friendly option, especially for buildings with good initial thermal performance supplied with low-carbon DH. On the other hand, increasing windows' energy-efficiency can help buildings with poor thermal properties in certain scenarios—such as colder climate zones and higher DH emission factors—reach net-zero LC GWP before the RSP ends.

Code: • Negative mechanism > Higher initial material use Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-03, p. 20
 Created: 24.6.2026 r. 12:41 by user, Modified: 24.6.2026 r. 12:41 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Non-energy-related renovations, such as maintaining original windows, can be the most climate-friendly option, especially for buildings with good initial thermal performance supplied with low-carbon DH. On the other hand, increasing windows' energy-efficiency can help buildings with poor thermal properties in certain scenarios—such as colder climate zones and higher DH emission factors—reach net-zero LC GWP before the RSP ends.

Code: • Circular strategy > Upgrade Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-03, p. 20
 Created: 24.6.2026 r. 12:41 by user, Modified: 24.6.2026 r. 12:41 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Non-energy-related renovations, such as maintaining original windows, can be the most climate-friendly option, especially for buildings with good initial thermal performance supplied with low-carbon DH. On the other hand, increasing windows' energy-efficiency can help buildings with poor thermal properties in certain scenarios—such as colder climate zones and higher DH emission factors—reach net-zero LC GWP before the RSP ends.

Code: • Circular strategy > Maintenance Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-03, p. 20
 Created: 24.6.2026 r. 12:41 by user, Modified: 24.6.2026 r. 12:41 by user
 Other codes assigned to segment:
 Circular strategy > Upgrade (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Non-energy-related renovations, such as maintaining original windows, can be the most climate-friendly option, especially for buildings with good initial thermal performance supplied with low-carbon DH. On the other hand, increasing windows' energy-efficiency can help buildings with poor thermal properties in certain scenarios—such as colder climate zones and higher DH emission factors—reach net-zero LC GWP before the RSP ends.

Code: • Mitigation > Maintenance planning Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-03, p. 20
 Created: 24.6.2026 r. 12:42 by user, Modified: 24.6.2026 r. 12:42 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Non-energy-related renovations, such as maintaining original windows, can be the most climate-friendly option, especially for buildings with good initial thermal performance supplied with low-carbon DH. On the other hand, increasing windows' energy-efficiency can help buildings with poor thermal properties in certain scenarios—such as colder climate zones and higher DH emission factors—reach net-zero LC GWP before the RSP ends.

Code: • Negative mechanism > Burden shifting Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-03, p. 20
 Created: 24.6.2026 r. 12:41 by user, Modified: 24.6.2026 r. 12:41 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Non-energy-related renovations, such as maintaining original windows, can be the most climate-friendly option, especially for

buildings with good initial thermal performance supplied with low-carbon DH. On the other hand, increasing windows' energy-efficiency can help buildings with poor thermal properties in certain scenarios—such as colder climate zones and higher DH emission factors—reach net-zero LC GWP before the RSP ends.

Code: • Mitigation > Lifecycle verification Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-03, p. 20
 Created: 24.6.2026 r. 12:41 by user, Modified: 24.6.2026 r. 12:41 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Negative mechanism > Burden shifting (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)

Moreover, it was evidenced that, destructive retrofitting techniques, such as windows replacement, not necessarily allow for the largest energy savings. Indeed the installation of single or double glazed internal windows may allow for similar or even more significant savings.

Code: • Circular strategy > Upgrade Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-10, p. 1
 Created: 24.6.2026 r. 12:52 by user, Modified: 24.6.2026 r. 12:52 by user
 Other codes assigned to segment:
 Circular strategy > Refurbishment (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Moreover, it was evidenced that, destructive retrofitting techniques, such as windows replacement, not necessarily allow for the largest energy savings. Indeed the installation of single or double glazed internal windows may allow for similar or even more significant savings.

Code: • Circular strategy > Refurbishment Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-10, p. 1
 Created: 24.6.2026 r. 12:53 by user, Modified: 24.6.2026 r. 12:53 by user
 Other codes assigned to segment:
 Circular strategy > Upgrade (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Moreover, it was evidenced that, destructive retrofitting techniques, such as windows replacement, not necessarily allow for the largest energy savings. Indeed the installation of single or double glazed internal windows may allow for similar or even more significant savings.

Code: • Negative mechanism > Premature replacement Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-10, p. 1
 Created: 24.6.2026 r. 12:53 by user, Modified: 24.6.2026 r. 12:53 by user
 Other codes assigned to segment:
 Circular strategy > Upgrade (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Moreover, it was evidenced that, destructive retrofitting techniques, such as windows replacement, not necessarily allow for the largest energy savings. Indeed the installation of single or double glazed internal windows may allow for similar or even more significant savings.

Code: • Negative mechanism > Higher initial material use Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-10, p. 1
 Created: 24.6.2026 r. 12:53 by user, Modified: 24.6.2026 r. 12:53 by user
 Other codes assigned to segment:
 Circular strategy > Upgrade (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

Moreover, it was evidenced that, destructive retrofitting techniques, such as windows replacement, not necessarily allow for the largest energy savings. Indeed the installation of single or double glazed internal windows may allow for similar or even more significant savings.

Code: • Mitigation > Lifecycle verification Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-10, p. 1
 Created: 24.6.2026 r. 12:53 by user, Modified: 24.6.2026 r. 12:53 by user
 Other codes assigned to segment:
 Circular strategy > Upgrade (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)

However, it should be noted that windows replacement is not the sole fenestration retrofitting measure allowing remarkable energy saving. Furthermore, the non-energy related drawbacks of windows replacement in historic buildings cannot be neglected. Next to the loss of building architectonic value, widely described in [3–5], a reduction of the property market value might occur [7]. Moreover, the environmental benefit of performing anyhow fenestration replacement in historic buildings is not always obvious from a life cycle environmental perspective because of the large initial embodied burdens [8].

For these reasons and for pursuing a more prudential approach when agreeing upon fenestration retrofitting measures in historic or heritage buildings, it is meaningful to explore the energy benefits obtainable by a wide set of retrofitting alternatives rather than confining

☐ Corresponding author.

Abbreviations: ERI, energy retrofitting interventions; EERI, energy environmental retrofitting interventions; EPBD, energy performance building directives; SL, service life; EOL, end of life; EOSL, end of service life; RSL, reference service life; BDT, blower door test; ACH, air change per hour; IRT, infrared thermography; X1, X2, physical properties with regard to building assemblies or components initial (1) and deteriorated (2) state of conservation; Nu, Nusselt number (-); λ , thermal conductivity (W/m K); R, thermal resistance (m² K/W); U, thermal transmittance (W/m² K); η_{50} , air infiltration rate at 50 Pa (h-1); η_{nat} , standardized air Infiltration rate (h-1)

E-mail address: giovanni.litti@uantwerpen.be (G. Litti).

T

© 2018. This manuscript version is made available under the CC-BY-NC-ND 4.0 license

<http://creativecommons.org/licenses/by-nc-nd/4.0/>

Published Journal Article available at: <https://doi.org/10.1016/j.job.2018.02.006>

Received 27 December 2017; Received in revised form 9 February 2018; Accepted 10 February 2018

Available online 17 February 2018 the decision making to the apparently most performing option.

Code: • Circular strategy > Upgrade Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-10, p. 1 - 2
 Created: 24.6.2026 r. 12:45 by user, Modified: 24.6.2026 r. 12:45 by user
 Other codes assigned to segment:
 Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)
Mitigation > Lifecycle verification (Weight: 0)

However, it should be noted that windows replacement is not the sole fenestration retrofitting measure allowing remarkable energy saving. Furthermore, the non-energy related drawbacks of windows replacement in historic buildings cannot be neglected. Next to the loss of building architectonic value, widely described in [3–5], a reduction of the property market value might occur [7]. Moreover, the environmental benefit of performing anyhow fenestration replacement in historic buildings is not always obvious from a life cycle environmental perspective because of the large initial embodied burdens [8].

For these reasons and for pursuing a more prudential approach when agreeing upon fenestration retrofitting measures in historic or heritage buildings, it is meaningful to explore the energy benefits obtainable by a wide set of retrofitting alternatives rather than confining

✉ Corresponding author.

Abbreviations: ERI, energy retrofitting interventions; EERI, energy environmental retrofitting interventions; EPBD, energy performance building directives; SL, service life; EOL, end of life; EOSL, end of service life; RSL, reference service life; BDT, blower door test; ACH, air change per hour; IRT, infrared thermography; X1, X2, physical properties with regard to building assemblies or components initial (1) and deteriorated (2) state of conservation; Nu, Nusselt number (-); λ , thermal conductivity (W/m K); R, thermal resistance (m² K/W); U, thermal transmittance (W/m² K); η_{50} , air infiltration rate at 50 Pa (h⁻¹); η_{nat} , standardized air Infiltration rate (h⁻¹)

E-mail address: giovanni.litti@uantwerpen.be (G. Litti).

T

© 2018. This manuscript version is made available under the CC-BY-NC-ND 4.0 license

<http://creativecommons.org/licenses/by-nc-nd/4.0/>

Published Journal Article available at: <https://doi.org/10.1016/j.job.2018.02.006>

Received 27 December 2017; Received in revised form 9 February 2018; Accepted 10 February 2018

Available online 17 February 2018the decision making to the apparently most performing option.

Code: ● Obsolescence > Functional Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-10, p. 1 - 2

Created: 24.6.2026 r. 12:46 by user, Modified: 24.6.2026 r. 12:46 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

However, it should be noted that windows replacement is not the sole fenestration retrofitting measure allowing remarkable energy saving. Furthermore, the non-energy related drawbacks of windows replacement in historic buildings cannot be neglected. Next to the loss of building architectonic value, widely described in [3–5], a reduction of the property market value might occur [7]. Moreover, the environmental benefit of performing anyhow fenestration replacement in historic buildings is not always obvious from a life cycle environmental perspective because of the large initial embodied burdens [8].

For these reasons and for pursuing a more prudential approach when agreeing upon fenestration retrofitting measures in historic or heritage buildings, it is meaningful to explore the energy benefits obtainable by a wide set of retrofitting alternatives rather than confining

✉ Corresponding author.

Abbreviations: ERI, energy retrofitting interventions; EERI, energy environmental retrofitting interventions; EPBD, energy performance building directives; SL, service life; EOL, end of life; EOSL, end of service life; RSL, reference service life; BDT, blower door test; ACH, air change per hour; IRT,

infrared thermography; X1, X2, physical properties with regard to building assemblies or components initial (1) and deteriorated (2) state of conservation; Nu, Nusselt number (-); λ , thermal conductivity (W/m K); R, thermal resistance (m² K/W); U, thermal transmittance (W/m² K); η_{50} , air infiltration rate at 50 Pa (h-1); η_{nat} , standardized air Infiltration rate (h-1)

E-mail address: giovanni.litti@uantwerpen.be (G. Litti).

T

© 2018. This manuscript version is made available under the CC-BY-NC-ND 4.0 license

<http://creativecommons.org/licenses/by-nc-nd/4.0/>

Published Journal Article available at: <https://doi.org/10.1016/j.job.2018.02.006>

Received 27 December 2017; Received in revised form 9 February 2018; Accepted 10 February 2018

Available online 17 February 2018 the decision making to the apparently most performing option.

Code: • Circular strategy > Maintenance Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-10, p. 1 - 2

Created: 24.6.2026 r. 12:45 by user, Modified: 24.6.2026 r. 12:45 by user

Other codes assigned to segment:

Circular strategy > Upgrade (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)

Obsolescence > Functional (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

However, it should be noted that windows replacement is not the sole fenestration retrofitting measure allowing remarkable energy saving. Furthermore, the non-energy related drawbacks of windows replacement in historic buildings cannot be neglected. Next to the loss of building architectonic value, widely described in [3–5], a reduction of the property market value might occur [7]. Moreover, the environmental benefit of performing anyhow fenestration replacement in historic buildings is not always obvious from a life cycle environmental perspective because of the large initial embodied burdens [8].

For these reasons and for pursuing a more prudential approach when agreeing upon fenestration retrofitting measures in historic or heritage buildings, it is meaningful to explore the energy benefits obtainable by a wide set of retrofitting alternatives rather than confining

✉ Corresponding author.

Abbreviations: ERI, energy retrofitting interventions; EERI, energy environmental retrofitting interventions; EPBD, energy performance building directives; SL, service life; EOL, end of life; EOSL, end of service life; RSL, reference service life; BDT, blower door test; ACH, air change per hour; IRT, infrared thermography; X1, X2, physical properties with regard to building assemblies or components initial (1) and deteriorated (2) state of conservation; Nu, Nusselt number (-); λ , thermal conductivity (W/m K); R, thermal resistance (m² K/W); U, thermal transmittance (W/m² K); η_{50} , air infiltration rate at 50 Pa (h-1); η_{nat} , standardized air Infiltration rate (h-1)

E-mail address: giovanni.litti@uantwerpen.be (G. Litti).

T

© 2018. This manuscript version is made available under the CC-BY-NC-ND 4.0 license

<http://creativecommons.org/licenses/by-nc-nd/4.0/>

Published Journal Article available at: <https://doi.org/10.1016/j.job.2018.02.006>

Received 27 December 2017; Received in revised form 9 February 2018; Accepted 10 February 2018

Available online 17 February 2018 the decision making to the apparently most performing option.

Code: • Negative mechanism > Premature replacement Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-10, p. 1 - 2

Created: 24.6.2026 r. 12:46 by user, Modified: 24.6.2026 r. 12:46 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

However, it should be noted that windows replacement is not the sole fenestration retrofitting measure allowing remarkable energy saving. Furthermore, the non-energy related drawbacks of windows replacement in historic buildings cannot be neglected. Next to the loss of building architectonic value, widely described in [3–5], a reduction of the property market value might occur [7]. Moreover, the environmental benefit of performing anyhow fenestration replacement in historic buildings is not always obvious from a life cycle environmental perspective because of the large initial embodied burdens [8].

For these reasons and for pursuing a more prudential approach when agreeing upon fenestration retrofitting measures in historic or heritage buildings, it is meaningful to explore the energy benefits obtainable by a wide set of retrofitting alternatives rather than confining

✉ Corresponding author.

Abbreviations: ERI, energy retrofitting interventions; EERI, energy environmental retrofitting interventions; EPBD, energy performance building directives; SL, service life; EOL, end of life; EOSL, end of service life; RSL, reference service life; BDT, blower door test; ACH, air change per hour; IRT, infrared thermography; X1, X2, physical properties with regard to building assemblies or components initial (1) and deteriorated (2) state of conservation; Nu, Nusselt number (-); λ , thermal conductivity (W/m K); R, thermal resistance (m² K/W); U, thermal transmittance (W/m² K); η_{50} , air infiltration rate at 50 Pa (h⁻¹); η_{nat} , standardized air Infiltration rate (h⁻¹)

E-mail address: giovanni.litti@uantwerpen.be (G. Litti).

T

© 2018. This manuscript version is made available under the CC-BY-NC-ND 4.0 license

<http://creativecommons.org/licenses/by-nc-nd/4.0/>

Published Journal Article available at: <https://doi.org/10.1016/j.job.2018.02.006>

Received 27 December 2017; Received in revised form 9 February 2018; Accepted 10 February 2018

Available online 17 February 2018 the decision making to the apparently most performing option.

Code: • Circular strategy > Refurbishment Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-10, p. 1 - 2

Created: 24.6.2026 r. 12:45 by user, Modified: 24.6.2026 r. 12:45 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Negative mechanism > Higher initial material use (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

However, it should be noted that windows replacement is not the sole fenestration retrofitting measure allowing remarkable energy saving. Furthermore, the non-energy related drawbacks of windows replacement in historic buildings cannot be neglected. Next to the loss of building architectonic value, widely described in [3–5], a reduction of the property market value might occur [7]. Moreover, the environmental benefit of performing anyhow fenestration replacement in historic buildings is not always obvious from a life cycle environmental perspective because of the large initial embodied burdens [8].

For these reasons and for pursuing a more prudential approach when agreeing upon fenestration retrofitting measures in historic or heritage buildings, it is meaningful to explore the energy benefits obtainable by a wide set of retrofitting alternatives rather than confining

✉ Corresponding author.

Abbreviations: ERI, energy retrofitting interventions; EERI, energy environmental retrofitting

interventions; EPBD, energy performance building directives; SL, service life; EOL, end of life; EOSL, end of service life; RSL, reference service life; BDT, blower door test; ACH, air change per hour; IRT, infrared thermography; X1, X2, physical properties with regard to building assemblies or components initial (1) and deteriorated (2) state of conservation; Nu, Nusselt number (-); λ , thermal conductivity (W/m K); R, thermal resistance (m² K/W); U, thermal transmittance (W/m² K); η_{50} , air infiltration rate at 50 Pa (h-1); η_{nat} , standardized air Infiltration rate (h-1)
E-mail address: giovanni.litti@uantwerpen.be (G. Litti).

T

© 2018. This manuscript version is made available under the CC-BY-NC-ND 4.0 license

<http://creativecommons.org/licenses/by-nc-nd/4.0/>

Published Journal Article available at: <https://doi.org/10.1016/j.job.2018.02.006>

Received 27 December 2017; Received in revised form 9 February 2018; Accepted 10 February 2018

Available online 17 February 2018 the decision making to the apparently most performing option.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-10, p. 1 - 2

Created: 24.6.2026 r. 12:46 by user, Modified: 24.6.2026 r. 12:46 by user

Other codes assigned to segment:

Circular strategy > Maintenance (Weight: 0)

Circular strategy > Upgrade (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)

Obsolescence > Functional (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Negative mechanism > Higher initial material use (Weight: 0)

However, it should be noted that windows replacement is not the sole fenestration retrofitting measure allowing remarkable energy saving. Furthermore, the non-energy related drawbacks of windows replacement in historic buildings cannot be neglected. Next to the loss of building architectonic value, widely described in [3–5], a reduction of the property market value might occur [7]. Moreover, the environmental benefit of performing anyhow fenestration replacement in historic buildings is not always obvious from a life cycle environmental perspective because of the large initial embodied burdens [8].

For these reasons and for pursuing a more prudential approach when agreeing upon fenestration retrofitting measures in historic or heritage buildings, it is meaningful to explore the energy benefits obtainable by a wide set of retrofitting alternatives rather than confining

☐ Corresponding author.

Abbreviations: ERI, energy retrofitting interventions; EERI, energy environmental retrofitting interventions; EPBD, energy performance building directives; SL, service life; EOL, end of life; EOSL, end of service life; RSL, reference service life; BDT, blower door test; ACH, air change per hour; IRT, infrared thermography; X1, X2, physical properties with regard to building assemblies or components initial (1) and deteriorated (2) state of conservation; Nu, Nusselt number (-); λ , thermal conductivity (W/m K); R, thermal resistance (m² K/W); U, thermal transmittance (W/m² K); η_{50} , air infiltration rate at 50 Pa (h-1); η_{nat} , standardized air Infiltration rate (h-1)
E-mail address: giovanni.litti@uantwerpen.be (G. Litti).

T

© 2018. This manuscript version is made available under the CC-BY-NC-ND 4.0 license

<http://creativecommons.org/licenses/by-nc-nd/4.0/>

Published Journal Article available at: <https://doi.org/10.1016/j.job.2018.02.006>

Received 27 December 2017; Received in revised form 9 February 2018; Accepted 10 February 2018

Available online 17 February 2018 the decision making to the apparently most performing option.

Code: • Negative mechanism > Higher initial material use Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-10, p. 1 - 2

Created: 24.6.2026 r. 12:46 by user, Modified: 24.6.2026 r. 12:46 by user

Other codes assigned to segment:

- Circular strategy > Maintenance (Weight: 0)
- Circular strategy > Upgrade (Weight: 0)
- Circular strategy > Refurbishment (Weight: 0)
- Obsolescence > Functional (Weight: 0)
- Negative mechanism > Premature replacement (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

Because materials and components within a building assembly deteriorate, any retrofitting intervention undergoes technical decay over time. We believe that this aspect should be taken into account within the Energy and Environmental Retrofitting Interventions (EERI) modelling and decision-making as it may strongly affect the results and even invalidate the taken decisions if based on the assumption that EERI keep constant technical performance until the assemblies End Of Service Life (EOSL).

Code: • Negative mechanism > Lock-in Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-10, p. 2

Created: 24.6.2026 r. 12:48 by user, Modified: 24.6.2026 r. 12:48 by user

Other codes assigned to segment:

- Obsolescence > Physical (Weight: 0)
- Obsolescence > Technical (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

Because materials and components within a building assembly deteriorate, any retrofitting intervention undergoes technical decay over time. We believe that this aspect should be taken into account within the Energy and Environmental Retrofitting Interventions (EERI) modelling and decision-making as it may strongly affect the results and even invalidate the taken decisions if based on the assumption that EERI keep constant technical performance until the assemblies End Of Service Life (EOSL).

Code: • Negative mechanism > Burden shifting Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-10, p. 2

Created: 24.6.2026 r. 12:48 by user, Modified: 24.6.2026 r. 12:48 by user

Other codes assigned to segment:

- Obsolescence > Physical (Weight: 0)
- Obsolescence > Technical (Weight: 0)
- Negative mechanism > Lock-in (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

Because materials and components within a building assembly deteriorate, any retrofitting intervention undergoes technical decay over time. We believe that this aspect should be taken into account within the Energy and Environmental Retrofitting Interventions (EERI) modelling and decision-making as it may strongly affect the results and even invalidate the taken decisions if based on the assumption that EERI keep constant technical performance until the assemblies End Of Service Life (EOSL).

Code: • Obsolescence > Technical Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-10, p. 2

Created: 24.6.2026 r. 12:47 by user, Modified: 24.6.2026 r. 12:47 by user

Other codes assigned to segment:

- Obsolescence > Physical (Weight: 0)
- Negative mechanism > Burden shifting (Weight: 0)
- Negative mechanism > Lock-in (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

Because materials and components within a building assembly deteriorate, any retrofitting intervention undergoes technical decay over time. We believe that this aspect should be taken into account within the Energy and Environmental Retrofitting Interventions (EERI) modelling and decision-making as it may strongly affect the results and even invalidate the taken decisions if based on the assumption that EERI keep constant technical performance until the assemblies End Of Service Life (EOSL).

Code: • [Obsolescence > Physical](#) Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-10, p. 2
 Created: 24.6.2026 r. 12:48 by user, Modified: 24.6.2026 r. 12:48 by user
 Other codes assigned to segment:
 [Obsolescence > Technical](#) (Weight: 0)
 [Negative mechanism > Burden shifting](#) (Weight: 0)
 [Negative mechanism > Lock-in](#) (Weight: 0)
 [Mitigation > Lifecycle verification](#) (Weight: 0)

Because materials and components within a building assembly deteriorate, any retrofitting intervention undergoes technical decay over time. We believe that this aspect should be taken into account within the Energy and Environmental Retrofitting Interventions (EERI) modelling and decision-making as it may strongly affect the results and even invalidate the taken decisions if based on the assumption that EERI keep constant technical performance until the assemblies End Of Service Life (EOSL).

Code: • [Mitigation > Lifecycle verification](#) Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-10, p. 2
 Created: 24.6.2026 r. 12:48 by user, Modified: 24.6.2026 r. 12:48 by user
 Other codes assigned to segment:
 [Obsolescence > Physical](#) (Weight: 0)
 [Obsolescence > Technical](#) (Weight: 0)
 [Negative mechanism > Burden shifting](#) (Weight: 0)
 [Negative mechanism > Lock-in](#) (Weight: 0)

Note that this study does not focus on the whole Life Cycle Assessment of the presented EERI, hence, initial and recurrent embodied energy is not accounted; only building operating energy for space heating is considered

Code: • [Mitigation > Lifecycle verification](#) Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-10, p. 2
 Created: 24.6.2026 r. 12:54 by user, Modified: 24.6.2026 r. 12:54 by user
 Other codes assigned to segment:
 [Negative mechanism > Burden shifting](#) (Weight: 0)

Note that this study does not focus on the whole Life Cycle Assessment of the presented EERI, hence, initial and recurrent embodied energy is not accounted; only building operating energy for space heating is considered

Code: • [Negative mechanism > Burden shifting](#) Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-10, p. 2
 Created: 24.6.2026 r. 12:54 by user, Modified: 24.6.2026 r. 12:54 by user
 Other codes assigned to segment:
 [Mitigation > Lifecycle verification](#) (Weight: 0)

By looking at Fig. 3.2.1, it can be observed that windows maintenance, namely the re-putty of glass bars, allowed to decrease infiltration rate by up to 21% compared with non-maintained windows (comparison space 2 VS space 1). The addition of drought-proofing reduced infiltration rate of extra 16% (comparison mean drought proofed VS space 2). In other words, the sealing of the window

sashes and edges allowed to reduce infiltration rates by up to 37% compared to non-maintained windows.

Code: • [Obsolescence > Physical](#) Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-10, p. 5
 Created: 24.6.2026 r. 12:49 by user, Modified: 24.6.2026 r. 12:49 by user
 Other codes assigned to segment:
 [Circular strategy > Maintenance](#) (Weight: 0)
 [Circular strategy > Repair](#) (Weight: 0)
 [Circular strategy > Upgrade](#) (Weight: 0)
 [Obsolescence > Functional](#) (Weight: 0)
 [Mitigation > Maintenance planning](#) (Weight: 0)
 [Mitigation > Lifecycle verification](#) (Weight: 0)

By looking at Fig. 3.2.1, it can be observed that windows maintenance, namely the re-putty of glass bars, allowed to decrease infiltration rate by up to 21% compared with non-maintained windows (comparison space 2 VS space 1). The addition of drought-proofing reduced infiltration rate of extra 16% (comparison mean drought proofed VS space 2). In other words, the sealing of the window sashes and edges allowed to reduce infiltration rates by up to 37% compared to non-maintained windows.

Code: • [Mitigation > Maintenance planning](#) Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-10, p. 5
 Created: 24.6.2026 r. 12:49 by user, Modified: 24.6.2026 r. 12:49 by user
 Other codes assigned to segment:
 [Circular strategy > Maintenance](#) (Weight: 0)
 [Circular strategy > Repair](#) (Weight: 0)
 [Circular strategy > Upgrade](#) (Weight: 0)
 [Obsolescence > Physical](#) (Weight: 0)
 [Obsolescence > Functional](#) (Weight: 0)
 [Mitigation > Lifecycle verification](#) (Weight: 0)

By looking at Fig. 3.2.1, it can be observed that windows maintenance, namely the re-putty of glass bars, allowed to decrease infiltration rate by up to 21% compared with non-maintained windows (comparison space 2 VS space 1). The addition of drought-proofing reduced infiltration rate of extra 16% (comparison mean drought proofed VS space 2). In other words, the sealing of the window sashes and edges allowed to reduce infiltration rates by up to 37% compared to non-maintained windows.

Code: • [Obsolescence > Functional](#) Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-10, p. 5
 Created: 24.6.2026 r. 12:49 by user, Modified: 24.6.2026 r. 12:49 by user
 Other codes assigned to segment:
 [Circular strategy > Maintenance](#) (Weight: 0)
 [Circular strategy > Repair](#) (Weight: 0)
 [Circular strategy > Upgrade](#) (Weight: 0)
 [Obsolescence > Physical](#) (Weight: 0)
 [Mitigation > Maintenance planning](#) (Weight: 0)
 [Mitigation > Lifecycle verification](#) (Weight: 0)

By looking at Fig. 3.2.1, it can be observed that windows maintenance, namely the re-putty of glass bars, allowed to decrease infiltration rate by up to 21% compared with non-maintained windows (comparison space 2 VS space 1). The addition of drought-proofing reduced infiltration rate of extra 16% (comparison mean drought proofed VS space 2). In other words, the sealing of the window sashes and edges allowed to reduce infiltration rates by up to 37% compared to non-maintained windows.

Code: • [Circular strategy > Upgrade](#) Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-10, p. 5

Created: 24.6.2026 r. 12:49 by user, Modified: 24.6.2026 r. 12:49 by user

Other codes assigned to segment:

- Circular strategy > Maintenance (Weight: 0)
- Circular strategy > Repair (Weight: 0)
- Obsolescence > Physical (Weight: 0)
- Obsolescence > Functional (Weight: 0)
- Mitigation > Maintenance planning (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

By looking at Fig. 3.2.1, it can be observed that windows maintenance, namely the re-putty of glass bars, allowed to decrease infiltration rate by up to 21% compared with non-maintained windows (comparison space 2 VS space 1). The addition of drought-proofing reduced infiltration rate of extra 16% (comparison mean drought proofed VS space 2). In other words, the sealing of the window sashes and edges allowed to reduce infiltration rates by up to 37% compared to non-maintained windows.

Code: • Circular strategy > Maintenance Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-10, p. 5

Created: 24.6.2026 r. 12:49 by user, Modified: 24.6.2026 r. 12:49 by user

Other codes assigned to segment:

- Circular strategy > Repair (Weight: 0)
- Circular strategy > Upgrade (Weight: 0)
- Obsolescence > Physical (Weight: 0)
- Obsolescence > Functional (Weight: 0)
- Mitigation > Maintenance planning (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

By looking at Fig. 3.2.1, it can be observed that windows maintenance, namely the re-putty of glass bars, allowed to decrease infiltration rate by up to 21% compared with non-maintained windows (comparison space 2 VS space 1). The addition of drought-proofing reduced infiltration rate of extra 16% (comparison mean drought proofed VS space 2). In other words, the sealing of the window sashes and edges allowed to reduce infiltration rates by up to 37% compared to non-maintained windows.

Code: • Circular strategy > Repair Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-10, p. 5

Created: 24.6.2026 r. 12:49 by user, Modified: 24.6.2026 r. 12:49 by user

Other codes assigned to segment:

- Circular strategy > Maintenance (Weight: 0)
- Circular strategy > Upgrade (Weight: 0)
- Obsolescence > Physical (Weight: 0)
- Obsolescence > Functional (Weight: 0)
- Mitigation > Maintenance planning (Weight: 0)
- Mitigation > Lifecycle verification (Weight: 0)

By looking at Fig. 3.2.1, it can be observed that windows maintenance, namely the re-putty of glass bars, allowed to decrease infiltration rate by up to 21% compared with non-maintained windows (comparison space 2 VS space 1). The addition of drought-proofing reduced infiltration rate of extra 16% (comparison mean drought proofed VS space 2). In other words, the sealing of the window sashes and edges allowed to reduce infiltration rates by up to 37% compared to non-maintained windows.

Code: • Mitigation > Lifecycle verification Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-10, p. 5

Created: 24.6.2026 r. 12:49 by user, Modified: 24.6.2026 r. 12:49 by user

Other codes assigned to segment:

- Circular strategy > Maintenance (Weight: 0)

Circular strategy > Repair (Weight: 0)
 Circular strategy > Upgrade (Weight: 0)
 Obsolescence > Physical (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Mitigation > Maintenance planning (Weight: 0)

The literature, often reports the usefulness of adding a secondary glazing (also termed storm glazing) on the inner or outer side of an existing window as alternative option to glass or window replacement

[63–65]. This alternative is especially valid in case of historic buildings as their operating energy can be reduced without threatening the existing fenestration

Code: • Circular strategy > Refurbishment Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-10, p. 11
 Created: 24.6.2026 r. 12:50 by user, Modified: 24.6.2026 r. 12:50 by user
 Other codes assigned to segment:
 Circular strategy > Upgrade (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Enabling condition > Component accessibility (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The literature, often reports the usefulness of adding a secondary glazing (also termed storm glazing) on the inner or outer side of an existing window as alternative option to glass or window replacement

[63–65]. This alternative is especially valid in case of historic buildings as their operating energy can be reduced without threatening the existing fenestration

Code: • Negative mechanism > Premature replacement Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-10, p. 11
 Created: 24.6.2026 r. 12:51 by user, Modified: 24.6.2026 r. 12:51 by user
 Other codes assigned to segment:
 Circular strategy > Upgrade (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Enabling condition > Component accessibility (Weight: 0)
 Mitigation > Design intervention (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The literature, often reports the usefulness of adding a secondary glazing (also termed storm glazing) on the inner or outer side of an existing window as alternative option to glass or window replacement

[63–65]. This alternative is especially valid in case of historic buildings as their operating energy can be reduced without threatening the existing fenestration

Code: • Mitigation > Design intervention Weight score: 0
 Search D - Window LCA and replacement decisions > D-GS-10, p. 11
 Created: 24.6.2026 r. 12:51 by user, Modified: 24.6.2026 r. 12:51 by user
 Other codes assigned to segment:
 Circular strategy > Upgrade (Weight: 0)
 Circular strategy > Refurbishment (Weight: 0)
 Obsolescence > Functional (Weight: 0)
 Enabling condition > Component accessibility (Weight: 0)
 Negative mechanism > Premature replacement (Weight: 0)
 Mitigation > Lifecycle verification (Weight: 0)

The literature, often reports the usefulness of adding a secondary glazing (also termed storm glazing) on the inner or outer side of an existing window as alternative option to glass or window replacement

[63–65]. This alternative is especially valid in case of historic buildings as their operating energy can be reduced without threatening the existing fenestration

Code: • Circular strategy > Upgrade Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-10, p. 11

Created: 24.6.2026 r. 12:50 by user, Modified: 24.6.2026 r. 12:50 by user

Other codes assigned to segment:

Circular strategy > Refurbishment (Weight: 0)

Obsolescence > Functional (Weight: 0)

Enabling condition > Component accessibility (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Mitigation > Design intervention (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The literature, often reports the usefulness of adding a secondary glazing (also termed storm glazing) on the inner or outer side of an existing window as alternative option to glass or window replacement

[63–65]. This alternative is especially valid in case of historic buildings as their operating energy can be reduced without threatening the existing fenestration

Code: • Mitigation > Lifecycle verification Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-10, p. 11

Created: 24.6.2026 r. 12:51 by user, Modified: 24.6.2026 r. 12:51 by user

Other codes assigned to segment:

Circular strategy > Upgrade (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)

Obsolescence > Functional (Weight: 0)

Enabling condition > Component accessibility (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Mitigation > Design intervention (Weight: 0)

The literature, often reports the usefulness of adding a secondary glazing (also termed storm glazing) on the inner or outer side of an existing window as alternative option to glass or window replacement

[63–65]. This alternative is especially valid in case of historic buildings as their operating energy can be reduced without threatening the existing fenestration

Code: • Obsolescence > Functional Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-10, p. 11

Created: 24.6.2026 r. 12:51 by user, Modified: 24.6.2026 r. 12:51 by user

Other codes assigned to segment:

Circular strategy > Upgrade (Weight: 0)

Circular strategy > Refurbishment (Weight: 0)

Enabling condition > Component accessibility (Weight: 0)

Negative mechanism > Premature replacement (Weight: 0)

Mitigation > Design intervention (Weight: 0)

Mitigation > Lifecycle verification (Weight: 0)

The literature, often reports the usefulness of adding a secondary glazing (also termed storm glazing) on the inner or outer side of an existing window as alternative option to glass or window replacement

[63–65]. This alternative is especially valid in case of historic buildings as their operating energy can be reduced without threatening the existing fenestration

Code: • Enabling condition > Component accessibility Weight score: 0

Search D - Window LCA and replacement decisions > D-GS-10, p. 11

Created: 24.6.2026 г. 12:51 by user, Modified: 24.6.2026 г. 12:51 by user

Other codes assigned to segment:

- Circular strategy > Upgrade (Weight: 0)

- Circular strategy > Refurbishment (Weight: 0)

- Obsolescence > Functional (Weight: 0)

- Negative mechanism > Premature replacement (Weight: 0)

- Mitigation > Design intervention (Weight: 0)

- Mitigation > Lifecycle verification (Weight: 0)